# The Architecture of Money

Research Dossier: Phase 1 Architecture and Research Streams R1-R7 (working draft, 7 November 2026)

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# The Architecture of Money: Phase 1 Research Architecture

*Monetary Power, Credit Creation, Fiat Currency, Financial Inequality, Social Trust, and the Future of Digital Money*

Authors: M. Bagus Prabowo and Hark (co-authored: M. Bagus Prabowo conceived and directs the project; Hark's agents research and draft). Document: Phase 1 deliverable (BRIEF Part XV, Phase 1, plus Part II). Research cut-off: November 2026. Drafted 7 November 2026. Companion file: `phase1/verification_log.md` (every [V] item below, with the URL opened and what it says).

Evidence tags (per FRAMEWORK.md): **[V]** = page opened in this phase and the claim is on it; **[E]** = established literature or search excerpt, not opened in this phase. Claim types are labelled where it matters: (fact), (empirical finding), (interpretation), (normative), (counterfactual).

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## Key findings and key decisions

1. **The book will be organised around one analytical distinction: the *issuer* of a monetary liability, the *legal and institutional promise* behind it, and the *social willingness* to accept it.** Every chapter asks the same three questions of each money-like instrument (who owes it, what backs the promise to convert it at par, why people take it). This lets the book compare a Mesopotamian silver debt, a goldsmith note, a bank deposit, an e-CNY wallet balance and a stablecoin without treating them as interchangeable. (interpretation)

2. **"Who creates money?" has a layered, verified answer that the book will defend with balance sheets rather than slogans.** Commercial banks create most broad money when they lend or buy assets (McLeay, Radia and Thomas 2014, Bank of England Quarterly Bulletin 2014 Q1 [V]); central banks create reserves and currency; governments create the demand for and legal status of the unit; and all three are constrained by profitability, capital and liquidity regulation, borrower demand, and the policy interest rate. Banks are not unlimited money creators. (fact + interpretation)

3. **Measurement is not neutral, and the book will not compare "M2" across countries without a definitions table.** The US M2 (Federal Reserve H.6) includes retail money market funds and since 2020 counts savings deposits in M1 [V]; the euro area's broad aggregate is M3, which adds repos, money market fund shares and short bank debt securities [V]; Indonesia's M2 (*uang beredar luas*) includes foreign-currency deposits within quasi money and short securities issued by the monetary system, including Bank Indonesia's own SRBI [V]. The United States has had zero reserve requirements since 26 March 2020 [V]; the euro area keeps a minimum reserve (average requirement EUR 174.7 billion in the July-September 2026 maintenance period, remunerated at 0%) [V]; Indonesia's rupiah reserve requirement (GWM) for conventional banks is being cut from 9% to 7% between 1 November and 1 December 2026 [V]. "Fractional reserve banking" therefore means materially different things in each system. (fact)

4. **CBDC status is volatile enough that every CBDC claim in the book must carry a date and a stage label.** As of November 2026: only the Bahamas, Jamaica and Nigeria have launched retail CBDCs

(Riksbank Payments Report 2026 [V]); China reclassified e-CNY balances held in commercial-bank wallets as interest-bearing *bank deposit liabilities* covered by deposit insurance from 1 January 2026 [V], which complicates the textbook definition of a CBDC as a central bank liability; the euro area is in legislative preparation, with possible issuance in 2029 if EU law is adopted in 2026 [V]; the UK remains in a design phase with a decision promised by end-2026 [E, official parliamentary answer]; the United States has statutorily prohibited the Federal Reserve from issuing a CBDC until 31 December 2030 (Public Law 119-101, Sec. 1101, 11 July 2026) [V]; Indonesia's Digital Rupiah (Project Garuda) has completed only a wholesale proof of concept (report dated 13 December 2024) and has not piloted or issued [V]. (fact)

5. **The theory chapters will separate three kinds of disagreement:** (a) genuine empirical disagreements (for example, how much inflation follows from money growth at different horizons and inflation levels); (b) genuine normative disagreements (who should hold the power to create money); and (c) pseudo-disagreements where schools talk past each other because they use different definitions of "money", different time horizons, or different counterfactuals (for example, monetarists and post-Keynesians on whether money is exogenous, which largely dissolves once the policy instrument is specified). (interpretation)
6. **The hypothesis register contains 24 hypotheses (H1-H24)**, each paired with a rival and with stated disconfirming evidence. Several are deliberately uncomfortable for every school, including the authors' working priors.
7. **The table of contents has 39 chapters in 9 parts plus 8 appendices**, mapped one-to-one to BRIEF Parts I-XIV and to Phases 2-10. Indonesia is both a dedicated part (BRIEF XII) and a recurring comparator in every other part, because the central research question is easiest to answer honestly in a country where the authors can read the primary legislation in the original language.
8. **Biggest research risks:** ideological capture (especially by the most rhetorically confident schools: Austrian, MMT, and "independent central bank" orthodoxy), survivorship and selection bias in case studies, definitional breaks in money and price data, and CBDC status volatility. Each has a written mitigation in Section 7.

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# 1. The central question and the five fundamental problems, as researchable questions

## 1.1 The central question (BRIEF Part I)

> How did humanity move from commodity-based monetary systems to a predominantly credit-based, fiat, and increasingly digital monetary architecture, and what makes societies accept, trust, criticize, and continue participating in this system?

The question contains a historical clause (*how did we move*) and a sociological clause (*why do people accept, trust, criticise and keep participating*). It also contains a hidden premise that the book must test rather than assume: that the movement was *from* commodity money *to* credit money. A substantial body of scholarship (Mitchell-Innes 1913; Graeber 2011; Hudson and the Assyriologists; Desan 2014) argues that credit and accounting money were present from the beginning and that commodity coinage was itself a state-and-credit institution. The book therefore restates the central question as three researchable sub-questions:

- **CQ-1 (historical-institutional):** In each major monetary regime examined (Mesopotamian temple accounting; Greek and Roman coinage; medieval bills of exchange; early modern deposit and note-issuing banks; the classical gold standard; Bretton Woods; post-1971 fiat; digital payment systems), what was the unit of account, what was the dominant means of payment, who issued it, what convertibility promise (if any) stood behind it, and what enforced that promise?
- **CQ-2 (functional):** Which properties of money (unit of account, final settlement, store of value, legal tender) migrated from commodities to institutions, and when? Specifically: when did the *anchor* of value shift from metal content to institutional credibility, and was that shift abrupt (1914, 1933, 1971) or gradual?
- **CQ-3 (sociological):** What combination of voluntary acceptance, network lock-in, legal obligation (taxes, legal tender, wage contracts), and perceived legitimacy explains continued participation, and how does the

mix differ between stable systems (Switzerland, Singapore), crisis systems (Argentina, Lebanon, Zimbabwe), and Indonesia?

## 1.2 The five fundamental problems restated

For each problem, the table gives the operational question, the unit of analysis, the evidence that would answer it, and what would change the book's answer.

**Problem 1. Who creates modern money?**

- *Operational question:* For a given economy and year, what share of the change in broad money (M2 or M3 by the local definition) is accounted for by (a) bank lending to the private sector, (b) bank purchases of government securities, (c) central bank asset purchases that reach non-bank holders, (d) net foreign asset inflows, and (e) offsetting leakages (longer-term liabilities, government deposits)?
- *Method:* Counterpart analysis of the consolidated monetary-system balance sheet. The ECB publishes M3 counterparts explicitly (claims on private sector, claims on general government, longer-term financial liabilities, net external assets, remaining counterparts) [V]; Bank Indonesia publishes M2 "affecting factors" as net foreign assets and net domestic assets, including net claims on central government and claims on other sectors [V]; the US requires reconstruction from the H.8 and Z.1 releases.
- *Evidence that would change the answer:* If counterpart analysis showed that broad money growth is dominated by central bank operations rather than bank credit in normal times, the "banks create most money" claim would have to be qualified to "in normal times". (2020-2021 in the US is the obvious test case: pandemic QE combined with Treasury transfers created deposits without new bank loans.)

**Problem 2. Who benefits from the creation and allocation of money?**

- *Operational question:* Decompose the benefit into (a) seigniorage and its recipients (the state via central bank remittances; banks via the net interest margin on deposit liabilities), (b) first-access advantages (Cantillon channels: who receives new credit or asset purchases first, and on what terms), (c) relative-price and asset-price effects, and (d) access to credit itself (who can borrow, at what rate, against what collateral).
- *Method:* Seigniorage accounting for central banks; bank profitability attribution (the "deposit franchise" literature); distributional national accounts (US Distributional Financial Accounts; Indonesian SUSENAS and OJK credit-distribution data); event studies of QE announcements; credit-registry evidence on loan allocation by firm size and region.
- *What would change the answer:* Evidence that the deposit franchise yields banks no excess return after the costs of deposit insurance, regulation, capital and payment services would weaken the "private privilege" reading. Evidence that the asset-price gains of QE are offset by employment gains for lower-wealth households would weaken the strong Cantillon reading.

**Problem 3. Why does money retain purchasing power without direct commodity convertibility?**

- *Operational question:* Which mechanisms empirically sustain the value of fiat money: (a) tax-driven demand (chartalist), (b) scarcity maintained by credible policy (monetarist, New Keynesian), (c) fiscal backing (fiscal theory of the price level), (d) network convention and expected future acceptance (search-theoretic and sociological accounts), (e) legal tender and contract enforcement?
- *Method:* Comparative evidence from cases where one mechanism fails while others survive: high-tax states that nonetheless suffered hyperinflation; currencies that kept value after the state that issued them collapsed (the "Swiss dinar" in Iraq after 1990-91 is the classic case [E]); dollarised economies where legal tender status and tax demand are decoupled.
- *What would change the answer:* If tax-driven demand were sufficient, no state with a functioning tax system would see its currency abandoned. If convention alone were sufficient, legal and fiscal reforms would not be needed to end hyperinflations (Sargent 1982 argues they were [E]).

**Problem 4. Why do people accept institutions they sometimes perceive as economically unfair?**

- *Operational question:* Separate four reasons for continued use: (a) genuine legitimacy (people believe the system is broadly fair or at least acceptable), (b) instrumental acceptance (it works for them despite perceived unfairness), (c) lock-in (switching costs and network effects), (d) legal compulsion (taxes, wages, legal tender). Measure each.

- *Method:* Survey evidence on trust in central banks, banks and governments (ECB Consumer Expectations Survey; Edelman Trust Barometer with caveats; World Values Survey; Indonesian surveys of financial literacy and inclusion by OJK); revealed-preference evidence (currency substitution, dollarisation, crypto adoption, cash hoarding) conditioned on legal and technical cost of exit.
- *What would change the answer:* If exit rises sharply whenever exit becomes cheap (for example after stablecoins become easily available), "lock-in" explains more than "legitimacy". If exit stays low even when cheap, legitimacy or convenience explains more.

**Problem 5. Will CBDCs fundamentally change the relationship between citizens, banks, governments and monetary authorities?**

- *Operational question:* For each design parameter (retail versus wholesale; account versus token; direct versus intermediated; holding limits; remuneration; offline capability; privacy tier; programmability), what is the predicted change in (a) bank deposit funding, (b) the central bank's balance sheet and credit-allocation role, (c) the state's access to transaction data, (d) the role of private payment firms, (e) citizens' access to a risk-free digital asset?
- *Method:* Design-comparison matrix across the ten jurisdictions named in BRIEF Part X, with each jurisdiction tagged by stage (research / experimentation / pilot / legislative preparation / issuance); quantitative evidence from launched systems (adoption, deposit substitution); simulation evidence from central bank papers (deposit outflow scenarios under holding limits).
- *What would change the answer:* The early evidence from launched retail CBDCs is low uptake (all three launched systems remain marginal relative to cash and bank money [E, multiple secondary trackers]); and China's 2026 move turned retail e-CNY balances into bank deposits [V]. If those patterns hold, the honest conclusion may be "evolution in payment infrastructure, not structural change", and the book must be willing to reach it.

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# 2. What is money? Sixteen concepts (BRIEF Part II)

## 2.1 Three axes for every definition

The literature mixes three different classifications that must be kept apart:

1. **Function** (what money does): medium of exchange, unit of account, store of value, means of payment (final settlement of debt).
2. **Form and backing** (what the claim is): commodity, representative (claim on a commodity), fiat (inconvertible), credit (an IOU).
3. **Issuer and legal status** (who owes it and under what law): central bank, commercial bank, non-bank private issuer, the state; legal tender or not.

The BIS "money flower" taxonomy (Bech and Garratt 2017, BIS Quarterly Review [E]) adds two further axes useful for the digital chapters: *universal versus restricted access*, and *account-based versus token-based*. The book adopts these five axes and will present each concept below by its position on them.

## 2.2 Definitions

**(1) Medium of exchange.** Anything generally accepted in exchange for goods and services, which removes the "double coincidence of wants" problem of barter (Jevons 1875; Menger 1892 [E]). The function is behavioural: an object becomes a medium of exchange because others accept it. Search-theoretic models (Kiyotaki and Wright 1989 [E]) show that a medium of exchange can arise as a self-confirming equilibrium even when the object has no intrinsic use. *Contested point:* the anthropological record does not support the textbook story that barter economies preceded money (Caroline Humphrey 1985; Graeber 2011 [E]); the book will treat the "barter origin" as a theoretical device, not a historical fact. *Measurement:* there is no aggregate of "media of exchange"; narrow money (M1) is the closest proxy, and its definition varies (see table 2.3).

**(2) Unit of account.** The abstract measure in which prices, debts and accounts are expressed. Historically the unit of account and the circulating coin often diverged: medieval European "money of account" (pounds, shillings

and pence as units) persisted while the coins in circulation varied in content and origin (Einaudi 1936; Bloch [E]). Keynes (*A Treatise on Money*, 1930, ch. 1 [E]) treats the money of account as the "primary concept", which places him close to the state and credit theories. *Why it matters for the book:* control of the unit of account is a different power from control of the means of payment. Dollarisation shows that the unit of account can move abroad while domestic law still designates the national currency.

**(3) Store of value.** An asset that transfers purchasing power through time. Money is a weak store of value under inflation but has the advantage of nominal certainty and liquidity. Many non-money assets (land, equities, gold) are better long-run stores of value; money's distinctive feature is that it is the *most liquid* store of value, which is why the demand for money rises in panics (Keynes's liquidity preference). *Measurement:* "real money balances" (M divided by a price index) and velocity; both are sensitive to the deflator chosen.

**(4) Means of payment.** The instrument that *discharges* a debt with finality. The distinction from medium of exchange matters in modern systems: a card swipe is an exchange instruction, but final payment occurs only when banks settle with each other in central bank money. Goodhart (1989) and the payments literature [E] use "means of payment" for final settlement assets. *Book's usage:* final settlement asset for a given tier of the payment hierarchy. For households, bank deposits are final; for banks, central bank reserves are final; for central banks in the international system, historically gold and now largely reserve-currency assets.

**(5) Legal tender.** A legal status: an instrument that a creditor cannot refuse in discharge of a money debt (scope varies by jurisdiction). United States: 31 U.S.C. 5103 makes United States coins and currency, including Federal Reserve notes, "legal tender for all debts, public charges, taxes, and dues" [V]. Euro area: euro banknotes are legal tender under the EU treaties (Article 128 TFEU [E]). Indonesia: the Currency Law (Law No. 7 of 2011 on Currency) requires use of the rupiah for transactions in Indonesian territory with specified exceptions [E, to verify text in Phase 12]; Bank Indonesia's statistical metadata defines currency as "notes and coins issued by Bank Indonesia as a legal tender" [V]. *Key distinction:* legal tender status attaches in most systems to *cash* only; bank deposits are not legal tender even though they are the dominant means of payment. This gap is central to the CBDC debate (a CBDC would be the first legal-tender digital money accessible to the public in most jurisdictions).

**(6) Commodity money.** Money whose value is (at least partly) the value of the material: metal coins at full weight, cowries, salt, cigarettes in prisoner-of-war camps (Radford 1945 [E]). *Contested point:* even "full-bodied" coins usually circulated at a premium over metal value because of minting, legal status and convenience (seigniorage and brassage); the gap between face and metal value is precisely what debasement exploited. The book will measure that gap where data exist (for example English and Roman debasements [E]).

**(7) Representative money.** A claim on a commodity held in reserve: goldsmith receipts, gold certificates, banknotes convertible into specie under the gold standard. Representative money is a *credit* instrument with a commodity redemption promise; its value depends on the credibility of the promise, not only the commodity. Partial reserve backing made such systems vulnerable to runs, which is why lender-of-last-resort doctrine (Bagehot 1873 [E]) arose inside commodity standards, not after them.

**(8) Fiat money.** Money that is not convertible into any commodity by the issuer's legal promise, and whose value rests on institutional arrangements: legal tender status, tax acceptance, central bank policy, and expectations. Fiat is not the same as "unbacked": a central bank's notes are liabilities matched on its balance sheet by assets (government bonds, foreign reserves, loans to banks). What is missing is a *convertibility right* for the holder. The global system has been fiat since the suspension of dollar-gold convertibility in August 1971 (fact [E]).

**(9) Bank deposits.** A liability of a deposit-taking institution to the depositor, repayable on demand (sight deposits) or at term. Deposits are created (a) when customers deposit cash or receive transfers, and (b) when banks lend or buy assets and credit the borrower's or seller's account (McLeay, Radia and Thomas 2014 [V]). Deposits are *private money*: the holder bears the credit risk of the bank, mitigated by deposit insurance (Indonesia: LPS; US: FDIC; China's scheme, which now covers e-CNY wallet balances at banks [V]).

**(10) Credit money.** In the broad (credit-theory) sense, all money is credit: a transferable claim of the holder on the issuer (Mitchell-Innes 1913, "What is Money?", *Banking Law Journal* [E]). In the narrower banking sense, credit money is money created as the counterpart of a loan. The book will use "credit money" in the narrow sense when discussing creation mechanics and flag explicitly when it uses the Mitchell-Innes sense.

**(11) Central bank money.** Liabilities of the central bank usable as money: banknotes held by the public, and

reserve balances held by banks (and in some systems by other eligible institutions) at the central bank. It is the system's settlement asset. US example (Federal Reserve H.6, September 2026, seasonally adjusted): currency in circulation USD 2,483.0 billion and reserve balances USD 2,966.8 billion, for a monetary base of USD 5,449.8 billion [V].

**(12) Commercial bank money.** Liabilities of commercial banks usable as money, chiefly demand and savings deposits. It is convertible at par into central bank money on demand (the "singleness" property the BIS identifies as the foundation of trust in the monetary system [V]). In the US, deposits make up the overwhelming majority of M2: in September 2026, currency was USD 2,397.2 billion of seasonally adjusted M2 of USD 23,413.9 billion, roughly 10% [V].

**(13) Base money (monetary base, M0, high-powered money, *uang primer*).** Currency plus bank reserves at the central bank. Definitions differ on whether vault cash, required versus excess reserves, and central bank liabilities to non-banks are included. The Fed's "total reserves" definition changed when vault cash ceased to count toward reserve requirements in April 2020 [V]; Bank Indonesia publishes "M0 adjusted", which nets out the effect of reserve-requirement incentives (*pengendalian moneter adjusted*) [V, M2 report]. *Interpretive warning:* the "money multiplier" between base and broad money is an accounting ratio, not a behavioural constant; its collapse after 2008 in the US (base money multiplied while broad money grew modestly) is the core empirical evidence against the textbook model (see H3).

**(14) Broad money.** A wider aggregate including less liquid deposits and money-like instruments: US M2, euro area M3, Indonesian M2, UK M4ex. Details in table 2.3.

**(15) Private money.** Any monetary liability issued by a private entity: bank deposits, historical private banknotes (US free banking era 1837-1863; Scottish free banking to 1845 [E]), e-money issued by non-banks, stablecoins, and cryptoassets that are not anyone's liability (Bitcoin is a bearer asset with no issuer, so it fits "private money" only loosely). The United States created a federal licensing regime for "payment stablecoins" through the GENIUS Act (P.L. 119-27, signed 18 July 2025 [E, CRS summary]), which defines them as digital assets redeemable at a fixed amount and prohibits issuers from paying interest [E].

**(16) Sovereign money.** Two meanings. (a) Money issued by the state or its central bank (as opposed to private bank money). (b) A reform proposal (Vollgeld; the Swiss "Sovereign Money" initiative rejected by referendum on 10 June 2018 [E]) under which only the central bank could create transaction money and banks would intermediate existing money only. The book uses meaning (a) by default and treats (b) as a reform option in Part IX (Scenario E).

## 2.3 How measurement differs across jurisdictions

| Item | United States (Fed H.6) | Euro area (ECB) | Indonesia (Bank Indonesia) | United Kingdom (BoE) |
|---|---|---|---|---|
| Narrow money | **M1** = currency outside Treasury, Fed and bank vaults + demand deposits + "other liquid deposits" incl. savings deposits and money market deposit accounts [V] | **M1** = currency in circulation + overnight deposits [V] | **M1** = currency held by domestic private sector + rupiah demand deposits incl. bank-issued e-money + rupiah savings withdrawable any time ("M1+") [V] | Notes and coin; M1 not a headline aggregate [E] |
| Intermediate | **M2** = M1 + small time deposits (< USD 100,000) + retail money market funds, less IRA/Keogh balances [V] | **M2** = M1 + deposits with agreed maturity up to 2 years + deposits redeemable at notice up to 3 months [V] | n.a. | n.a. |

| Item | United States (Fed H.6) | Euro area (ECB) | Indonesia (Bank Indonesia) | United Kingdom (BoE) |
|---|---|---|---|---|
| Broad (headline) | M2 (M3 discontinued in 2006 [E]) | **M3** = M2 + repurchase agreements + money market fund shares + MFI debt securities up to 2 years [V] | **M2** (*uang beredar luas*) = M1 + quasi money (time and other savings deposits in rupiah *and foreign currency*, plus FX demand deposits) + securities other than shares issued by the monetary system held by the domestic private sector with remaining maturity up to 1 year (incl. SRBI, SVBI, CDs) [V] | **M4ex** = M4 excluding intermediate other financial corporations [E] |
| Money-holding sector | Non-bank public excl. US government and foreign official institutions (for demand deposits) [V] | Euro area non-MFI residents excl. central government [V] | Domestic private sector, excl. central government and non-residents [V] | UK private sector excl. intermediate OFCs [E] |
| Foreign-currency deposits | Excluded from M2 [E] | Included (all currencies held by residents) [E] | Included in quasi money [V] | Sterling only in M4ex headline [E] |
| Latest level | M2 USD 23,413.9 bn (Sep 2026, SA) [V] | to pull from ECB Data Portal (Phase 3) | M2 Rp10,448.1 trillion, +8.2% y/y (Aug 2026, preliminary) [V] | to pull (Phase 3) |
| Reserve requirement | 0% since 26 Mar 2020 [V] | Minimum reserve (1% ratio [E]); avg requirement EUR 174.7 bn, remuneration 0% (Jul-Sep 2026) [V] | Rupiah GWM conventional banks 9% -> 8% (1 Nov 2026) -> 7% (1 Dec 2026), with macroprudential liquidity incentive (KLM) offsets [V] | None (voluntary reserves scheme) [E] |
| Known breaks | Savings deposits moved into M1 from May 2020 data [E]; vault-cash change Apr 2020 [V] | Membership changes (Croatia 2023, Bulgaria 2026 [V, ECB changeover list]) | Sharia rural banks added July 2013 [V]; M1+ redefinition ~2021 [V metadata dated Oct 2021] | M4 to M4ex emphasis [E] |

*Methodological limitation:* even with identical definitions, broad money/GDP ratios differ for structural reasons (financial depth, use of non-bank finance, informality, dollarisation). Cross-country comparisons of money growth will therefore be made in growth rates and with definitions shown, not in levels.

**Note on China (definitional break, 2026).** From 1 January 2026, e-CNY held in commercial-bank wallets is classified as bank deposit liabilities, earns interest at deposit rates, is covered by deposit insurance, and counts in the reserve-requirement base; non-bank payment institutions must hold 100% reserves against e-CNY they

manage [V]. Secondary reporting states that e-CNY moved from the M0 (currency in circulation) classification into deposits [E, Caixin]. This is a live definitional break for anyone comparing Chinese monetary aggregates across 2025-2026, and a conceptual challenge for the CBDC chapters (Part VIII).

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# 3. Competing theories of money

Each framework is presented in the same five-part format required by BRIEF Part II (foundational assumptions, strongest arguments, empirical support, limitations, major criticisms), followed by a comparative section on where schools genuinely disagree and where they talk past each other. The book's rule: each school is stated in the form its best current proponents would accept, before any criticism.

## 3.1 Commodity theory of money (Part II-A)

*Representative authors:* Aristotle (*Politics* I) on the origin of coinage in trade [E]; Adam Smith (*Wealth of Nations* I.4) [E]; Carl Menger, "On the Origin of Money" (1892) [E]; the metallist tradition; modern gold-standard advocates; Mises (1912) for the regression theorem.

- **Foundational assumptions.** Money emerges spontaneously from barter as traders converge on the most saleable commodity; value is anchored in the commodity's non-monetary use and scarcity; the state at most certifies weight and fineness. Mises's *regression theorem* adds that today's purchasing power of money can be traced back to a point where the money good had non-monetary value.
- **Strongest arguments.** (i) Commodity convertibility constrains issuers: a convertible currency cannot be expanded indefinitely without reserve losses. (ii) The classical gold standard (c. 1870s-1914) coincided with very low long-run average inflation and with large cross-border capital flows (Bordo and Schwartz eds. 1984; Eichengreen 1992 [E]). (iii) The theory explains why precious metals dominated across unconnected civilisations: divisibility, durability, recognisability, value-density.
- **Empirical support.** Long-run price-level stability under gold (price levels in Britain and the US in 1914 were close to their mid-nineteenth-century levels [E]); Rolnick and Weber (1997, *Journal of Political Economy*) find inflation and money growth higher under fiat than commodity standards across a large sample [E].
- **Limitations.** Long-run price stability coexisted with large short-run price volatility and extended deflations (for example 1873-1896 [E]); supply of the anchor depended on mining discoveries (California 1848, Witwatersrand 1886 [E]); balance-of-payments adjustment required domestic deflation and unemployment. Rolnick and Weber also report *higher* output growth under fiat [E], which the book must report alongside the inflation result.
- **Major criticisms.** (i) Historical: the barter-to-money sequence lacks ethnographic support (Humphrey 1985; Graeber 2011 [E]); early Mesopotamian money was a temple/palace unit of account for debts, with silver as measure rather than circulating coin (Hudson; Van De Mieroop [E]). (ii) Theoretical: coins routinely circulated above metal value, so the "commodity" was partly a token even under metallism. (iii) Macroeconomic: Eichengreen (*Golden Fetters*, 1992) and Eichengreen and Sachs (1985) show that countries leaving gold earlier recovered earlier from the Great Depression [E], implying gold transmitted and deepened the crisis.

## 3.2 Credit theory of money (Part II-B)

*Representative authors:* Henry Dunning Macleod (1850s-80s); Alfred Mitchell-Innes, "What is Money?" (1913) and "The Credit Theory of Money" (1914), both in the *Banking Law Journal* [E]; Schumpeter (1911, 1954); Keynes (*Treatise*, 1930) in part; Hyman Minsky; Geoffrey Ingham (*The Nature of Money*, 2004); Perry Mehrling (hierarchy of money); Graeber (2011).

- **Foundational assumptions.** Money is a social relation of credit and debt, recorded in a unit of account; a "purchase" is an exchange of a commodity for a credit; the value of money depends on the issuer's ability to accept its own IOUs back in settlement; there is a *hierarchy* of credit, with the most widely accepted IOUs (state and central bank liabilities) at the top.

- **Strongest arguments.** (i) Matches modern balance sheets exactly: every unit of money is someone's liability (central bank or bank), so credit theory is the natural accounting description of today's system (McLeay et al. 2014 [V] is essentially a credit-theoretic account). (ii) Matches historical evidence of tallies, bills of exchange and clearing fairs where debts were netted without coin (medieval Champagne fairs; English exchequer tallies [E]). (iii) Explains why bank money can be created by lending: it is the creation of a matched pair of claims.
- **Empirical support.** Werner (2014, *International Review of Financial Analysis*) documents, through a live bank-loan experiment, that the lending bank did not transfer pre-existing deposits but credited a new deposit [E]. Central bank publications (Bank of England 2014; Bundesbank Monthly Report April 2017 [E]) now describe money creation in credit-theoretic terms.
- **Limitations.** By itself the theory does not explain *which* credits become money (why deposits circulate and corporate bonds do not), nor what anchors the *price level*. It needs a theory of the hierarchy (state, law, central bank backstop) to close.
- **Major criticisms.** (i) From monetarists and New Keynesians: describing money as credit does not change the macro result that excessive money growth causes inflation. (ii) From Austrians: the credit view normalises fiduciary media created "out of nothing" which they see as the source of cycles. (iii) From historians of coinage: the credit theory underplays periods where bullion genuinely circulated by weight and where creditworthiness was unknowable (long-distance trade among strangers).

### 3.3 State theory of money: chartalism (Part II-C)

*Representative authors:* Georg Friedrich Knapp, *Staatliche Theorie des Geldes* (1905; English translation *The State Theory of Money*, 1924) [E]; Keynes (*Treatise*, ch. 1); Abba Lerner ("Money as a Creature of the State", 1947 [E]); modern neo-chartalists (Wray, Tcherneva); legal historians (Desan 2014).

- **Foundational assumptions.** Money is a "creature of law": the state defines the unit of account and declares what it will accept in payment of taxes and dues; the acceptance by the state at "pay offices" gives money its validity (Knapp's *proclamation* and acceptance). Note that Knapp's own emphasis is on state *acceptance* and the definition of the unit, not primarily on taxation; the strong "taxes drive money" thesis is a later neo-chartalist development (Wray 1998 [E]).
- **Strongest arguments.** (i) Explains why inconvertible fiat retains value in states with functioning fiscal systems. (ii) Explains why colonial administrations imposed hut and poll taxes payable in colonial currency to monetise subsistence economies (British East and Central Africa [E]); the Dutch colonial use of taxes in the Netherlands Indies is a candidate Indonesian case for Phase 2/12 (to verify). (iii) Explains why currency unions follow political unions far more often than the reverse.
- **Empirical support.** Historical association of monetary unit with jurisdiction; colonial tax-monetisation episodes; the persistence of national currencies despite their costs for small open economies.
- **Limitations.** States with strong tax demand still experience currency collapse (Weimar Germany had a tax system), so tax demand is not sufficient. Private and foreign currencies circulate widely where states are weak (dollarisation in Ecuador, Zimbabwe, Lebanon), so state proclamation is not necessary either.
- **Major criticisms.** Mengerian critics argue the state adopted money that already existed; historians note long periods when monies of foreign states (Spanish dollars, Maria Theresa thalers) dominated circulation in places where those states had no tax power [E].

### 3.4 Quantity theory of money (Part II-D)

*Representative authors:* Bodin, Hume ("Of Money", 1752 [E]), Irving Fisher (*The Purchasing Power of Money*, 1911), the Cambridge cash-balance approach (Marshall, Pigou), Friedman's restatement (1956).

- **Foundational assumptions.** The equation of exchange $MV = PY$ is an identity; it becomes a theory when one assumes velocity (or the demand for real balances) is stable or predictable, real output is determined by real factors in the long run, and money is exogenous (controlled by the monetary authority). Then sustained money growth in excess of real output growth produces proportional inflation in the long run (neutrality).
- **Strongest arguments.** The cross-country long-run association between money growth and inflation is one of the most robust correlations in macroeconomics. McCandless and Weber (1995, FRB Minneapolis *Quarterly Review*) report a correlation close to one between long-run average money growth and inflation

across 110 countries over three decades [E]; all known hyperinflations involved extreme money growth (Cagan 1956 [E]).

- **Empirical support.** Strong at long horizons and high inflation; weak at short horizons and low inflation. Teles, Uhlig and Valle e Azevedo (2016, *Economic Journal*) argue the quantity relation survives once one accounts for disinflation periods and interest-rate changes [E].
- **Limitations.** Velocity has been unstable in low-inflation economies, especially after financial innovation and after 2008; in Japan after 2013 and in the US after 2008, large base-money expansions did not produce proportional inflation (data to be pulled from FRED and BoJ in Phase 3/4). Direction of causality is disputed (money may accommodate prices).
- **Major criticisms.** Post-Keynesians: money is endogenous, so M responds to P and Y rather than causing them (Kaldor 1982, Moore 1988 [E]). New Keynesians: the policy instrument is the interest rate, so the money stock is demand-determined and has no independent causal role in the standard model (Woodford 2003 [E]).

## 3.5 Monetarism (Part II-D)

*Representative authors:* Milton Friedman (1956, 1968 presidential address), Friedman and Schwartz (*A Monetary History of the United States, 1867-1960*, 1963), Karl Brunner and Allan Meltzer, Phillip Cagan.

- **Foundational assumptions.** Money demand is a stable function of a few variables; the central bank can control the money stock (through the monetary base and a reasonably stable multiplier); money affects nominal income with "long and variable lags"; there is a natural rate of unemployment, so there is no long-run trade-off between inflation and unemployment (Friedman 1968 [E]); discretion is destabilising, so rules (constant money growth) are preferable.
- **Strongest arguments.** (i) Friedman-Phelps natural-rate hypothesis was vindicated by the 1970s stagflation. (ii) Friedman and Schwartz's reinterpretation of the Great Depression as a monetary contraction (the money stock fell by about a third from 1929 to 1933 [E]) became the mainstream view, endorsed by Bernanke (2002 speech: "we did it" [E]). (iii) The emphasis on central bank responsibility for inflation is now universal.
- **Empirical support.** Strong for high-inflation episodes and for the 1970s; supportive evidence on the Depression.
- **Limitations.** Monetary targeting failed in practice in the late 1970s and 1980s (US 1979-82 experiment; UK medium-term financial strategy) because velocity broke down and money proved hard to control; Goodhart's Law (1975) was coined in this context [E]. Central banks abandoned money targets for interest-rate operating procedures.
- **Major criticisms.** Endogenous-money critics (Kaldor 1982 *The Scourge of Monetarism*) argued the money stock cannot be controlled; New Keynesians retained the natural rate but dropped money; Austrians argue monetarism focuses on the price level and ignores relative-price distortions from credit expansion.

## 3.6 Keynesian economics (Part II-D)

*Representative authors:* Keynes (*A Treatise on Money*, 1930; *The General Theory of Employment, Interest and Money*, 1936); Hicks (IS-LM, 1937); the neoclassical synthesis (Samuelson, Tobin, Modigliani).

- **Foundational assumptions.** Aggregate demand can be deficient for long periods; money is held for transactions, precautionary and speculative motives (liquidity preference); the interest rate is a monetary phenomenon set in the market for liquidity; expectations are subject to fundamental uncertainty and "animal spirits"; money is not neutral in the short run and possibly not in the long run.
- **Strongest arguments.** (i) Explains liquidity traps and why monetary expansion at the zero lower bound may fail to raise spending ("pushing on a string"), consistent with Japan post-1990s and much of the advanced world 2009-2015. (ii) Explains why fiscal policy can matter for output when monetary policy is constrained. (iii) Keynes's distinction between the money of account and money "proper" makes him one of the bridges between state theory and modern macro.
- **Empirical support.** Fiscal multipliers larger in recessions and at the zero lower bound (Auerbach and Gorodnichenko 2012; Ramey's surveys argue the evidence is mixed [E]); Great Depression and 2008-15 experience.
- **Limitations.** Postwar Keynesian models did not anticipate stagflation; the original Phillips-curve trade-off failed in the 1970s.

- **Major criticisms.** Monetarist critique (natural rate), rational-expectations and Lucas critique (1976), Austrian critique of aggregate demand reasoning, and post-Keynesian critique that the "neoclassical synthesis" discarded Keynes's emphasis on uncertainty and endogenous money ("bastard Keynesianism", Joan Robinson [E]).

## 3.7 Post-Keynesian endogenous money theory (Part II-D)

*Representative authors:* Nicholas Kaldor; Basil Moore (*Horizontalists and Verticalists*, 1988); Marc Lavoie (*Post-Keynesian Economics: New Foundations*, 2014); Victoria Chick (stages of banking); Hyman Minsky; Charles Goodhart (a central banker's version).

- **Foundational assumptions.** Loans create deposits; banks extend credit to creditworthy borrowers at a mark-up over the policy rate; the central bank sets the price of reserves (the interest rate) and must supply the reserves the system demands at that price to avoid payment system failure; the money supply is therefore credit-driven and demand-determined. "Horizontalists" (Moore) model the reserve supply as flat at the policy rate; "structuralists" (Pollin, Palley) argue liquidity preference and balance-sheet constraints can make the curve upward-sloping.
- **Strongest arguments.** (i) Describes actual central bank operating procedures since the 1980s (interest-rate targeting) more accurately than base-multiplier models; this is now acknowledged in official publications (McLeay et al. 2014 [V]; Bundesbank 2017 [E]). (ii) Explains the post-2008 decoupling of reserves from lending: banks did not lend out excess reserves because reserves are not lent to the public. (iii) Minsky's financial instability hypothesis gives a credit-cycle account of crises (stable periods breed fragile balance sheets).
- **Empirical support.** Operating-procedure evidence; Granger-causality studies showing loans leading deposits and base money in several economies (mixed results [E]); post-2008 US data (base money up several-fold, M2 and lending much less) to be documented in Phase 3.
- **Limitations.** "Endogenous" does not mean "unconstrained": profitability, capital requirements, borrower demand and liquidity regulation bind; the horizontalist view understates cases where the central bank restricts reserves or collateral (emerging-market sudden stops). The theory of inflation is often cost-push and conflict-based, which is harder to test.
- **Major criticisms.** New Keynesians respond that endogenous money is fully compatible with their interest-rate models, so the "revolution" is semantic. Monetarists respond that the central bank could still control money if it chose to target reserves. Austrians accept the mechanism but treat it as the problem.

## 3.8 New Keynesian monetary economics (Part II-D)

*Representative authors:* Clarida, Gali and Gertler (1999); Michael Woodford (*Interest and Prices*, 2003); Jordi Gali (*Monetary Policy, Inflation, and the Business Cycle*); Ben Bernanke.

- **Foundational assumptions.** Optimising households and firms with rational expectations; nominal rigidities (sticky prices and wages) create short-run non-neutrality; the central bank sets a short-term nominal interest rate according to a rule (Taylor 1993 [E]); inflation is determined by expected future inflation and the output gap (the New Keynesian Phillips curve); money aggregates are typically absent ("cashless" limit).
- **Strongest arguments.** (i) It is the framework that central banks actually use for inflation targeting, and inflation targeting coincided with low and stable inflation from the early 1990s to 2020 in many countries. (ii) It explains why credibility and expectations management (forward guidance) matter. (iii) Its extensions incorporate financial frictions (Bernanke-Gertler-Gilchrist 1999 financial accelerator [E]).
- **Empirical support.** Disinflation and anchored expectations in inflation-targeting countries; estimated DSGE models fit postwar data reasonably.
- **Limitations.** Pre-2008 models largely lacked banks and could not capture the financial crisis; the 2021-2023 inflation surge was under-forecast by most central banks. Bernanke and Blanchard (2023) attribute the initial US surge mainly to price shocks (energy, food, sectoral shortages) with labour-market tightness becoming more important later [E] - which is consistent with New Keynesian logic but exposed weaknesses in forecasting.
- **Major criticisms.** Post-Keynesians and MMT: the model has no real money or banks, treats the natural rate as knowable, and ignores distribution. Austrians: interest-rate manipulation distorts intertemporal

prices. Fiscal theorists (Sims 1994; Cochrane 2023 [E]): the price level is ultimately pinned down by fiscal backing, not by an interest-rate rule alone.

## 3.9 Modern Monetary Theory (Part II-D)

*Representative authors:* L. Randall Wray (*Understanding Modern Money*, 1998; *Modern Money Theory*, 2012); Stephanie Kelton (*The Deficit Myth*, 2020); Warren Mosler; Bill Mitchell; Pavlina Tcherneva (job guarantee).

- **Foundational assumptions.** A government that issues its own non-convertible currency, borrows in it, and floats its exchange rate cannot be forced into involuntary default on its own-currency debt; taxes create demand for the currency and regulate aggregate demand rather than "fund" spending; government spending creates reserves and deposits, and bond issuance is a reserve-management operation; the binding constraint is real resources and inflation, not financing; a job guarantee can anchor prices and employment.
- **Strongest arguments.** (i) Accounting description of government payments through the central bank is broadly correct (Treasury spending credits bank reserves). (ii) Explains why Japan has sustained very high public debt in its own currency with low interest rates and no default. (iii) Sharpens the distinction between currency-issuing sovereigns and currency users (euro-area members, US states, countries with foreign-currency debt), which explains the euro-area sovereign debt crisis.
- **Empirical support.** Japan's experience; absence of own-currency defaults by monetary sovereigns in floating regimes (with caveats: Russia 1998 defaulted on ruble GKOs [E]).
- **Limitations.** The inflation constraint is acknowledged but the political-institutional mechanism for raising taxes or cutting spending quickly enough to control inflation is underspecified; the 2021-23 inflation after large fiscal transfers is a test case critics cite; emerging-market currency issuers face exchange-rate and capital-flight constraints that MMT's "monetary sovereignty" category handles only partially.
- **Major criticisms.** Mankiw (2020, *AEA Papers and Proceedings*) argues MMT's claims either restate standard macro or are wrong on inflation and interest-rate dynamics [E]; Epstein (2019) and others criticise its applicability to the Global South [E]; Austrian and monetarist critics see it as a rationale for monetary financing.

## 3.10 Austrian monetary economics (Part II-D)

*Representative authors:* Ludwig von Mises (*The Theory of Money and Credit*, 1912); Friedrich Hayek (*Prices and Production*, 1931; *Denationalisation of Money*, 1976); Murray Rothbard (*Man, Economy, and State*, 1962; *The Mystery of Banking*, 1983); Jesus Huerta de Soto (*Money, Bank Credit, and Economic Cycles*, 2006); the free-banking wing (Lawrence White, George Selgin).

- **Foundational assumptions.** Money originated as a market commodity (Menger, Mises regression theorem); interest reflects time preference; credit expansion not backed by real saving lowers interest rates below the natural rate, induces malinvestment in long production processes, and must end in a bust (Austrian business cycle theory); new money enters at specific points and changes relative prices (Cantillon effects); the "100% reserve" wing (Rothbard, Huerta de Soto) regards fractional-reserve demand deposits as a legal-ethical violation, while the free-banking wing (White, Selgin) argues competitive fractional-reserve banking without a central bank is stable and efficient.
- **Strongest arguments.** (i) Emphasis on credit-fuelled booms anticipated aspects of 2008 (housing overinvestment, interest rates held low). (ii) The Cantillon insight that money is non-neutral in distribution is now taken seriously in the mainstream heterogeneous-agent literature. (iii) Hayek's knowledge problem gives a serious argument against central planning of credit allocation, which is relevant to CBDC proposals that would expand central bank lending.
- **Empirical support.** Case studies of credit booms followed by busts (Jorda, Schularick and Taylor 2013 show credit booms predict deeper recessions [E] - a finding Austrians cite, though the authors are not Austrian); limited formal econometric work within the school, by methodological choice (praxeology).
- **Limitations.** The theory predicts that busts follow credit-driven booms but offers weak quantitative guidance on timing and magnitude; the claim that liquidation is curative is hard to square with the Great Depression evidence on early exit from gold; the internal split on fractional reserves shows the school does not have one answer on banking.
- **Major criticisms.** Mainstream: methodological rejection of empirical testing makes core claims unfalsifiable; the regression theorem is a theory of origin, not of current value; Friedman (1993, "plucking model") found

little evidence that boom size predicts bust size in US data [E]. The BRIEF explicitly forbids treating gold as a perfect solution; Austrian proposals will be evaluated against the gold standard's documented deflationary and crisis record.

## 3.11 A rival not named in the BRIEF but required for honesty: the fiscal theory of the price level

Sims (1994), Woodford (1995) and Cochrane (*The Fiscal Theory of the Price Level*, 2023) argue that the price level adjusts so that the real value of government debt equals the present value of expected primary surpluses [E]. It belongs in the book because it gives a distinct answer to Problem 3 (fiat money keeps value because it is backed by expected fiscal surpluses) and because it reframes Sargent and Wallace's "unpleasant monetarist arithmetic" (1981). It will be presented in chapter 4 alongside the seven modern frameworks the BRIEF names.

## 3.12 Where the schools genuinely disagree, and where they talk past each other

**Genuine empirical disagreements (testable):**

1. *The strength and horizon of the money-inflation link at low inflation.* Quantity theorists and monetarists say long-run proportionality holds; New Keynesians and post-Keynesians say low-inflation correlations are weak and causality runs both ways. Testable with long panels split by inflation regime (H8).
2. *Whether credit booms cause busts beyond what fundamentals explain.* Austrians, Minskyans and the Jorda-Schularick-Taylor literature say yes; standard NK models historically said no. Testable (H9).
3. *Whether QE raises inequality net of employment effects.* Austrian/Cantillon and some post-Keynesian views say yes; central bank research often finds small or offsetting effects (Bunn, Pugh and Yeates 2018 for the UK [E]). Testable (H11, H12).
4. *Whether fiscal deficits financed in own currency are inflationary when the economy is below capacity.* MMT says not until capacity binds; FTPL and monetarists say expectations of future financing can move prices earlier. Partly testable (H7).

**Genuine normative disagreements (not settled by data):**

5. Whether private banks *should* hold the power to create deposits through lending (Austrian 100%-reserve wing, sovereign-money reformers, versus mainstream and post-Keynesians). The book will lay out the consequentialist and rights-based arguments separately (Part VI).
6. How much discretion an unelected central bank should have, and whether distributional effects should enter its mandate.

**Pseudo-disagreements (definitional or framing):**

7. *"Is money exogenous or endogenous?"* Monetarists mean the central bank *could* control the money stock by controlling reserves; post-Keynesians mean that under interest-rate targeting it *does not*. Both can be true. Once one specifies the operating procedure, much of the dispute dissolves.
8. *"Do banks lend out deposits?"* At the level of the individual bank, a loan creates a deposit that typically leaves the bank and must be funded or settled in reserves; so a single bank behaves *as if* it needs deposits (funding), while the system as a whole creates deposits by lending. Textbook intermediation and endogenous-money accounts often describe different levels of aggregation.
9. *"Taxes do / do not fund spending."* MMT describes the order of operations in the payments system; critics describe the intertemporal budget constraint. The question that matters - what limits spending - gets the answer "inflation and real resources" from MMT and "inflation, interest rates and solvency expectations" from critics, which overlap more than the rhetoric suggests.
10. *"Fiat money is backed by nothing."* Austrians mean "no convertibility right"; chartalists and FTPL theorists mean "backed by the taxing power and fiscal surpluses". These are different claims about different things.

**Summary matrix (direction of answer to the five fundamental problems):**

| School | Who creates money? | Who benefits? | Why does fiat keep value? | Why accept unfair system? (implied) | CBDC view (typical) |
|---|---|---|---|---|---|
| Commodity / metallist | Miners and mints; banks issue claims on metal | Issuers via debasement / seigniorage | It shouldn't; value from commodity | Lack of alternatives; legal tender compulsion | Sceptical |
| Credit theory | Whoever's IOUs are accepted; banks and state at top of hierarchy | Those with access to credit at the top of the hierarchy | Issuer accepts own IOUs in settlement | Embedded in debt relations | Neutral: another IOU |
| Chartalist | State defines and accepts; banks as franchisees | State (seigniorage) and its agents | Taxes and state acceptance | Legal obligation (taxes) | Neutral to positive |
| Quantity theory / monetarist | Central bank via base; multiplier | Debtors and the state under inflation | Controlled scarcity, credible rules | Instrumental (stable prices) | Neutral if rule-bound |
| Keynesian | Banks and central bank; liquidity preference | Depends on demand management | Convention and policy | Convention | Pragmatic |
| Post-Keynesian | Banks via lending; central bank sets price | Banks and asset holders; distribution via interest | Wage-price institutions, state | Institutional dependence | Mixed; worry about banks |
| New Keynesian | Central bank sets rate; quantity endogenous | Small distributional focus (changing) | Credible inflation target | Instrumental | Cautious; design-dependent |
| MMT | State spending creates; banks via loans | Depends on fiscal choice | Taxes plus real resource management | Legal obligation | Positive if public |
| Austrian | Banks and central banks "out of nothing" | First recipients (Cantillon): state, banks, asset owners | Habit and coercion; fragile | Coercion and illusion | Strongly negative |
| FTPL | Treasury and central bank jointly | Debt holders vs taxpayers | Expected primary surpluses | Instrumental | Neutral |

(The matrix is interpretive and simplified; each cell will be defended with citations in Part I of the book.)

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## 4. Hypothesis register

Rules for the register: every hypothesis has (a) a rival, (b) evidence that would disconfirm it, (c) the data or method that will test it, (d) the phase and chapter responsible, and (e) a prior status (*working prior*, *agnostic*, or *deliberately adversarial*, meaning stated in its strongest form so that the book can try to break it). A hypothesis is never "confirmed" by agreement among the authors; it is either supported, weakened, or unresolved by evidence logged in `research/`.

## Part IV: How money is created

**H1. Bank lending creates deposits; the dominant source of broad money growth in normal times is bank credit to the private sector.**

- Rival H1-R: Broad money growth is dominated by central bank and fiscal operations (QE, Treasury spending from central bank accounts) and by net foreign inflows, especially in emerging markets.
- Disconfirming evidence: counterpart decompositions in which claims on the private sector explain a minority of M2/M3 growth over full cycles (not only crisis years).
- Test: ECB M3 counterparts; BI M2 affecting factors (NFA, net claims on central government, claims on other sectors); US reconstruction from H.8/Z.1. Phase 3, ch. 10. Prior: working prior for normal times; agnostic for 2020-21 and for Indonesia.

**H2. Banks' money creation is constrained primarily by profitability, capital and borrower demand, not by reserves.**

- Rival H2-R: Reserve availability (and reserve requirements where they exist, as in Indonesia) materially constrains lending.
- Disconfirming evidence: lending responds significantly to exogenous reserve-requirement changes holding rates and demand constant (Indonesia's GWM cuts of November-December 2026 [V] and earlier changes are natural experiments; China's frequent RRR changes are another).
- Test: event studies of GWM/RRR changes on loan growth; Senior Loan Officer Survey; OJK bank-level data. Phase 3, ch. 10-11. Prior: working prior for zero-requirement systems; agnostic for Indonesia.

**H3. The textbook money multiplier is an ex-post accounting ratio with no stable behavioural content in post-2008 advanced economies.**

- Rival H3-R: The multiplier is stable once one controls for interest on reserves and regulatory liquidity demand; it "collapsed" only because the Fed paid interest on reserves from October 2008.
- Disconfirming evidence: a stable, predictable relationship between base money and broad money after controlling for interest on excess reserves and liquidity regulation.
- Test: FRED series (BOGMBASE, M2SL, TOTRESNS) 1959-2026; similar for Japan and euro area. Phase 3, ch. 11. Prior: working prior; rival given a fair test.

**H4. Non-bank financial institutions create money-like liabilities (MMF shares, repo, stablecoins) whose "moneyness" depends on an explicit or implicit public backstop.**

- Rival H4-R: Market discipline and full collateralisation make non-bank money self-standing; public backstops are incidental.
- Disconfirming evidence: runs on non-bank money that resolve without public support at par; stablecoins holding par through severe stress without access to central bank liquidity.
- Test: 2008 Reserve Primary Fund; March 2020 MMF and repo stress; 2022 TerraUSD collapse; March 2023 USDC de-peg during the Silicon Valley Bank failure [E]; BIS AER 2025 tests of singleness, elasticity and integrity [V]. Phase 3, ch. 14. Prior: working prior.

## Part V: Inflation and distribution

**H5. No single mechanism explains modern inflation; the relative contribution of money, credit, fiscal, supply and expectations channels varies systematically with the inflation regime.**

- Rival H5-R (monetarist strong form): sustained inflation is always and everywhere driven by excess money growth; other factors only change relative prices.
- Disconfirming evidence for H5: a single variable (money growth, or alternatively supply shocks) explains the bulk of inflation variance across regimes and horizons.
- Test: variance decompositions; Bernanke-Blanchard (2023) decomposition method applied to the US, euro area and Indonesia; long-run panels. Phase 4, ch. 15-16. Prior: working prior; rival tested in its strongest form.

**H6. The 2021-2023 inflation in advanced economies was initially dominated by supply and energy shocks, with demand and labour-market tightness growing in importance over time.**

- Rival H6-R: The inflation was fundamentally monetary/fiscal, caused by the 2020-21 surge in money and transfers; supply shocks were the trigger, not the cause.
- Disconfirming evidence: countries with similar supply shocks but much smaller monetary and fiscal expansions showing similar inflation would support H6; the reverse would support the rival. Japan and Switzerland are important comparators.
- Test: cross-country panel of M2 growth 2020-21, fiscal impulse, energy dependence, and core inflation 2021-23. Phase 4, ch. 16. Prior: agnostic.

**H7. Own-currency fiscal deficits are not inflationary at the margin when output is well below capacity and expectations are anchored, but become inflationary once capacity binds or credibility weakens.**

- Rival H7-R1 (FTPL/monetarist): expectations of future monetisation can raise inflation even below capacity. H7-R2 (strong MMT): deficits are inflationary only at full employment.
- Disconfirming evidence: inflation responding to deficits independently of slack; or no inflation response even at full capacity.
- Test: state-dependent local projections; historical episodes (Japan 1990s-2010s; US 2021; Indonesia's 2020-2022 burden-sharing with Bank Indonesia under Law 2/2020 [E, to verify]). Phases 3-4, ch. 13 and 15. Prior: working prior.

**H8. The money-inflation correlation is strong in high-inflation regimes and long horizons, and weak in low-inflation regimes and short horizons.**

- Rival H8-R: The relation is uniformly strong once velocity shifts are modelled.
- Disconfirming evidence: equally strong low-inflation correlations after controlling for velocity.
- Test: replication of McCandless-Weber (1995) and Teles-Uhlig-Valle e Azevedo (2016) with data to 2025, split by inflation level. Phase 4, ch. 15. Prior: working prior.

**H9. Rapid private credit expansion predicts financial crises and deeper recessions more reliably than broad money growth predicts consumer price inflation.**

- Rival H9-R: Credit booms are symptoms of good fundamentals (productivity) and only predict crises when combined with external shocks.
- Disconfirming evidence: credit-to-GDP gaps with no predictive power out of sample.
- Test: BIS credit-to-GDP gap data; Jorda-Schularick-Taylor Macrohistory database. Phases 4 and 8, ch. 15 and 34. Prior: working prior.

**H10. Unanticipated inflation redistributes wealth from net nominal creditors (older, wealthier households holding bonds and deposits, and pension savers) to net nominal debtors (younger mortgage holders and governments); anticipated inflation's main regressive channel is cash holding and incomplete wage indexation among low-income households.**

- Rival H10-R: Inflation is broadly regressive in all forms because the poor hold more cash and have less bargaining power.
- Disconfirming evidence: distributional studies showing the poor losing more from unanticipated inflation than older creditors.
- Test: Doepke and Schneider (2006) method; Easterly and Fischer (2001) survey evidence; Indonesian SUSENAS consumption baskets and BPS CPI by expenditure decile where available. Phase 4, ch. 17. Prior: agnostic; both may hold in different regimes.

**H11. Cantillon effects are real but conditional: first-round recipients of newly created money gain measurably only when the new money is concentrated (in a sector, asset class or group), large relative to the economy, and not fully anticipated.**

- Rival H11-R1 (strong Austrian): Cantillon effects are a primary driver of long-run wealth inequality. H11-R2 (mainstream sceptic): effects are transitory and second-order.
- Disconfirming evidence: no measurable differential gain for first recipients in QE or credit episodes; or, against the sceptic, persistent gains that do not dissipate.
- Test: QE event studies by asset class; credit allocation studies; historical cases (Spanish New World silver price revolution; Weimar inflation's distributional effects). Phase 4, ch. 18. Prior: deliberately adversarial

(strong form stated to be tested).

**H12. Low policy rates and QE raised asset prices (equities, housing, long bonds) more than consumer prices after 2008, but most of the rise in wealth concentration in the US since the 1980s is attributable to structural causes (technology, globalisation, tax policy, housing supply restrictions, returns to capital) rather than monetary policy.**

- Rival H12-R: Monetary policy is the dominant driver of post-2008 wealth inequality.
- Disconfirming evidence: wealth shares rising mainly during QE windows and stalling otherwise; housing price growth uncorrelated with supply constraints.
- Test: US Distributional Financial Accounts (Federal Reserve) quarterly wealth shares; national housing affordability indices; cross-country comparison with economies that did not do QE. Phase 4, ch. 19. Prior: agnostic.

## Part VI and VII: Psychology, sociology and legitimacy

**H13. Trust in a currency is distinct from trust in banks and governments: people can distrust banks and governments while continuing to trust and use the national currency, until a threshold of inflation or convertibility failure is crossed.**

- Rival H13-R: Currency trust and institutional trust move together.
- Disconfirming evidence: survey data showing currency use falls in step with institutional trust in non-crisis periods.
- Test: survey trust series (ECB, Eurobarometer, World Values Survey, Indonesian surveys) versus currency substitution indicators. Phase 5, ch. 20 and 22. Prior: working prior.

**H14. Money illusion and anchoring make populations tolerant of moderate inflation but intensely averse to visible price increases in salient goods, so that political backlash correlates more with food and fuel prices than with headline CPI.**

- Rival H14-R: Public inflation perceptions track headline CPI.
- Disconfirming evidence: perceptions tracking headline CPI more closely than salient-goods inflation.
- Test: Shafir, Diamond and Tversky (1997); Shiller (1997); Stantcheva (2024) surveys [E]; Indonesian fuel-subsidy episodes (2005, 2013-14, 2022) [E]. Phase 5, ch. 21. Prior: working prior.

**H15. Continued participation in a monetary system perceived as unfair is explained more by institutional dependence and legal obligation (wages paid in national currency, taxes, contracts, infrastructure) than by genuine legitimacy.**

- Rival H15-R: Participation reflects genuine, if critical, legitimacy: people judge the system acceptable on balance.
- Disconfirming evidence: exit remaining low when exit is cheap and legal (for example stablecoin availability in high-inflation countries) suggests legitimacy or convenience; rapid exit when cheap suggests dependence.
- Test: comparison of stablecoin/crypto adoption across countries by inflation level and legal access; dollarisation histories. Phase 5, ch. 26. Prior: deliberately adversarial.

**H16. Dissatisfaction leads to partial exit (dollar deposits, gold, real assets, stablecoins) far more often than to mass abandonment, because full exit is blocked by tax and wage structures.**

- Rival H16-R: Above a threshold of inflation, abandonment is rapid and total (hyperinflation dynamics).
- Disconfirming evidence: episodes of full abandonment at moderate inflation; or partial exit persisting during hyperinflation.
- Test: Argentina, Lebanon, Zimbabwe, Venezuela, Indonesia 1997-98 deposit and currency data. Phases 5 and 8, ch. 22 and 34. Prior: working prior; both may be regime-dependent.

## Part VIII: International monetary order

**H17. Dollar dominance rests more on financial depth, legal predictability and network effects than on US trade share, and therefore changes slowly even as the US share of world output falls.**

- Rival H17-R: Dominance follows geopolitical and trade power, so a shift toward multipolarity (renminbi, euro, gold) is under way.
- Disconfirming evidence: sustained, non-valuation-driven decline in the dollar's share of reserves, invoicing and FX turnover.
- Test: IMF COFER (USD share 56.70% of allocated reserves in 2026Q2 versus 57.18% in Q1 [E, IMF data brief]); BIS Triennial Survey (2025); SWIFT RMB Tracker; invoicing data (Gopinath et al. 2020 [E]). Phase 6, ch. 27. Prior: working prior; rival tested with gold reserve accumulation data.

**H18. US sanctions use and weaponisation of the dollar system accelerate the build-out of alternative settlement infrastructure but have not yet produced a meaningful currency substitute.**

- Rival H18-R: Alternatives (CIPS, mBridge, bilateral local-currency settlement) are already displacing the dollar in significant trade corridors.
- Disconfirming evidence: large, persistent shifts in invoicing currency in sanctioned or BRICS corridors.
- Test: CIPS volumes; Russia-China trade invoicing; BIS reports on mBridge (BIS exited mBridge in 2024 [E, verify]). Phase 6, ch. 28. Prior: agnostic.

## Part X: CBDC

**H19. Retail CBDCs launched to date have had low adoption because they do not offer users a clear advantage over existing digital bank money and fast payment systems.**

- Rival H19-R: Low adoption reflects poor design and weak rollout (wallets, merchant incentives), not a structural lack of demand.
- Disconfirming evidence: a well-designed retail CBDC achieving high voluntary adoption in an economy with a mature fast-payment system.
- Test: Sand Dollar, JAM-DEX, eNaira, e-CNY data; comparison with UPI (India), PIX (Brazil), QRIS (Indonesia). Phase 7, ch. 30 and 32. Prior: working prior; the rival is plausible and must be tested.

**H20. Under realistic design parameters (holding limits, zero remuneration, intermediated distribution), a retail CBDC causes modest deposit substitution in normal times but can accelerate digital bank runs in stress.**

- Rival H20-R1: CBDCs cause large structural disintermediation of banks. H20-R2: CBDCs reduce run risk by giving a safe outlet that keeps money inside the regulated perimeter.
- Disconfirming evidence: simulation or live evidence of large normal-time outflows under limits; or run-amplification failing to appear in stress tests.
- Test: ECB, BoE and BIS simulations; China's 2026 reclassification as a natural experiment in the opposite direction (CBDC folded into deposits) [V]. Phase 7, ch. 31. Prior: working prior.

**H21. Programmability and surveillance risk depend on legal design and architecture, not on the CBDC form as such; existing account-based bank money already permits extensive transaction surveillance.**

- Rival H21-R: CBDCs uniquely concentrate transaction data in the state and create an unprecedented capacity for control.
- Disconfirming evidence: evidence that state access to transaction data under CBDC designs exceeds what AML/KYC rules already allow under bank money.
- Test: legal comparison of data-access provisions (EU digital euro regulation drafts; PBoC rules; US P.L. 119-101 definitions which exempt "open, permissionless, and private" currency [V]); BIS and IMF privacy design papers. Phase 7, ch. 32. Prior: agnostic; the rival is taken seriously.

**H22. Whether a CBDC shifts monetary power toward the central bank depends on who bears the liability: if retail balances remain central bank liabilities with direct claims, power shifts toward the central bank; if they are re-intermediated as bank deposits (as in China from 2026), the CBDC is primarily a payment-infrastructure upgrade.**

- Rival H22-R: Any CBDC structurally strengthens the state relative to banks regardless of liability structure.
- Disconfirming evidence: a re-intermediated CBDC that nonetheless shifts credit allocation or deposit pricing power to the central bank.

- Test: e-CNY 2026 framework [V]; digital euro holding-limit design; Bahamas and Jamaica tiered wallets. Phase 7, ch. 32. Prior: working prior.

### Part XIV: Synthesis

**H23. Monetary stability without institutional trust is possible only in narrow or transitional forms (commodity standards with costly enforcement, currency boards, dollarisation), all of which import trust from elsewhere rather than eliminating the need for it.**

- Rival H23-R: Trust-minimised systems (rule-bound algorithms, cryptoassets with fixed supply) can deliver monetary stability without institutional trust.
- Disconfirming evidence: a trust-minimised money achieving stable purchasing power and wide use as a unit of account for a large economy.
- Test: Bitcoin price volatility versus fiat; El Salvador's legal-tender experiment (adopted 2021; legal-tender obligation removed in 2025 under an IMF programme [E, verify]); currency boards (Hong Kong, Bulgaria pre-euro). Phase 9, ch. 39. Prior: working prior; deliberately exposed to the strongest crypto arguments.

**H24. Feasible institutional reforms (transparency of seigniorage, distributional reporting by central banks, public payment rails such as QRIS/FedNow/PIX, deposit insurance, and narrow-bank or tokenised-deposit options) can improve fairness and inclusion with small efficiency costs, whereas radical redesigns (100% reserve banking, return to gold, full retail CBDC substitution for deposits) carry large transition and stability costs.**

- Rival H24-R1: Only radical redesign can address structural unfairness. H24-R2: Even modest reforms impose significant efficiency costs.
- Disconfirming evidence: evidence that incremental reforms leave distributional outcomes unchanged; or that a radical regime (for example a full-reserve system) operated stably at scale.
- Test: comparative institutional analysis, simulation literature (Chicago Plan revisited, Benes and Kumhof 2012 [E]), payment-system adoption data. Phase 9, ch. 38-39. Prior: working prior, explicitly flagged as the authors' current lean so reviewers can attack it.

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## 5. Detailed table of contents (mapped to BRIEF parts and research phases)

Each chapter lists: BRIEF part(s) covered; research phase that produces it; core contents; hypotheses tested; required exhibits (T = T-account balance sheet, D = diagram, C = chart from data/, TL = timeline, DS = dataset).

**Front matter.** Preface and authorship statement (M. Bagus Prabowo and Hark); How to read this book (evidence tags; claim types; how CBDC statuses are dated); Abbreviations (spelled out); Notes on data definitions.

### Part I. Foundations (BRIEF I, II; Phase 1)

**Ch. 1. The Question, the Method, and the Rules of Evidence.** BRIEF I, XV. Central question; five fundamental problems as operational questions (Section 1 of this file); intellectual independence rules; the five claim types; how the book handles disagreement. Exhibits: D (map of the book's argument).

**Ch. 2. What Is Money? Functions, Forms, Issuers, Measures.** BRIEF II (16 concepts). The five-axis taxonomy; the 16 definitions; jurisdictional measurement table (US, euro area, Indonesia, UK, Japan, China incl. the 2026 e-CNY break); the hierarchy of money. Exhibits: T (central bank, bank, household balance sheets showing where each money sits); D (money flower adapted); DS (definitions table as CSV).

**Ch. 3. Commodity, Credit and State: The Three Classical Theories.** BRIEF II-A, B, C. Menger and metallism; precious-metal standards and debasement; intrinsic value and acceptance; Mitchell-Innes and Macleod; Knapp's chartalism (with the distinction between Knapp's acceptance thesis and the later tax-driven thesis);

debts, obligations and public/private credit institutions. Exhibits: TL (debasement episodes); C (coin face value vs metal value where data exist).

**Ch. 4. The Modern Schools Compared.** BRIEF II-D. Quantity theory; monetarism; Keynesian; post-Keynesian endogenous money; New Keynesian; MMT; Austrian; plus FTPL. Five-part assessment for each; the disagreement typology (genuine empirical, genuine normative, pseudo). Exhibits: summary matrix; D (operating procedures under each school).

## Part II. The History of Monetary Systems (BRIEF III; Phase 2)

**Ch. 5. Before Central Banks: Accounting, Coinage and Credit.** BRIEF III.1. Mesopotamian temple and palace accounting, silver as measure; Lydian and Greek coinage; Roman debasement; Chinese paper money (Song-Yuan) as an early fiat experiment; Islamic and Mediterranean trade credit; bills of exchange; Italian merchant banking; early deposit institutions (Barcelona Taula de Canvi, Venetian banks, Amsterdam Wisselbank 1609). How trust, enforcement and accounting worked without central banks. Exhibits: TL; T (a bill of exchange cycle).

**Ch. 6. From Goldsmiths to the Lender of Last Resort.** BRIEF III.2. Goldsmith banking and banknotes; fractional reserve arrangements; Bank of England 1694 (founded to lend to the Crown); Bank Charter Act 1844; Bagehot and the LOLR (1873); private claims integrated into national systems; the US path (free banking, National Banking Acts, Federal Reserve Act 1913). Exhibits: T (goldsmith receipts to notes; LOLR operation).

**Ch. 7. The Classical Gold Standard and Its Discontents.** BRIEF III.3. Convertibility; price-specie flow and its actual operation (central bank "rules of the game" often violated); deflation 1873-96; crises 1890, 1907; interwar restoration and collapse; Great Depression and exit timing. Central test: does gold prevent instability or relocate it? Exhibits: C (price levels under gold vs fiat); DS.

**Ch. 8. Bretton Woods, the Triffin Dilemma and the Nixon Shock.** BRIEF III.4, XI (Bretton Woods, Nixon). 1944 conference; IMF and IBRD; dollar-gold at USD 35 per ounce; fixed but adjustable pegs and capital controls; Triffin (1960); London Gold Pool; US payments deficits; 15 August 1971; Smithsonian and the 1973 float. Exhibits: TL; C (US gold reserves vs foreign dollar liabilities).

**Ch. 9. The Post-1971 Architecture.** BRIEF III.5. Floating rates; eurodollars and offshore dollar creation; capital account liberalisation; financialisation; derivatives and securitisation growth; payment networks (SWIFT 1973, card networks, RTGS, fast payments). Exhibits: C (BIS OTC derivatives notional; securitisation volumes - datasets already being pulled under data/r1 by a parallel agent); D (offshore dollar creation T-accounts).

## Part III. How Money Is Created (BRIEF IV; Phase 3)

**Ch. 10. How Commercial Banks Create (and Destroy) Money.** BRIEF IV.A. Balance sheets; loan and deposit creation; deposit destruction on repayment; interbank settlement in reserves; liquidity management (LCR, NSFR); capital requirements (Basel III); credit risk; prudential limits. Why "banks lend out deposits" is incomplete. Indonesian worked example in rupiah. Exhibits: T (step-by-step: loan, payment to another bank, settlement, repayment); numerical capital-constraint example. Hypotheses: H1, H2.

**Ch. 11. Fractional Reserves, the Multiplier, and Endogenous Money.** BRIEF IV.B. Historical fractional reserves; reserve requirements across systems (US zero since 2020; euro area minimum reserves at 0% remuneration; Indonesia GWM 9% -> 7% with KLM incentives; China RRR); textbook multiplier versus operations; corridor and floor systems. Explicit separation of the fractional-reserve framework from the endogenous-money framework. Exhibits: C (US base vs M2 vs loans 1959-2026); T. Hypotheses: H2, H3.

**Ch. 12. Central Bank Money: Currency, Reserves, QE and QT.** BRIEF IV.C. Currency issuance (demand-driven); open market operations; QE mechanics (reserves created; deposits created only when the seller is a non-bank); QT; emergency facilities (2008, 2020, 2023); transmission; central bank losses and the Fed's deferred asset. Exhibits: T (QE with bank vs non-bank seller); C (Fed balance sheet).

**Ch. 13. Deficits, Monetary Financing, Seigniorage and Fiscal Dominance.** BRIEF IV.D. Treasury financing; ordinary borrowing versus direct monetary financing; central bank independence; fiscal dominance; seigniorage measures; Indonesia's 2020-22 burden-sharing; historical inflationary finance (Weimar, Zimbabwe,

Indonesia 1960s). Exhibits: T (Treasury spending via the TGA; direct purchase in primary market). Hypothesis: H7.

**Ch. 14. Shadow Money: Non-Banks, Repo, Money Funds and Stablecoins.** BRIEF IV.E. Shadow banking; MMFs; repo; securitisation; private credit; investment funds; stablecoins (GENIUS Act regime; reserve composition; BIS singleness test); why non-bank credit is not deposit creation. Exhibits: T (MMF, repo, stablecoin issuer balance sheets). Hypothesis: H4.

## Part IV. Inflation and the Distribution of Purchasing Power (BRIEF V; Phase 4)

**Ch. 15. Inflations Plural: Mechanisms and Measures.** BRIEF V (12 drivers), V.A. Money growth, credit, demand, capacity, supply shocks, energy, wages, exchange rates, deficits, policy, expectations, asset prices; distinctions between monetary aggregates growth, credit growth, CPI, PPI, asset inflation and currency depreciation; why they diverge. Exhibits: C (US M2, CPI, PPI 2015-2026 - parallel agent output in research/r3-charts); C (cross-country money-inflation scatter). Hypotheses: H5, H8, H9.

**Ch. 16. The 2021-2023 Inflation as a Test of Theories.** BRIEF V, XIV.3. US, euro area, Japan, Switzerland, Indonesia, Argentina comparison. Hypothesis: H6.

**Ch. 17. Who Wins and Who Loses from Inflation.** BRIEF V.B. Implicit tax; fixed incomes; cash holders; debtors and creditors; asset owners; wage earners; pensioners; governments; financial institutions; anticipated vs unanticipated inflation. Indonesian inflation by income group. Hypothesis: H10.

**Ch. 18. Cantillon Effects: Theory, Evidence, Limits.** BRIEF V.C. Cantillon (*Essai*, published 1755); Hume; Austrian and modern heterogeneous-agent interpretations; conditions under which it applies; evidence strength. Explicitly not a universal theory of inequality. Hypothesis: H11.

**Ch. 19. Asset Prices, Housing and Wealth Concentration.** BRIEF V.D. Housing affordability; equity valuations; bonds; wealth concentration (US DFA); intergenerational inequality; low rates and valuations; monetary vs structural causes. Exhibits: C (DFA wealth shares; housing affordability - parallel agent outputs in data/r3). Hypothesis: H12.

## Part V. Trust, Legitimacy and Fairness (BRIEF VI, VII, VIII; Phase 5)

**Ch. 20. Why Fiat Money Holds Value: Collective Trust and Coordination.** BRIEF VI.A, VI.D, XIV.6. Conventions; network effects; shared expectations; institutional credibility; repeated interaction; coordination games; trust in money vs trust in banks vs trust in government. Hypothesis: H13.

**Ch. 21. The Behavioural Economics of Money and Inflation.** BRIEF VI.B, VI.C. Loss aversion; status quo bias; money illusion; anchoring; present bias; risk perception; social conformity; financial literacy (OJK national surveys); intergenerational habits. Hypothesis: H14.

**Ch. 22. Distrust: Runs, Crises, Dollarisation and Exit.** BRIEF VI.E. Currency crises; bank runs (including 2023 digital-speed runs); hyperinflation; scandals; sovereign defaults; currency substitution; dollarisation; crypto and stablecoins as exit. Why dissatisfaction rarely becomes abandonment. Hypotheses: H13, H16.

**Ch. 23. Who Has the Power to Create Money? Debt and Society.** BRIEF VII.A, VII.B. Allocation of monetary power among central banks, commercial banks, governments, markets and IFIs; household, mortgage, student, sovereign and corporate debt; interest-bearing obligations; mobility and political relationships.

**Ch. 24. Philosophers of Money: A Structured Disagreement.** BRIEF VII.C. Aristotle (*Politics* I; *Nicomachean Ethics* V), Smith, Marx, Simmel, Keynes, Hayek, Polanyi, Graeber, Schumpeter, Minsky - organised not as summaries but around five axes of disagreement: (1) is money a thing or a relation; (2) is money neutral; (3) is credit productive or extractive; (4) who should govern money; (5) what social obligations does money create or dissolve. Exhibit: disagreement matrix.

**Ch. 25. The Fairness Problem.** BRIEF VII.D, XIV.4, XIV.8. Public function vs private privilege; bank profits from deposit creation; legitimate central bank independence; inflation and the asset-poor; wealth concentration; stability vs unequal credit access; criteria for a fair monetary system (efficiency, stability, distributive fairness, procedural legitimacy, freedom). Normative claims labelled.

**Ch. 26. Why Societies Keep a System They Criticise.** BRIEF VIII, XIV.5. The ten interacting factors (network effects, switching costs, institutional dependence, legal obligations, taxes, wage structures, infrastructure, expectations, political legitimacy, lack of alternatives); distinction between voluntary acceptance, institutional dependence and legal obligation; the book's conceptual model (provisional name: the *Participation-Legitimacy Matrix*, crossing exit cost with perceived fairness to yield four regimes: endorsement, reluctant dependence, contested use, flight). Hypothesis: H15.

## Part VI. The International Monetary Order (BRIEF IX; Phase 6)

**Ch. 27. The Hierarchy of Currencies and the Dollar System.** BRIEF IX. Dollar roles (reserves, invoicing, funding, FX turnover); Treasury market; dollar debt; swap lines; global liquidity and the global financial cycle (Rey 2013); determinants of dominance (credibility, depth, infrastructure, trade networks, geopolitics, legal predictability, convertibility). Exhibits: C (COFER shares; BIS triennial). Hypothesis: H17.

**Ch. 28. Sustainability and Alternatives.** BRIEF IX, XIII-D. Euro; renminbi (internationalisation limits under capital controls); gold accumulation; Triffin in modern form; sanctions and fragmentation; CIPS, mBridge, local-currency settlement; scenarios for 2035. Hypothesis: H18.

## Part VII. Central Bank Digital Currencies and Digital Money (BRIEF X; Phase 7)

**Ch. 29. CBDC Fundamentals and Architectures.** BRIEF X.A. Definition and central bank liability (including the definitional challenge from China's 2026 reclassification); retail vs wholesale; account vs token; direct, intermediated and hybrid architectures; two-tier systems; offline payments; interoperability. Exhibits: D (architecture diagrams); T (CBDC conversion from deposits).

**Ch. 30. Ten Jurisdictions: From Research to Issuance.** BRIEF X.B. China, EU, US, UK, India, Nigeria, Bahamas, Jamaica, Sweden, Indonesia - each tagged by stage with date and official source (see the status snapshot in Section 6.2). Exhibit: DS (CBDC status table as CSV, with "last verified" column). Hypothesis: H19.

**Ch. 31. CBDC, Banks and Financial Stability.** BRIEF X.C. Deposit substitution; disintermediation; run risk; central bank balance sheets; credit allocation; transmission. Hypothesis: H20.

**Ch. 32. CBDC, Freedom and Monetary Power; Alternative Digital Monies Compared.** BRIEF X.D, X.E, X.F, XIV.9. Privacy; surveillance; data ownership and access; cybersecurity; inclusion; offline access; resilience; legal protections; due process; accountability. Power shifts among central banks, banks, governments, fintechs and citizens. Comparison of CBDC, stablecoins, tokenised deposits, cryptocurrencies, existing e-money and bank money, and wholesale settlement systems on trust, governance, backing, convertibility, privacy and stability. Hypotheses: H21, H22.

## Part VIII. Comparative Cases and Indonesia (BRIEF XI, XII; Phases 8 and 7)

**Ch. 33. Stable Systems.** BRIEF XI. United States, Switzerland, Singapore, Japan, Germany, Sweden - each on the nine dimensions (monetary structure, banking architecture, fiscal conditions, inflation history, credibility, public confidence, distribution, political responses, long-term consequences). Note: "stable" is relative; Japan's deflation and Sweden's 1990s banking crisis are included.

**Ch. 34. Crises.** BRIEF XI. Weimar Germany, Zimbabwe, Venezuela, Argentina, Lebanon, Indonesia 1997-98. Hypotheses: H9, H16.

**Ch. 35. Institutional Transformations.** BRIEF XI. US after the Great Depression (Banking Acts 1933/1935, FDIC, gold nationalisation 1933-34); Bretton Woods; the Nixon shock; the euro; China's financial system; post-2008 monetary policy; the rise of digital payments.

**Ch. 36. Indonesia I: From Colonial Money to the 1998 Crisis.** BRIEF XII. Colonial arrangements (De Javasche Bank 1828; Netherlands Indies guilder); Japanese occupation money; the revolution's ORI (1946); nationalisation of De Javasche Bank and founding of Bank Indonesia (1953); Sukarno-era deficits and the 1965-66 hyperinflation and redenomination; Suharto stabilisation and the 1983 and 1988 (PAKTO) liberalisations; the 1997-98 crisis, BLBI, bank closures, IBRA restructuring. Exhibits: TL; C (Indonesian inflation history - parallel agent data in data/indonesia_inflation_history.csv).

**Ch. 37. Indonesia II: Independence, Credit, QRIS and the Digital Rupiah.** BRIEF XII. UU 23/1999 and independence; amendments (2004, 2009); UU P2SK 4/2023 (mandate broadened); UU 4/2026 amending P2SK (job-creation clause; independence debate - statute text to be verified); BI-Ministry of Finance relationship, including 2020-22 burden-sharing; banking structure and credit creation (OJK, LPS); how BI influences credit (BI-Rate, GWM and KLM incentives [V]); distribution of credit (UMKM, regional); inflation by income group (BPS); financial inclusion; digital payments; QRIS (including cross-border links); Project Garuda status (wholesale PoC only, no pilot or issuance as of the latest official page [V]). Answers to the six Indonesia questions in BRIEF XII. Exhibits: T (rupiah money creation); DS (Indonesia key indicators).

### Part IX. Futures and Synthesis (BRIEF XIII, XIV; Phase 9)

**Ch. 38. Five Scenarios.** BRIEF XIII. Continuity; digital central bank money; tokenised infrastructure; fragmented order; institutional reform - each evaluated on feasibility, institutional requirements, stability, inclusion, privacy, distribution, geopolitics, sustainability. Analytical, not predictive. Hypothesis: H24.

**Ch. 39. Answers to the Eleven Questions, and an Ideal Architecture under Constraints.** BRIEF XIV.1-11. Direct answers, each tagged with confidence level and the hypotheses that support or weaken it. Hypotheses: H23, H24.

### Appendices (Phase 10)

- A. Historical timeline of monetary systems (3000 BCE to 2026).
- B. Balance-sheet workbook: all T-accounts with numerical examples.
- C. Monetary aggregates definitions by jurisdiction, with known series breaks.
- D. Comparative case-study dataset (CSV) with variable dictionary.
- E. CBDC status table with verification dates and sources.
- F. Hypothesis register outcomes (supported / weakened / unresolved, with evidence links).
- G. Methodology: source hierarchy, evidence tags, data pulls and reproducibility scripts.
- H. Bibliography and glossary.

*Note:* historical dates in this table of contents (for example De Javasche Bank 1828, Bank of England 1694, Wisselbank 1609, ORI 1946, Bank Indonesia 1953) are [E] and will be verified against primary or archival sources in Phase 2 and Phase 12 work.

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## 6. Source map

Source hierarchy follows the BRIEF: peer-reviewed research; central bank publications; BIS; IMF; World Bank; national statistical agencies; legislation; archives; academic books; reputable investigative journalism. URLs marked [V] were opened in Phase 1; others are institutional landing pages or DOIs to be opened by the responsible phase. “Classic” means cited from established bibliographic knowledge.

### 6.1 By chapter

| Ch. | Primary data and official sources (URLs) | Core literature (all schools) |
|---|---|---|
| 1-2 | Fed H.6 https://www.federalreserve.gov/releases/h6/; BoE MFSD https://www.bankofengland.co.uk/statistics; ECB monetary aggregates https://www.ecb.europa.eu/stats/money-credit-banking/html/index.en.html [V]; BI SEKI metadata https://www.bi.go.id/en/statistik/Metadata/SEKI/Documents/Broad_Money_and_Liquidity_Aggregates.pdf [V]; BoE Bankstats https://www.bankofengland.co.uk/statistics; IMF MFSMCG 2016 https://www.imf.org/en/Data/Statistics/Methodology; 31 USC 5103 https://www.law.cornell.edu/uscode/text/31/5103 [V] | McLeay-Radia-Thomas 2014 https://www.bankofengland.co.uk/quarterly-bulletin/2014/q1/money-creation-in-the-modern-economy [V]; Bagehot 1873; Hicks 1937; Tobin 1963; Friedman 1968; Lucas 1972; Kydland-Prescott 1982; Bernanke 1983; Diamond-Dybvig 1983; Gorton-Pennacchi 1990; Holmstrom-Tirole 1997; Diamond-Rajan 2001; Brunnermeier-Pedersen 2009; Gorton-Metrick 2012; Adrian-Shin 2010; Krishnamurthy-Vissing-Jorgensen 2012; Gertler-Kiyotaki 2010; Bernanke-Gertler-Gilchrist 1999; Kiyotaki-Moore 1997; Jorda-Schularick-Taylor 2011; Schularick-Taylor 2012; Reinhart-Rogoff 2009; Borio 2014; Minsky 1986; Kindleberger-Aliber 2005; Gorton 2010; Gorton-Ordoñez 2014; Caballero-Farhi-Gourinchas 2008; Caballero-Krishnamurthy 2009; Caballero-Simsek 2020; Caballero-Farhi-Gourinchas 2017; Caballero-Farhi-Gourinchas 2021; Caballero-Farhi-Gourinchas 2015; Caballero-Farhi-Gourinchas 2016

| Ch. | Primary data and official sources (URLs) | Core literature (all schools) |
|---|---|---|
| 10-11 | Fed reserve requirements https://www.federalreserve.gov/monetarypolicy/reservereq.htm [V]; ECB minimum reserves https://www.ecb.europa.eu/mopo/implement/mr/index.en.html [V]; BI GWM regulations https://www.bi.go.id/id/publikasi/peraturan/; BI press release 21 Oct 2026 https://www.bi.go.id/en/publikasi/ruang-media/news-release/Pages/sp_2823326.aspx [V]; OJK banking statistics https://ojk.go.id/id/kanal/perbankan/data-dan-statistik/; Basel Committee https://www.bis.org/bcbs/ | McLeay et al. 2014; Bundesbank Monthly Report April 2017; Werner 2014 (IRFA); Jakab-Kumhof 2015 (BoE Staff WP 529); https://www.bankofengland.co.uk/working-paper/2015/banks-are-not-intermediaries-of-loanable-funds-and-why-this-matters; Disyatat 2011 (JMCB); Borio-Disyatat 2010; Mishkin; Rothbard 1983 (for the 100%-reserve critique) |
| 12 | Fed H.4.1 https://www.federalreserve.gov/releases/h41/; ECB weekly financial statements; BoJ accounts https://www.boj.or.jp/en/statistics/; BI balance sheet | Bernanke 2022 (*21st Century Monetary Policy*); Borio-Disyatat 2010; Bailey et al. on QE; Gagnon |
| 13 | US Treasury Fiscal Data https://fiscaldata.treasury.gov; Indonesia Ministry of Finance https://www.kemenkeu.go.id and DJPPR https://www.djppr.kemenkeu.go.id; Law 2/2020 via https://peraturan.bpk.go.id | Sargent-Wallace 1981; Sargent 1982; Cagan 1956; Buiter on seigniorage; Wray 2012 |
| 14 | ICI MMF data; Fed financial stability reports https://www.federalreserve.gov/publications/financial-stability-report.htm; FSB NBFI monitoring https://www.fsb.org; BIS AER 2025 ch. III https://www.bis.org/publ/arpdf/ar2025e3.htm [V]; GENIUS Act CRS IN12553 https://www.congress.gov/crs-product/IN12553; OCC final rule release https://www.occ.gov/news-issuances/news-releases/2026/nr-occ-2026-92.html | Gorton 2010; Gorton-Metrick 2012; Pozsar 2014; Ricks 2016; Gorton-Zhang on stablecoins |
| 15-16 | FRED; ECB HICP; BPS CPI https://www.bps.go.id; IMF WEO https://www.imf.org/en/Publications/WEO; World Bank WDI API https://api.worldbank.org/v2/ | McCandless-Weber 1995; Teles-Uhlig-Valle e Azevedo 2016; Bernanke-Blanchard 2023 (NBER/Brookings); Gali; Borio on asset prices |
| 17 | Fed SCF https://www.federalreserve.gov/econres/scfindex.htm; BPS SUSENAS | Doepke-Schneider 2006 (JPE); Easterly-Fischer 2001 (JMCB); Erosa-Ventura 2002; Albanesi 2007 |

| Ch. | Primary data and official sources (URLs) | Core literature (all schools) |
|---|---|---|
| 18-19 | Fed Distributional Financial Accounts https://www.federalreserve.gov/releases/z1/; Zillow/Haver national housing indices; BIS property prices https://data.bis.org | Cantillon (*Essai*, 1755); Coibion-Gorodnichenko-Kueng-Silvia 2017 (JME); Bunn-Pugh-Yeates 2018 (BoE SWP); Mian-Sufi; Piketty 2014; Saez-Zucman |
| 20-22 | ECB Consumer Expectations Survey; World Values Survey https://www.worldvaluessurvey.org; OJK SNLIK financial literacy survey https://ojk.go.id; Reinhart-Rogoff data https://carmenreinhart.com | Simmel 1900; Zelizer 1994; Kahneman-Tversky 1979; Shafir-Diamond-Tversky 1997 (QJE); Shiller 1997; Stantcheva 2024; Diamond-Dybvig 1983; Reinhart-Rogoff 2009; Calvo-Reinhart; Hirschman *Exit, Voice, and Loyalty* 1970 |
| 23-25 | Household debt (NY Fed Household Debt and Credit https://www.newyorkfed.org/microeconomics/hhdc); IIF; OJK credit by sector | Aristotle; Smith; Marx *Capital* I, III; Simmel; Keynes; Hayek; Polanyi 1944; Graeber 2011; Schumpeter 1911; Minsky 1986; Pistor 2019; Tucker *Unelected Power* 2018 |
| 26 | Stablecoin and crypto adoption (Chainalysis, with caveats); dollarisation data (IMF); legal-tender and tax law | Hirschman 1970; Searle (social ontology); Aglietta-Orlean; Levi (legitimacy and compliance) |
| 27-28 | IMF COFER https://data.imf.org (USD share 56.70% 2026Q2 [E]); BIS Triennial Survey https://www.bis.org/statistics/rpfx25.htm; SWIFT RMB Tracker https://www.swift.com/our-solutions/compliance-and-shared-services/business-intelligence/renminbi/rmb-tracker; Fed swap lines; US Treasury TIC https://home.treasury.gov/data/treasury-international-capital-tic-system | Eichengreen 2011; Eichengreen-Mehl-Chitu 2017; Gopinath 2015 and Gopinath et al. 2020 (AER); Rey 2013; Farhi-Maggiori; Prasad *Gaining Currency* |

| Ch. | Primary data and official sources (URLs) | Core literature (all schools) |
|---|---|---|
| 29-32 | Atlantic Council CBDC tracker https://www.atlanticcouncil.org/cbdctracker/ (cross-check only); ECB digital euro progress https://www.ecb.europa.eu/euro/digital_euro/progress/html/index.en.html [V]; BoE digital pound https://www.bankofengland.co.uk/the-digital-pound; PBoC http://www.pbc.gov.cn and Xinhua/gov.cn 29 Dec 2025 https://english.www.gov.cn/news/202512/29/content_WS69526d4ec6d00ca5f9a08511.html [V]; P.L. 119-101 https://www.congress.gov/119/plaws/publ101/PLAW-119publ101.htm [V]; RBI e-rupee https://www.rbi.org.in; CBN eNaira https://enaira.gov.ng and https://www.cbn.gov.ng; Central Bank of The Bahamas https://www.centralbankbahamas.com and https://www.sanddollar.bs; Bank of Jamaica https://boj.org.jm; Riksbank https://www.riksbank.se (Payments Report 2026 [V]); BI Digital Rupiah https://www.bi.go.id/en/rupiah/digital-rupiah/default.aspx [V] | Auer-Boehme 2020 (BIS QR); Bindseil 2020 (ECB WP 2351); Brunnermeier-James-Landau 2019; Fernandez-Villaverde et al. 2021; Keister-Sanches 2023 (JME); BIS AER 2021 ch. III and 2023 ch. III; IMF rCBDC financial-integrity papers |
| 33-35 | National central banks (SNB, MAS, BoJ, Bundesbank, Riksbank, Fed); Hanke-Krus hyperinflation table; IMF Article IV reports; Banque du Liban and World Bank Lebanon Economic Monitor; BCRA (Argentina); RBZ (Zimbabwe); BCV (Venezuela) | Sargent 1982; Bernholz 2003; Feldman on Weimar; Eichengreen on the euro; Lardy and Prasad on China; Reinhart-Rogoff 2009 |
| 36-37 | Bank Indonesia https://www.bi.go.id (SEKI statistics https://www.bi.go.id/en/statistik/ekonomi-keuangan/seki/; history https://www.bi.go.id/id/tentang-bi/museum/); OJK https://ojk.go.id; LPS https://lps.go.id; BPS https://www.bps.go.id; Ministry of Finance https://www.kemenkeu.go.id; JDIH BI https://jdih.bi.go.id and BPK legislation database https://peraturan.bpk.go.id (UU 23/1999; UU 7/2011; UU 2/2020; UU 4/2023; UU 4/2026); QRIS (BI and ASPI) | Booth; Lindblad; Dick et al. *The Emergence of a National Economy* 2002; Grenville; Hill; Enoch et al. (IMF) on 1997-98 bank restructuring; Sato; Boediono |

| Ch. | Primary data and official sources (URLs) | Core literature (all schools) |
|---|---|---|
| 38-39 | All of the above | Synthesis literature; BIS AER 2025; Hayek 1976; Benes-Kumhof 2012 *The Chicago Plan Revisited* (IMF WP) |

## 6.2 CBDC status snapshot (as of 7 November 2026; to be re-verified at each later phase)

| Jurisdiction | Stage | Basis |
|---|---|---|
| China (e-CNY) | Large-scale pilot since 2019-20; from 1 Jan 2026 bank-wallet balances reclassified as interest-bearing bank deposit liabilities with deposit insurance and reserve requirements | [V] gov.cn/Xinhua 29 Dec 2025; "not formally issued" per JRI 2026 [E] |
| Euro area (digital euro) | Legislative preparation; preparation phase closed Oct 2025; possible issuance 2029 if EU regulation adopted in 2026 | [V] ECB progress page |
| United States | Prohibited: Federal Reserve may not issue a CBDC until 31 Dec 2030 (P.L. 119-101 Sec. 1101); EO 14178 bars agencies from promoting CBDCs | [V] statute; [E] EO |
| United Kingdom (digital pound) | Design phase through 2026; no decision; decision promised by end-2026; any issuance requires primary legislation | [E] BoE pages and UK Parliament written answer Oct 2026 |
| India (e-rupee) | Pilot (wholesale from 1 Nov 2022, retail from 1 Dec 2022) | [E]; Riksbank 2026 lists India among pilots [V] |
| Nigeria (eNaira) | Issued (Oct 2021); low adoption; reported repositioning | [V] Riksbank 2026 (launched); details [E] |
| The Bahamas (Sand Dollar) | Issued (Oct 2020) | [V] Riksbank 2026; date [E] |
| Jamaica (JAM-DEX) | Issued (2022) | [V] Riksbank 2026; date [E] |
| Sweden (e-krona) | Research/experimentation concluded; no issuance decision; Riksbank says it "previously explored" an e-krona | [V] Riksbank Payments Report 2026 |
| Indonesia (Digital Rupiah, Project Garuda) | Experimentation (wholesale): White Paper 30 Nov 2022; consultative paper 31 Jan 2023; PoC report 13 Dec 2024; 2026 work on tokenised securities ledger and cross-border experiments reported; no pilot, no issuance | [V] BI page; 2026 developments [E] |

# 7. Research risks and mitigations

**7.1 Ideological capture.** Monetary economics is unusually polarised, and several schools produce confident popular literatures (Austrian and "sound money" writing; MMT; central-bank-orthodox communication). Risks: (a) adopting a school's definitions and thereby its conclusions (for example, defining inflation as money growth makes H5 true by definition); (b) asymmetric scepticism (scrutinising heterodox claims more harshly than official ones, or the reverse); (c) quoting central banks as neutral when they are interested parties on independence and CBDC. *Mitigations:* every chapter uses the five claim-type labels; every school is stated in its strongest form before criticism (steelman rule); central bank sources are treated as primary for *facts about their operations* but as interested parties for *evaluative claims*; each Phase 10 chapter gets an adversarial review (reviews/) by an agent instructed to argue the opposing school; the hypothesis register includes "deliberately adversarial" entries; banned framings from the BRIEF (conspiracy, "fiat is worthless", "gold is perfect", "CBDC is inherently authoritarian", unlimited bank money creation) are checked in review.

**7.2 Survivorship and selection bias in case selection.** The BRIEF's case lists mix success and failure, but "successful" systems are partly defined by survival, and crisis cases are chosen because they were dramatic. Risks: inferring that independence or gold or dollarisation "causes" stability from cases selected on the outcome. *Mitigations:* code all cases on the same nine dimensions (BRIEF XI) in a dataset; add counter-cases where the same institution coexisted with the opposite outcome (independent central banks that still had crises, for example Sweden early 1990s; dollarised economies with stagnation; gold-standard countries with banking panics, for example the US 1907); report base rates from Reinhart-Rogoff and Laeven-Valencia crisis databases rather than anecdote; state explicitly when a case is illustrative rather than evidential.

**7.3 Data definitional breaks.** Documented breaks already identified: US M1 redefinition (savings deposits from May 2020 data [E]) and vault-cash treatment (April 2020 [V]); Fed reserve requirements to zero (March 2020 [V]); ECB membership changes (Bulgaria 2026 [V]); BI coverage change (July 2013 [V]) and M1+ redefinition (~2021 [V]); China's e-CNY reclassification (Jan 2026 [V]); CPI basket and base-year changes (BPS rebasing; US methodological changes); Indonesia's 1965 redenomination (1,000 old rupiah = 1 new rupiah [E]). *Mitigations:* Appendix C logs every break; every chart states series ID, unit, seasonal adjustment and break dates; growth rates are computed within consistent definitions or spliced with documented methods; raw pulls saved under data/ with README lines (FRAMEWORK rule).

**7.4 CBDC status volatility.** Within 2025-2026 alone: the US moved from executive-order policy to statutory prohibition [V]; China changed the legal nature of retail e-CNY balances [V]; the euro area moved to a new phase [V]; the UK signalled no urgency [E]. *Mitigations:* every CBDC claim carries a stage label (research / experimentation / pilot / legislative preparation / issuance) and an "as of" date; the CBDC dataset has a "last verified" column and an official-source URL; secondary trackers (Atlantic Council, CBDC registries, news) are used only to discover changes, never as the sole basis; the snapshot in 6.2 is re-verified at Phases 7, 9 and 10.

**7.5 Further risks.** (a) *Language and source access:* Indonesian legislation and BI documents are partly in Bahasa Indonesia only, and some official downloads block automated access (the BPK copy of UU 4/2026 could not be retrieved in Phase 1); mitigation: use JDIH BI, DPR and Setneg mirrors, and mark unverified statutory text. (b) *Recency bias:* 2021-23 inflation and 2025-26 policy news can crowd out the long view; mitigation: every contemporary claim is paired with a historical comparator. (c) *Quantitative overreach:* distributional effects of monetary policy are model-dependent; mitigation: report ranges and model assumptions, label counterfactuals. (d) *Parallel-agent inconsistency:* other agents are already pulling data into data/r1-r5 and data/indonesia_*.csv; mitigation: Phase 10 reconciles every figure against the verification log, and conflicting numbers are resolved in favour of the opened primary source.

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# 8. Disputes and uncertainties (Phase 1)

1. **Definition of a CBDC after China's 2026 reform.** If retail e-CNY wallet balances at banks are now bank deposit liabilities [V], is e-CNY still a CBDC in the BIS sense (a central bank liability)? Working answer: it is now a hybrid (central-bank-issued digital currency architecture with bank-liability retail balances); the book will define CBDC by liability and treat e-CNY as a boundary case. Unresolved pending PBoC primary documents.

2. **Bank Indonesia's mandate and independence after UU 4/2026.** Multiple reports [E] describe a new clause directing BI's policy mix toward real-sector growth and job creation; the Jakarta Post headline frames this as loss of independence. The statute text was not opened in Phase 1. Status: unverified; kept out of conclusions.
3. **Bank Indonesia leadership.** 2026 reports [E] name Destry Damayanti as Governor; the succession details were not verified in Phase 1 and are not needed for any conclusion here.
4. **US legal status details.** The ROAD to Housing Act CBDC section is verified [V]; reported vote counts and the "without signature" detail are [E]. The broader Anti-CBDC Surveillance State Act (H.R. 1919) passed the House in July 2025 but is not law [E].
5. **Launch dates and adoption metrics for Sand Dollar, eNaira and JAM-DEX** are [E] from secondary trackers; official central bank figures will be pulled in Phase 7.
6. **Whether e-CNY counts as "issued".** Some analyses say it has not been formally issued despite years of pilots [E, JRI 2026]; others treat it as live. The book will classify it as "large-scale pilot / reclassified 2026" unless PBoC documents state otherwise.
7. **The strength of the quantity-theory evidence at low inflation** and **the size of Cantillon effects** are genuinely open questions; they are registered as H8 and H11 rather than asserted.

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# 9. Sources (Phase 1; tags as defined above)

Opened in Phase 1 [V] (details in `phase1/verification_log.md`):


1. Board of Governors of the Federal Reserve System, Money Stock Measures H.6, release of 27 Oct 2026. https://www.federalreserve.gov/releases/h6/current/default.htm [V]
2. Federal Reserve, Reserve Requirements. https://www.federalreserve.gov/monetarypolicy/reservereq.htm [V]
3. European Central Bank, Monetary aggregates. https://www.ecb.europa.eu/stats/money_credit_banking/monetary_aggreg [V]
4. European Central Bank, Minimum reserves. https://www.ecb.europa.eu/mopo/implement/mr/html/index.en.html [V]
5. European Central Bank, Progress on the digital euro. https://www.ecb.europa.eu/euro/digital_euro/progress/html/index.en [V]
6. Bank Indonesia, SEKI metadata: Broad Money and Its Affecting Factors (Oct 2021). https://www.bi.go.id/en/statistik/Meta [V]
7. Bank Indonesia, Analisis Perkembangan Uang Beredar (M2) Agustus 2026. https://www.bi.go.id/id/publikasi/laporan/Docu Perkembangan-Uang%20Beredar-(M2)-Agustus-2026.pdf [V]
8. Bank Indonesia, press release No. 28/233/DKom, 21 Oct 2026. https://www.bi.go.id/en/publikasi/ruang-media/news-release/Pages/sp_2823326.aspx [V]
9. Bank Indonesia, Project Garuda: Navigating the Architecture of Digital Rupiah. https://www.bi.go.id/en/rupiah/digital-rupiah/default.aspx [V]
10. McLeay, M., Radia, A. and Thomas, R. (2014), "Money creation in the modern economy", Bank of England Quarterly Bulletin 2014 Q1. https://www.bankofengland.co.uk/quarterly-bulletin/2014/q1/money-creation-in-the-modern-economy [V]
11. 31 U.S. Code 5103, Legal tender. https://www.law.cornell.edu/uscode/text/31/5103 [V]
12. Public Law 119-101, 21st Century ROAD to Housing Act (11 July 2026), Sec. 1101. https://www.congress.gov/119/plaws/pu 119publ101.htm [V]
13. Xinhua via english.www.gov.cn, "China to enhance digital yuan management with deposit features starting 2026" (29 Dec 2025). https://english.www.gov.cn/news/202512/29/content_WS69526d4ec6d00ca5f9a08511.html [V]
14. Sveriges Riksbank, Payments Report 2026, "Retail CBDC - a whole new kind of money". https://www.riksbank.se/en-gb/payments–cash/payments-in-sweden/payments-report-2026/article–money-in-transition–traditional-and-new-forms/retail-cbdc–a-whole-new-kind-of-money/ [V]
15. BIS Annual Economic Report 2025, ch. III "The next-generation monetary and financial system". https://www.bis.org/publ/arpdf/ar2025e3.htm [V]


Search excerpts or established knowledge [E] (to be opened by the responsible phase):

16. IMF COFER Data Brief, 29-30 Sept 2026. https://data.imf.org/en/news/imf%20data%20brief%20september%2030 [E]
17. Bank of England, The digital pound; progress update (2026). https://www.bankofengland.co.uk/the-digital-pound [E]; UK Parliament written answer 2026-10-13/34410 https://questions-statements.parliament.uk/written-questions/detail/2026-10-13/34410/ [E]
18. Congressional Research Service, IN12553 (GENIUS Act overview, updated 20 Aug 2026). https://www.congress.gov/crs-product/IN12553 [E]; CRS R49354 (ROAD to Housing Act). https://www.congress.gov/crs_external_products/R/PDF/R4 [E]
19. Bank Indonesia, Kebanksentralan: Konsep dan Kebijakan (mandate before and after UU P2SK). https://www.bi.go.id/id/bi-institute/policy-mix/Documents/Kebijakan-Moneter.pdf [E]; Jakarta Post, 4 June 2026, "BI gets job creation mandate under newly passed law". https://www.thejakartapost.com/business/2026/06/04/b gets-job-creation-mandate-under-newly-passed-law [E]
20. CSIS Indonesia (Fauri), "Refocusing the Purpose of Indonesia's CBDC", 2 Nov 2026. https://csis.or.id/publication/refocusin the-purpose-of-indonesias-cbdc [E]
21. Katsurada, K. (2026), "The Shift in China's CBDC (Digital Yuan) Policy and Key Implications", JRI Research Journal 9(13). https://www.jri.co.jp/en/reports/jrirj/2026/katsurada/13 [E]
22. Classic and academic literature cited by author-year in Sections 2-5 (Mitchell-Innes 1913, 1914; Knapp 1905; Menger 1892; Keynes 1930, 1936; Friedman 1956, 1968; Friedman-Schwartz 1963; Moore 1988; Lavoie 2014; Woodford 2003; Wray 1998, 2012; Kelton 2020; Mises 1912; Hayek 1931, 1976; Rothbard 1962, 1983; Huerta de Soto 2006; Sims 1994; Cochrane 2023; McCandless-Weber 1995; Teles-Uhlig-Valle e Azevedo 2016; Rolnick-Weber 1997; Eichengreen 1992, 2011; Eichengreen-Sachs 1985; Cagan 1956; Sargent 1982; Sargent-Wallace 1981; Werner 2014; Jakab-Kumhof 2015; Bernanke-Blanchard 2023; Coibion et al. 2017; Doepke-Schneider 2006; Easterly-Fischer 2001; Bunn-Pugh-Yeates 2018; Jorda-Schularick-Taylor 2013; Rey 2013; Gopinath et al. 2020; Auer-Boehme 2020; Bindseil 2020; Mankiw 2020; Graeber 2011; Humphrey 1985; Simmel 1900; Polanyi 1944; Minsky 1986; Schumpeter 1911; Shafir-Diamond-Tversky 1997; Shiller 1997; Stantcheva 2024; Benes-Kumhof 2012). [E] Bibliographic details to be verified and expanded with DOIs in Phase 10.

# R1 – The History of Monetary Systems (BRIEF Part III)

Research stream R1 for the monograph *The Architecture of Money*. Authors of the monograph: M. Bagus Prabowo and Hark (he conceived and directs the project; Hark's agents research and draft). This file is a research dossier, not final manuscript prose.

Status: complete draft, Nov 7 2026. Data files: `data/r1/` (see `data/r1/README_r1.md`). Working notes with verification log: `work/r1/verified_notes.md`.

Evidence tags (per FRAMEWORK.md): **[V]** = page opened in this research run and the claim is on it; **[E]** = from a search excerpt or established scholarly knowledge, not opened in this run. Claim types are marked where it matters: (fact), (empirical finding: Author year), (interpretation: school/author), (normative), (counterfactual).

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## Key findings

1. **Credit and accounting precede coinage; "barter first" is not supported as a historical sequence, but neither is a pure "credit only" story.** (fact) Mesopotamian temple and palace administrations used silver shekels and barley measures as units of account with an administered norm of 1 shekel of silver = 1 gur (300 sila, roughly litres) of barley, and standard interest of 20% on silver and 33% on barley loans, more than a millennium before the first coins [V: CDLJ 2017:2]. Coinage appears only around 650-600 BC in Lydia and Ionia [V: British Museum]. (interpretation: Graeber 2011; Humphrey 1985) Ethnography has found no economy organised around barter between neighbours. (interpretation: Menger 1892; Selgin; Murphy) The Mengerian argument is about the logic of a medium of exchange emerging from saleability, not necessarily about a datable barter era; absence of evidence is partly explained by how quickly media of exchange arise. The defensible synthesis: units of account and credit relations were usually organised by institutions (temples, palaces, states, merchant networks) before generalised commodity media of exchange, but commodity metals then became the dominant settlement and store-of-value asset for two millennia.

2. **Commodity money never removed the sovereign's temptation; it changed the technique.** Roman denarius fineness fell from near-pure silver before AD 64 to about 80% under Nero, about 70% after AD 155-157, and about 46% after AD 194 [V: Butcher and Ponting 2012]; by the 260s the antoninianus was a near-base coin and prices rose roughly tenfold in a decade [V: ANS]. Debasement was, in Butcher and Ponting's words, "a form of increased taxation". Modern metallurgical work also shows that some famous "debasement crises" in older data (Walker 1976-78) were measurement artefacts – a warning about over-reading single numbers.

3. **Paper and book-entry money appeared early and failed or succeeded for institutional reasons.** Song China's state jiaozi (from 1023) worked while issue was capped, renewed every three years, receivable for taxes and supported by reserve interventions, and degenerated under 13th-century war finance; Ming paper collapsed by the early 15th century and China moved to silver [V: von Glahn 2025]. The Bank of Amsterdam (1609) ran a fiat-like bank florin after 1683 for about a century and failed in 1780-1795 after opaque lending to the Dutch East India Company without fiscal backing (backing fell from 97% to 28%) [V: Bolt, Frost, Shin and Wierts 2023]. The shared lesson (interpretation: BIS authors; von Glahn): fiat

money is not "backed by nothing"; it is backed by the issuer's balance sheet, the fiscal capacity behind it, tax receivability, and governance.

4. **Modern banking grew out of private credit and was progressively nationalised in function.** Goldsmith notes (1660s), the Bank of England (1694) as a private lender to the Crown, the Bank Charter Act (1844) monopolising note issue in England and Wales, and Bagehot's lender-of-last-resort doctrine (1873) show private claims being integrated into a public hierarchy: commercial bank deposits promise convertibility into central bank money, which itself was for a long time convertible into gold. The US arrived late and unevenly (free banking 1837-1863; National Banking Acts 1863-64; panics in 1873, 1893, 1907; Federal Reserve Act 23 Dec 1913) [V: FRH].

5. **The classical gold standard delivered long-run price-level stability, not short-run stability.** Our computation from Minneapolis Fed and Bank of England data: UK prices changed by -1.1% cumulatively over 1822-1913, but annual inflation had a standard deviation of 4.3 percentage points with 42 deflation years out of 92; real GDP growth volatility (s.d. 2.5 pp) was higher than under Bretton Woods (2.0) or post-1993 inflation targeting (1.7). US prices fell 31% and UK prices 23% between 1873 and 1896. Gold did not prevent panics (1873, 1890, 1893, 1907). (interpretation) Gold changes the mechanism of instability – from inflation risk to deflation, banking-panic and adjustment-through-output risk – rather than eliminating instability.

6. **The interwar gold standard transmitted the Great Depression.** (empirical finding: Eichengreen and Sachs 1985; Bernanke 1995) Countries that left gold earlier recovered sooner; "no country exhibited significant economic recovery while remaining on the gold standard" (Bernanke 1995, via FRASER text). (interpretation: Eichengreen 1992, *Golden Fetters*) The gold-exchange standard's asymmetry, loss of credibility and lack of cooperation turned a US downturn into a global deflation.

7. **Bretton Woods failed for structural and political reasons, not because of one decision.** (fact) The 1944 design pegged currencies to a dollar convertible at $35 an ounce; it became fully operational only in 1958; by 1961 foreign dollar claims exceeded the US gold stock; the London Gold Pool (Nov 1961-Mar 1968) and Fed swap lines (from Mar 1962) were stop-gaps; Nixon closed the gold window on 15 Aug 1971; the Smithsonian realignment (Dec 1971) lasted about fifteen months; generalized floating came in March 1973; Jamaica (Jan 1976) and the IMF Second Amendment (1 Apr 1978) legalised it [V: FRH, IMF]. (interpretation: Triffin 1960) Supplying world reserves through US deficits undermined confidence in the convertibility those reserves relied on; (interpretation: Bordo, Meltzer, Hetzel) expansionary US policy after 1965 accelerated the breakdown.

8. **The post-1973 order is a dollar-based credit system run through private balance sheets and public backstops.** Eurodollar banking (from the late 1950s), petrodollar recycling, Volcker's 1979-82 disinflation, liberalization, and the growth of derivatives (BIS notional outstanding $72tn in 1998-H1, peak $673tn in 2008-H1, $845tn at end-2025; gross market value $22.8tn) and securitization (US agency pools plus ABS reached about 60% of GDP in 2007 before an accounting break) [V: BIS, Fed Z.1 data pulled]. The US dollar is on one side of 89% of FX trades and FX turnover is $9.6tn/day (April 2025) [E: official BIS release, read via search excerpt]. Payment infrastructure moved from telegraph and end-of-day netting to real-time gross settlement (Fedwire origins 1918; 49 of 176 central banks with RTGS by 2000) and SWIFT messaging (1973/1977) [V: FRH, BIS, SWIFT].

9. **Fiat money has a measurable cost and a measurable benefit.** US prices rose about 33-fold from 1913 to 2025 and about 8-fold from 1971 to 2025 (Minneapolis Fed index); under inflation targeting (UK 1993-2016) inflation averaged about 2.0% with a standard deviation under 1 pp, the most stable price behaviour in the 800-year UK record. (interpretation) The fiat era combines persistent moderate inflation with lower inflation volatility, fewer deflations and (in the UK data) lower output volatility; which regime is "better" depends on whom one weighs – holders of nominal assets, debtors, workers exposed to recessions – a question for Parts V and VII.

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## Scope, method and limitations

- Coverage follows BRIEF Part III subsections 1-5. Cross-references: theoretical schools (Part II) are handled by another stream; this dossier records only where history bears on them. Indonesia (Part XII) and CBDCs (Part X) are out of scope here.
- Numbers come from datasets pulled in this run (Minneapolis Fed CPI 1800-2026; Bank of England *A Millennium of Macroeconomic Data* v3.1; BIS OTC derivatives bulk file, May 2026 release; FRED/Federal Reserve Z.1 series) or from opened pages. Pre-1913 price indices are reconstructions with narrow baskets; ancient "prices" are often administered equivalences. Each table states its limitation.
- Neutrality: the dossier treats commodity, credit and state accounts of money as partial lenses. It avoids treating gold as a perfect anchor or fiat as worthless; both claims fail on the evidence below.

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# 1. Ancient and pre-modern monetary systems

## 1.1 The barter debate: Menger versus Graeber

**The commodity/Mengerian account.** (interpretation: Menger 1892, "On the Origin of Money", *Economic Journal* 2: 239-255 [E, bibliographic]) Money emerges spontaneously because some goods are more saleable than others; traders accept the most saleable good indirectly, and convergence on it (typically precious metals) solves the double coincidence of wants. Adam Smith (WN I.4) gives a similar conjectural history. Its strength is theoretical: it explains why a medium of exchange can emerge without central design, and network-effect models of money (Kiyotaki and Wright 1989) formalise it. Its claim is a logic of emergence; Austrian defenders (e.g., Murphy 2011, Mises Institute [E]) argue it was never meant as a dated history of a "barter economy".

**The credit/anthropological critique.** (interpretation: Graeber 2011, *Debt: The First 5,000 Years*; Mitchell-Innes 1913) Graeber argues that the textbook sequence barter -> money -> credit is backwards: credit and debt accounting came first, coins later, and barter appears mainly among strangers or when money breaks down. He leans on Caroline Humphrey's survey, "Barter and Economic Disintegration", *Man* NS 20(1), 1985, pp. 48-72 [V, bibliographic], which concludes that "No example of a barter economy, pure and simple, has ever been described, let alone the emergence from it of money; all available ethnography suggests that there never has been such a thing" [E: quotation as reproduced by Graeber and by *The Atlantic* 2016; the scanned PDF had no text layer in this run]. Humphrey's own argument is narrower than often cited: barter occurs in specific conditions – low currency supply, atomised exchange, economies that already know money.

**Assessment.** (interpretation) The two positions answer different questions. Graeber is strongest on historical sequence and on the institutional (temple, palace, state) origin of units of account; Menger is strongest on why, once exchange widens beyond trust networks, a highly saleable commodity becomes a settlement asset. The evidence below supports a hybrid: administered units of account and credit inside institutions, coexisting with weighed silver used as a means of payment in trade, with coinage arriving late as a technology for certifying weight and fineness. Selgin's "myth of the myth of barter" point (Cato 2016 [E]) – absence of barter economies is consistent with money appearing quickly – is valid as logic but cannot by itself turn conjecture into history. Neither side supports the strong slogans "money is just debt" or "money is just a commodity".

## 1.2 Mesopotamian accounting and the shekel

- (fact) In Ur III Mesopotamia (c. 2100-2000 BC), "the shekel of silver was the basis of valuing all commodities in the Ur III accounts with the prescribed norm of 1 shekel = 300 sila3 ('liters') of barley" [V: CDLJ 2017:2, citing Englund 2012]. Hudson calls this a "bi-monetary price ratio that was the first step in administering prices" (Hudson 2004:112, as quoted in CDLJ 2017:2) [V].
- (fact) "Most discussions of interest rates in the Ur III period conclude that the usual interest rates were 33% for barley loans and 20% for silver loans" [V: CDLJ 2017:2]. The Laws of Eshnunna (s.18A, c. 1800 BC) express these as 36 barleycorns per shekel (20%) and 100 sila per gur (33 1/3%) [E: Roth 1995 translation via encyclopedia excerpt]. A shekel weighed about 8.3 g [E: Englund 2012; Banca d'Italia museum note].
- (empirical finding: CDLJ 2017:2 [V]) Actual loan-repayment ratios in Ur III texts range from 360 to 600 sila of barley per shekel, i.e., market-like variation around the administered norm. This is the core of a scholarly

dispute: Polanyi/Englund/Hudson read equivalences as administered transfer prices; Snell and others read them as prices responsive to supply and demand. Both elements appear present.

- (interpretation: Hudson 2000, "How Interest Rates Were Set, 2500 BC-1000 AD" [E]) Interest rates were set by arithmetic convenience of the sexagesimal system (1/60 per month = 20% per year) rather than by productivity; royal debt cancellations (misharum acts) periodically reset agrarian debts. This matters for the later Fairness debate (Part VII): the earliest monetary systems already embedded creditor-debtor politics.
- (fact) The British Museum's money gallery states that "In both Egypt and Mesopotamia, extensive and sophisticated accounting and payment systems existed before the introduction of coins" [V].

**Trust and enforcement before central banks (Mesopotamia).** Value was anchored by (i) the institutions that kept the ledgers and accepted payment (temples and palaces), (ii) law codes setting rates and remedies, (iii) the physical silver that could settle balances, and (iv) sovereign power to cancel debts. This is an early example of the state-and-credit layering that later reappears with central banks.

## 1.3 Lydian coinage and precious-metal standards

- (fact) "The first coins were probably produced and used in Lydia, now central Turkey. They were made from electrum ... made to particular weights" (British Museum, Room 68 guide; earliest coins dated "about 650 BC") [V]. The dating rests largely on foundation deposits of the Artemision at Ephesus, including a pot hoard of 19 electrum coins and 93 coins in total from the 1904-05 excavation [E: Konuk; Macquarie ACANS]; dates in the literature range from c. 650 to c. 600 BC.
- (fact) Croesus (561-546 BC) introduced separate gold and silver coinages – bimetallism [E: Austrian Academy of Sciences project page; BM guide says Lydia was "the first place to mint coins in gold and in silver separately" [V]].
- (empirical finding: Craddock and Cahill 2020, via ANU Classics Museum [E]) The electrum of early coins may have been a deliberately prepared alloy (around 55% gold), not natural river electrum – implying that the earliest coins already carried a standardised, authority-fixed composition.
- (interpretation) Coinage solved a verification problem: a stamped piece of known weight and fineness reduced the cost of assaying metal in every transaction. It also created a fiscal opportunity – the issuer could tariff coins above metal value (seigniorage) or reduce metal content. Whether early coinage served markets (Mengerian reading) or state payments to soldiers and fiscal obligations (chartalist reading, e.g., Kurke; Graeber's military-coinage-slavery complex [E]) is unresolved; both uses are documented in later Greek evidence.

## 1.4 Rome: the denarius and debasement with silver-content data

Rome offers the best long series of coin fineness in antiquity. Table 1 summarises the modern metallurgical evidence (full data and source per row in `data/r1/roman_silver_content.csv`).

**Table 1. Roman silver coin fineness (approximate % silver bullion)**

| Period (AD) | Ruler / issue | Fineness | Tag / source |
|---|---|---|---|
| before c.64 | Tiberius-Nero (pre-reform) | ~97-100 (pure bullion) | [V] Butcher and Ponting 2012 |
| c.64 | Nero reform (weight also cut to ~3.45 g) | ~80 | [V] same |
| 68 | Nero (late) | ~90 | [V] same |
| 69-82 | Otho to Titus | ~80 | [V] same |
| 82-85 | Domitian | pure (raised) | [V] same |
| 85-99 | Domitian-Nerva | ~90 | [V] same |
| 99-117 | Trajan | ~80 (79.6 mean, AD 116-117, n=5) | [V] same, table 2 |
| 118 | Hadrian (temporary) | 71.4 (n=9) | [V] same |
| 138-155 | Antoninus Pius | ~77-81 | [V] same, table 3 |

| Period (AD) | Ruler / issue | Fineness | Tag / source |
|---|---|---|---|
| 156-161 | Antoninus Pius after two-step debasement | ~70-73 | [V] same |
| 161-192 | Marcus Aurelius, Commodus | mostly ~66-72 | [V] same, tables 4-5 |
| c.194 | Septimius Severus | ~46 | [V] same, citing Gitler and Ponting 2003 |
| 215 | Caracalla's antoninianus (1.5x weight, tariffed at 2 denarii) | effective extra ~25% debasement | [V] ANS |
| 253-268 | Valerian, Gallienus | better coins ~40-50; poor Gallienus coins ~6 | [E] ANS Digital Library (older analyses) |
| 260-270 | antoninianus | silver "a trace element"; prices x10 in the decade (~25%/yr) | [V] ANS |
| 274 | Aurelian reform | ~5 commonly cited | [E] unverified in this run |
| 301 | Diocletian: nummus | billon ~5%; 1 aureus = 1,200 denarii (unit of account) | [V] ANS |

Limitation: fineness is only half of silver content; weight also fell and surviving coin weights are biased by corrosion (Butcher and Ponting estimate an average loss of about 0.3 g of copper in late-second-century denarii). Older surface analyses (Walker 1976-78) overstated fineness because Roman mints deliberately depleted copper at the surface ("surface enrichment") [V].

Findings from the series:

- (empirical finding: Butcher and Ponting 2012 [V]) The supposed accession-donative debasements under Marcus Aurelius (AD 161) and Commodus (AD 180), and the "reform" of Pertinax (AD 193), are "phantoms" produced by faulty older data. The pivotal second-century debasement is AD 155-157 under Antoninus Pius (78-80% -> ~70%), in a period of fiscal stability; the period AD 156/7-194 was one of "extreme stability" at the lower standard. Debasement was not a simple one-way slide: Nero (68) and Domitian (82) raised fineness again, possibly in response to public dissatisfaction.
- (interpretation: Butcher and Ponting 2012, p. 66 n.19 [V]) "the distinction between debasement and taxation is rather strained, since debasement *is* a form of increased taxation." This is the ancient prototype of seigniorage and the inflation tax discussed in Part V.
- (fact) The ANS dates inflation's appearance to the third century: the late-third-century collapse saw prices multiply about tenfold in AD 260-270; after Diocletian's reforms the gold price of the denarius unit went from 72,000 per pound (AD 301) to about 3 billion by the end of the fourth century (~12.5% per year compounded), while gold (the solidus, 1/72 pound from AD 312) became the system's anchor [V: ANS].
- (interpretation) Rome shows three mechanisms that recur in fiat systems: fiscal pressure (army pay) as the driver; Gresham-type displacement of good coin by bad; and a shift from a degraded circulating medium to a narrow, high-quality reserve asset (the solidus) used by elites – a distributional split between those paid in base coin and those holding gold.

## 1.5 Song China and the jiaozi

- (fact) Private firms in Sichuan – an iron-coin zone where 1,000 iron coins were heavy and of low value – began issuing paper bills (jiaozi) against deposits "at the end of the tenth century"; many became insolvent; "After efforts to regulate private paper bills failed, in 1023 the Song government took over the issue of jiaozi as state-issued paper currency" [V: von Glahn, Hoover Institution, 30 Apr 2025]. Around 1005 issue was restricted to sixteen merchant houses [E: Horesh 2012, *Provincial China* 4(1)].
- (fact) Under the state office set up in 1023 (first series issued 1024), each issue was capped at about 1.25-1.26 million strings (guan/min) with a metallic reserve of 360,000 strings, a 3% conversion fee, and a three-year life after which notes were exchanged [E: Horesh 2012 search excerpt; chinaknowledge.de; corroborating

each other].

- (fact) Success conditions identified by von Glahn [V]: the state "intervened to buy up excess bills using its reserve of silver bullion"; "most money taxes were collected half in coin and half in paper money"; and the state "tightly controlled the amount of paper bills it issued and withdrew old and worn bills by requiring the public to redeem them every three years." In the 13th century, war finance against northern states and the Mongols led to "exorbitant quantities" of notes and severe depreciation, yet paper "became even more deeply entrenched". The Yuan banned bronze coins and silver in trade; the Ming's paper money, undermined by "the failure to collect taxes in paper money", was "largely defunct" by the early fifteenth century, after which China shifted to bullion silver [V].
- (interpretation: von Glahn; chartalist reading) Tax receivability, quantity discipline and reserve operations – not convertibility as such – sustained the jiaozi. (interpretation: Austrian reading [E]) The eventual over-issue confirms that unconstrained state issue degrades money. Both readings fit the record; they disagree about whether the failure was avoidable within a fiat design.

## 1.6 Trade credit, bills of exchange and merchant banking

- (fact) Medieval Italian merchants developed the bill of exchange (lettera di cambio): a merchant in city A pays local currency to a banker, who draws an order on a correspondent in city B to pay a named party in B's currency at a future date [E: de Roover 1963; Kohn 1999]. The instrument combined remittance, foreign exchange and short-term credit, and allowed the Church's usury prohibition to be navigated because profit was embedded in exchange-rate differentials and time.
- (fact) Bills were cleared at the Champagne fairs (12th-13th centuries) and later at Lyon, Besancon/Piacenza and Antwerp, where reciprocal obligations were netted and only residuals settled in coin [E: established literature]. The BoE Millennium dataset includes a bills-of-exchange table (sheet A25a) for the UK's later period, showing the instrument's longevity into the 19th century [V: dataset sheet list].
- (fact) The Medici Bank (1397-1494), headquartered in Florence with branches in Rome, Venice, Geneva/Lyon, Bruges, London and elsewhere, operated as a partnership network handling papal finance, wool and cloth trade, and bills of exchange; its decline involved loans to princes (e.g., Edward IV) and weak branch governance [E: de Roover 1963, *The Rise and Decline of the Medici Bank, 1397-1494*].
- (interpretation: BIS QR 2017 citing Kohn 1999 [V]) "The use of a simple ledger in the Middle Ages allowed the transfer of credit on the books of a money changer – the precursor of deposit banks." Deposit transfer on a banker's book is the institutional ancestor of today's bank deposits: a private liability used as means of payment.

**How trust and enforcement worked before central banks (merchant era).** Repeated dealing within family and partnership networks, notarial records, merchant courts (lex mercatoria) and fair-court jurisdiction, reputational exclusion, and city-state backing of exchange banks. Their limits showed in periodic failures (the Bardi and Peruzzi in the 1340s after royal defaults [E]) – sovereign borrowers could default on private bankers without recourse, a recurrent problem that public banks and later central banks were partly designed to manage.

## 1.7 Early deposit institutions and the Bank of Amsterdam (1609)

- (fact) Public exchange/giro banks appeared in the Mediterranean (Barcelona's Taula de Canvi 1401; Venice's Banco della Piazza di Rialto 1587 [E]). The Bank of Amsterdam (Wisselbank) was "a public giro or payments bank owned by the municipality of Amsterdam", founded in 1609 amid coin debasement, with a mandate to "check all agio (of the current money) and confusion of coin" [V: Bolt, Frost, Shin and Wierts, BIS WP 1065, rev. Nov 2023]. It originally did not lend; bills of exchange above a threshold had to be settled in it [E: Quinn and Roberds 2007, AEA paper excerpt].
- (fact) In 1683 the Bank "ended the policy of redeemability of deposits into coin ... In this sense, the Bank started issuing fiat money", combined with a receipt system allowing coin depositors to repurchase their coin after about six months for a fee of 1/4% (silver) or 1/2% (gold). It conducted open-market purchases and sales of coin to keep the agio of bank money over current money in a 4-5% range [V: BIS WP 1065].
- (fact) Downfall: during the Fourth Anglo-Dutch War (1780-84) the Bank lent heavily and non-transparently to the Dutch East India Company (VOC); loans peaked at 7.8 million guilders in February 1783 (71% of assets); metal fell from 17.6 million guilders (1776) to 7.8 million (1783); backing fell from 97% to 28% by summer 1783; the agio turned negative in October 1790-February 1791 and fell to about -30% after the

1795 disclosure of the accounts; De Nederlandsche Bank was founded in 1814 and the Bank closed in 1820 [V: BIS WP 1065].

- (interpretation: Bolt et al. 2023 [V]) "Money is a social convention" sustained by network effects, but fiat issuers need fiscal backing: "The Bank of Amsterdam did not receive fiscal support from a sovereign with (adequate) power to tax." Their global-game model implies a discrete break point – trust in fiat money can collapse as a step function once equity is deeply negative and assets illiquid. This links Part III to Parts VI (trust) and X (CBDC governance).

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# 2. The emergence of modern banking

## 2.1 Goldsmith bankers and the logic of fractional reserves

- (fact) London goldsmiths accepted deposits of coin and plate in the mid-17th century and issued receipts and promissory notes; the British Museum dates the first UK banknotes to "goldsmith's notes in the 1660s, allowing the bearer to exchange the sum of the note for gold coin" [V]. Goldsmiths lent to merchants and heavily to the Crown; Charles II's Stop of the Exchequer (1672) suspended repayment of Crown debts held by goldsmiths and ruined several houses [E].
- (interpretation) The key innovation was that a *claim* on a banker circulated as a means of payment in place of the coin it promised. Once holders treat claims as money, a banker can issue claims beyond the coin held, as long as redemption demand is predictable. This is "fractional reserve" banking in its historical, convertible-claim form. It is distinct from the modern endogenous-money description (Part IV), where reserves are central bank money and the binding constraints are profitability, capital and liquidity regulation rather than a metal ratio.

**T-account 1. A goldsmith's balance sheet (illustrative numbers, GBP).**

Step A – deposits of coin, receipts issued (100% reserve):

| Assets | | Liabilities | |
|---|---|---|---|
| Coin in vault | 1,000 | Notes/receipts payable on demand | 1,000 |

Step B – the goldsmith lends by issuing new notes to a merchant borrower (no coin leaves):

| Assets | | Liabilities | |
|---|---|---|---|
| Coin in vault | 1,000 | Notes payable on demand | 1,800 |
| Loan to merchant | 800 | | |
| Total | 1,800 | Total | 1,800 |

Coin reserve ratio = 1,000 / 1,800 = 56%. New means of payment (800 of notes) was created by the act of lending. (fact, arithmetic) If holders present more than 1,000 of notes at once, the goldsmith cannot pay even if the loan is sound: a liquidity failure. If the merchant defaults on 300, liabilities exceed assets: an insolvency. The goldsmith in this sketch has no equity; real banks held some capital, and the absence or presence of capital is exactly what later regulation targets. (interpretation, contested) Rothbard (1983, *The Mystery of Banking*) and some Austrians regard issuing claims beyond reserves as inherently fraudulent when claims are presented as warehouse receipts; free-banking scholars (Selgin, White) and mainstream historians treat note-holders as informed creditors of a debtor institution. The legal history mostly followed the second view (deposits as loans to the bank), confirmed in English law in *Foley v Hill* (1848) [E].

## 2.2 Early deposit banks, banknotes and the Swedish precedent

- (fact) Stockholms Banco issued Europe's first banknotes (1661) partly because Swedish copper plate money was impractically heavy (an eight-daler plate weighed about 15 kg); it over-issued and failed, and the Riksens

Standers Bank (later Sveriges Riksbank) was founded in 1668 [V: British Museum guide on copper plate money and 1660s paper money; E for founding dates].

- (interpretation) Sweden reproduces the Chengdu pattern: heavy base-metal currency creates demand for paper claims; private issue over-expands; the state takes over or charters a public bank. Horesh (2012) draws this comparison explicitly ("From Chengdu to Stockholm") [V: abstract].

## 2.3 The Bank of England, 1694

- (fact) "The Bank of England began as a private bank that would act as a banker to the Government. It was primarily founded to fund the war effort against France." Founded 27 July 1694; opened 1 August 1694 in Mercers' Hall with 17 clerks and two gatekeepers; William and Mary were original stockholders [V: BoE history page]. Its founding loan to the Crown is conventionally given as GBP 1.2 million at 8% [E].
- (fact) Early business: government finance, recoinage, and conventional deposit banking; partially printed notes from 1725 for GBP 20 upwards [V: BoE history].
- (interpretation: North and Weingast 1989 [E]) The Bank was one element of the post-1688 "financial revolution" in which parliamentary control of taxation made sovereign debt credible; (interpretation: Ingham, Desan) it fused private banking credit with public debt, so that Bank notes were backed by claims on the state – a hybrid of credit and state theories of money. This hybrid structure, rather than metal alone, is the template for later central banks.
- (fact) Crises shaped its role: the South Sea Bubble (1720); the 1797 run when "The amount of gold held by the Bank dropped from GBP 16 million to just GBP 2 million" and the Privy Council ordered it to stop paying notes in gold – the Restriction Period 1797-1821, during which it issued GBP 1 and GBP 2 notes for the first time [V: BoE history].
- (fact, computed) UK prices during the Restriction (1797-1821) were extremely volatile: annual inflation s.d. 10.9 pp, but cumulative change only +7% because prices fell sharply after 1814 and convertibility was restored in 1821 at the old parity [V data: `regime_inflation_stats_us_uk.csv` from BoE Millennium dataset]. The Bullionist debate (Thornton 1802; the 1810 Bullion Report; Ricardo) over whether note over-issue caused the gold premium is the origin of the quantity-theory versus banking-school controversy [E].

## 2.4 The Bank Charter Act 1844

- (fact) "The Bank Charter Act of 1844 gave the Bank of England a range of new powers and formalised the issuance of banknotes in the UK. This Act ... placed restrictions on any banks, companies or persons in England and Wales that issued their own banknotes, and stopped any new banks from starting to issue notes" [V: BoE history; statute at legislation.gov.uk 7 & 8 Vict c.32]. It separated the Bank into an Issue Department and a Banking Department; notes beyond a fixed fiduciary issue (GBP 14 million at the outset) had to be backed by gold [E].
- (interpretation) The Act embodied the Currency School (Overstone, Norman, Torrens): only notes are money and their quantity should vary one-for-one with gold, mimicking a pure metallic currency. The Banking School (Tooke, Fullarton) replied that deposits and bills of exchange also function as money and that the quantity of credit is demand-determined (law of reflux) – an early version of endogenous-money arguments (Part IV). (empirical record [E]) The Act had to be suspended in the crises of 1847, 1857 and 1866 to allow the Bank to lend beyond its gold limits – evidence that a rigid quantity rule for notes did not control deposit-based credit and could not withstand panics. Note-issue monopoly did not prevent crises because the system's main money had become bank deposits and bills.

**T-account 2. The 1844 structure (stylised, GBP mn).**

Issue Department:

| Assets | | Liabilities | |
|---|---|---|---|
| Government securities (fiduciary issue) | 14 | Notes issued | 34 |
| Gold coin and bullion | 20 | | |

Banking Department:

| Assets | | Liabilities | |
|---|---|---|---|
| Notes held as reserve ("the reserve") | 8 | Bankers' deposits | 10 |
| Bills discounted and advances | 20 | Other deposits (public, private) | 18 |
| Government securities | 6 | Capital and rest | 6 |

Every note beyond 14 required gold. In a panic, commercial banks drew on their deposits at the Banking Department, which could pay only from its note reserve; when that reserve ran low the government issued a "letter" indemnifying the Bank for exceeding the fiduciary limit. (illustrative numbers; the structure, not the figures, is the point)

## 2.5 Bagehot and the lender of last resort

- (fact) Overend Gurney, "the largest discount house of the time", suspended payments on 10 May 1866 after the Bank of England refused assistance on 9 May "on the basis of the broker's insolvency"; the ensuing debate "helped shape the Bank of England as a lender of last resort" [V: BoE history].
- (fact, quotation) Walter Bagehot, *Lombard Street* (1873), Ch. II: "the holders ... of the final Bank reserve must lend freely. Very large loans at very high rates are the best remedy for the worst malady of the money market when a foreign drain is added to a domestic drain." Ch. VII sets two rules: "First. That these loans should only be made at a very high rate of interest. This will operate as a heavy fine on unreasonable timidity ... " and "Secondly. That at this rate these advances should be made on all good banking securities, and as largely as the public ask for them" [V: Project Gutenberg #4359 text].
- (interpretation) The doctrine is usually summarised as "lend freely, at a penalty rate, against good collateral, to solvent institutions". The "penalty rate" and "solvent" components are interpretive glosses debated by modern scholars (Goodhart 1999; Bordo 1990 [E]); Bagehot's own text emphasises high rates and good security. Thornton (1802) anticipated the argument [E].
- (interpretation: Kindleberger, Minsky) LOLR transforms bank deposits into near-sovereign money in a crisis, which stabilises payments but creates moral hazard. (interpretation: Austrian school) LOLR and later deposit insurance socialise losses and encourage credit expansion. (empirical finding: Bordo 1990; Jalil 2015 [E]) Panics became rarer in Britain after the Bank adopted LOLR behaviour (after 1866) than in the US, which lacked a central bank until 1913.

## 2.6 The United States: from free banking to the Federal Reserve

- (fact) Before the Civil War, US currency "consisted of coins and notes issued by state-chartered banks ... notes issued by banks in one state were often accepted only at a discount, if at all, in other states" [V: FRH, National Banking Acts]. The First (1791-1811) and Second (1816-1836) Banks of the United States had federal charters and branches but were not renewed [V/E: FRH references].
- (fact) Free banking (1837-1863): states such as Michigan (1837) and New York (1838) allowed entry for any bank posting state bonds as backing for notes [E]. (empirical finding: Rolnick and Weber 1984 [E]) Many free-bank failures were driven by falls in the value of the state bonds backing notes, not by fraudulent "wildcat" banking per se; (empirical finding: Gorton 1996 [E]) note discounts priced issuer risk. The episode is contested: free-banking advocates read it as reasonably functional private money; critics stress non-par notes and transaction costs.
- (fact) Civil War: the Legal Tender Act of 1862 authorized USD 150 million of non-interest-bearing notes ("greenbacks"), not convertible into gold or silver but usable for taxes – "the first real paper money ever issued by the United States government" (Studenski and Krooss, quoted by FRH) [V].
- (fact) National Currency Act, 25 Feb 1863 (Office of the Comptroller of the Currency; 2% tax on state notes); National Bank Act, 3 June 1864 (capital minima; reserve requirements of 25% for central reserve city and reserve city banks and 15% for country banks; note issue up to USD 500,000 per bank backed by Treasury bonds; single-office restriction); 683 national banks by end-1864; an 1865 10% tax cut state-bank note circulation from USD 143 million (1865) to USD 4 million (1867) [V: FRH]. (fact) Reserve requirements applied to notes and deposits until 1874, then to deposits only [V: FRH footnote].

- (interpretation) The Acts nationalised the *note* while leaving the *deposit* private and fragmented. Pyramiding of reserves in New York correspondent balances and inelastic bond-backed currency produced seasonal stringency and recurrent panics (1873, 1884, 1890, 1893, 1907) [V: FRH lists Gilded Age panics; E for specific years other than 1907].
- (fact) Panic of 1907: trust companies held cash reserves around 5% of deposits versus 25% for national banks; Knickerbocker Trust suspended on 22 October 1907 after nearly USD 8 million of withdrawals; the call-money rate jumped from 9.5% to 70% and then 100%; the New York Clearing House issued loan certificates (26 October) and banks restricted cash convertibility; New York trusts lost over 36% of deposits between 22 August and 19 December 1907; "Industrial output fell 17 percent in 1908, and real GNP fell by 12 percent" [V: FRH, Moen and Tallman]. (interpretation: Moen and Tallman) The trusts were the "shadow banks" of 1907, outside the private lender of last resort – a direct parallel to 2007-09.
- (fact) Aldrich-Vreeland Act (30 May 1908) created emergency currency and the National Monetary Commission; the secret Jekyll Island meeting (Nov 1910) drafted the Aldrich plan; Wilson signed the Federal Reserve Act on 23 December 1913, creating 8-12 regional Reserve Banks under a Board of presidential appointees with ten-year staggered terms [V: FRH]. The Fed began operating in 1914-15 with a wire-transfer settlement system (Gold Settlement Fund) [V: FRH Fedwire].
- (interpretation) The Fed's founding compromise – private member banks owning regional Reserve Banks, public Board oversight – encodes the public-private hybrid that recurs in the Fairness debate (Part VII). Conspiracy readings of Jekyll Island are not supported: the meeting's secrecy is documented, but its plan was publicly debated, rejected and replaced by a different statute.

### 2.7 How private claims became national money: a synthesis

| Stage | Example | Private claim | Public integration step | Residual fragility |
|---|---|---|---|---|
| 1. Private notes against coin | Goldsmiths 1660s; Sichuan c.1000 | note promises coin | none / licensing | over-issue, runs |
| 2. Chartered public or semi-public bank | Bank of Amsterdam 1609; BoE 1694; Riksbank 1668 | bank money / notes | charter, city/crown ownership, link to public debt | sovereign borrowing from the bank |
| 3. Note monopoly | Bank Charter Act 1844; US 10% tax 1865 | deposits remain private | notes centralised | deposit runs, inelasticity |
| 4. Lender of last resort | BoE after 1866; Fed 1913 | deposits convertible into central bank money | central bank backs liquidity | moral hazard |
| 5. Deposit insurance and supervision | US FDIC 1933 [E] | deposits | state guarantee of small deposits | regulatory arbitrage to shadow banks |
| 6. End of metal convertibility | 1931 UK; 1933-34 US domestic; 1971 international | all money is credit money | money anchored by policy and fiscal capacity | inflation and credibility risk |

(interpretation) At every stage, the system kept private banks as the main creators of money for daily use (deposits) while moving the ultimate settlement asset and the crisis backstop into public hands. This is the historical root of the modern two-tier structure analysed in Part IV.

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## 3. The gold standard

### 3.1 From bimetallism to the classical gold standard

- (fact) Britain was on a de facto gold standard from Newton's 1717 rating of the guinea, which undervalued silver; the Coinage Act of 1816 and resumption in 1821 made it formal [E]. Most other countries were

bimetallic or on silver until the 1870s. The switch cascaded after Germany used the French war indemnity to adopt gold (1871-73); the Latin Monetary Union (France, Belgium, Italy, Switzerland; 1865) suspended free coinage of silver in 1873-78; the US Coinage Act of 1873 ended free silver coinage ("Crime of '73" to its critics) and the US resumed specie convertibility in 1879, legally adopting gold by the Gold Standard Act of 1900 [E: Eichengreen 2019 *Globalizing Capital*; Friedman 1990 "The Crime of 1873"].

- (interpretation: Eichengreen; Flandreau) Adoption was driven by network effects (Britain was the dominant trading and lending centre), silver discoveries that made bimetallic ratios unstable, and the prestige of sterling. The classical gold standard is usually dated c. 1880-1914 and was never universal: much of Asia and Latin America remained on silver or inconvertible paper for long periods.

## 3.2 How it was supposed to work: price-specie flow and the "rules of the game"

**Price-specie flow (Hume 1752) [E].** (interpretation, classical theory) A country with a trade deficit loses gold; its money supply contracts and prices fall; surplus countries gain gold and prices rise; relative price changes restore balance automatically. Numerical sketch: if country A runs a deficit of 2% of GDP settled in gold and its money supply is 40% of GDP with a 50% gold cover, its gold stock falls by $2/20 = 10\%$ in a year unless offset; with a fixed cover ratio, money falls 10% and (under a strict quantity theory with fixed velocity and output) the price level falls 10%. Real economies did not adjust this way: output and employment absorbed much of the shock.

**The "rules of the game".** (interpretation: Keynes 1925 coined the phrase [E]) Central banks were expected to reinforce gold flows – raising discount rates and contracting domestic credit when gold left, and the reverse when it arrived. (empirical finding: Bloomfield 1959, *Monetary Policy under the International Gold Standard* [E]) Central banks frequently violated the rules: domestic and foreign assets moved in opposite directions in a majority of years for most central banks studied, i.e., they sterilised gold flows. (empirical finding: McCloskey and Zecher 1976 [E]) Prices of traded goods were arbitraged internationally (law of one price), so adjustment worked more through capital flows and interest rates than through Hume's national price levels.

**What actually made it work.** (interpretation: Eichengreen 1992; Bordo and Kydland 1995 [E])

1. *Credibility*: markets believed parities would be restored after suspensions (the "contingent rule" – suspension in war, resumption at the old parity afterwards), so capital flows were stabilising rather than speculative.
2. *Bank Rate and London*: the Bank of England operated on a thin gold reserve and moved Bank Rate frequently; short-term sterling bills attracted capital from abroad. UK Bank Rate (BoE dataset, end-period) ranged mostly between 2% and 6% in 1880-1913 [V data].
3. *Cooperation in crises*: e.g., the Banque de France lending gold to the Bank of England in the 1890 Baring crisis [E]; and in 1907 gold inflows to New York ended the panic (Gorton and Tallman, cited by FRH) [V].
4. *Political economy*: limited suffrage and weak labour organisation made deflationary adjustment politically tolerable (interpretation: Eichengreen 1992; Polanyi 1944).

## 3.3 Price stability in the long run, volatility in the short run

**Table 2. Inflation and output by monetary regime, US and UK (computed for this project).**

Source: `data/r1/regime_inflation_stats_us_uk.csv`, built from Minneapolis Fed CPI 1800-2025 [V] and BoE *Millennium of Macroeconomic Data* v3.1 sheet A1 (CPI and real GDP growth, to 2016) [V]. Inflation = annual change in log CPI (x100).

| Country | Regime | Years | Mean inflation % | S.d. inflation pp | Deflation years | Cumulative price change % | Mean real GDP growth % | S.d. GDP growth pp |
|---|---|---|---|---|---|---|---|---|
| US | 1801-1861 specie/bimetallic | 61 | -1.04 | 5.41 | 25 | -47.1 | n/a | n/a |
| US | 1862-1878 greenback (inconvertible) | 17 | 0.42 | 9.33 | 10 | +7.4 | n/a | n/a |

| Country | Regime | Years | Mean inflation % | S.d. inflation pp | Deflation years | Cumulative price change % | Mean real GDP growth % | S.d. GDP growth pp |
|---|---|---|---|---|---|---|---|---|
| US | 1879-1913 classical gold standard | 35 | 0.07 | 2.22 | 6 | +2.4 | n/a | n/a |
| US | 1914-1945 wars and interwar | 32 | 1.87 | 7.01 | 10 | +82.2 | n/a | n/a |
| US | 1946-1971 Bretton Woods | 26 | 3.12 | 3.21 | 2 | +125.0 | n/a | n/a |
| US | 1972-2025 fiat/floating | 54 | 3.84 | 2.70 | 1 | +695.0 | n/a | n/a |
| US | 1983-2019 post-Volcker, pre-Covid | 37 | 2.63 | 1.16 | 1 | +164.9 | n/a | n/a |
| UK | 1700-1796 specie | 97 | 0.29 | 7.62 | 45 | +32.2 | 0.94 | 4.82 |
| UK | 1797-1821 Restriction (inconvertible) | 25 | 0.27 | 10.90 | 12 | +7.0 | 1.52 | 4.09 |
| UK | 1822-1913 gold standard | 92 | -0.01 | 4.32 | 42 | -1.1 | 2.14 | 2.53 |
| UK | 1914-1945 wars and interwar | 32 | 3.14 | 8.31 | 12 | +173.6 | 1.43 | 4.65 |
| UK | 1946-1971 Bretton Woods | 26 | 4.37 | 2.41 | 0 | +211.5 | 2.87 | 2.01 |
| UK | 1972-2016 fiat/floating | 45 | 5.00 | 4.64 | 0 | +849.9 | 2.21 | 2.16 |
| UK | 1993-2016 inflation targeting | 24 | 1.98 | 0.97 | 0 | +60.7 | 2.20 | 1.67 |

Limitations: (1) pre-1913 US indices are integer-rounded reconstructions with narrow baskets (Vermont farm purchases; Hoover; Rees), so their measured volatility partly reflects noise; (2) pre-1850 UK output data are reconstructions; (3) regimes coincide with wars, industrialisation and changes in economic structure (agriculture's weather shocks inflate early volatility), so the table describes associations, not causal effects of the monetary regime; (4) the US had no central bank 1836-1913.

Readings (empirical findings from the table, with interpretation flagged):

- (fact, computed) The gold standard delivered near-zero average inflation and near-unchanged price levels over decades: UK prices were 1.6 times their 1700 level in 1914, whereas they rose about 79-fold from 1914 to 2016; US prices fell 42% from 1800 to 1913 and rose about 33-fold from 1913 to 2025 (Minneapolis Fed index).
- (fact, computed) Year-to-year price volatility under gold was *higher* than under post-1983 US policy or UK inflation targeting (UK s.d. 4.3 pp versus 1.0 pp; US 2.2 versus 1.2), and deflation was frequent (UK: 42 of

92 years). (interpretation) Long-run predictability of the price *level* coexisted with short-run unpredictability of *inflation*. For long-term creditors (bondholders in consols), the gold standard was excellent; for indebted farmers facing falling crop prices, it was harsh.
- (fact, computed) UK real GDP growth volatility was higher under gold (2.5 pp) than under Bretton Woods (2.0) or inflation targeting (1.7), and negative-growth years were more frequent (18 of 92). Confounded by structural change, but inconsistent with the claim that gold produced macroeconomic stability in the output sense.

### 3.4 The long deflation of 1873-1896

- (fact, computed) US CPI fell 31% between 1873 and 1896 and UK CPI fell 23% (Minneapolis Fed; BoE dataset) [V data].
- (interpretation: Friedman and Schwartz 1963; Bordo, Landon-Lane and Redish 2004/2009 [E]) Much of this was "good deflation": output grew strongly as railways, steamships and agricultural expansion lowered costs, while the world gold stock grew slowly relative to output and to the demand for gold as more countries adopted it. Bordo et al. find that supply shocks dominated and deflation had little negative effect on output growth in the US, UK and Germany.
- (interpretation: debtor-politics reading [E]) Falling prices raised the real burden of nominal debts, fuelling the US Populist and Free Silver movements (W. J. Bryan's "Cross of Gold" speech, 1896). (counterfactual: Friedman 1990 [E]) Had the US stayed bimetallic after 1873, the price fall might have been milder. Deflation ended after 1896 as South African (Witwatersrand) and Klondike gold and the cyanide process expanded supply – i.e., the price level under gold depended on mining discoveries and technology, an exogenous and arbitrary anchor.

### 3.5 Financial crises under gold

- (fact) Major crises occurred during the classical gold standard: 1873 (US, Central Europe), 1884 (US), 1890 (Baring/Argentina), 1893 (US, Australia), 1907 (US, global) [E for dates other than 1907 [V]]. (empirical finding: Bordo, Eichengreen, Klingebiel and Martinez-Peria 2001 [E]) Crisis frequency in 1880-1913 was lower than after 1973 but higher than in 1945-71 (when capital controls and repressed finance prevailed).
- (fact) In 1907 the US was on gold and the panic still produced a 12% fall in real GNP (FRH) [V]; it ended partly through gold imports and clearing-house loan certificates, not through any automatic gold mechanism.
- (interpretation) Gold convertibility constrained the note issue but not deposit creation by banks, nor maturity transformation, nor runs. It also made the central bank's crisis response conditional on its gold reserve – in a combined external and internal drain (Bagehot's case) the central bank had to raise rates sharply exactly when domestic credit was collapsing.

### 3.6 Core and periphery

- (empirical finding: Triffin 1964 "The Evolution of the International Monetary System"; Bordo and Schwartz 1996; Eichengreen 2019 [E]) The adjustment burden fell disproportionately on periphery countries (Latin America, Southern Europe, primary producers), which faced capital-flow reversals, terms-of-trade swings and frequent suspensions (Argentina, Brazil, Italy, Portugal and others left and re-joined). Capital flowed from London to the periphery in booms and returned in busts – an early version of the "global financial cycle" (Rey 2013) discussed in Part IX.
- (interpretation) The gold standard was therefore hierarchical: sterling and London's bill market sat at the centre; gold was the nominal anchor, but much international settlement was in sterling credit.

### 3.7 The interwar gold-exchange standard and "Golden Fetters"

- (fact) Gold convertibility was suspended in World War I; Britain returned at the pre-war parity in April 1925; the UK suspended gold again in September 1931 after a run that "meant that the Bank of England lost much of its reserves" [V: BoE history]. The US ended domestic gold convertibility in 1933-34 ("For domestic purposes, the US dollar was separated from gold in 1934" [V: FRH Great Inflation footnote]); the Gold Reserve Act of 1934 set the price at $35 per ounce [E].
- (fact, computed) US CPI fell 24.6% from 1929 to 1933 (Minneapolis Fed) [V data].

- (empirical finding: Eichengreen and Sachs 1985, *Journal of Economic History* 45(4): 925-946 [E: Cambridge abstract via search excerpt]) "depreciation benefited the initiating countries", there was no presumption that it was beggar-thy-neighbour, and coordinated depreciation "would have hastened recovery from the Great Depression."
- (empirical finding: Bernanke 1995, *JMCB* 27(1), via FRASER full text [E: excerpt]) "To an overwhelming degree, the evidence shows that countries that left the gold standard recovered from the Depression more quickly than countries that remained on gold. Indeed, no country exhibited significant economic recovery while remaining on the gold standard."
- (interpretation: Eichengreen 1992, *Golden Fetters: The Gold Standard and the Great Depression, 1919-1939*) The interwar system lacked the classical system's credibility and cooperation: wartime debts and reparations, the political rise of labour, gold maldistribution (France and the US accumulated and sterilised gold), and the use of foreign-exchange reserves (gold-exchange standard) made the system fragile. When the US tightened in 1928-29, the gold standard transmitted deflation worldwide; central banks defending parities raised rates into collapsing economies; banking crises spread (Austria's Creditanstalt, May 1931 [E]). Leaving gold removed the constraint and allowed reflation.
- (interpretation: Friedman and Schwartz 1963) The Fed's failure to prevent the collapse of the US money stock (bank failures 1930-33) was the primary cause of the US Depression; Eichengreen's gold-standard view and Friedman-Schwartz's monetarist view are complementary on the role of monetary contraction but differ on whether the Fed could have acted while defending gold (Eichengreen: gold constrained it; Friedman-Schwartz/Hsieh-Romer 2006 [E]: the Fed had ample gold and could have expanded in 1932).
- (interpretation: Austrian school; Rothbard 1963 *America's Great Depression*; Hayek [E]) The Depression was the bust following a credit-fuelled boom made possible by the managed gold-exchange standard and Fed easing in the 1920s; the cure was liquidation. Empirical support is weaker: recoveries began after exits from gold and monetary expansion, not after liquidation.

## 3.8 Does gold backing prevent monetary instability, or change its mechanism?

Evidence summary for the BRIEF's question:

| Dimension | Gold standard record | Evidence |
|---|---|---|
| Long-run price level | Very stable over decades (UK 1822-1913 cum. -1.1%) | Table 2 [V data] |
| Short-run inflation volatility | High; frequent deflation (UK s.d. 4.3 pp) | Table 2 |
| Output volatility | Higher than post-war regimes (UK s.d. 2.5 pp vs 1.7 under IT) | Table 2 (confounded) |
| Banking panics | Frequent (1873, 1890, 1893, 1907) | FRH; literature |
| Anchor determinant | Gold discoveries and mining technology; demand for gold by adopting states | 1873-96 vs post-1896 |
| Fiscal discipline | Strong in peace, suspended in war (UK 1797, US 1862, all 1914) | BoE; FRH |
| Crisis response | Constrained by reserves; LOLR possible but rationed | Bagehot; 1847/57/66 suspensions |
| Distribution | Favoured creditors and holders of nominal assets in deflations; burdened debtors and farmers | 1873-96 politics |
| International adjustment | Asymmetric burden on deficit and periphery countries; output adjustment | Eichengreen; Triffin |

**Conclusion (interpretation, held with moderate confidence).** Gold backing *changes the mechanism* of instability rather than preventing it. It largely removes chronic inflation in peacetime and constrains sovereign monetary financing, which is a real benefit for long-horizon contracts. But it relocates instability into (a)

deflationary adjustment through output and employment, (b) banking panics in a credit system that gold does not control, (c) dependence of the price level on mining luck, and (d) suspension exactly when shocks are largest (wars, depressions). Its success in 1880-1914 depended on institutions – credibility of resumption, central bank cooperation, London's credit network, limited democratic pressure – that were not reproducible after 1918. This undercuts both the claim that gold is a perfect solution and the opposite claim that the gold standard was mere folly: it was a credible commitment technology suited to a particular political order.

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# 4. Bretton Woods (1944-1973)

## 4.1 The 1944 conference: Keynes versus White

- (fact) The United Nations Monetary and Financial Conference met in July 1944 at the Mount Washington Hotel, Bretton Woods, New Hampshire: 730 delegates from 44 allied nations [V: FRH, Ghizoni, "Creation of the Bretton Woods System"; State Dept Office of the Historian]. Its two creations were the International Monetary Fund and the International Bank for Reconstruction and Development (World Bank). Congress passed the Bretton Woods Agreements Act in July 1945 [V: Office of the Historian]; the IMF Articles entered into force on 27 December 1945, with 29 initial signatory members [V: FRH; IMF Articles front matter E].
- (fact) Both plans assumed fixed but adjustable exchange rates and financing for temporary payments deficits so that countries would not resort to the competitive devaluations and trade restrictions of the 1930s [V: Office of the Historian]. They differed in scale and in who bore adjustment:
  - **Keynes's International Clearing Union**: an overdraft-style clearing bank with its own unit, the *bancor*; large resources; symmetric pressure on surplus as well as deficit countries (charges on persistent creditor balances). [V: FRH, names and outline]
  - **White's Stabilization Fund**: a contributory fund of members' gold and currencies with quotas and voting by quota, much smaller resources, and adjustment pressure mainly on deficit countries; the dollar (convertible into gold) at the centre. [V: FRH]
  - Resources: the FRH essay states Keynes's plan at "$26 million", White's at "$5 million", and the final figure as "$8.5 million". The orders of magnitude in the standard literature are billions (unverified in this run; treat the FRH figures as likely typographical and do not cite the numbers without checking Steil 2013 or the IMF's official history).
  - The final agreement followed White's structure, with a "scarce currency" clause as a concession to Keynes allowing members to restrict exchange with a persistent surplus country whose currency the Fund ran short of [V: FRH]. It was never invoked [E].
- (fact) Par values: members declared a par in gold or dollars and kept spot rates within 1% of parity; the US fixed the dollar at $35 per troy ounce for official holders; parities could change only to correct a "fundamental disequilibrium" (Article IV, never precisely defined) [V: FRH for 1% band and $35; E for article wording].
- (interpretation: Steil 2013; Skidelsky [E]) The outcome reflected US creditor power: the US held, by Ghizoni's account, about three-quarters of the world's official gold reserves at the outset [V: FRH]. (interpretation: Keynesian/post-Keynesian critics) The loss of symmetric surplus-country adjustment is a design flaw that resurfaces in later imbalances (Japan 1980s, China 2000s, euro area 2010s).

## 4.2 IMF and World Bank roles

- (fact) The IMF oversaw the par value system, provided short-term balance-of-payments financing from quota subscriptions (25% in gold, the rest in members' currencies [E]) and served as a forum for consultation; the IBRD financed reconstruction and later development [V: Office of the Historian]. The GATT (1947) handled trade [V: same].
- (fact) The system became fully operational only in 1958, when Western European currencies achieved current-account convertibility [V: FRH, Hetzel]. Before then Europe relied on Marshall Plan aid (1948-52) and the European Payments Union (1950-58) [E], a multilateral clearing arrangement closer in spirit to Keynes's plan.
- (fact) The IMF's First Amendment (effective 28 July 1969) created the Special Drawing Right (SDR) as a supplementary reserve asset; Triffin had first proposed centralising reserves in an enlarged Fund in March

1959 [V: IMF *F&D* June 1978 chronology].

## 4.3 The Triffin dilemma

- (interpretation: Triffin 1960, *Gold and the Dollar Crisis*) If the world's growing demand for reserves is supplied by US dollar liabilities, the US must run balance-of-payments deficits; as foreign dollar claims grow relative to US gold, confidence that the dollar can be converted at $35 erodes; if the US stops running deficits to protect confidence, world liquidity becomes inadequate and deflationary. Either way the system is unstable.
- (fact) "By 1961, outstanding US dollar claims held abroad exceeded US gold holdings" [V: FRH, Humpage, Smithsonian essay]; FRH's Great Inflation essay states the same logic: "Eventually, the supply of dollar reserves held abroad exceeded the US stock of gold, implying that the United States could not maintain complete convertibility at the existing price of gold" [V].

**T-account 3. The Triffin mechanism (illustrative, USD bn; not historical values).**

United States, official external position vis-a-vis foreign central banks:

| Assets | | Liabilities | |
|---|---|---|---|
| Monetary gold (at $35/oz) | 20 | Dollar claims held by foreign official institutions (deposits, T-bills) | 12 |
| | | Net external position (official) | +8 |

After a decade of payments deficits financed by foreigners accumulating dollar reserves, with some gold conversion:

| Assets | | Liabilities | |
|---|---|---|---|
| Monetary gold | 14 | Dollar claims held by foreign official institutions | 24 |
| | | Net official position | -10 |

Coverage of official claims by gold falls from 167% to 58%. Once below 100%, every holder knows that not all can convert; the incentive is to convert first (a run on the gold window) – the same coordination logic as a bank run (and as the Bank of Amsterdam's break point, Section 1.7). Foreign central banks that hold back do so out of cooperation or political leverage (e.g., Germany's 1967 undertaking not to convert, the "Blessing letter" [E]), not because of the system's design.

(interpretation, opposing schools)

- Triffin and Keynesian internationalists: the problem is structural – a national currency cannot be the world's reserve asset without a confidence-liquidity trade-off; the solution is a supranational reserve asset (bancor, SDR).
- Monetarist and "discipline" view (e.g., Meltzer, Bordo; Hetzel [V partly]): the system could have survived longer had US monetary policy not become expansionary after 1965 (Great Society spending, Vietnam, Fed accommodation); Triffin's dilemma made it fragile but US inflation broke it.
- French critique (de Gaulle and Rueff, 1965 [E]): the dollar enjoyed an "exorbitant privilege" (Giscard d'Estaing's phrase [E]): the US could finance deficits by issuing its own liabilities; Rueff advocated a return to a full gold standard with a higher gold price. France converted dollars into gold in the mid-1960s and withdrew from the Gold Pool.
- (interpretation: "Bretton Woods II" view, Dooley, Folkerts-Landau and Garber 2003 [E]) Dollar-reserve accumulation by export-oriented economies can be a stable bargain for long periods; this is relevant to the post-2000 era rather than to 1960s gold convertibility.

## 4.4 Defending the system: Gold Pool, swaps, controls

- (fact) The London gold price reached $40 on 20 October 1960 [V: FRH Ghizoni, Nixon essay]. The US Treasury's Exchange Stabilization Fund began intervening in March 1961 and the Fed opened reciprocal currency (swap) lines from March 1962 – the first with the Bank of France – reaching nine central banks and $900 million by end-1962 [V: FRH].
- (fact) The London Gold Pool, formed 1 November 1961 by eight central banks, sold gold to hold the London price near $35; France withdrew in 1967 [E for year; FRH says France withdrew], and the Pool collapsed in March 1968, replaced by a two-tier market in which official transactions stayed at $35 while the private price floated [V: FRH]. Hetzel: after 1968 the system effectively became a "dollar standard" [V].
- (fact) The US also used capital controls – the Interest Equalization Tax (1963) and voluntary, then mandatory, restraints on foreign lending and direct investment [E; FRH Hetzel notes US capital controls [V]] – which helped push dollar banking offshore into the eurodollar market (Section 5.3).
- (interpretation) Each measure substituted cooperation, controls or two-tier pricing for the automatic convertibility promised in 1944. The system increasingly depended on the willingness of surplus central banks to hold dollars rather than gold.

## 4.5 The Nixon Shock, 15 August 1971

- (fact) After a meeting at Camp David on 13-15 August 1971 (participants included Fed Chairman Arthur Burns, Treasury Secretary John Connally and Under Secretary Paul Volcker), President Nixon announced on the evening of 15 August the suspension of the dollar's convertibility into gold for foreign official holders ("closing the gold window"), a 90-day freeze on wages and prices, and a 10% import surcharge [V: FRH, Ghizoni]. US inflation was about 5.4% and unemployment about 6% after the 1969-70 recession [V: FRH].
- (fact) Germany and the Netherlands had already floated in May 1971 [V: IMF chronology]. The IMF records that the US decision "initiated a period of serious exchange market disorder" [V].
- (interpretation) The legal anchor of the world monetary system to gold ended for official holders; domestic convertibility had already ended in 1933-34. From this point the dollar's value rested on US monetary policy, fiscal capacity and the depth of US financial markets – what FRH's Bryan calls the first time that "most of the monies of the industrialized world were on an irredeemable paper money standard" outside crisis periods [V].

## 4.6 The Smithsonian Agreement (December 1971)

- (fact) On 18 December 1971 the Group of Ten agreed at the Smithsonian Institution in Washington a realignment: the official gold price rose from $35 to $38 (a dollar devaluation of about 8.5% against gold) and other currencies revalued so that the dollar fell about 10.7% on average against key currencies; margins widened from 1% to 2.25% [V: FRH Humpage for $38 and 10.7%; IMF chronology for "central rates and wider margins" V; 2.25% E]. Nixon called it "the most significant monetary agreement in the history of the world" [E].
- (fact) The dollar remained inconvertible. The private gold price rose to about $60 by mid-1972 and $90 by early 1973 [V: Humpage]. On 12 February 1973 the US devalued again, to $42 per ounce (about 10%); major exchange markets in Europe and Japan closed from 2 to 19 March 1973; the EEC introduced a joint float against the dollar; "within a month" nearly all major currencies floated [V: IMF chronology; FRH Humpage].

## 4.7 The Jamaica Accords (1976) and the Second Amendment (1978)

- (fact) The Committee of Twenty abandoned full reform in January 1974 after the oil shock and published an *Outline of Reform* in June 1974; the Interim Committee was established in October 1974 [V: IMF chronology]. At Rambouillet (November 1975) France and the US reached a compromise on a new Article IV [V].
- (fact) At the Interim Committee meeting in Kingston, Jamaica, 7-8 January 1976, members agreed that "floating would be legalized; the Fund would be responsible for surveillance of members' exchange rate arrangements"; quotas would rise (total from SDR 29.2 billion to SDR 39 billion); the Fund would sell one-sixth of its gold (25 million ounces) by auction for a Trust Fund for developing countries and restitute

another sixth to members [V: IMF *F&D* 1978]. The Board of Governors approved the Second Amendment on 30 April 1976; it entered into force on 1 April 1978 [V], ending, in the IMF historian's words, the "extraordinary period ... that began on August 15, 1971" [E: IMF History 1972-1978 excerpt].
- (fact) The Second Amendment established members' right to choose their exchange arrangements, abolished the official price of gold and aimed to make the SDR the principal reserve asset [E: IMF publications excerpts]. That last aim was not realised: the dollar remained dominant (Section 5).

### 4.8 Assessment of Bretton Woods

- (fact, computed, Table 2) Bretton Woods years combined moderate average inflation (US 3.1%, UK 4.4%), few deflation years, and the lowest output volatility before inflation targeting (UK real GDP growth s.d. 2.0 pp). Financial crises were rare in 1945-71 (Bordo et al. 2001 [E]), under widespread capital controls and financial repression (interest ceilings such as Regulation Q, directed credit).
- (interpretation) Its successes came from a combination that cannot easily be reassembled: capital controls, a dominant creditor (US), reconstruction growth, and domestic financial repression. Its failure combined Triffin's structural problem, asymmetric adjustment (no pressure on surplus countries), and an inflationary US policy mix after 1965. Proposals for a "new Bretton Woods" (Part XIV) must specify how they would handle these three issues.

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## 5. The post-Bretton Woods system (1973-2026)

### 5.1 Floating exchange rates

- (fact) Generalized floating among major currencies began in March 1973 and was legalised by the IMF Second Amendment (1 April 1978) [V: IMF]. Many countries continued to peg, crawl or manage floats (to the dollar, the Deutsche Mark, later the euro), and Europe built its own fixed-rate arrangements (the "snake", 1972; the European Monetary System and ERM, 1979; the euro, 1999 [E]).
- (fact, data) Exchange-rate moves since 1971 have been large: the yen went from 358 per dollar (January 1971) to about 158 (October 2026); the Swiss franc from 4.31 to about 0.83 per dollar [V data: FRED EXJPUS, EXSZUS]. Limitation: nominal rates; real effective rates move far less over long periods because of inflation differentials.
- (interpretation: Friedman 1953 "The Case for Flexible Exchange Rates" [E]) Floating rates allow independent monetary policy and absorb real shocks. (interpretation: Mundell's trilemma/"impossible trinity" [E]) A country cannot have fixed rates, free capital mobility and independent monetary policy at once; Bretton Woods chose controls, the post-1973 world chose capital mobility. (empirical finding: Rey 2013 Jackson Hole [E]) With a global financial cycle driven by US monetary policy and risk appetite, floating does not fully insulate small open economies – the trilemma becomes a "dilemma" unless capital flows are managed. This is relevant to Part IX (dollar hegemony) and Part XII (Indonesia; out of scope here).
- (fact) Coordinated interventions continued: the Plaza Accord (September 1985) to weaken the dollar and the Louvre Accord (February 1987) to stabilise it [E].

### 5.2 Petrodollar recycling

- (fact) The oil embargo began in October 1973 and "crude oil prices quadrupled"; the Iranian revolution in 1979 "tripled the cost of oil" [V: FRH Great Inflation]. The IMF records that after OPEC's January 1974 price rise "massive surpluses by oil exporting countries and massive deficits by oil importing countries were expected", leading the Committee of Twenty to abandon its reform timetable and to propose an oil facility [V: IMF chronology].
- (fact) Much of the surplus was deposited with international banks, especially in the eurodollar market, and lent on to oil-importing developing countries, notably in Latin America [E: Brookings "Enter Finance: The 1970s" chapter excerpt; IMF WP 08/180 excerpt]. Brookings: "By the time the oil crisis hit in 1973, the eurodollar market was a fully established financial infrastructure that could happily absorb the flow of petrodollars" [E: excerpt].

- (fact) The recycling ended in the Latin American debt crisis, beginning with Mexico's moratorium in August 1982, after US interest rates rose under Volcker [E].
- (interpretation; contested) The "petrodollar system" is sometimes described as a formal 1974 US-Saudi agreement that oil be priced only in dollars in exchange for security guarantees. Documented elements include US-Saudi economic cooperation arrangements of 1974 and Saudi purchases of US Treasury securities through special arrangements [E]; claims of a secret binding "oil only in dollars" treaty are not supported by the documents surfaced in this run and should be treated as unverified. Oil's dollar invoicing is better explained by the dollar's liquidity and network effects (Part IX).

## 5.3 The eurodollar market

- (fact) Eurodollars are dollar-denominated deposits at banks outside the United States [V: St. Louis Fed 2022]. Origins: Soviet-bloc banks placing dollars in London and Paris in the 1950s for fear of US seizure [E: Boston Fed 2001; BIS 2005]; UK exchange controls of 1957 (Suez crisis), which "banned the use of sterling to finance third-party transactions", so "London banks started using dollar deposits as credit instruments for nonresidents"; and interest arbitrage around US Regulation Q – Midland Bank bid 1.875% on 30-day dollar deposits (above the Regulation Q ceiling), swapped them into sterling and obtained sterling at about 4% when Bank Rate was 4.5% (Schenk 1998) [V: St. Louis Fed]. Until July 1963 Regulation Q capped US 30-day deposits at 1% and 90-day at 2.5% [E: Schenk 1998 excerpt].
- (fact) Estimated net size (BIS eight-country reporting): from about $75 billion to $264 billion in 2020 dollars between 1964 and 1969, a 252% increase [V: St. Louis Fed]. US banks borrowed heavily from their London branches when domestic CD rates hit Regulation Q ceilings (1966, 1969) [E: IMF F&D 1975 excerpt].
- (interpretation: Boston Fed 2001 [E]) The eurodollar market's "freedom from national banking regulation ... led to major changes in the U.S. banking system, including the end of interest rate ceilings on bank deposits, a diminished role for reserve requirements, and the creation of money market accounts." Offshore dollar creation outside US reserve requirements is the historical root of today's offshore dollar system and of the Fed's swap-line role as international lender of last resort (2008, 2020) – see Part IX.

**T-account 4. Creating a eurodollar deposit (illustrative, USD mn).**

A European exporter moves $10 from a New York bank to a London bank, which holds the dollars at its New York correspondent and lends $9 to a Brazilian borrower.

London bank:

| Assets | | Liabilities | |
|---|---|---|---|
| Deposit at NY correspondent | 1 | Eurodollar deposit (exporter) | 10 |
| Loan to Brazilian borrower | 9 | | |

New York correspondent (US banking system), after the borrower spends the loan in the US:

| Assets | | Liabilities | |
|---|---|---|---|
| (unchanged reserves at Fed) | – | Deposit of London bank | 1 |
| | | Deposit of US seller (paid by borrower) | 9 |

US-resident deposits total the same $10 as before; in addition, a $10 offshore dollar claim now exists. (interpretation) Whether eurodollars "multiply" money is disputed: early analyses (Friedman 1969 [E]) used a multiplier analogy; later work stresses that eurobanks are mostly intermediaries funded in wholesale markets. Either way, dollar liabilities expanded outside US regulation and outside the Fed's balance sheet – a key feature of the modern system.

## 5.4 Volcker and the end of the Great Inflation

- (fact) US inflation rose from about 1% in 1964 to "near 14.5 percent" by summer 1980 [V: FRH Bryan]; CPI-U year-on-year peaked at 14.8% in March 1980 (our computation from FRED CPIAUCNS) [V data].

Causes debated: Phillips-curve full-employment policy, the collapse of the Bretton Woods anchor, fiscal imbalances, oil shocks, and mismeasured potential output (Orphanides) [V: FRH Bryan].

- (fact) Volcker became Fed chairman in August 1979. On Saturday 6 October 1979, after an unscheduled FOMC meeting, he announced that the FOMC would target the volume of bank reserves rather than the day-to-day federal funds rate [V: FRH Medley]. The federal funds rate reached a record of about 20% in late 1980 [V: FRH]; the monthly average peaked at 19.1% in June 1981 [V data: FRED FEDFUNDS]. Recessions followed in January-July 1980 and July 1981-November 1982; unemployment peaked at 10.8% in late 1982; inflation fell to 3.7% in 1983 [V: FRH].
- (interpretation: Goodfriend and King 2005; FRH) The disinflation "re-established credibility for low inflation" and laid the ground for the Great Moderation [V: FRH]. (interpretation: monetarists) The reserves-targeting episode vindicated controlling money growth; (interpretation: critics, including post-Keynesians and later Fed researchers) the money targets were hit only loosely and what mattered was the willingness to raise real rates sharply; (normative, distributional) the costs – farm and manufacturing distress, the 1982 debt crisis in Latin America – fell heavily on debtors and on the periphery, while creditors gained from high real rates. Farmers protested at the Fed and car dealers mailed coffins of keys [V: FRH].
- (fact) The Depository Institutions Deregulation and Monetary Control Act of 1980 phased out deposit-rate ceilings (Regulation Q) and required the Fed to price its payment services and open them to non-member institutions [V: FRH Fedwire essay for pricing and access; E for Reg Q phase-out].

## 5.5 Liberalization and financialization

- (fact) Major liberalization steps [E]: removal of UK exchange controls (1979); London's "Big Bang" (27 October 1986); US interstate banking (Riegle-Neal 1994); repeal of the Glass-Steagall separation of commercial and investment banking by the Gramm-Leach-Bliley Act (1999); capital-account liberalization across advanced and many emerging economies in the 1980s-1990s; Basel I (1988) capital standards as the main prudential constraint.
- (fact, data) Finance and insurance value added was 9.0% of US GDP in 2026Q2 (finance, insurance, real estate, rental and leasing: 22.6%) [V data: FRED VAPGDPFI, VAPGDPFIRL, BEA]. (empirical finding: Greenwood and Scharfstein 2013, *JEP* [E]) The US financial sector's share of GDP rose from about 2.8% in 1950 to about 8.3% in 2006, driven by asset management and household credit. Limitation: definitions differ between these sources; the 2026 figure is not directly comparable with the 1950 estimate.
- (interpretation: Epstein 2005 [E]) Financialization is "the increasing role of financial motives, financial markets, financial actors and financial institutions in the operation of domestic and international economies." (interpretation: Krippner 2005 [E]) Profits shifted toward financial channels. (interpretation: mainstream/efficient-finance view) Deeper finance lowered the cost of capital, improved risk-sharing and supported growth up to a point; (empirical finding: Arcand, Berkes and Panizza 2015, IMF/JEG [E]) the finance-growth relationship turns negative when private credit exceeds roughly 100% of GDP. Both views are live; Part VII treats distribution.
- (fact) Crises in the liberalized era [E]: US savings and loan crisis (1980s), 1987 equity crash, ERM crisis (1992-93), Mexico (1994-95), Asia (1997-98), Russia and LTCM (1998), dot-com (2000-01), the Global Financial Crisis (2007-09), euro-area sovereign crisis (2010-12), March 2020 dash for cash, UK LDI episode (2022), US regional bank failures (2023).

## 5.6 Derivatives and securitization growth (BIS and Fed data)

**Table 3. Global OTC derivatives outstanding (USD trillion, end of half-year).** Source: BIS OTC derivatives statistics, bulk file WS_OTC_DERIV2 (May 2026 release), all reporting dealers, all counterparties [V data: `bis_otc_derivatives_global_usd_trn.csv`].

| Period | Notional total | of which interest rate | FX | Equity | Commodity | CDS | Gross market value total |
|---|---|---|---|---|---|---|---|
| 1998-H1 | 72.1 | 48.1 | 22.0 | 1.3 | 0.5 | 0.1 | 2.6 |
| 2007-H2 | 585.9 | 432.4 | 66.6 | 9.5 | 9.3 | 68.1 | 15.7 |
| 2008-H1 | 672.6 | 503.4 | 74.9 | 11.4 | 14.1 | 68.7 | 20.2 |

| Period | Notional total | of which interest rate | FX | Equity | Commodity | CDS | Gross market value total |
|---|---|---|---|---|---|---|---|
| 2008-H2 | 598.1 | 472.7 | 60.6 | 7.6 | 5.2 | 51.9 | 34.9 |
| 2013-H2 | 710.1 | 600.8 | 78.5 | 6.7 | 2.5 | 21.5 | 18.8 |
| 2016-H2 | 482.4 | 385.5 | 78.8 | 6.3 | 1.7 | 10.1 | 15.0 |
| 2020-H2 | 582.1 | 466.5 | 97.6 | 7.1 | 2.1 | 8.7 | 15.8 |
| 2024-H2 | 699.5 | 548.3 | 130.1 | 8.9 | 2.4 | 9.2 | 17.6 |
| 2025-H2 | 844.6 | 669.6 | 148.7 | 11.9 | 2.8 | 11.3 | 22.8 |

Definitions and limitations: *notional* = face value of contracts, which overstates risk because payments are computed on it rather than exchanged; *gross market value* = sum of absolute replacement values before netting, a better measure of exposure (and still before collateral). Totals are not fully comparable over time: trade compression, central clearing after 2009 (G20 reforms) and USD exchange-rate movements alter notionals. Risk categories do not sum exactly to totals because of unallocated items. The 2025 rise partly reflects higher interest-rate activity and a weaker dollar raising USD values of non-USD contracts (interpretation, not decomposed here).

Readings:

- (fact) Notional outstanding grew about 9-fold from 1998-H1 to the 2008-H1 pre-crisis peak; gross market value peaked at $34.9tn in 2008-H2, when crisis price moves raised replacement values even as notionals fell.
- (fact) Credit default swaps fell from a $68.7tn notional peak (2008-H1) to $11.3tn (2025-H2) – the clearest post-crisis shrinkage, consistent with compression, clearing and reduced appetite for single-name protection.
- (fact) Interest-rate derivatives dominate (79% of notional at end-2025). (interpretation) Derivatives redistribute and reshape risk rather than create money directly; but collateral and margin flows (variation margin in March 2020; UK LDI 2022) link them to liquidity and central-bank backstops.

**Table 4. US securitization: agency/GSE mortgage pools and ABS issuers (USD bn, Q4).** Source: Federal Reserve Z.1 via FRED [V data: `us_securitization_z1_q4.csv`].

| Year | Agency/GSE mortgage pools | ABS issuers: debt securities and loans held | ABS issuers: mortgages | Nominal GDP (SAAR) | Pools + ABS as % GDP |
|---|---|---|---|---|---|
| 1970 | 4.8 | 0.0 | 0.0 | 1,088.6 | 0.4 |
| 1980 | 114.0 | 0.0 | 0.0 | 2,985.6 | 3.8 |
| 1990 | 1,019.9 | 250.3 | 66.0 | 6,004.7 | 21.2 |
| 2000 | 2,493.2 | 1,413.6 | 604.2 | 10,435.7 | 37.4 |
| 2007 | 4,464.4 | 4,394.3 | 2,970.0 | 14,715.1 | 60.2 |
| 2009 | 5,376.7 | 3,250.8 | 2,273.1 | 14,651.2 | 58.9 |
| 2010 | 1,146.8 | 2,223.3 | 1,961.6 | 15,309.5 | 22.0 |
| 2025 | 3,209.9 | 1,570.3 | 1,100.0 | 31,462.5 | 15.2 |

Limitations: (1) series break in 2010 – accounting rules FAS 166/167 moved Fannie Mae and Freddie Mac pools onto the GSEs' own balance sheets, so the post-2010 "pools" line mainly reflects Ginnie Mae; agency MBS did not actually shrink by three-quarters (Fannie and Freddie remained in conservatorship from September 2008 [E]). (2) The ABS line is the asset side of private-label issuers. (3) Percentages divide stocks by annual flow GDP.

Readings:

- (fact) Agency pooling, a 1970 innovation (Ginnie Mae pass-throughs [E]), grew to $1.0tn by 1990 and $4.5tn by 2007; private ABS issuers held $4.4tn of assets in 2007, of which $3.0tn mortgages.
- (fact) Private-label securitization fell from $4.4tn (2007) to $1.6tn (2025), and from 30% to 5% of GDP.
- (interpretation: Gorton 2010; Pozsar et al. 2010 [E]) Securitization plus repo and money-market funds formed a "shadow banking" system that performed maturity transformation outside deposit insurance and

central-bank access, reproducing the 1907 trust-company problem at larger scale. (interpretation: originate-to-distribute critique) Securitization weakened screening incentives (Keys et al. 2010 [E]). (interpretation: defenders) It broadened access to credit and diversified funding; most agency securitization performed as designed.

**T-account 5. A private-label securitization (illustrative, USD mn).**

Originating bank sells 100 of mortgages to a special-purpose vehicle (SPV) for cash:

| Bank: Assets | | Bank: Liabilities |
|---|---|---|
| Mortgages | -100 | (no change) |
| Cash/reserves | +100 | |

SPV (trust):

| Assets | | Liabilities | |
|---|---|---|---|
| Mortgage pool | 100 | Senior tranche (AAA) | 80 |
| | | Mezzanine tranche | 15 |
| | | Equity/residual tranche | 5 |

Investors (money funds, pensions, foreign banks, or other SPVs such as CDOs and ABCP conduits) buy the tranches. The bank can originate again with the cash. (interpretation) If the senior tranche is funded short-term (ABCP or repo), the structure performs bank-like maturity transformation; a fall in house prices that erodes the junior 20 exposes the senior holders, who may run, as in 2007.

## 5.7 The institutional architecture of payments

**SWIFT (1973).** (fact) "239 ambitious banks, 15 countries" formed the cooperative in 1973 to replace telex; it went live in 1977 with 518 institutions from 22 countries; central banks first connected in 1983; by the 2000s it carried 3.76 billion messages a year among about 9,000 institutions in 200 countries and territories; it is migrating to ISO 20022 [V: SWIFT "Our story"]. (fact) SWIFT is a messaging network: it does not hold accounts or settle funds; settlement happens in correspondent accounts and payment systems such as CHIPS, Fedwire, CHAPS and TARGET2 [E: standard description; consistent with BIS QR 2017's description of correspondent chains [V]]. (interpretation) Because cross-border dollar payments run through SWIFT messages and US correspondent accounts, access to SWIFT and dollar clearing became an instrument of sanctions (Iran 2012; Russia 2022 [E]) – relevant to Part IX.

**CHIPS (1970).** (fact) The New York Clearing House Association brought CHIPS online in 1970, used especially for the dollar side of foreign-exchange and international transactions; since the late 1980s CHIPS volume has averaged about two-thirds of Fedwire's [V: FRH Fedwire]. It operated as a deferred net settlement system until moving to a real-time "hybrid" design on 22 January 2001 [V: BIS QR 2017; E: GAO-02-303 for date]. (empirical finding: Humphrey 1986, cited by BIS [V]) Under the old rules, the failure of a major CHIPS participant could trigger system-wide unwinding.

**Fedwire.** (fact) The Fed has operated a wire payment system since 1915 (Gold Settlement Fund, renamed Interdistrict Settlement Fund in 1935), leasing a dedicated telegraph network from 1918; it computerised in 1970; volume rose from about 7 million transfers (1970) to 52 million (1980), 100 million (1998) and 200 million (2022), with an average transfer of about $5.4 million in 2022; the Monetary Control Act required pricing from 1981; daylight-overdraft fees began in 1994; FedNow instant payments launched in July 2023 [V: FRH Fedwire]. The BIS calls Fedwire Funds "the world's oldest RTGS system" [V: BIS QR 2017].

**RTGS diffusion.** (fact) Before the 1980s, wholesale payments were typically settled by deferred net settlement at end of day, creating settlement risk; "In 1985, three central banks had RTGS systems; by 1990, the number was eight"; RTGS was made a prerequisite for joining EMU; "By 2000, 49 out of 176 central banks worldwide had RTGS systems; and by 2005, all advanced economy central banks had adopted RTGS with the exception

of Canada" (which used an equivalent hybrid) [V: Bech, Shimizu and Wong, BIS QR March 2017]. The Bank of England introduced RTGS in 1996 [V: BoE history]. The CPSS published its RTGS report in March 1997 (CPMI Papers No 22) [E: BIS publication page excerpt]. The 1974 failure of Bankhaus Herstatt – which left counterparties unpaid on the dollar leg of FX trades – named "Herstatt risk" and motivated both the Basel Committee (1974) and later CLS Bank (2002) payment-versus-payment settlement [E].

**Fast payments.** (fact) Korea's Electronic Banking System (2001) was the first system meeting the CPMI fast-payment definition; by end-2016 there were 20 such systems covering over 40% of world population [V: BIS QR 2017]. Later systems include India's UPI (2016), Brazil's Pix (2020) and FedNow (2023) [E except FedNow V].

**Cards.** (fact) The Diners Club charge card (1950), American Express card (1958), BankAmericard (1958; renamed Visa in 1976) and Master Charge (1966; later Mastercard) built four-party card networks in which the network sets rules and interchange fees while banks issue cards and acquire merchants [E: established history; not opened in this run]. BIS notes that before the 1980s card transactions required phone authorisation and paper imprints [V: BIS QR 2017].

**FX turnover and the dollar.** (fact) OTC FX turnover averaged $9.6 trillion per day in April 2025, up 28% from $7.5 trillion in 2022; the US dollar was on one side of 89% of trades, the euro 28.9%, the yen 16.8% (shares sum to 200%) [E: BIS Triennial Survey press release, 30 Sep 2025, read via search excerpt]. Limitation: April 2025 coincided with tariff-related volatility, inflating turnover.

**T-account 6. Interbank settlement in an RTGS system (illustrative, USD mn).**

Customer of Bank A pays 5 to a customer of Bank B. Both banks hold reserve accounts at the central bank.

Central bank:

| Assets | Liabilities | |
|---|---|---|
| (unchanged) | Reserves of Bank A | -5 |
| | Reserves of Bank B | +5 |

Bank A: deposits of payer -5; reserves -5. Bank B: deposits of payee +5; reserves +5. Under RTGS each payment is final when the central-bank entries are made; under deferred net settlement, entries would be made only for net positions at day's end, and Bank B's customer would bear Bank A's failure risk until then. (interpretation) The shift to RTGS moved settlement risk into intraday liquidity demand, which central banks supply through collateralised intraday credit – another public backstop embedded in private money.

### 5.8 The current architecture in one sentence and a balance-sheet hierarchy

(interpretation: Mehrling's "hierarchy of money" [E]) At the top sits central-bank money (reserves and cash) whose value rests on the state's fiscal capacity and the central bank's policy credibility; beneath it, bank deposits promise par convertibility into central-bank money and are protected by deposit insurance, supervision and LOLR; beneath those, money-market fund shares, repo, eurodollars and stablecoins promise convertibility into bank deposits, with partial or no public backstop. The Fed's balance sheet illustrates the scale of the public layer: total assets rose from $0.72tn (18 December 2002) to a peak of $8.97tn (13 April 2022) and stood at $6.75tn on 4 November 2026 [V data: FRED WALCL]. Limitation: nominal values; the 2008 and 2020 expansions reflect crisis lending and asset purchases (Part VI).

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## 6. Cross-cutting synthesis: trust, enforcement and who bears adjustment

| Era | Unit of account set by | Settlement asset | Trust/enforcement mechanism | Typical failure mode | Who bore losses |
|---|---|---|---|---|---|
| Mesopotamia (3rd-2nd mill. BC) | Temple/palace (shekel, gur) | Weighed silver, barley | Institutional ledgers, law codes, royal decrees | Debt bondage; harvest failure | Debtors (relieved by periodic cancellations) |
| Greek/Roman coinage | State mint | Silver/gold coin | Stamp of authority; legal tender; army pay | Debasement, inflation (3rd c AD) | Holders of base coin; soldiers and wage earners |
| Song/Yuan/Ming paper | State | Paper notes, coin, silver | Tax receivability, quotas, reserves | Over-issue for war | Note holders |
| Medieval merchant credit | Merchant networks, money of account | Bills of exchange, coin | Reputation, fairs, merchant courts | Sovereign default on bankers | Bankers (Bardi, Peruzzi, Medici) |
| Public banks (Amsterdam 1609) | City bank (bank florin) | Bank money | City ownership, agio management | Hidden lending, no fiscal backstop | Depositors (1795 agio -30%) |
| Gold standard (1880-1914) | National unit defined in gold | Gold, sterling bills | Credible convertibility, Bank Rate, cooperation | Deflation, panics, periphery suspensions | Debtors, farmers, periphery, workers |
| Bretton Woods (1944-73) | National units pegged to $; $ to gold | Dollars, gold (official) | IMF rules, capital controls, US commitment | Triffin; US inflation | Holders of dollars (devaluation 1971-73) |
| Fiat/floating (1973-) | Central bank inflation objective | Central-bank reserves; dollar globally | Central-bank independence, fiscal backing, regulation, LOLR | Inflation episodes; credit booms and banking crises | Savers in inflations; taxpayers/bail-outs; borrowers in disinflations |

(interpretation) No monetary order has eliminated the trade-off between *elasticity* (ability to expand money in a crisis) and *discipline* (protection against over-issue). Commodity standards favoured discipline and paid in crises; fiat systems favour elasticity and pay in inflation and moral hazard. Institutions – not the material of money – decide how the trade-off is managed.

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## 7. Master timeline

Full machine-readable version with evidence tags and URLs: `data/r1/r1_master_timeline.csv` (80+ rows). Condensed table:

| Date | Event | Tag |
|---|---|---|
| c.2100-1750 BC | Mesopotamian shekel/gur accounting norm; 20% silver and 33% barley interest | V |
| c.650-600 BC | First coins, electrum, Lydia/Ionia | V |
| 561-546 BC | Croesus: separate gold and silver coins | E |
| c.211 BC | Roman denarius introduced | E |
| AD 64 | Nero cuts denarius weight and fineness (~80%) | V |

| Date | Event | Tag |
|---|---|---|
| AD 155-157 | Antoninus Pius debasement to ~70% | V |
| c.AD 194 | Septimius Severus: ~46% fine | V |
| AD 260-270 | Antoninianus near-base; prices x10 | V |
| AD 301 | Diocletian's Price Edict | V |
| AD 312 | Constantine's solidus | V |
| 1023-1024 | Song state jiaozi | V |
| 1260-1368 | Yuan paper-only money | V |
| early 1400s | Ming paper collapses; move to silver | V |
| 1397-1494 | Medici Bank | E |
| 1609 | Bank of Amsterdam | V |
| 1661-1668 | Stockholms Banco notes; Riksbank founded | E |
| 1660s | London goldsmiths' notes | V |
| 1683 | Amsterdam bank money becomes inconvertible (receipt system) | V |
| 27 Jul 1694 | Bank of England founded | V |
| 1717 | UK de facto gold standard (Newton) | E |
| 1780-1795 | Bank of Amsterdam's downfall | V |
| 1797-1821 | BoE Restriction Period | V |
| 1837-1863 | US free banking | E |
| 1844 | Bank Charter Act | V |
| 1862 | US Legal Tender Act (greenbacks) | V |
| 1863-1865 | US National Banking Acts; 10% tax on state notes | V |
| 10 May 1866 | Overend Gurney fails | V |
| 1871-1879 | Gold standard spreads (Germany, LMU, US resumption 1879) | E |
| 1873 | Bagehot's *Lombard Street* | V |
| 1873-1896 | Long deflation (US -31%, UK -23%) | V |
| 1890 | Baring crisis | E |
| Oct 1907 | Panic of 1907 | V |
| 23 Dec 1913 | Federal Reserve Act | V |
| Apr 1925 | UK returns to gold at pre-war parity | E |
| Sep 1931 | UK leaves gold | V |
| 1933-1934 | US ends domestic gold convertibility; $35/oz | V/E |
| Jul 1944 | Bretton Woods conference | V |
| 27 Dec 1945 | IMF Articles in force | V |
| 1950-1958 | Diners Club; BankAmericard | E |
| 1957-1963 | Eurodollar market emerges | V |
| 1958 | European current-account convertibility | V |
| 1960 | Triffin's *Gold and the Dollar Crisis*; London gold $40 | V/E |
| 1 Nov 1961 | London Gold Pool | V |
| Mar 1962 | Fed swap network begins | V |
| Mar 1968 | Two-tier gold market | V |
| 28 Jul 1969 | SDR created (IMF First Amendment) | V |
| 1970 | CHIPS launched | V |
| 15 Aug 1971 | Nixon closes the gold window | V |

| Date | Event | Tag |
|---|---|---|
| 18 Dec 1971 | Smithsonian Agreement ($38/oz) | V |
| Feb-Mar 1973 | Second devaluation ($42); generalized floating | V |
| May 1973 | SWIFT founded (live 1977) | V |
| Oct 1973 | Oil embargo; oil price x4 | V |
| 1974 | Herstatt failure; Basel Committee | E |
| 7-8 Jan 1976 | Jamaica: floating legalised | V |
| 1 Apr 1978 | IMF Second Amendment in force | V |
| 6 Oct 1979 | Volcker's reserves-targeting shift | V |
| 1980 | Monetary Control Act (DIDMCA) | V |
| Aug 1982 | Mexico default; LDC debt crisis | E |
| 1985 / 1990 | 3 / 8 central banks with RTGS | V |
| 1986 | London Big Bang | E |
| 1988 | Basel I | E |
| 1996 | BoE RTGS | V |
| 1999 | Euro; Gramm-Leach-Bliley | E |
| 2000 | 49 of 176 central banks with RTGS | V |
| 2001 | CHIPS hybrid real-time; Korea fast payments | V |
| 2007-2009 | Global Financial Crisis; OTC notional peak $673tn | V |
| Apr 2022 | Fed balance sheet peak $8.97tn | V |
| Jul 2023 | FedNow | V |
| Apr 2025 | FX turnover $9.6tn/day; USD 89% | V |
| Dec 2025 | OTC derivatives notional $845tn | V |

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## 8. Register of numerical series and key numbers

Each entry: value | unit | year | definition | source | limitation.

| # | Value | Unit | Year/period | Definition | Source (URL or file) | Limitation |
|---|---|---|---|---|---|---|
| N1 | 1 shekel = 300 sila barley | equivalence | Ur III, c.2100-2000 BC | Administered norm used in institutional accounts | https://cdli.earth/articles/cdlj/2017-2 | Variable; ranged 360-600 sila/shekel in loans |
| N2 | 20 / 33.3 | % per year | Ur III-Old Babylonian | Standard interest on silver / barley loans | same; Laws of Eshnunna s.18A (Roth 1995) | Norms; actual rates varied |
| N3 | c.650 | year BC | 7th c BC | Date of earliest electrum coins | https://www.britishmuseum.org/sites/default/files/2020-05/Money_Gallery_LPG_2020_Room_68.pdf | Dating disputed (c.650-600 BC) |
| N4 | ~80 / ~70 / ~46 | % silver | AD 64 / 157 / 194 | Denarius fineness after reforms | http://royalnumismaticsociety.org/NC_Offprints/data/r1/roman_silver_content.csv | Fineness only; weights also fell; small samples |

| # | Value | Unit | Year/period | Definition | Source (URL or file) | Limitation |
|---|---|---|---|---|---|---|
| N5 | x10 | price multiple | AD 260-270 | Rise in prices in a decade | https://numismatics.org/ny-a-thousand-years-of-monetary-history/ | Fragmentary price evidence |
| N6 | 1.25m / 0.36m | strings (guan) | 1023 onward | Jiaozi issue cap / metallic reserve per issue | Horesh 2012 https://doi.org/10.5130/pie.v4i1.2844 (excerpt) | [E]; Chinese sources give 1,256,340 |
| N7 | 97 -> 28 | % metal backing | 1779 -> 1783 | Bank of Amsterdam guilders backed by coin | https://www.bis.org/publ/work1065.pdf | Reconstructions from archives (Quinn and Roberds) |
| N8 | 16 -> 2 | GBP mn | 1797 | BoE gold reserve before Restriction | https://www.bankofengland.co.uk/about/history | Round numbers on institutional page |
| N9 | 25 / 15 | % reserve requirement | 1864 | US national banks: reserve cities / country | https://www.federalreservehistory.org/essays/national-banking-acts | Statutory, not actual holdings |
| N10 | 143 -> 4 | USD mn | 1865 -> 1867 | State bank notes in circulation | same | – |
| N11 | -17 / -12 | % change | 1908 | US industrial output / real GNP after 1907 panic | https://www.federalreservehistory.org/essays/panic-of-1907 | Historical GNP estimates uncertain |
| N12 | -31 / -23 | % change | 1873-1896 | US / UK CPI change | data/r1/us_cpi_1800_2026_minneapolisfed.csv; uk_boe_millennium_basket_headline_extract.csv | Annual reconstructions |
| N13 | -1.1 | % cumulative | 1822-1913 | UK CPI change under gold | data/r1/regime_inflation_stats_us_uk.csv | Deflation confounded with structure |
| N14 | 4.32 / 0.97 | pp (s.d.) | 1822-1913 / 1993-2016 | UK annual inflation volatility, gold vs inflation targeting | same | Measurement error larger pre-1914 |
| N15 | 2.53 / 2.01 / 1.67 | pp (s.d.) | gold / BW / IT | UK real GDP growth volatility | same (BoE A1 col. real GDP growth) | Reconstructed GDP pre-1950 |
| N16 | -24.6 | % change | 1929-1933 | US CPI | data/r1/us_cpi_1800_2026_minneapolisfed.csv | Annual averages |
| N17 | 44 / 730 | nations / delegates | Jul 1944 | Bretton Woods conference | https://www.federalreservehistory.org/essays/bretton-woods-created | |

| # | Value | Unit | Year/period | Definition | Source (URL or file) | Limitation |
|---|---|---|---|---|---|---|
| N18 | 35 -> 38 -> 42 | USD per troy oz | 1944 / Dec 1971 / Feb 1973 | Official gold price | FRH Smithsonian essay; IMF F&D 1978 | Official, not market price |
| N19 | 10.7 | % average | Dec 1971 | Dollar devaluation vs key currencies (Smithsonian) | https://www.federalreservehistory.org/essays/ agreement | Trade weighting method per de Vries |
| N20 | 900 | USD mn | end-1962 | Fed swap lines with 9 central banks | https://www.federalreservehistory.org/essays/ convertibility-ends | |
| N21 | 75 -> 264 | USD bn (2020 dollars) | 1964 -> 1969 | Net eurodollar market size (BIS 8 countries) | https://www.stlouisfed.org/on-the-economy/2022/january/ bretton-woods-growth-eurodollar-market | Estimate; deflated by author's |
| N22 | 14.8 | % y/y | Mar 1980 | US CPI-U inflation peak | data/r1/CPIAUCNS.csv (FRED) | NSA index |
| N23 | 19.1 | % (monthly avg) | Jun 1981 | Effective federal funds rate peak | data/r1/FEDFUNDS.csv (FRED) | Daily highs were higher (~20%) |
| N24 | 72.1 / 672.6 / 844.6 | USD tn | 1998-H1 / 2008-H1 / 2025-H2 | OTC derivatives notional outstanding | data/r1/bis_otc_derivatives_global_usd_tr https://data.bis.org/topics/OTC_DER | Notional exposure |
| N25 | 34.9 / 22.8 | USD tn | 2008-H2 / 2025-H2 | OTC derivatives gross market value | same | Before netting and collateral |
| N26 | 68.7 -> 11.3 | USD tn | 2008-H1 -> 2025-H2 | CDS notional | same | Compression, clearing |
| N27 | 60.2 / 15.2 | % of GDP | 2007 / 2025 | US agency pools + ABS issuer assets | data/r1/us_securitization_z1_q4.csv (Fed Z.1 via FRED) | 2012 accounting break |
| N28 | 239 / 15 ; 518 / 22 | banks / countries | 1973 ; 1977 | SWIFT founders ; live membership | https://www.swift.com/about-us/history | Company history page |
| N29 | 7 / 52 / 100 / 200 | mn transfers per year | 1970/1980/1998/2022 | Fedwire funds volume | https://www.federalreservehistory.org/essays/ | |
| N30 | 3 / 8 / 49 of 176 | central banks with RTGS | 1985 / 1990 / 2000 | RTGS adoption | https://www.bis.org/publ/qtrpdf/r_qt1703g. | Dfj/pitb/qf RTGS per CPMI |

| # | Value | Unit | Year/period | Definition | Source (URL or file) | Limitation |
|---|---|---|---|---|---|---|
| N31 | 9.6 | USD tn per day | Apr 2025 | Global OTC FX turnover (net-net) | https://www.bis.org/media releases/20250930-global-fx-trading-hits-96-trillion-day-april-2025-and-otc-interest-rate-derivatives-surge-79 | Org-wide snapshot in volatile month |
| N32 | 89 | % of trades | Apr 2025 | USD on one side of FX trades | same | Shares sum to 200% |
| N33 | 0.72 / 8.97 / 6.75 | USD tn | Dec 2002 / Apr 2022 / Nov 2026 | Fed total assets | data/r1/WALCLNSA (FRED) | Nominal |
| N34 | 9.0 | % of GDP | 2026Q2 | US finance and insurance value added | data/r1/VAPGDFIRE (FRED/BEA) | Excludes real estate |
| N35 | x33 / x8 | price multiple | 1913-2025 / 1971-2025 | US CPI level change | data/r1/us_cpi_index_2026_minneapolisfed.csv | 1800–2026; quality change |

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## 9. Disputes and uncertainties

1. **Origins of money (barter vs credit vs state).** Unresolved as a matter of theory; historically, institutional credit and units of account predate coinage in Mesopotamia. The exact Humphrey quotation could not be checked against the scanned article in this run (text layer missing); it is reproduced from Graeber and *The Atlantic*. Whether Menger's account is "refuted" depends on reading it as history or as logic.
2. **Administered prices vs market prices in Mesopotamia.** Englund/Hudson (administered equivalences) vs Snell and others (prices). The CDLJ evidence of 360-600 sila per shekel in loans suggests both.
3. **Dating of the earliest coins.** c.650 BC (British Museum) to c.600 BC (LBMA, ANU); depends on the Artemision stratigraphy.
4. **Roman fineness data.** Older XRF surface analyses (Walker) are unreliable for the second century; later-century figures (Aurelian ~5%, Gallienus low points) are partly from older analyses and partly unverified in this run.
5. **"First paper money".** The British Museum attributes the first paper money to Tang China (remittance certificates, "flying cash"); most monetary historians date the first state-issued circulating paper currency to the Song jiaozi (1023/1024). Treat "first" as definition-dependent.
6. **Jiaozi quantitative parameters** (1.25m strings cap, 360k reserve, 3% fee) come from search excerpts of Horesh (2012) and chinaknowledge.de, not from a primary source opened here.
7. **Bretton Woods resource figures.** FRH reports Keynes $26 million / White $5 million / final $8.5 million; these appear to be typographical (billions in the literature) and are left unverified.
8. **Causes of the Great Depression and the Fed's room for manoeuvre under gold.** Eichengreen (gold constraint) vs Friedman-Schwartz and Hsieh-Romer (Fed could have expanded) vs Austrian (liquidation of 1920s malinvestment).
9. **The 1873-96 deflation: good or bad?** Bordo-Landon-Lane-Redish find mostly benign supply-driven deflation; debtor-politics historians emphasise distributional harm. Both can hold simultaneously.
10. **Rules of the game.** Bloomfield (1959) found widespread violation; some later work argues that credibility rather than rule-following was what mattered. Specific quantitative shares from Bloomfield were not opened

here.

11. **Why Bretton Woods collapsed.** Structural Triffin dilemma vs US policy failure after 1965 vs surplus-country refusal to adjust. Weights are contested; this dossier treats all three as contributory.
12. **"Petrodollar agreement".** Dollar oil invoicing is a fact; a binding secret 1974 dollar-only oil treaty is not verified. Avoid in the monograph unless primary documents are cited.
13. **Regime comparisons (Table 2).** Associations, not causal effects. Pre-1913 US CPI is integer-rounded; UK pre-1850 GDP reconstructed; wars dominate 1914-45. A more rigorous comparison would use Bordo (1981), Bordo and Schwartz, or Cooper (1982) estimates and multiple countries.
14. **Derivatives notionals and securitization stocks.** Series breaks (central clearing, compression; FAS 166/167 in 2010) limit time comparisons; our tables flag them.
15. **Cards and some payment dates** (Diners Club 1950, BankAmericard 1958, Visa 1976, CHIPS 2001 date, CLS 2002, Herstatt 1974) are [E] from established history, not opened in this run.
16. **Medici Bank, bills of exchange, Champagne fairs, Stop of the Exchequer, Bank of England's GBP 1.2m founding loan at 8%, Bank Charter Act's GBP 14m fiduciary issue** are standard [E] facts; verify against de Roover (1963), Clapham (1944) and the statute text before final publication.

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*End of R1 dossier. Prepared by Hark's research agent for the monograph by M. Bagus Prabowo and Hark.*

# R2. How Money Is Created: Commercial Banks, Central Banks, Governments and Non-Banks

Research stream R2 for the monograph *The Architecture of Money* (BRIEF Part IV, subsections A-E). Authors of the monograph: M. Bagus Prabowo and Hark (he conceived and directs the project; Hark's research agents research and draft). This stream was researched and drafted by a Hark research agent on 7 November 2026 (WIB).

Status: research draft, not final manuscript prose. Evidence tags follow FRAMEWORK.md: **[V]** = the page or file was opened and the claim is on it; **[E]** = from a search excerpt or from established knowledge, not opened in this session. Claim-type labels are used where they matter: (fact), (empirical finding: author year), (interpretation: school/author), (normative), (counterfactual).

Companion files:

- Data: `data/r2/` (FRED pulls, derived monthly panel, two hand-compiled CSVs; see `data/r2/README.md`).
- Figures: `research/figs/r2_fig1_reserves_m2_loans.png`, `r2_fig2_multiplier.png`, `r2_fig3_fed_balance_sheet.p` `r2_fig4_fed_deferred_asset.png` (built by `work/r2_charts.py`).

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## Key findings

1. **(fact) Most money held by the public is a bank liability created when banks lend or buy assets.** The Bank of England states that bank deposits made up "97% of the amount currently in circulation" of UK broad money and that "whenever a bank makes a loan, it simultaneously creates a matching deposit in the borrower's bank account" (McLeay, Radia and Thomas 2014) [V]. The T-accounts in Part A show the mechanics: the loan and the deposit are booked in the same instant, against each other, with no prior deposit "lent out".

2. **(fact) Repayment destroys money; bank purchases of securities from non-banks create it; bank sales destroy it.** Money creation is two-way. Any transaction in which a bank acquires a non-bank asset by crediting a deposit adds to broad money; any transaction in which a non-bank pays a bank from a deposit (loan principal, bond purchase from the bank, bank share issue) subtracts from it.

3. **(fact) Reserves are a different kind of money that circulates only among banks and the central bank.** Households and firms cannot hold reserves. Reserves settle interbank payments; they are not "lent out" to the public. A loan does not use up reserves; a *payment* to another bank does.

4. **(interpretation, now mainstream at central banks) The textbook money multiplier is an accounting ratio, not a causal mechanism, in modern ample-reserve systems.** US reserve balances rose roughly 60-fold between August 2008 and October 2014 (USD 45.8bn to USD 2,760bn) while M2 rose 48.5% and commercial bank loans and leases 13.0% (FRED TOTRESNS, M2SL, TOTLL) [V]. The M2/monetary-base ratio fell from about 9.2 to 2.9 (Figure 2) [V]. A stable multiplier would have implied an explosion of lending that never happened.

5. **(fact) Reserve requirements differ by jurisdiction and are not a binding quantity constraint in the US or euro area.** The Federal Reserve cut reserve requirement ratios to zero effective 26 March 2020 [V]; the ECB has a 1% minimum reserve coefficient since 18 January 2012 (2% before), remunerated at 0% [V]; Bank Indonesia still runs an active reserve requirement (GWM) for conventional banks, being cut from 9% to 8% on 1 November 2026 and to 7% on 1 December 2026, alongside a bank-specific liquidity incentive (KLM) and a loan-to-funding band (RIM 84-94%) [V]. Indonesia is a case where reserve and liquidity requirements are still used as active policy levers, which the brief explicitly asks us not to ignore.

6. **(interpretation) Banks are constrained, not unconstrained, money creators.** The binding limits are: the policy rate and the funding cost it implies; profitability and competition; borrower creditworthiness and credit demand; capital requirements (Basel III); liquidity requirements (LCR, NSFR) and the risk of losing deposits and reserves to other banks; and, in some countries, quantitative macroprudential rules. Endogenous money theory does not say banks can create unlimited money; it says the constraint is price, risk and regulation rather than a fixed stock of reserves.

7. **(fact) Central banks create reserves and currency; they create broad money (bank deposits) only when they buy from, or lend to, a non-bank.** A QE purchase from a pension fund creates both a reserve balance and a customer deposit; the same purchase from a bank only swaps one bank asset (a bond) for another (reserves). QT reverses this. These T-accounts are worked in Part C.

8. **(fact) Since 2008 the Fed and most advanced-economy central banks steer rates with a floor system: they pay interest on reserves.** On 28 October 2026 the Fed maintained interest on reserve balances (IORB) at 3.90% with a federal funds target range of 3.75-4.00%, an overnight reverse repo offering rate of 3.75% and a standing repo rate of 4.00% [V]. The Fed ended QT (balance-sheet runoff) on 1 December 2025 and began "reserve management purchases" of Treasury bills in mid-December 2025 [E, NY Fed annual report excerpt].

9. **(fact) Seigniorage is not a free lunch when reserves are remunerated.** The Federal Reserve Banks recorded an operating loss of USD 77.6bn in 2024 and USD 18.7bn in 2025, mainly because interest paid to depository institutions and others (USD 186.5bn in 2024; USD 147.7bn in 2025) exceeded interest earned on securities [V, combined financial statements 2025]. The accumulated "deferred asset" (losses to be recouped before remittances to the Treasury resume) peaked at about USD 245.9bn in late January 2026 and had declined to about USD 231.9bn by 4 November 2026 [V, FRED RESPPLLOPNWW]. Before 2022, remittances were large (about USD 109bn in 2021 and USD 76bn in 2022 [E, Reuters excerpt]).

10. **(fact) Normal government deficit financing does not by itself "print money"; direct monetary financing does something different, and most legal systems prohibit it in normal times.** Article 123 TFEU prohibits ECB/national central bank credit facilities to governments and "the purchase directly from them ... of debt instruments" [V]. Indonesia's Bank Indonesia bought government securities (SBN) in the primary market worth about IDR 1,104.85 trillion in 2020-2022 under three joint decrees (SKB/KB I-III) authorized by Law 2/2020 [E, consistent across BI and press sources; component figures partly V]. Law 4/2023 (P2SK) added Article 36A to the crisis-management law, letting BI buy long-term SBN in the primary market only in a crisis declared by the President, on a decision of the financial stability committee (KSSK) [V via CNN Indonesia quoting the statute; BI crisis-protocol page E]. In September 2025, BI and the Ministry of Finance agreed to share interest costs on SBN funding two government programmes, implemented as additional interest paid by BI on government accounts, while BI's SBN buying was in the secondary market (IDR 200 trillion Jan-Aug 2025) [V, BI joint press release].

11. **(fact) Most non-bank lenders do not create money; they move it.** A finance company, a private credit fund or a pension fund lends deposits it already holds; the borrower's deposit rises and the lender's falls. Non-banks can, however, issue *money-like liabilities* (MMF shares, repo, stablecoins) that the public treats as near-money, and they can make bank balance sheets more elastic (securitization). The FSB counts non-bank financial intermediation at USD 256.8 trillion in 2024, 51% of global financial assets, with a "narrow measure" of bank-like activities of USD 76.3 trillion [V, FSB press release 16 Dec 2025].

12. **(fact) Stablecoins are private, non-bank, 1:1-reserved liabilities, not bank deposits and not central bank money.** The US GENIUS Act (P.L. 119-27) was signed on 18 July 2025; it requires at least 1:1 reserves in cash, Fed balances, demand deposits, Treasuries of 93 days or less remaining maturity, and overnight repo/reverse repo against Treasuries [V statute text excerpt and CRS]. The Act's effective date is

the earlier of 18 January 2027 or 120 days after final regulations; the OCC issued its final rule on 21 October 2026, effective 18 January 2027 [E]. Total stablecoin supply was roughly USD 290-305bn in late October to early November 2026 depending on the data provider [E]. Whether issuance adds to broad money depends on who sells the T-bills to the issuer (Part E T-accounts).

13. **(interpretation) The "loans create deposits" view is well established for the mechanics but does not settle the big questions.** It does not by itself show that banks cause inflation, that credit creation is unfair, or that full-reserve or sovereign-money reform would work better. Those are separate empirical and normative questions handled in R3 (inflation) and later chapters.

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# 0. Scope, definitions and notation

## 0.1 What this chapter is for

The brief calls Part IV "one of the most important sections" and asks for "a technically accurate explanation of money creation". The single most common confusion in public debate is a confusion of *levels*: the statement "banks create money" is true of one kind of money (deposits), false of another (reserves), and silent about a third (currency). This chapter therefore keeps a strict vocabulary.

## 0.2 Three tiers of money in a two-tier system

| Tier | Instrument | Whose liability | Who can hold it | How created | How destroyed |
|---|---|---|---|---|---|
| 1. Central bank money (base money, M0) | Banknotes (currency) | Central bank | Anyone | Banks buy notes from the central bank by giving up reserves | Notes returned to central bank (reserves credited back) |
| 1. Central bank money | Reserve balances (settlement accounts) | Central bank | Banks and some other financial institutions with accounts (and the government via the treasury account) | Central bank buys an asset or lends | Central bank sells an asset, loan repaid, or currency withdrawal / treasury inflow |
| 2. Commercial bank money | Bank deposits (demand, savings, time) | Commercial bank | Households, firms, non-bank financial institutions, governments (sometimes) | Bank lends, or buys an asset from a non-bank | Non-bank repays a loan, buys an asset from a bank, or deposit converted to cash or to a treasury account |
| 3. Near-money / money-like claims | MMF shares, repo, commercial paper, stablecoins, e-money | Non-bank institutions | Anyone with access | Investor exchanges a deposit (or reserves) for the claim | Redemption back into deposits |

(fact; standard central bank taxonomy, e.g., McLeay, Radia and Thomas 2014 [V]; Mehrling's "hierarchy of money" [E].)

Two definitional warnings:

- **Base money** = currency in circulation + reserve balances (US: FRED BOGMBASE). It is *not* a component of M2 except for currency held by the public.
- **Broad money** (M2 in the US; M3 in the euro area; M2 in Indonesia per BI definitions) = currency held by the public + deposits (with differing maturity cut-offs) + sometimes retail money market funds. The US M2 definition was changed in May 2020 when savings deposits were reclassified into M1; M2 itself was not broken by that change but M1 was (fact) [E].

### 0.3 T-account conventions

Every example uses simplified balance sheets ("T-accounts") in markdown tables. Assets left, liabilities and equity right. Only *changes* (marked with + or -) are shown unless stated. All numbers are illustrative; real-world magnitudes are given separately with sources. Currency units are "units" (u) so the examples apply to dollars, euros or rupiah. Every transaction must leave each balance sheet balanced; checking that is the simplest guard against fallacies.

Sectors used: **Bank A**, **Bank B** (commercial banks), **CB** (central bank), **Gov** (Treasury / Ministry of Finance), **HH** (households), **Firm**, **PF** (pension fund or other non-bank investor), **MMF**, **SC** (stablecoin issuer), **SPV** (securitization vehicle), **Dealer**.

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## A. Commercial bank money creation

### A.1 Anatomy of a bank balance sheet

A stylised commercial bank (numbers are illustrative but in proportions typical of a large deposit-funded bank):

**Bank A, opening balance sheet (u)**

| Assets | | Liabilities and equity | |
|---|---|---|---|
| Reserves at central bank | 80 | Customer deposits | 750 |
| Vault cash | 5 | Wholesale funding (repo, interbank, bonds) | 150 |
| Government securities | 165 | Other liabilities | 20 |
| Loans to households and firms | 700 | Equity (capital) | 80 |
| Other assets | 50 | | |
| **Total** | **1,000** | **Total** | **1,000** |

Reading it:

- **Deposits are the bank's debts.** A deposit is a promise by the bank to pay, on demand or at maturity, either in currency or by transferring reserves to another bank on the customer's instruction (fact).
- **Loans are the bank's claims** on borrowers, carrying credit risk.
- **Reserves** are the bank's own deposit at the central bank, used to settle payments with other banks and to meet any reserve requirement.
- **Equity** is the residual: assets minus liabilities. It absorbs losses. Capital is *not* a pile of cash the bank "lends out"; it is a source of funding on the right-hand side (fact; a frequent public confusion, discussed by Admati and Hellwig 2013 [E]).

### A.2 Worked example 1: a loan creates a deposit

A household (HH) borrows 100 from Bank A to buy a car. Bank A approves the loan and credits HH's current account.

**Bank A**

| Assets | | Liabilities | |
|---|---|---|---|
| Loan to HH | +100 | Deposit of HH | +100 |

**Household**

| Assets | | Liabilities | |
|---|---|---|---|
| Deposit at Bank A | +100 | Loan from Bank A | +100 |

What changed:

- Broad money (deposits held by the non-bank public) rose by 100. Nobody else's deposit fell (fact, accounting identity).
- Bank A's reserves did not change. No reserve was "used" to make the loan.
- The household's net worth did not change: it has a new asset and an equal new liability. **Money was created; wealth was not** (fact). Wealth changes only if the borrowed money finances something that produces income or if asset prices move.

This is the core claim of McLeay, Radia and Thomas (2014): "Rather than banks receiving deposits when households save and then lending them out, bank lending creates deposits" [V]. It is also the claim of the older credit theory of banking (Macleod 1856; Wicksell 1898; Schumpeter 1911/1934; Keynes 1930) [E].

## A.3 Worked example 2: the borrower spends; interbank settlement in reserves

HH pays 100 to a car dealer (Firm) that banks with Bank B. The payment instruction goes through the interbank payment system (in the US, Fedwire or a clearing house that settles on Fed accounts; in the euro area, T2; in Indonesia, BI-RTGS or BI-FAST).

**Household**

| Assets | | Liabilities |
|---|---|---|
| Deposit at Bank A | -100 | |
| Car | +100 | |

**Firm (car dealer)**

| Assets | | Liabilities |
|---|---|---|
| Deposit at Bank B | +100 | |
| Inventory (car) | -100 | |

**Bank A**

| Assets | | Liabilities | |
|---|---|---|---|
| Reserves | -100 | Deposit of HH | -100 |

**Bank B**

| Assets | | Liabilities | |
|---|---|---|---|
| Reserves | +100 | Deposit of Firm | +100 |

**Central bank**

| Assets | Liabilities | |
|---|---|---|
| | Reserves of Bank A | -100 |
| | Reserves of Bank B | +100 |

What changed:

- Broad money is unchanged in total (still +100 relative to before the loan); it moved from HH at Bank A to Firm at Bank B.
- Total reserves in the system are unchanged. They moved from A to B.
- **Bank A now has a funding problem it did not have when it made the loan.** It still holds the loan of 100 but has lost 100 of reserves. This is the precise sense in which banks are "constrained" by reserves and deposits: not in the act of lending, but in the settlement of the payments that follow (fact; McLeay et al. 2014, Figure 2 discussion [V]).

## A.4 Liquidity management after settlement

Bank A has four ways to restore its reserve position (all fact; standard bank treasury practice):

1. **Attract deposits** from other banks' customers (offer a higher deposit rate). Reserves flow back as those customers move money in.
2. **Borrow in the interbank or repo market**: Bank B lends its surplus reserves to Bank A overnight.

   **Bank A**

   | Assets | | Liabilities | |
   |---|---|---|---|
   | Reserves | +100 | Interbank borrowing from B | +100 |

   **Bank B**

   | Assets | | Liabilities |
   |---|---|---|
   | Reserves | -100 | |
   | Interbank loan to A | +100 | |

3. **Sell or repo securities** to the central bank or the market.
4. **Borrow from the central bank** at a standing facility (US discount window or standing repo facility at 4.00% as of 29 October 2026 [V]; ECB marginal lending facility; BI lending facility at 6.50% as of the 20-21 October 2026 board meeting [V]).

The price of these options is anchored by the policy rate. That is why, as McLeay et al. put it, central banks "implement monetary policy by setting the price of reserves - that is, interest rates" rather than their quantity [V]. In aggregate, the banking system cannot lose reserves through customer payments among banks: one bank's outflow is another bank's inflow. Reserves leave the banking system as a whole only through (i) currency withdrawals, (ii) payments into the government's account at the central bank, (iii) central bank asset sales or loan repayments, and (iv) flows into other central-bank liabilities such as the Fed's overnight reverse repo facility (fact).

**Why banks still want deposits.** Endogenous-money accounts are sometimes caricatured as saying deposits do not matter. They do. A bank whose lending outpaces its deposit-taking will steadily lose reserves to other banks and must fund the gap in wholesale markets at a higher and less stable cost. Retail deposits are the cheapest and, under liquidity regulation, the most "stable" funding (they receive favourable treatment in the Net Stable Funding Ratio and low run-off assumptions in the Liquidity Coverage Ratio) [E: Basel Committee standards]. So the micro-level statement "a bank needs deposits to lend" is a statement about *funding cost and liquidity risk*

*after the loan*, not about *a prerequisite for creating the loan* (interpretation, consistent with McLeay et al. 2014 [V] and Jakab and Kumhof 2015 [E]).

**Keynes's "banks in step".** If all banks expand lending at the same pace, payments between them roughly net out and none loses reserves on balance (Keynes, *A Treatise on Money*, 1930, vol. I) [E]. This is why system-wide credit booms are possible even when each individual bank believes it is cautious: the constraint each bank feels (reserve outflow) disappears when everyone expands together. It is a central reason why prudential regulation focuses on the system, not just the individual bank (interpretation).

## A.5 Worked example 3: repayment destroys money; interest does something different

A year later HH repays the 100 principal plus 6 interest from wages that have accumulated in its Bank A deposit.

Principal repayment:

**Household**

| Assets | | Liabilities | |
|---|---|---|---|
| Deposit at Bank A | -100 | Loan from Bank A | -100 |

**Bank A**

| Assets | | Liabilities | |
|---|---|---|---|
| Loan to HH | -100 | Deposit of HH | -100 |

Broad money falls by 100. The money created by the loan is extinguished when the loan is repaid (fact).

Interest payment of 6:

**Household**

| Assets | | Liabilities and net worth | |
|---|---|---|---|
| Deposit at Bank A | -6 | Net worth | -6 |

**Bank A**

| Assets | Liabilities and equity | |
|---|---|---|
| | Deposit of HH | -6 |
| | Equity (retained profit) | +6 |

The deposit of 6 also disappears from broad money, becoming bank equity (fact). It does not stay out of circulation forever: when the bank pays wages, rent, taxes, dividends or buys assets, it credits deposits again and money re-enters the public's hands.

This bookkeeping answers a perennial popular argument: "if banks create only the principal, where does the money to pay interest come from? Debt must therefore grow forever." The answer (interpretation, widely accepted in the monetary literature; e.g., the flow-of-funds treatment in stock-flow consistent models of Godley and Lavoie 2007 [E]) is that interest income is spent back into the economy by the bank and its shareholders and staff; interest is a flow out of a stock of money that circulates many times, not a stock that must be created once. There is no accounting necessity that the stock of debt must grow to pay interest. Whether debt ratios *in fact* tend to grow is an empirical question (they have, in most advanced economies since 1950 [E: BIS credit statistics]) with institutional explanations (tax treatment of interest, housing finance, financial liberalisation), not an arithmetic one.

## A.6 Worked example 4: a bank buys a bond (from a non-bank vs from another bank)

Bank A buys a 100 government bond from a pension fund (PF) that holds its account at Bank A.

**Bank A**

| Assets | | Liabilities | |
|---|---|---|---|
| Government bond | +100 | Deposit of PF | +100 |

**Pension fund**

| Assets | | Liabilities |
|---|---|---|
| Government bond | -100 | |
| Deposit at Bank A | +100 | |

Broad money rises by 100, exactly as with a loan (fact). The difference from a loan is that the counterparty is a bondholder rather than a borrower; money creation by banks is about *banks acquiring non-bank assets by issuing deposits*, not about lending as such.

If PF banks at Bank B instead, Bank A pays with reserves (Bank A reserves -100, Bank B reserves +100, Bank B deposit of PF +100). Broad money still rises by 100; reserves are redistributed.

If Bank A buys the bond from **Bank B**:

**Bank A**

| Assets | | Liabilities |
|---|---|---|
| Government bond | +100 | |
| Reserves | -100 | |

**Bank B**

| Assets | | Liabilities |
|---|---|---|
| Government bond | -100 | |
| Reserves | +100 | |

No deposit is created; broad money is unchanged (fact). Interbank trades in assets shuffle reserves and assets within the banking system.

The symmetric case also holds: when a bank *sells* a bond to a non-bank, or a non-bank buys newly issued bank shares or long-term bank bonds with deposits, broad money shrinks (fact). This matters for interpreting money growth: in 2022-2023 US M2 fell in nominal terms (peak USD 21,770.7bn in March 2022; trough year-on-year change of -4.66% in April 2023, the first negative year-on-year reading in the FRED M2 series since 1959) [V, FRED M2SL computed in `data/r2/r2_monthly_panel.csv`], partly because QT and Treasury bill issuance pulled deposits into Treasury securities and money market funds (interpretation, see Part C.5 and E.3).

## A.7 Worked example 5: a customer withdraws cash

HH withdraws 10 in banknotes from an ATM of Bank A.

**Household**

| Assets | | Liabilities |
|---|---|---|
| Deposit at Bank A | -10 | |
| Banknotes | +10 | |

**Bank A**

| Assets | | Liabilities | |
|---|---|---|---|
| Vault cash | -10 | Deposit of HH | -10 |

Broad money is unchanged (one form of money swapped for another). When Bank A replenishes its vault cash from the central bank:

**Bank A**

| Assets | | Liabilities |
|---|---|---|
| Vault cash | +10 | |
| Reserves | -10 | |

**Central bank**

| Assets | Liabilities | |
|---|---|---|
| | Reserves of Bank A | -10 |
| | Banknotes in circulation | +10 |

Base money is unchanged in total, but its composition shifts from reserves to currency. Currency is demand-driven: central banks supply whatever notes the public wants to hold and offset the reserve drain with open-market operations if needed (fact). US currency in circulation was about USD 2,416bn in October 2025 [V, FRED CURRCIR]; Indonesia's currency in circulation (uang kartal yang diedarkan, UYD) grew 11.77% year-on-year to IDR 1,341 trillion in Q3 2026 [V, BI Monetary Policy Report Q3 2026].

**The bank-run case.** If many depositors withdraw at once, banks lose reserves to currency (or, in a digital run, to other banks), which is why lender-of-last-resort facilities exist (Part C.6). A bank can be solvent (assets exceed liabilities) yet illiquid (cannot meet withdrawals quickly), the distinction at the heart of Bagehot (1873) [E] and Diamond and Dybvig (1983) [E].

## A.8 Capital, credit risk and the loss case

Suppose the car loan defaults and the bank recovers only 40 from selling the car.

**Bank A**

| Assets | | Liabilities and equity | |
|---|---|---|---|
| Loan to HH | -100 | | |
| Reserves (recovery proceeds) | +40 | Equity | -60 |

Equity absorbs the loss of 60. Depositors are unaffected while equity is positive (fact). If Bank A's equity had been only 50, it would be insolvent; depositors (beyond deposit insurance) and other creditors would face losses, and resolution would follow.

This is why **capital requirements** limit money creation. Under Basel III, banks must hold Common Equity Tier 1 capital of at least 4.5% of risk-weighted assets (RWA), plus a capital conservation buffer of 2.5%, plus

any countercyclical buffer (0-2.5%) and systemic surcharges; total capital at least 8% plus buffers; and a non-risk-weighted leverage ratio of at least 3% of exposure [E: Basel Committee, Basel III framework]. Numerical illustration:

- Bank A has CET1 equity of 80 and RWA of 800, a CET1 ratio of 10%.
- Its effective target (regulatory minimum + buffers + internal management buffer) is 9.5%.
- Maximum RWA consistent with the target = 80 / 0.095 = about 842. Headroom = 42 of RWA.
- A new mortgage with a 35% risk weight consumes 0.35 of RWA per unit of loan, so headroom allows about 120 of new mortgages; an unsecured corporate loan at 100% risk weight allows only about 42.

So capital constrains *how much and what kind* of lending a bank can do, and therefore how much money it creates, unless it can raise new equity or retain profits (fact for the arithmetic; risk weights illustrative). Note that issuing new equity to non-bank investors itself *destroys* deposits (investors pay with deposits that become equity), another reminder that bank capital is not "money the bank lends".

## A.9 Liquidity regulation

Two Basel III liquidity standards (fact; [E] Basel Committee):

- **Liquidity Coverage Ratio (LCR)**: stock of high-quality liquid assets (reserves, government bonds and similar) at least 100% of projected net cash outflows over a 30-day stress scenario. Fully phased in from 1 January 2019.
- **Net Stable Funding Ratio (NSFR)**: available stable funding (equity, long-term debt, stable retail deposits weighted) at least 100% of required stable funding (which depends on asset liquidity and maturity). In force from 2018 in the Basel timetable, with national implementation dates varying (US final rule effective July 2021 [E]).

A bank that creates deposits by lending creates a liability that can run (run-off assumptions in LCR) and an illiquid asset that needs stable funding (NSFR). Both ratios therefore tighten as lending expands unless the bank raises stable funding or holds more liquid assets, which costs money (interpretation of the mechanism; fact for the rules).

In the floor-system world after 2008, liquidity regulation is one reason banks *demand* large reserve balances: reserves count as Level 1 HQLA. That demand is part of why the Fed's "ample" level of reserves is large and why it ended QT in December 2025 when money-market rates signalled scarcity (interpretation consistent with NY Fed annual report on 2025 operations [E]).

## A.10 Why "banks lend out existing deposits" is incomplete (and the sense in which it is partly right)

The intermediation (loanable funds) story says: savers deposit money; banks keep a fraction and lend the rest. Its problems, established above:

1. **Sequence.** A loan is booked against a new deposit; no pre-existing deposit is debited (examples 1 and 4). The Bank of England calls the textbook sequence "the reverse" of reality [V].
2. **Aggregation.** For the system, deposits are mostly *created by* lending and asset purchases, so they cannot be the prior source of lending in aggregate (fact).
3. **Reserves are not lent to the public.** The public cannot hold reserves; "Reserves are an IOU from the central bank to commercial banks ... they cannot 'lend' them on to consumers in the economy" [V, McLeay et al. 2014].
4. **Saving is not a deposit inflow.** When a household saves (spends less), its deposit rises only because someone else's falls or because it repays less debt; saving is a decision about consumption, not a source of new bank deposits for the system (interpretation, Borio and Disyatat 2011 [E] and 2015 [E] on "saving vs financing").

What the intermediation story gets right (interpretation, presented at its strongest):

- **At the level of the individual bank**, deposits *are* a funding source: a bank that attracts deposits gains reserves and can expand more cheaply (A.4).

- **For non-banks**, the loanable-funds description is accurate: a finance company or fund can only lend money it has raised (Part E).
- **In the long run**, real resources (saving in the national-accounts sense, i.e., output not consumed) are what finance real investment. Credit creation allocates *purchasing power*; it does not create the machines. If credit-financed spending outruns productive capacity, the result is inflation or imports, not more real investment (interpretation shared by Austrian, monetarist and many Keynesian writers; see R3).
- **Banks need equity and funding** that are, at the margin, sourced from savers (shareholders and bondholders), so saving behaviour still affects lending through prices.

A fair summary (interpretation): "loans create deposits" is a statement about accounting and causation *within* the banking system; "saving finances investment" is a statement about *real* resources in the national accounts. Confusing the two produces both the false claim that banks merely relend savings and the equally false claim that banks can create real resources by keystroke.

## A.11 The empirical and theoretical literature

**McLeay, Radia and Thomas (2014), "Money creation in the modern economy", Bank of England Quarterly Bulletin 2014 Q1** [V]. Key propositions on the page:

- "the principal way is through commercial banks making loans. Whenever a bank makes a loan, it simultaneously creates a matching deposit in the borrower's bank account, thereby creating new money."
- "In normal times, the central bank does not fix the amount of money in circulation, nor is central bank money 'multiplied up' into more loans and deposits."
- Limits: "there are three main sets of constraints": (i) banks' own limits from market forces (profitability in a competitive market), risk management, and regulatory policy; (ii) the behaviour of households and companies, who may "destroy" money by repaying debt; and (iii) "Monetary policy acts as the ultimate limit on money creation."
- On QE: "QE boosts broad money without directly leading to, or requiring, an increase in lending" and "the newly created reserves do not, by themselves, meaningfully change the incentives for the banks to create new broad money by lending."
- The theory of the money multiplier "operates in the reverse way to that normally described": more lending and spending leads banks and customers to demand more reserves and currency.

Assessment: an official central bank publication, not peer-reviewed in the journal sense, but its accounting claims are uncontroversial among central bankers and consistent with the Deutsche Bundesbank (2017 Monthly Report, "The role of banks, non-banks and the central bank in the money creation process") [E] and with Federal Reserve staff writing (e.g., Ihrig, Weinbach and Wolla 2021, "Teaching the linkage between banks and the Fed: R.I.P. money multiplier", St. Louis Fed) [E]. Limitation: the article describes the UK framework in 2014; reserve-regime details differ across countries and time.

**Werner (2014), "Can banks individually create money out of nothing? - The theories and the empirical evidence", International Review of Financial Analysis 36: 1-19** [E, abstract seen in search results; bibliographic data confirmed on ScienceDirect listing]. Werner distinguishes three theories: (1) financial intermediation theory (banks are intermediaries like other non-banks); (2) fractional reserve theory (individual banks are intermediaries, but the system creates money collectively); (3) credit creation theory (each bank creates money when it lends). He borrowed money (reported in the paper as EUR 200,000 from Raiffeisenbank Wildenberg, a small German cooperative bank, in 2013 [E: detail from established knowledge of the paper, not re-verified this session]) while observing the bank's internal accounts, and found no transfer of funds from other accounts or reserves to fund the loan; the loan was booked as a new customer deposit. He concludes the credit creation theory is the only one consistent with the evidence. A follow-up, Werner (2016), "A lost century in economics: Three theories of banking and the conclusive evidence", IRFA 46: 361-379 [E], repeats the test with tighter controls and argues that capital-adequacy regulation is ineffective at limiting credit creation and that "credit guidance" is needed.

Assessment (interpretation): the single-bank test convincingly documents the bookkeeping, which central banks do not dispute; its novelty lies more in rhetoric and in the history-of-thought argument than in overturning official doctrine after 2014. Its policy conclusion (capital rules do not constrain credit) is contested: capital requirements bind through the *cost* of lending (equity is expensive), which a one-transaction test cannot detect. A small

cooperative bank in 2013 with excess liquidity is not the marginal case where funding and capital constraints bite.

**Jakab and Kumhof (2015), "Banks are not intermediaries of loanable funds - and why this matters", Bank of England Working Paper 529; revised as "... facts, theory and evidence", Staff Working Paper 761 (2018, updated June 2019)** [E, abstracts seen in search results]. They build otherwise-identical DSGE models with (a) intermediation-of-loanable-funds banks and (b) "financing through money creation" banks. Abstract (WP 529): banks "create deposits of new money through lending, and in doing so are mainly constrained by profitability and solvency considerations ... financing models predict changes in bank lending that are far larger, happen much faster, and have much greater effects on the real economy." Contribution: shows the modelling choice matters for crisis dynamics (fast credit contractions, procyclical leverage). Limitation: model results depend on calibration; the "intermediation" benchmark is somewhat a straw model; empirical validation is by stylised facts (procyclicality of bank leverage, fast declines in lending).

**Borio and Disyatat** [E abstracts seen in search results]:

- (2009) "Unconventional monetary policies: an appraisal", BIS Working Paper 292 (published Manchester School 2010). Key concept: the **decoupling principle**: when reserves are remunerated at or near the policy rate (or the rate is at zero), "the size and structure of [central bank] balance sheets can be managed separately from the policy rate". They argue the emphasis on bank reserves in discussions of QE is "misplaced" because reserves become "just another type of liquid asset", and the "same amount of bank reserves" can coexist with very different policy rates.
- (2011) "Global imbalances and the financial crisis: link or no link?", BIS WP 346: criticises explaining market interest rates through saving-investment balances and stresses gross *financing* flows and credit, not net saving.
- (2015) "Capital flows and the current account: taking financing (more) seriously", BIS WP 525 [E]. Contribution: provides the operational-framework argument that, in modern systems, the quantity of reserves is a residual of balance-sheet policy and has no direct mechanical link to lending.

**Other relevant positions (for balance):**

- *Post-Keynesian horizontalists* (Kaldor 1982; Moore 1988 *Horizontalists and Verticalists*) [E]: the central bank sets the price and supplies reserves "horizontally" on demand; money is credit-driven and demand-determined. *Structuralists* (Pollin 1991; Palley) [E]: the supply curve of credit slopes upward because liquidity preference and bank balance-sheet risk rise with lending; this is a more qualified endogenous view and fits the constraints discussed above.
- *Monetarists* (Friedman and Schwartz 1963; Brunner and Meltzer) [E]: did not deny banks create deposits; they argued the central bank *could* control the money stock through the base and a reasonably stable multiplier, and that it *should* target money growth. The empirical instability of money demand and the multiplier after the 1980s weakened the practical case (empirical finding: e.g., Friedman and Kuttner 1992 [E]).
- *Austrian school* (Mises 1912; Rothbard 1983 *The Mystery of Banking*; Huerta de Soto 1998) [E]: accept that fractional-reserve banks create "fiduciary media" (unbacked deposits) and argue this is the root of credit-driven booms and busts (Austrian business cycle theory). Some (Rothbard, Huerta de Soto) regard it as a legal-ethical problem (deposit contracts that are not fully backed); others in the "free banking" tradition (Selgin 1988; White) accept fractional reserves under competition without a central bank. Their mechanics overlap strongly with the credit creation view; their disagreement is about consequences and legitimacy, which belong to later chapters.
- *Critics of the "out of thin air" phrase* (interpretation): deposits are created against an asset (the borrower's promise to repay, usually collateralised), and the bank bears the credit risk with its equity. "Out of nothing" is accurate for the *ledger entry* but misleading as to *economics*: the bank exchanges its liability for the borrower's liability, and the trade can lose money. A VoxEU/CEPR column titled "Banks do not create money out of thin air" makes this argument (author not verified in this session) [E].

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# B. Fractional reserve banking

## B.1 The historical fractional reserve model

The classic narrative (fact for the institutional history; details in R1/Phase 2): London goldsmiths in the seventeenth century accepted deposits of coin and bullion and issued receipts (goldsmith notes); finding that only a fraction of deposits was withdrawn on any day, they lent part of the deposited metal or, more importantly, issued *more notes than they held in metal*, earning interest on the excess (Quinn 1997 on goldsmith banking [E]; Richards 1929 [E]). Earlier deposit banks in medieval and Renaissance Italy and Catalonia already created credit by book transfer, with a reserve fraction of coin (Usher 1943; de Roover 1948) [E]. The essential feature is a **convertible** promise (a note or deposit payable in a commodity or in a higher-tier money) backed by a **fraction** of that commodity.

Two points matter for this chapter:

1. Even in the goldsmith case, the bank's main creation of means of payment was by *issuing* claims (notes, deposit credits), not by physically handing out other people's coins. The credit theory of banking is therefore not a modern discovery; it is a recovery of how bankers described their business before the mid-twentieth century textbook model (interpretation; Werner 2016 [E]; Jakab and Kumhof 2015 [E]).
2. The reserve was a **liquidity buffer against convertibility runs**, not a lending base in the textbook sense. Under the gold standard, the binding reserve was the central bank's or the banking system's gold, defended through the discount rate (Bagehot 1873; Bloomfield 1959) [E].

## B.2 Reserve requirements

A reserve requirement is a legal rule obliging banks to hold a minimum of reserve balances (and sometimes vault cash) against certain liabilities. Historically it served three purposes, in order of their twentieth-century importance (fact; Feinman 1993, "Reserve requirements: history, current practice, and potential reform", Federal Reserve Bulletin [E, the Fed page links to it]):

- **Prudential/liquidity**: in the US National Banking era (1863-1913), reserve requirements were meant to guarantee convertibility of notes and deposits. They failed as a liquidity tool in panics because required reserves cannot be used without breaching the requirement.
- **Monetary control**: after 1913, and especially in the 1970s-80s, requirements were treated as a lever on the money multiplier. The Fed's record shows dozens of ratio changes; for example, in 1960 the requirement against net demand deposits was 16.5-18% for central reserve city banks, and from 2 April 1992 the top ratio on transaction accounts was cut from 12% to 10% [V, Fed reserve requirements page, items 3, 6 and 81].
- **Interest-rate control and demand for reserves**: in corridor systems, a predictable required demand for reserves, averaged over a maintenance period, helped the central bank hit an overnight rate target with small daily operations.

Current regimes (all fact):

| Jurisdiction | Requirement | Remuneration | Source |
|---|---|---|---|
| United States | 0% on all liabilities since 26 March 2020 (was 10% on net transaction accounts above the low-reserve tranche, 3% within it); the March 2020 cut "reduced required reserves by an estimated $200 billion" | IORB 3.90% (effective 29 Oct 2026) | Fed [V] |

| Jurisdiction | Requirement | Remuneration | Source |
|---|---|---|---|
| Euro area | 1% on overnight deposits, deposits up to 2 years, short debt securities and money market paper (since 18 Jan 2012; 2% from 1999); 0% on longer liabilities | 0% (minimum reserves unremunerated); average requirement EUR 174.7bn vs current-account holdings EUR 338.0bn in the July-September 2026 maintenance period | ECB [V] |
| Indonesia (conventional banks) | GWM rupiah (average): 9% until 31 Oct 2026, 8% from 1 Nov 2026, 7% from 1 Dec 2026; reduced further for individual banks by the macroprudential liquidity incentive KLM (max 6% of third-party funds, adjusted to 5% from 1 Dec 2026); part of the GWM remunerated (up to 5% of average GWM from 1 Dec 2026) | Partial | BI Monetary Policy Report Q3 2026 [V] |
| Indonesia (sharia banks and units) | 7.5% -> 6.5% (1 Nov 2026) -> 5.5% (1 Dec 2026) | | BI [V] |
| United Kingdom, Canada, Australia, New Zealand, Sweden | No binding reserve requirements for monetary policy (UK has a cash ratio deposit scheme to fund the Bank, not a monetary tool) | Varies | [E] |
| China | Reserve requirement ratio actively used (system average around 6-7% in 2025 after repeated cuts) | Low remuneration | [E, not verified this session; mark unverified] |

The brief's instruction "Do not assume that all modern banking systems operate under identical reserve requirements or monetary arrangements" is borne out: the US has none, the euro area a small unremunerated one, and Indonesia a substantive, actively managed one used together with credit-steering incentives.

## B.3 The textbook money multiplier, derived

Assumptions: banks hold reserves R equal to a ratio r of deposits D (required ratio plus a constant excess ratio e); the public holds currency C equal to a ratio c of deposits; the central bank controls base money B = C + R.

Then B = cD + (r+e)D, so D = B / (c + r + e), and broad money M = C + D = (1 + c)D, giving

**M = [(1 + c) / (c + r + e)] x B = m x B**

Numerical example (the classroom case): r = 10%, e = 0, c = 0. Then m = 1 / 0.10 = 10. A central bank injection of 100 of reserves allows deposits to expand to 1,000 through the "deposit expansion" sequence:

| Round | Bank receiving deposit | New deposit | Required reserve (10%) | "Excess" lent on |
|---|---|---|---|---|
| 1 | Bank A | 100.0 | 10.0 | 90.0 |

| Round | Bank receiving deposit | New deposit | Required reserve (10%) | "Excess" lent on |
|---|---|---|---|---|
| 2 | Bank B | 90.0 | 9.0 | 81.0 |
| 3 | Bank C | 81.0 | 8.1 | 72.9 |
| 4 | Bank D | 72.9 | 7.3 | 65.6 |
| ... | ... | ... | ... | ... |
| Sum (limit) | | 1,000.0 | 100.0 | 900.0 |

With a currency drain c = 10% and excess reserves e = 2%: m = 1.10 / 0.22 = 5.0.

The model is an internally consistent *accounting* relationship. Its causal reading requires two assumptions: (i) the central bank chooses B exogenously, and (ii) banks are reserve-constrained, so that extra reserves lead to extra lending. McLeay et al. (2014) state precisely this: "For the theory to hold, the amount of reserves must be a binding constraint on lending, and the central bank must directly determine the amount of reserves" [V].

## B.4 Why the multiplier fails as a causal account in modern operations

1. **Central banks set the price, not the quantity, of reserves** (fact for current frameworks). Under a corridor system (pre-2008 Fed, BI's interest-rate framework, the pre-2008 ECB), the central bank supplies whatever quantity of reserves keeps the overnight rate at target. If banks lend more and need more required reserves, the central bank supplies them, otherwise the overnight rate would rise above target. Causation runs from loans to deposits to required reserves (Goodhart 1984/2009; Moore 1988; Holmes 1969 of the New York Fed: "in the real world banks extend credit, creating deposits in the process, and look for the reserves later") [E].
2. **Under a floor system, reserves are deliberately abundant** (fact). Since October 2008 the Fed has paid interest on reserves; since 2019 it has explicitly operated an "ample reserves" regime in which administered rates (IORB, ON RRP) set the floor and the quantity of reserves is set by balance-sheet policy. The October 2026 directive tells the Desk to buy bills "to maintain an ample level of reserves" [V]. Reserve quantity and lending are decoupled (Borio and Disyatat 2009 [E]).
3. **Zero requirements** remove even the accounting link in the US (since March 2020) [V].
4. **The data**: Figures 1 and 2.

*Figure R2-1. Reserve balances (TOTRESNS), M2 (M2SL) and loans and leases at all US commercial banks (TOTLL, monthly mean of weekly data), indexed to August 2008 = 100, log scale, 2007-2026. Source: FRED (Federal Reserve Board H.3, H.6, H.8) [V]. Limitation: TOTLL covers commercial banks only; M2 definitions and seasonal adjustment differ across series; reserves are not seasonally adjusted.*

| Period | Reserve balances | M2 | Bank loans and leases | Fed total assets |
|---|---|---|---|---|
| Aug 2008 -> Oct 2014 (QE1-QE3) | +5,927% (USD 45.8bn -> 2,760bn) | +48.5% | +13.0% | +393% |
| Feb 2020 -> Apr 2022 (pandemic QE) | +109% | +40.4% | +9.9% | +115% |
| Apr 2022 -> Sep 2026 (QT and after) | -17.9% | +7.6% | +27.3% | -24.7% |

*Source: computed from FRED series in* `data/r2/r2_monthly_panel.csv` *[V]. Percent changes of monthly levels (weekly series averaged to months).*

What the data show (interpretation):

- **2008-2014**: a 60-fold rise in reserves coincided with modest growth in M2 and almost flat lending until 2011. A multiplier of even 2 or 3 applied to the reserve increase would have implied trillions of dollars of extra lending that did not occur. Banks were constrained by capital losses, weak credit demand and tightened standards, not by reserves (consistent with the Fed's Senior Loan Officer surveys [E]).

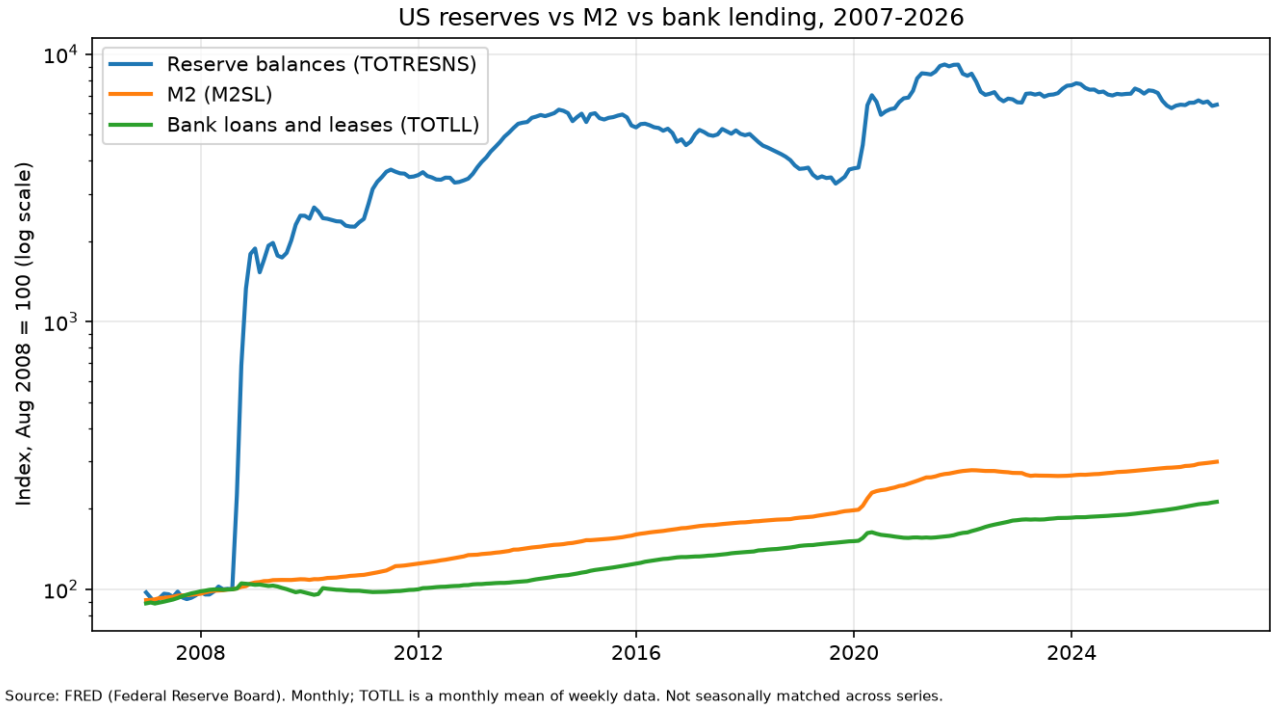



Figure 1: Figure R2-1: US reserves vs M2 vs bank lending

- **2020-2022**: M2 rose 40% in two years, the fastest since the 1940s (peak year-on-year growth 26.8% in February 2021 [V, computed]). Lending rose only about 10%. The M2 surge came mainly from *government deficit spending financed by bond sales that the Fed bought* (Part C.4 and D.1) and from firms drawing credit lines, i.e., from QE-cum-deficits rather than from a multiplier on reserves.
- **2022-2026**: reserves fell while lending rose 27%. Lending growth was not held back by falling reserves; it was shaped by rates, credit demand and the 2023 regional bank stress.

*Figure R2-2. Ratio of M2 to the monetary base, monthly, 1959-2026. Source: FRED M2SL, BOGMBASE [V]. The ratio drifted between about 6 and 12 before 2008, then collapsed from about 9.2 (August 2008) to about 2.8-2.9 (2014) and was about 4.3 in September 2026. Limitation: the ratio is an ex-post quotient that mixes changes in reserve supply, currency demand and deposit growth; it is not a behavioural parameter.*

5. **Causation reversed, not abolished** (interpretation; McLeay et al. 2014 [V]): a looser policy stance increases lending and broad money, and "a larger stock of broad money ... may cause banks and customers to demand more reserves and currency." In a corridor system, the central bank accommodates that demand. The multiplier relationship is observed in the data in quiet times (the ratio is fairly stable from 1995 to 2007 in Figure 2) precisely because the central bank was supplying reserves to match deposits, not the other way round.

## B.5 Where reserve and quantity constraints still matter

It would be a mistake to replace one universal story with another. Reserve quantities and requirements still matter in specific settings (interpretation with factual examples):

- **Emerging and developing economies** often use reserve requirements as a *macroprudential and liquidity-management* tool, to sterilise foreign-exchange inflows cheaply, or to support the exchange rate (Federico, Vegh and Vuletin 2014, "Reserve requirement policy over the business cycle", NBER WP 20612 [E] find that about two-thirds of developing countries used them countercyclically). Bank Indonesia's 2026 package is an example: cutting GWM and steering a liquidity incentive (KLM) towards job-creating sectors and money-market deepening, with KLM incentives reaching IDR 458.5 trillion by early October 2026 (IDR 336.4 trillion "financing channel", IDR 122.1 trillion money-market deepening) [V]. When a requirement

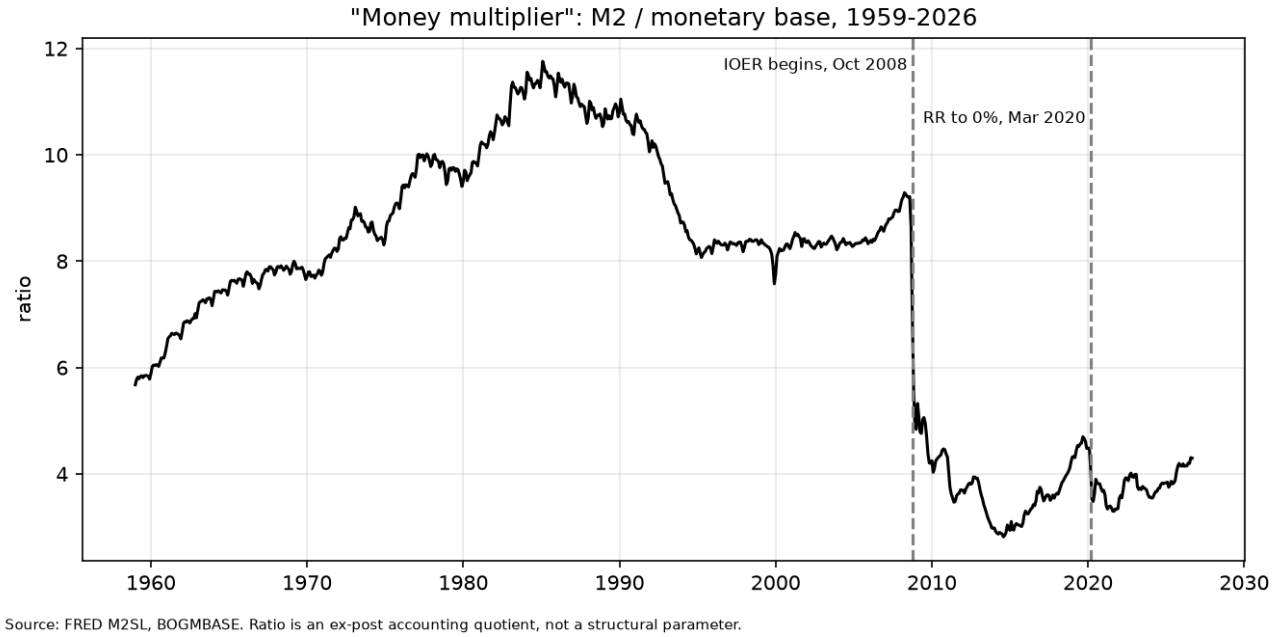



Figure 2: Figure R2-2: M2 / monetary base

is *unremunerated*, it acts like a tax on deposits that widens the spread between lending and deposit rates, which does affect credit volumes through price.

- **The RIM band** (loans plus certain securities over third-party funds, 84-94% [V]) penalises banks that intermediate too little or too much relative to their funding, a quantitative credit-steering rule that has no analogue in the US or euro area.
- **Corridor systems with scarce reserves** (US before 2008, many EMs today): the central bank still supplies reserves on demand to hit its rate, but it can create temporary quantity pressure, and banks' marginal funding costs depend on access to the interbank market.
- **Liquidity regulation** (LCR) creates a structural demand for reserves and other HQLA that interacts with balance-sheet size (A.9).
- **Gold-standard or currency-board systems**: base money is constrained by external reserves, and the central bank cannot freely accommodate. Under such regimes the multiplier logic has more causal bite, though banks still create deposits by lending (interpretation; see R1 on the classical gold standard).
- **China**: the People's Bank of China actively uses the RRR alongside window guidance and quantity targets; whether a multiplier mechanism or administrative credit guidance does the work is debated [E; flagged unverified].

## B.6 Distinguishing frameworks

| Framework | Who creates broad money? | Role of reserves | Role of saving/deposits | Main limit on bank money | Proponents |
|---|---|---|---|---|---|
| Financial intermediation / loanable funds | Not banks; banks relend savings | Minor | Prior source of loans | Supply of saving | Many 1950s-2000s textbooks; most DSGE models before 2008 [E] |

| Framework | Who creates broad money? | Role of reserves | Role of saving/deposits | Main limit on bank money | Proponents |
|---|---|---|---|---|---|
| Fractional reserve + money multiplier | Banking system collectively (each bank lends excess reserves) | Binding quantity constraint; CB controls base | Deposit inflows give banks excess reserves | Reserve ratio and base | Classic textbooks (e.g., Mishkin, *Economics of Money, Banking and Financial Markets*; Samuelson) [E]; Friedman and Schwartz [E] |
| Credit creation / endogenous money | Each bank, when it lends or buys assets | Supplied by CB on demand at its policy rate; settlement asset | Deposits are a consequence; matter for funding cost | Policy rate, profitability, credit demand, risk, capital, liquidity | McLeay et al. 2014 [V]; Moore 1988; Goodhart; Werner 2014 [E]; Jakab and Kumhof [E]; Borio and Disyatat [E]; MMT writers (Wray, Kelton) [E] |
| Austrian critique (fiduciary media) | Banks, through fractional reserves | Should be 100% for demand deposits (Rothbard, Huerta de Soto) or set by free-banking competition (Selgin, White) | Genuine saving should finance investment; credit beyond it distorts | In current system: the central bank and regulation, ineffectively | Mises 1912; Rothbard 1983; Huerta de Soto [E] |
| Sovereign / full-reserve reform proposals | The state (central bank) only; banks become intermediaries | Deposits backed 100% by reserves | Lending funded by investment accounts | Central bank money growth rule | Fisher 1935 *100% Money*; the 1933 Chicago Plan; Benes and Kumhof 2012 "The Chicago Plan Revisited" IMF WP [E]; Swiss Vollgeld referendum rejected 10 June 2018 [E] |

The brief requires distinguishing the fractional reserve framework from the endogenous-money framework. The key differences (interpretation): (i) the *unit* that creates money (system vs individual bank); (ii) the *direction of causation* between reserves and deposits; (iii) what the central bank *controls* (quantity vs price); and (iv) whether the binding constraint is reserves or the cluster of price, risk and regulatory constraints. The two agree on the accounting identities and on the fact that banks' liabilities are most of broad money.

# C. Central bank money creation

## C.1 Anatomy of a central bank balance sheet

**Federal Reserve, simplified (approximate orders of magnitude, late 2026)**

| Assets | | Liabilities and capital | |
|---|---|---|---|
| Treasury securities | majority of assets | Federal Reserve notes (currency) | about 2.4 trillion (CURRCIR, Oct 2025) |
| Agency mortgage-backed securities | large, declining | Reserve balances of depository institutions | about 3.0 trillion (TOTRESNS, Sep 2026) |
| Loans, repo, swap lines | small | Treasury General Account (TGA) | about 0.96 trillion (WTREGEN, early Nov 2026) |
| Deferred asset (cumulative losses to be recouped) | about 0.23 trillion (Nov 2026) | Overnight reverse repo (ON RRP) | near zero (about 1bn, Nov 2026) |
| | | Capital | about 46bn (end-2025) |
| **Total assets** | **about 6.75 trillion (WALCL, 4 Nov 2026)** | | |

*Sources: FRED WALCL, TOTRESNS, CURRCIR, WTREGEN, RRPONTSYD, RESPPLLOPNWW [V]; capital from Combined Financial Statements 2025 (total Reserve Bank capital USD 45.9bn at 31 Dec 2025) [V]. Units USD. Figures are from different dates and do not sum exactly.*

Two points that surprise many readers (fact):

- **A central bank can operate with negative equity or a large "deferred asset"** and still function, because its liabilities are the economy's ultimate settlement asset and are not redeemable in anything else. The Fed books losses as a deferred asset rather than reducing capital; this is an accounting convention with political-economy implications (D.4).
- **The government's account at the central bank (TGA) is not part of the money supply.** When money moves from bank deposits into the TGA (tax payments, bond purchases), reserves and deposits both fall.

## C.2 Currency issuance

Covered in A.7. Currency enters circulation when banks buy it with reserves to meet public demand. The central bank does not "push" notes into the economy; supply is perfectly elastic at par (fact). The seigniorage on currency comes from the fact that the central bank holds interest-earning assets against a zero-interest liability (D.4).

## C.3 Open market operations (OMO): creating reserves, not deposits

The central bank buys 50 of Treasury bills from Bank A (an outright purchase), or lends 50 against Treasury collateral (a repo).

**Central bank**

| Assets | | Liabilities | |
|---|---|---|---|
| Treasury bills (or repo loan to Bank A) | +50 | Reserves of Bank A | +50 |

**Bank A**

| Assets | | Liabilities | |
|---|---|---|---|
| Treasury bills (or nothing if repo) | -50 | (Repo borrowing from CB if repo) | (+50) |
| Reserves | +50 | | |

What changed (fact): reserves (base money) rose by 50; **no deposit held by the non-bank public changed**; broad money unchanged. This is the generic answer to the brief's question "the difference between creating reserves and creating commercial bank deposits": central bank operations with banks change the *composition* of bank assets and the quantity of settlement money; they change the public's money holdings only indirectly, through the effect of interest rates and asset prices on borrowing and spending.

**Corridor vs floor (fact).**

- In a **corridor** system, the central bank sets a lending rate (ceiling), a deposit rate (floor), and a target in between, and uses small daily OMOs to keep the overnight rate near target by adjusting the scarce supply of reserves. Bank Indonesia operates a corridor around the BI-Rate: on 20-21 October 2026 it held the BI-Rate at 5.75% and the lending facility at 6.50% and raised the deposit facility to 5.00% [V]. BI also absorbs liquidity by issuing its own short-term securities (SRBI); the SRBI position fell from IDR 923 trillion at the start of 2025 to IDR 715 trillion by end-August 2025 as BI expanded liquidity [V, BI joint press release Sept 2025].
- In a **floor** (ample reserves) system, reserves are abundant, so the overnight rate settles near the rate the central bank pays on reserves (or, for non-banks, the ON RRP rate). The Fed adopted this explicitly in January 2019 [E] and reaffirmed it ("maintaining ample reserves") throughout 2025-2026 [V]. Current US settings (28 Oct 2026): IORB 3.90%; fed funds target 3.75-4.00%; ON RRP 3.75%; standing repo 4.00% [V].

In a floor system, OMOs do not set the rate; they manage the *quantity* of reserves for liquidity and financial-stability reasons (the "decoupling principle", Borio and Disyatat 2009 [E]).

## C.4 Quantitative easing: purchase from a non-bank vs from a bank

QE is a large-scale purchase of longer-term securities paid for with newly created reserves. Whether it also creates broad money depends on the seller.

**Case 1: CB buys 100 of government bonds from a pension fund (PF) that banks with Bank A.**

**Central bank**

| Assets | | Liabilities | |
|---|---|---|---|
| Government bonds | +100 | Reserves of Bank A | +100 |

**Bank A** (acts as the PF's settlement bank)

| Assets | | Liabilities | |
|---|---|---|---|
| Reserves | +100 | Deposit of PF | +100 |

**Pension fund**

| Assets | | Liabilities |
|---|---|---|
| Government bonds | -100 | |
| Deposit at Bank A | +100 | |

Result: **reserves +100 and broad money +100.** The central bank has created both base money and, via the commercial bank, a deposit. This is the Bank of England's description of QE (McLeay et al. 2014, Figure 3: QE

"first increasing the deposits of" asset holders) [V] and why QE in 2009-2014 and 2020-2021 raised M2 even when lending was flat.

**Case 2: CB buys 100 of government bonds from Bank A's own portfolio.**

**Central bank**

| Assets | | Liabilities | |
|---|---|---|---|
| Government bonds | +100 | Reserves of Bank A | +100 |

**Bank A**

| Assets | | Liabilities |
|---|---|---|
| Government bonds | -100 | |
| Reserves | +100 | |

Result: **reserves +100, broad money unchanged.** An asset swap within the banking system. Any effect on the wider economy runs through prices (lower long-term yields, portfolio rebalancing, signalling), not through deposit quantities.

In practice primary dealers intermediate, and the ultimate seller is a mix of banks, foreign official holders, funds and households; the broad-money effect is roughly the share sold by non-banks (fact for mechanism; magnitude depends on data, e.g., Butt et al. "QE and the bank lending channel in the United Kingdom" (Bank of England working paper, 2012/2014; number to be verified) found most UK gilt sales came from non-bank financial companies and that deposits rose but bank lending did not [E]).

**Channels of QE (interpretation, with contested magnitudes):**

- *Portfolio balance / duration*: removing duration from the market lowers term premia (Gagnon, Raskin, Remache and Sack 2011 estimated QE1 lowered 10-year yields by roughly 30-100 basis points [E]).
- *Signalling*: commits the central bank to low policy rates for longer.
- *Liquidity / market functioning*: especially in crisis (March 2020 Treasury market dysfunction).
- *Bank lending*: weak and indirect; reserves themselves do not raise banks' incentives to lend (McLeay et al. [V]).
- *Fiscal interaction*: when QE coincides with large deficits, the combination finances the deficit at short-term rates and adds deposits to the private sector (Part D).

## C.5 Quantitative tightening: the reverse, with complications

QT occurs when the central bank lets securities mature without reinvesting (passive) or sells them (active).

**Passive runoff: a Treasury bond held by the Fed matures and the Treasury refinances it by issuing a new bond to a money market fund (MMF) that banks with Bank A.**

Step 1, Treasury issues a new bond to the MMF and receives deposits:

**MMF**

| Assets | | Liabilities |
|---|---|---|
| Deposit at Bank A | -100 | |
| Treasury bond | +100 | |

**Bank A**

| Assets | | Liabilities | |
|---|---|---|---|
| Reserves | -100 | Deposit of MMF | -100 |

**Central bank**

| Assets | Liabilities | |
|---|---|---|
| | Reserves of Bank A | -100 |
| | TGA | +100 |

Step 2, Treasury repays the maturing bond held by the Fed out of the TGA:

**Central bank**

| Assets | | Liabilities | |
|---|---|---|---|
| Treasury bonds | -100 | TGA | -100 |

Net result: **Fed assets -100, reserves -100, broad money (deposits of the MMF) -100.** QT that is absorbed by non-banks destroys both reserves and deposits, mirroring Case 1 QE (fact).

**The ON RRP buffer.** In 2021-2023 the Fed's overnight reverse repo facility absorbed more than USD 2 trillion of cash, mostly from MMFs, peaking at USD 2,553.7bn on 30 December 2022 (FRED RRPONTSYD) [V, Figure 3]. When MMFs pulled cash from the ON RRP to buy newly issued Treasury bills during QT, the Fed's liabilities fell through ON RRP rather than reserves. This is why reserves stayed around USD 3.0-3.5 trillion through much of QT even as total assets fell by about USD 2.2 trillion from the April 2022 peak (USD 8.97 trillion on 13 April 2022 [V, WALCL max]) (interpretation from Figure 3). Once the ON RRP was drained (near zero in 2025-2026 [V]), further QT would have come out of reserves; repo-rate pressure in late 2025 led the FOMC to end runoff on 1 December 2025 and to start reserve management purchases of Treasury bills in mid-December 2025 (initially about USD 40bn per month) [E, NY Fed annual report on 2025 operations; Reuters 10 Dec 2025 excerpt]. The NY Fed planned no reserve management purchases for the 15 October-13 November 2026 period, only reinvestments (about USD 14.6bn) [E, NY Fed operational details page excerpt].

*Figure R2-3. Fed total assets (WALCL), reserve balances (TOTRESNS, monthly), Treasury General Account (WTREGEN) and overnight reverse repo (RRPONTSYD), USD billions, 2008-2026. Source: FRED (H.4.1, H.3) [V]. The chart shows QE1-QE3 (2008-2014), QT1 (2017-2019), the September 2019 repo stress and restart of bill purchases, pandemic QE (2020-2022), QT2 (2022-2025), the ON RRP absorbing and then releasing about USD 2 trillion, and the small rise in total assets in 2026 from reserve management purchases. Limitation: TGA and ON RRP swings move reserves independently of Fed asset policy, so asset changes alone do not determine reserves.*

## C.6 Emergency liquidity facilities and the lender of last resort

Emergency lending creates reserves against collateral, usually at a penalty rate and for a short term:

**Central bank lends 30 to Bank A against loan collateral**

| Central bank assets | | Central bank liabilities | |
|---|---|---|---|
| Loan to Bank A | +30 | Reserves of Bank A | +30 |

| Bank A assets | | Bank A liabilities | |
|---|---|---|---|
| Reserves | +30 | Borrowing from CB | +30 |

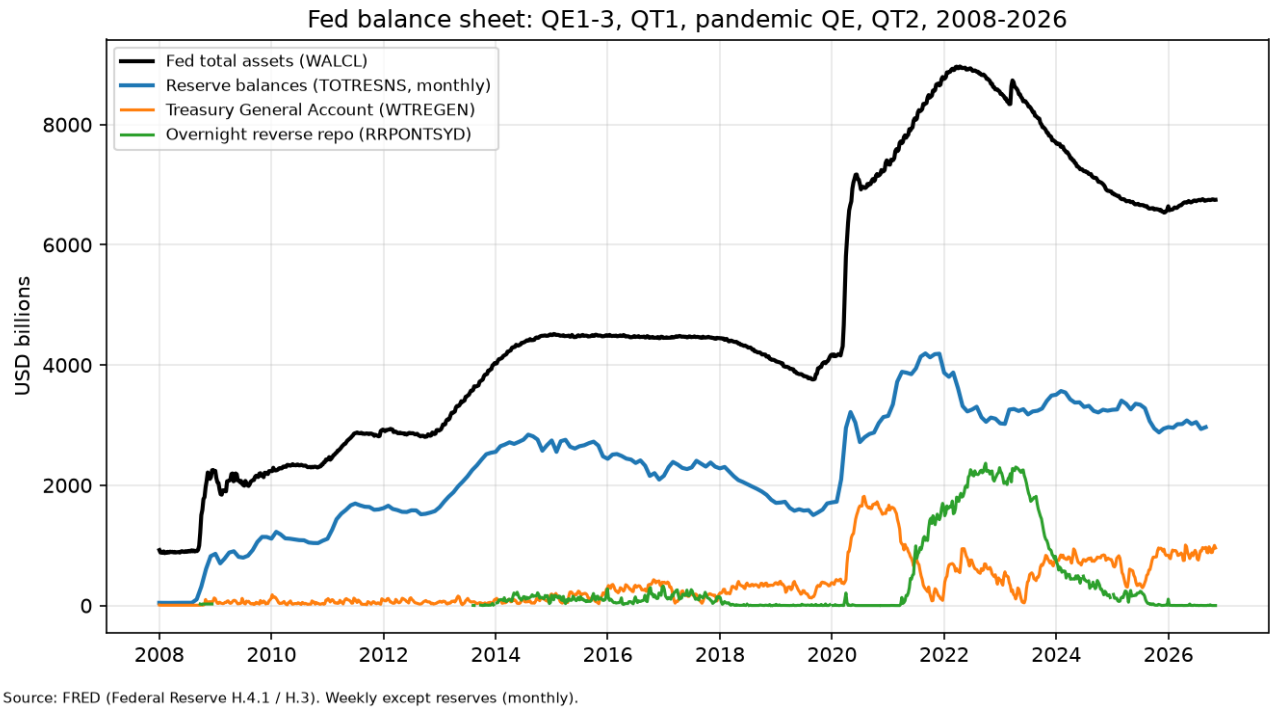



Figure 3: Figure R2-3: Fed balance sheet

Bank A uses the reserves to pay depositors who are leaving (reserves -30, deposits -30). Broad money falls as depositors move to other banks or to currency; the central bank has replaced flighty private funding with its own credit (fact).

Doctrinal anchor: Bagehot (1873), *Lombard Street*: lend freely, at a high rate, against good collateral, to solvent institutions [E]. Modern variants (fact, [E]): the Fed's discount window and standing repo facility; the Bank Term Funding Program of March 2023 (one-year loans against Treasuries and agency securities valued *at par*, a departure from market valuation, created after Silicon Valley Bank's failure); the ECB's Emergency Liquidity Assistance provided by national central banks at their own risk; Bank Indonesia's short-term liquidity facilities and, under Article 36A P2SK, crisis repo access for corporations via banks [V via BI crisis protocol excerpt and CNN]. Moral hazard is the standing critique: if lenders of last resort lend too freely or at too low a rate, they reward risk-taking (interpretation; Austrian and public-choice critics; also mainstream literature on "too big to fail").

## C.7 Creating reserves vs creating deposits: summary table

| Operation | Reserves (base money) | Broad money (deposits + currency held by public) | Example section |
|---|---|---|---|
| CB buys bond from bank (OMO/QE) | + | 0 | C.3, C.4 case 2 |
| CB buys bond from non-bank (QE) | + | + | C.4 case 1 |
| CB lends to bank (repo, LOLR) | + | 0 (directly) | C.3, C.6 |
| Bank lends to non-bank | 0 | + | A.2 |
| Bank buys bond from non-bank | 0 (system) | + | A.6 |

| Operation | Reserves (base money) | Broad money (deposits + currency held by public) | Example section |
|---|---|---|---|
| Payment between customers of different banks | 0 (redistributed) | 0 | A.3 |
| Cash withdrawal | - (reserves -> currency) | 0 | A.7 |
| Loan repaid | 0 | - | A.5 |
| Government spends from TGA | + | + | D.1 |
| Taxes paid into TGA | - | - | D.1 |
| Treasury sells bond to non-bank (proceeds to TGA) | - | - | D.1 |
| Treasury sells bond to bank (proceeds to TGA) | - | 0 | D.1 |
| QT absorbed by non-banks | - | - | C.5 |
| MMF moves cash from bank deposit into ON RRP | - | - | E.3 |
| Stablecoin issuer buys T-bills from a non-bank | 0 | 0 net (deposit moves); stablecoin added outside M2 | E.7 |

### C.8 Monetary policy transmission (brief)

Transmission works mainly through prices (fact for mechanisms; magnitudes contested): the policy rate moves short-term market rates, bank lending and deposit rates, long-term yields via expectations and term premia, asset prices and wealth, the exchange rate, and credit supply through bank balance sheets (the bank lending and balance-sheet channels; Bernanke and Gertler 1995 [E]; Disyatat 2011 "The bank lending channel revisited", JMCB [E], who argues the channel works through bank balance-sheet strength and external finance premia, not reserve quantities). Lags are long and variable (Friedman 1961 [E]); estimates of peak effects on inflation typically arrive 1-2 years after a policy change in advanced economies [E]. Transmission in Indonesia additionally runs strongly through the exchange rate, which is why BI's 2026 package pairs rate decisions with FX intervention and hedging incentives [V].

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## D. Government deficits and monetary financing

### D.1 Treasury plumbing: what deficit spending does to money

Governments in most modern systems hold their main account at the central bank (the US Treasury General Account; Indonesia's government accounts at BI under Article 52 of the BI Act as amended by P2SK [V, BI press release]; national treasuries in the euro area hold accounts at national central banks). This placement has a mechanical consequence: **money paid to the government leaves the banking system; money paid by the government enters it.**

**Step 1. Government spends 100 (e.g., wages to a teacher who banks with Bank A).**

**Central bank**

| Assets | Liabilities | |
|---|---|---|
| | TGA | -100 |
| | Reserves of Bank A | +100 |

**Bank A**

| Assets | | Liabilities | |
|---|---|---|---|
| Reserves | +100 | Deposit of teacher | +100 |

Spending creates deposits and reserves (fact).

**Step 2. Government taxes 80.** Exactly the reverse: deposits -80, reserves -80, TGA +80.

The deficit (20 in this example) has added 20 of deposits and 20 of reserves to the private sector *before* any bond is issued. How the deficit is financed determines the net effect.

**Financing option (a): sell 20 of bonds to non-banks (households, pension funds, MMFs).**

| Non-bank investor | |
|---|---|
| Deposit at Bank A -20 | Bond +20 |

| Bank A | |
|---|---|
| Reserves -20 | Deposit -20 |

| Central bank | |
|---|---|
| | Reserves -20; TGA +20 |

Net of spending, taxes and bond sales: **deposits 0, reserves 0, private bond holdings +20.** Normal bond-financed deficits leave the money stock unchanged; the private sector ends up holding more government bonds (fact, sectoral accounting). The deficit can still raise spending and, under some conditions, prices, through income and wealth effects, but it is not "money printing" in the accounting sense.

**Financing option (b): sell 20 of bonds to banks.**

| Bank A |
|---|
| Reserves -20; Bond +20 |

Net: **deposits +20 (from the deficit), reserves 0, bank bond holdings +20.** Bank purchases of government debt *do* add to broad money, in the same way as any bank asset purchase (A.6) (fact).

**Financing option (c): direct monetary financing: the central bank buys 20 of new bonds directly from the Treasury (primary market) or lends via an overdraft.**

| Central bank | |
|---|---|
| Government bonds +20 | TGA +20 |

Net after spending: **deposits +20, reserves +20, central bank holdings of government debt +20.** Broad money and base money both rise and no private investor had to be persuaded to hold the debt at a market price (fact).

**Option (d): bonds sold to non-banks, then bought by the central bank in the secondary market (QE).** The end state is identical to (c) in quantities. This is why the line between "QE" and "monetary financing" is drawn by *law, intent, permanence and control*, not by balance-sheet arithmetic (interpretation; Buiter 2014; Turner 2015 *Between Debt and the Devil* [E]). The distinguishing criteria usually proposed are:

1. **Who decides**: an independent central bank pursuing its mandate vs a fiscal authority instructing it.
2. **Market price discipline**: secondary-market purchases at market prices vs primary-market purchases at government-set terms.
3. **Reversibility**: holdings the central bank can sell or let run off (QT) vs permanent monetisation.
4. **Remuneration**: reserves that pay the policy rate (so the "free" financing is only as free as the policy rate is low) vs unremunerated money.

The 2020-2021 US case illustrates the economics without the legal form: Treasury issued large amounts of debt to fund pandemic transfers while the Fed bought Treasuries and MBS on a large scale; M2 rose about 40% between February 2020 and April 2022 [V, FRED]. The Fed bought in the secondary market under its own mandate (the Federal Reserve Act allows purchases of Treasury obligations "in the open market", i.e., not directly from the Treasury [E, Federal Reserve Act Section 14]), so legally it was not monetary financing, but the combined fiscal-monetary operation had similar *short-run* monetary effects. Whether that combination explains a large share of the 2021-2023 inflation is examined in R3; the evidence assigns roles to both demand (including fiscal transfers) and supply shocks (interpretation).

## D.2 Legal prohibitions on monetary financing

| Jurisdiction | Rule | Status/notes | Tag |
|---|---|---|---|
| European Union | **Article 123(1) TFEU**: "Overdraft facilities or any other type of credit facility with the European Central Bank or with the central banks of the Member States ... in favour of Union institutions ... central governments, regional, local or other public authorities ... shall be prohibited, as shall the purchase directly from them by the European Central Bank or national central banks of debt instruments." Article 123(2) exempts publicly owned credit institutions in the supply of reserves. | The Court of Justice upheld OMT (Gauweiler, C-62/14, 2015) and PSPP (Weiss, C-493/17, 2018) as compatible with Art. 123 given safeguards (secondary market, blackout periods, limits) [E]; the German Constitutional Court's PSPP judgment of 5 May 2020 challenged proportionality [E]. | [V] text; [E] case law |
| United States | Federal Reserve Act Section 14(b): purchases of Treasury obligations "in the open market"; direct purchase authority from Treasury lapsed in 1981 [E] | No overdraft facility for the Treasury | [E] |

| Jurisdiction | Rule | Status/notes | Tag |
|---|---|---|---|
| Indonesia | BI Act (Law 23/1999 as amended) prohibits BI from buying SBN in the primary market for its own account, with exceptions for short-term securities for monetary operations and for crisis financing [E, exact article wording not verified this session]; Law 2/2020 (originally Perppu 1/2020) temporarily allowed primary-market purchases for 2020-2022 [E]; Law 4/2023 (P2SK) Article 36A permanently allows crisis-time purchases of long-term SBN in the primary market | Article 36A(1)(a): BI may buy "Surat Berharga Negara berjangka panjang di pasar perdana" to handle financial-system problems that endanger the national economy; 36A(2): the crisis condition "ditetapkan oleh presiden"; purchases on a KSSK decision; details in a joint decision of the Governor and Minister of Finance | [V] (CNN Indonesia quoting the statute); BI crisis protocol page [E] |
| Japan | Public Finance Act Article 5 prohibits BOJ underwriting of government bonds except with Diet approval for refinancing maturing holdings [E] | | [E] |

## D.3 Indonesia's burden-sharing episodes, 2020-2022 and 2025

**Pandemic burden sharing (2020-2022).** Under Law 2/2020, BI and the Ministry of Finance signed three joint decrees (Keputusan Bersama, often called SKB or KB I-III) [E for dates; amounts below]:

- **KB I (16 April 2020)**: BI as "standby buyer"/non-competitive bidder of long-term SBN in primary auctions at market prices, for 2020-2022 budgets.
- **KB II (7 July 2020)**: direct private placement for 2020 "public goods" spending, IDR 397.56 trillion, with BI bearing the full interest cost (coupon returned to the government), plus sharing of interest costs on "non-public goods" financing for SMEs (IDR 114.81 trillion) and corporates (IDR 62.22 trillion) [E: BI Monetary Policy Review December 2020 excerpt].
- **KB III (23 August 2021)**: direct private placement for health and humanitarian spending, IDR 215 trillion for 2021 and IDR 224 trillion for 2022, at a coupon equal to BI's 3-month reverse repo rate, with BI returning part of the coupon [E: BI LPI 2021 appendix excerpt; Bisnis.com].

| Year | KB I (primary market, market terms) | KB II / KB III (private placement) | Total | Tag |
|---|---|---|---|---|
| 2020 | 75.86 | 397.56 (KB II) | 473.42 | [E] concordant: BI MPR Dec 2020; Katadata; CNBC |
| 2021 | 143.32 | 215.00 (KB III) | 358.32 | [V] CNN Indonesia; [E] BPK note |

| Year | KB I (primary market, market terms) | KB II / KB III (private placement) | Total | Tag |
|---|---|---|---|---|
| 2022 | 49.11 | 224.00 (KB III) | 273.11 | [E] BI Annual Financial Statements 2022 excerpt |
| **2020-2022** | **268.29** | **836.56** | **1,104.85** | [E] Governor Warjiyo, CNBC Indonesia 20 Mar 2023 |

*Units: IDR trillion. Limitation: press sources sometimes report "KB I" totals differently because of differing cut-off dates (e.g., IDR 974.09 trillion "as of 15 November 2022"); the table uses year-end figures. Not all component sources were opened directly; treat totals as [E] pending BI annual report verification.*

Monetary meaning (interpretation): these were textbook cases of direct monetary financing (option c above), on a large scale relative to Indonesia's base money, executed by an independent central bank under an emergency statute with a sunset. Inflation stayed low in 2020-2021 (output gap, weak demand) and rose in 2022 with global commodity prices, then fell back within BI's target band; the rupiah weakened moderately. Proponents cite this as evidence that time-limited, transparent monetary financing in a deflationary shock need not cause inflation; critics (and the IMF in its 2022 Article IV, per press reports [E]) stressed exit and the precedent for central bank independence. Both readings are defensible; the episode cannot by itself identify the counterfactual (counterfactual).

**The 2025 "interest-cost sharing" agreement.** On 4 September 2025 BI and the Ministry of Finance announced a Joint Decree "concerning Additional Interest in Supporting the Implementation of Government's Asta Cita Programs for a People-Centered Economy" [V, BI joint press release]. Key elements as stated by BI [V]:

- Interest costs on SBN issued for the Public Housing and Merah Putih Village Cooperatives programmes are split equally between government and BI, after deducting returns on government placements of those funds in domestic financial institutions; press reports quantify the shares as 2.9% each for housing and 2.15% each for cooperatives [E, Katadata/IBP].
- Implemented as "additional interest to Government accounts held at Bank Indonesia", invoking BI's state-treasury function (Art. 52 BI Act as amended by P2SK, Art. 22; Art. 23 of Law 1/2004 on State Treasury).
- BI's SBN purchases in 2025 were in the secondary market: IDR 200 trillion through end-August 2025, including IDR 150 trillion through the secondary market and government debt switching.
- Effective "starting in 2025 until the completion of the relevant Government programs."

Analysis (interpretation): this is not primary-market monetary financing in the legal sense, and BI was explicit that "there is no printing of new money because purchases are in the secondary market" [E, CNBC Indonesia quoting BI]. Economically, however, paying above-market interest on government deposits transfers central bank income to the budget, which is a quasi-fiscal subsidy financed by BI's seigniorage; and large secondary-market purchases coordinated with fiscal programmes blur the line in D.1 criterion 1 (who decides). Economists quoted in Reuters, Kompas and Tempo questioned BI's independence [E]. The episode is directly relevant to Part VII (fairness) and Part XII (Indonesia) and to the hypothesis that legal prohibitions on direct financing can be circumvented by functionally equivalent arrangements (interpretation; needs follow-up in R-Indonesia stream).

## D.4 Seigniorage: what governments get from money creation

Definitions (fact; standard in the literature, e.g., Buiter 2007 "Seigniorage", Economics e-journal [E]):

- **Monetary seigniorage**: the real value of the change in base money, (delta B)/P.
- **Opportunity-cost seigniorage** (central bank income view): interest earned on assets backing non-interest-bearing liabilities, i, times currency (and unremunerated reserves).
- **Fiscal seigniorage**: the central bank's profit remitted to the treasury.

In a floor system with remunerated reserves, only currency and unremunerated liabilities generate seigniorage. Reserves paid IORB are, economically, overnight government debt. When the policy rate rises above the average yield on the central bank's (long-duration) asset portfolio, the central bank makes losses.

**Federal Reserve evidence:**

| Year | Fed net income / (loss) from operations | Remittances to Treasury | Tag |
|---|---|---|---|
| 2011-2021 cumulative | | over USD 920bn | [E] St. Louis Fed blog excerpt |
| 2021 | | about USD 109bn (press release table: 107.4 incl. adjustments) | [E] |
| 2022 | net income USD 58.8bn | USD 76bn (remittances suspended from September 2022; the gap between 76 and 58.8 is roughly the deferred asset of about USD 16.6bn opened in late 2022, since 133.0 - 116.4 = 16.6) | [V] Fed press release 12 Jan 2024 (net income); [E] Reuters (remittances) |
| 2023 | expenses exceeded earnings by USD 114.3bn (preliminary) | 0; deferred asset USD 133.0bn at end-2023 | [V] Fed press release 12 Jan 2024 |
| 2024 | operating loss USD 77.6bn; interest expense to depository institutions and others USD 186.5bn | 0 (net); deferred asset increase USD 82.6bn | [V] Combined FS 2025 |
| 2025 | operating loss USD 18.7bn; interest expense to depository institutions and others USD 147.7bn; total interest income USD 155.3bn | some Reserve Banks remitted USD 6.3bn; system deferred asset increase USD 27.5bn | [V] Combined FS 2025 |
| 2026 YTD | deferred asset peaked about USD 245.9bn (28 Jan 2026), about USD 231.9bn on 4 Nov 2026, i.e., positive net earnings in 2026 | | [V] FRED RESPPLLOPNWW |

*Figure R2-4. Federal Reserve "earnings remittances due to the U.S. Treasury" (negative = deferred asset), USD billions, 2020-2026. Source: FRED RESPPLLOPNWW (H.4.1) [V]. Limitation: weekly accounting series; reflects Reserve Bank accounting conventions, not a market valuation of the portfolio.*

Interpretation: (i) QE lent the central bank's balance sheet to the government at short-term rates. When rates were near zero the Treasury received large remittances; when rates rose, the Fed paid banks and MMFs large sums in IORB and ON RRP interest, and remittances stopped. Over the full cycle the fiscal gain or loss depends on the counterfactual debt-management path, which is disputed (counterfactual). (ii) **Distribution**: IORB payments flow to banks (and ON RRP payments to MMFs and their investors). Critics argue this is a transfer to banks (normative; e.g., proposals to tier or reduce reserve remuneration, which the ECB partly adopted by setting minimum reserves at 0% remuneration in 2023 [V: ECB page shows 0% remuneration]); defenders reply that IORB is the price of controlling interest rates in a floor system and that banks pass on most of it through deposit rates (interpretation; contested empirically).

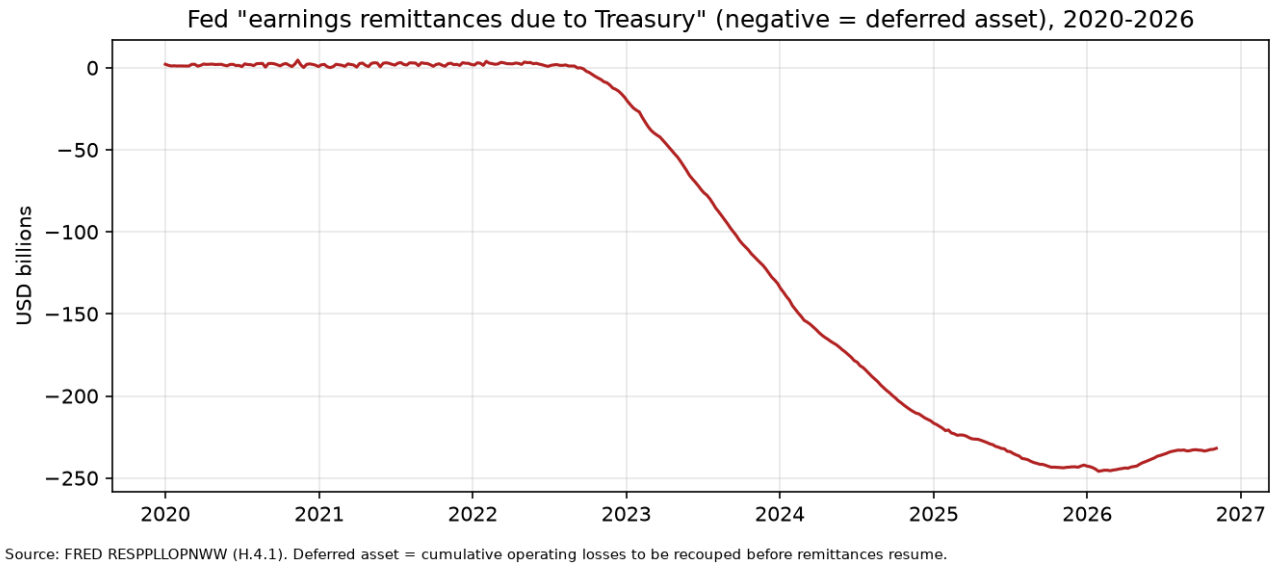



Figure 4: Figure R2-4: Fed deferred asset

## D.5 Central bank independence and fiscal dominance

**Concepts (fact for definitions):**

- *Monetary dominance*: the central bank sets policy for its mandate; fiscal policy must adjust to keep debt sustainable.
- *Fiscal dominance*: fiscal needs force the central bank to accommodate (keep rates low, buy debt), so inflation becomes the residual that keeps debt sustainable.

**Theory:**

- Sargent and Wallace (1981), "Some unpleasant monetarist arithmetic" [E]: if fiscal policy is set independently and debt cannot grow faster than the economy indefinitely, a tight-money central bank may only postpone, and enlarge, later inflation.
- The fiscal theory of the price level (Leeper 1991; Sims 1994; Woodford 1995; Cochrane 2023 *The Fiscal Theory of the Price Level*) [E]: the price level adjusts so that the real value of government debt equals the present value of expected primary surpluses.
- MMT (Wray; Kelton 2020) [E]: for a currency issuer, the binding constraint on deficits is inflation (real resources), not financing; it treats central bank independence as largely an illusion because the central bank must accommodate Treasury operations to hit its rate target. Critics (e.g., Mankiw 2020; Palley) argue MMT understates the inflation and exchange-rate risks of abandoning the institutional separation, especially for emerging economies with foreign-currency debt [E].

**Evidence (empirical findings, with limitations):**

- Cross-country correlations between legal central bank independence indices and lower inflation in advanced economies (Alesina and Summers 1993; Cukierman, Webb and Neyapti 1992) [E]. Limitations: legal indices measure statutes, not practice; causality may run from inflation-averse societies to independent central banks (Posen 1993) [E].
- High inflations and hyperinflations are almost always associated with large fiscal deficits financed by the central bank (Cagan 1956; Sargent 1982 "The ends of four big inflations"; Fischer, Sahay and Vegh 2002) [E]. Ending them required fiscal reform plus a credible new monetary regime. (See R-cases on Weimar, Zimbabwe, Venezuela, Argentina, Lebanon.)
- The reverse is not true: many large deficits financed by market borrowing (Japan for three decades) have not produced high inflation (fact) [E].

**Inflationary financing in brief (interpretation):** monetary financing tends to be inflationary when (i) it is

large relative to money demand and output capacity, (ii) it is expected to persist, (iii) the central bank cannot or will not raise rates/remunerate reserves to keep the new money held willingly, and (iv) the currency's credibility (and external constraint) is weak. It tends not to be inflationary in a deep-deflation or liquidity-trap setting, if temporary and reversible. This is a conditional, not a universal, claim, consistent with the brief's warning against "simplistic claims that all inflation is caused by money printing".

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# E. Non-bank financial institutions

## E.1 The test: does the institution's liability serve as money, and is it created when it lends?

A bank is special because (i) its liability (the deposit) is accepted as a means of payment at par, settled in central bank reserves; and (ii) it acquires assets by issuing that liability. A non-bank lender acquires assets by *spending* means of payment it already holds or has raised. That is the core of the brief's instruction to "explain why not every financial institution that extends credit creates money in the same way as a commercial bank" (interpretation, consistent with Bundesbank 2017 [E] and McLeay et al. 2014 [V]).

**Worked example: a finance company lends 50 to a firm.** The finance company holds a deposit at Bank A (raised earlier by issuing bonds to investors).

**Finance company**

| Assets | | Liabilities |
|---|---|---|
| Deposit at Bank A | -50 | |
| Loan to Firm | +50 | |

**Firm**

| Assets | | Liabilities | |
|---|---|---|---|
| Deposit at Bank A | +50 | Loan from finance co. | +50 |

**Bank A**

| Assets | Liabilities | |
|---|---|---|
| | Deposit of finance co. | -50 |
| | Deposit of Firm | +50 |

Broad money unchanged; credit increased by 50 (fact). Non-bank credit *redistributes* existing money and can raise its velocity (money moves from an idle holder to a spender), which can raise spending even without money creation (interpretation; the "Gurley-Shaw" point, Gurley and Shaw 1960 [E]).

How non-banks nevertheless affect money and credit (interpretation):

- **Money-like liabilities**: MMF shares, repo and stablecoins are close substitutes for deposits for some holders. Most are outside M2 (US M2 includes retail but not institutional MMF balances; see note), yet they function as near-money.
- **Balance-sheet elasticity**: securitization and repo let banks originate more loans per unit of capital and funding.
- **Bank lending to non-banks**: when a bank lends to a fund or a finance company, *that* bank loan creates deposits. Much "non-bank" credit growth is ultimately funded by bank credit lines and repo (interconnectedness, FSB 2025 [V]).

*Note on US M2 definition*: US M2 includes retail money market fund balances (FRED WRMFNS, about USD 3.04 trillion in early October 2026 [V]) but excludes institutional MMF shares; total MMF financial assets were about USD 8.44 trillion in Q2 2026 (FRED MMMFFAQ027S) [V]. Definitions differ across countries; euro-area M3 includes MMF shares and repos.

## E.2 Shadow banking: size and definitions

The FSB's monitoring framework (created after the G20 Seoul Summit request of 2010) measures "non-bank financial intermediation" (NBFI) [V]:

- **Broad measure**: all non-bank financial institutions (insurers, pension funds, other financial intermediaries, public financial institutions, financial auxiliaries).
- **Narrow measure**: entities "involved in credit intermediation activities that may pose bank-like financial stability risks" (maturity/liquidity transformation, leverage) [V].

FSB Global Monitoring Report 2025 (data for end-2024, 29 jurisdictions with over 90% of world GDP) [V, press release 16 Dec 2025]:

- NBFI assets USD 256.8 trillion, 51% of total global financial assets; grew 9.4% in 2024 vs 4.7% for banks.
- Other financial intermediaries (MMFs, hedge funds, other investment funds, trust companies, structured finance vehicles): USD 169.4 trillion, +11%.
- Narrow measure: USD 76.3 trillion, +12% (12.7% in the report), 15.4% of total financial assets (up from 14.5%) [V press release; report excerpt E for 12.7% and 15.4%].
- "Severe limitations in the availability of regulatory data for private credit" [V].

Limitations: national classification differs; the broad measure includes many entities (pension funds, insurers) that perform little maturity transformation; valuation effects drive much of the growth.

## E.3 Money market funds and the ON RRP

**Step 1. A household moves 100 from its bank deposit into a government MMF.**

| Household |
|---|
| Deposit -100; MMF shares +100 |

| MMF | |
|---|---|
| Deposit at Bank A +100 | Shares +100 |

So far broad money (narrowly, deposits held by households and firms) is unchanged in total; ownership moved to the MMF (fact).

**Step 2a. The MMF buys 100 of T-bills from a dealer that is itself a bank (Bank D).** The MMF's deposit at Bank A is extinguished; Bank A pays 100 of reserves to Bank D; Bank D's T-bill holdings fall by 100 and its reserves rise by 100 (an asset swap, no new deposit at Bank D). Net: deposits -100 (the MMF's deposit is gone), reserves redistributed; the public now holds MMF shares backed by T-bills instead of deposits. Broad money in the M2 sense falls unless the MMF shares are retail (and thus in M2) (fact).

**Step 2b. The MMF instead places 100 in the Fed's ON RRP.**

| Central bank | |
|---|---|
| | Reserves of Bank A -100; ON RRP (MMF) +100 |

| Bank A | |
|---|---|
| Reserves -100 | Deposit of MMF -100 |

Reserves and deposits both fall; the MMF holds a claim directly on the central bank (fact). This is how MMFs can drain bank reserves and why the ON RRP acted as a buffer during QT (C.5).

**Run risk.** MMFs promising stable net asset values ran in September 2008 (Reserve Primary Fund "broke the buck") and prime funds suffered outflows in March 2020, requiring Fed facilities (MMLF) [E]. US reforms in 2014 and 2023 (liquidity fees, removal of redemption gates) followed [E].

## E.4 Repo markets

A repurchase agreement is a collateralised loan: the cash borrower sells a security and agrees to buy it back at a higher price.

**MMF lends 100 of cash overnight to a dealer against Treasury collateral (tri-party repo).**

| MMF |
|---|
| Deposit -100; Reverse repo claim on dealer +100 |

| Dealer | |
|---|---|
| Deposit (or reserves via its bank) +100 | Repo borrowing +100 |

No new deposits are created by the repo itself; cash moves from the MMF to the dealer, who uses it to finance securities inventory (fact). If the dealer is a bank subsidiary using its bank's balance sheet, or if a bank provides the repo cash, deposit or reserve effects follow the general rules above.

Why repo matters for "money-like" claims (interpretation): for institutional cash pools too large for insured deposits, a short repo claim backed by Treasuries functions like a safe deposit (Gorton and Metrick 2012, "Securitized banking and the run on repo", JFE [E]; Pozsar 2014 on "shadow money" [E]). In 2007-2008, rising haircuts on private-label collateral acted as a run, forcing deleveraging. In September 2019, an overnight repo rate spike (to around 10% intraday in some trades [E]) signalled reserve scarcity and led the Fed to restart bill purchases; the Fed created a standing repo facility in July 2021 [E]; its current minimum bid rate is 4.00% [V].

## E.5 Securitization

**Step 1. Bank A originates 1,000 of mortgages** (creates 1,000 of deposits, as in A.2).

**Step 2. Bank A sells the mortgages to an SPV.** The SPV funds the purchase by issuing mortgage-backed securities (MBS) to a pension fund, which pays with deposits.

| Pension fund |
|---|
| Deposit -1,000; MBS +1,000 |

| SPV | |
|---|---|
| Mortgages +1,000 | MBS +1,000 |

| Bank A | |
|---|---|
| Mortgages -1,000 | Deposit of pension fund -1,000 |

Net after both steps: the deposits created by origination (held by home sellers) remain; the pension fund's deposits of 1,000 are destroyed; Bank A's balance sheet returns to its original size, *freeing capital and funding* to originate again (fact). Securitization does not itself create money; it recycles bank balance-sheet capacity, so the same capital base can support more cumulative origination (interpretation; this "originate-to-distribute" model was central to the pre-2008 credit boom, Brunnermeier 2009 [E]). If a bank (or the central bank, via QE in agency MBS) buys the MBS, deposits are not destroyed and money is higher accordingly.

## E.6 Private credit and investment funds

Private credit funds raise committed capital from institutional investors and lend directly to mid-sized firms. They do not create deposits (E.1 logic), but they (i) often use bank subscription lines and back-leverage (bank loans to the fund create deposits), (ii) extend credit outside bank capital and liquidity rules, and (iii) are hard to monitor, which the FSB flags (fact for FSB statement [V]). Estimates of global private credit assets vary widely by definition (commonly cited at around USD 1.5-2 trillion for 2024 by IMF and industry sources [E, not verified this session; treat as unverified]).

## E.7 Stablecoins

A fiat-backed stablecoin is a token on a blockchain representing a claim on an issuer that promises redemption at par (e.g., USD 1) and holds reserve assets. Under the US GENIUS Act (P.L. 119-27, signed 18 July 2025) a permitted payment stablecoin issuer must "maintain identifiable reserves backing the outstanding payment stablecoins ... on an at least 1 to 1 basis", limited to: US coins and currency or balances at a Federal Reserve Bank; demand deposits at insured depository institutions; Treasury bills, notes or bonds with a remaining maturity of 93 days or less (or issued with 93 days or less); overnight repos backed by such bills; overnight reverse repos collateralised by Treasuries; government MMFs investing in those assets; and similar tokenised forms [V, statute text excerpt via govinfo/congress.gov and CRS IN12553]. The Act's effective date is the earlier of 18 months after enactment (18 January 2027) or 120 days after final regulations [E, Federal Register and law-firm summaries]. The OCC issued a final implementing rule on 21 October 2026, effective 18 January 2027, including liquidity and concentration requirements (e.g., at least 30% of reserves in highly liquid instruments receivable within five business days, a 40% cap at any one financial institution, a weighted average maturity of no more than 20 days) [E, WilmerHale summary]. Treasury proposed rules on issuance under section 3 in August 2026 and the Federal Reserve proposed its rule in September 2026 [E].

**Worked example: a user buys 100 of a stablecoin; the issuer invests in T-bills.**

Step 1. User pays the issuer 100 from a bank deposit:

| User |
|---|
| Deposit at Bank A -100; Stablecoin +100 |

| Issuer (SC) | |
|---|---|
| Deposit at Bank B +100 | Stablecoin liabilities +100 |

(Bank A pays reserves to Bank B; deposits move from user to issuer.)

Step 2a. Issuer buys 100 of T-bills from an MMF (a non-bank):

| SC |
|---|
| Deposit -100; T-bills +100 |

| MMF |
|---|
| T-bills -100; Deposit +100 |

Net: bank deposits unchanged in total (now held by the MMF); a new private money-like liability (the stablecoin) of 100 exists outside M2; T-bills moved from MMF to issuer. Measured broad money unchanged; "money-like" claims +100 (fact for accounting).

Step 2b. Issuer instead buys 100 of newly issued T-bills at a Treasury auction:

Deposits -100 and reserves -100 (proceeds go to the TGA, C.1), until the Treasury spends them. Net effect identical to any Treasury bill sale to a non-bank (D.1 option a) (fact).

Step 2c. Issuer keeps the reserves as a demand deposit at a bank: bank deposits unchanged (held by issuer), stablecoin +100, so the stablecoin is effectively a claim on a bank deposit (a "narrow" layered claim).

Market facts (all [E], third-party aggregators, not official statistics; ranges reflect different coverage): total stablecoin supply about USD 290-306bn in late October to early November 2026 (CoinDesk research: USD 305.5bn in October 2026; Stablecoin Beat "base supply": USD 290.4bn on 31 October 2026; Glassnode: about USD 291.4bn on 3 November 2026); USDT about 60% and USDC about 24% of supply. Tether's Q3 2026 attestation (BDO, 30 September 2026) reported total assets of about USD 191.5bn against liabilities of about USD 183.5bn, with about USD 132.4bn in direct and indirect US Treasury bill holdings, about USD 19.1bn in gold and about USD 8.4bn in bitcoin [E, multiple news reports of the attestation]. Tether is not a GENIUS-permitted issuer as of this writing [E, unverified].

Interpretation (contested):

- *Why stablecoins are not bank money*: the issuer cannot create stablecoins by lending (under GENIUS, reserves must be 1:1 and issuers are restricted in activities); a stablecoin is "narrow" money backed by bank deposits, central bank money, or T-bills. It does not expand credit to the private sector except through its reserve investments (BIS Annual Economic Report 2025, chapter on stablecoins, argues they fail tests of "singleness, elasticity and integrity" [E]).
- *Why they still matter for monetary architecture*: (i) demand for T-bills (a large issuer holds more T-bills than some sovereigns, per news coverage [E]); (ii) possible deposit outflows from banks if stablecoins become payment media (deposit substitution reduces bank funding, and potentially bank credit, depending on where reserves are held); (iii) run risk if reserve quality is doubted (TerraUSD, an algorithmic coin, collapsed in May 2022 [E]); (iv) dollarisation pressure in emerging markets (relevant to Indonesia and R-international).

## E.8 A hierarchy of money-like liabilities

| Layer | Example | Backed by / settles in | Par guarantee | Run risk | Created by lending? |
|---|---|---|---|---|---|
| Central bank money | Currency, reserves | Itself (state liability) | Definitional | None in own currency | Yes, CB lending/asset purchases |
| Insured bank deposits | Retail current accounts | Reserves; deposit insurance; LOLR | Up to insurance limit | Low | Yes |
| Uninsured bank deposits | Corporate balances | Reserves; bank capital | Contractual | Moderate (SVB 2023) | Yes |

| Layer | Example | Backed by / settles in | Par guarantee | Run risk | Created by lending? |
|---|---|---|---|---|---|
| Government MMF shares, repo | Institutional cash | T-bills, repo, ON RRP | Not guaranteed | Low-moderate | No |
| Prime MMF shares, commercial paper | | Private short-term debt | Not guaranteed | Moderate-high (2008, 2020) | No |
| Fiat-backed stablecoins | USDC, USDT | Deposits, T-bills, repo (rules vary) | Contractual (GENIUS: 1:1 reserves) | Depends on reserves and redemption access | No (under 1:1 rules) |
| Crypto assets | Bitcoin | Nothing (no issuer) | None | n/a (price risk) | No |

(Interpretation building on Mehrling 2012 "The inherent hierarchy of money" [E] and Gorton 2017 [E].)

---

## F. What limits bank money creation: synthesis

The brief's synthesis question 2 asks: "Why does commercial bank lending create money, and what limits that process?" The *why* is the accounting of A.2 combined with the acceptance of bank deposits as means of payment (which rests on convertibility into central bank money, deposit insurance, the lender of last resort and legal-tender/tax arrangements, covered in R1 and the trust chapters). The *limits* are summarised here.

| Limit | Mechanism | How it binds | Strength of evidence | Where it binds most |
|---|---|---|---|---|
| Policy rate | Sets marginal cost of reserves and wholesale funding; transmits to loan rates and credit demand | Price | Strong (decades of transmission evidence) [E] | Everywhere; McLeay et al. call it "the ultimate limit" [V] |
| Profitability and competition | Loan must earn more than funding cost + expected loss + capital cost + operating cost; competitors limit pricing | Price/margin | Strong (industry practice) | Competitive markets |
| Credit demand | Banks cannot lend to borrowers who do not want to borrow at offered terms | Quantity, demand side | Strong (2009-2014 US: reserves exploded, lending flat) [V data] | Recessions, balance-sheet recessions |
| Borrower creditworthiness | Expected loss and collateral rules; lending standards | Risk | Strong (SLOOS tightening predicts lending) [E] | Downturns |

| Limit | Mechanism | How it binds | Strength of evidence | Where it binds most |
|---|---|---|---|---|
| Capital requirements | Equity per unit of risk-weighted assets; leverage ratio | Balance-sheet | Strong for undercapitalised banks; debated at margin (Werner 2016 vs mainstream) | After losses; small banks |
| Liquidity requirements (LCR, NSFR) | Need HQLA and stable funding against new deposits/loans | Balance-sheet | Moderate-strong | Banks with flighty funding |
| Interbank outflows ("banks in step") | A bank expanding faster than peers loses reserves to them | Individual-bank funding | Strong at bank level; weak at system level | Individual banks; disappears in system-wide booms |
| Customer behaviour | Households/firms repay debt or buy non-money assets (bank bonds, equity) | Money destruction | Moderate | Deleveraging periods |
| Reserve requirements and macroprudential quantity rules | GWM, RIM, LTV caps, sectoral limits | Quantity/price | Varies; meaningful in Indonesia, China [V for BI rules] | EMs; macroprudential episodes |
| Central bank balance-sheet policy | Can add (QE) or drain (QT) deposits directly | Quantity of broad money | Strong for broad money; weak for lending | Floor systems |
| Fiscal operations | TGA movements, bond issuance | Quantity of reserves and deposits | Strong mechanically | Large deficits/tax dates |

**A numerical illustration of the price constraint.** A bank's marginal loan pricing (illustrative, annual):

- Funding cost (deposits plus wholesale at margin): 4.0% (anchored by policy rate 3.75-4.00% in the US case [V])
- Expected credit loss: 1.0%
- Capital charge: equity of 10% of the loan x required return on equity 12% = 1.2%, minus the funding saved on that equity (10% x 4.0% = 0.4%) = 0.8%
- Operating cost: 1.0%
- Minimum loan rate to break even: about 6.8%.

If the policy rate rises by 1 percentage point, the minimum loan rate rises roughly 1 point; marginal borrowers drop out; new lending and deposit creation slow. Nothing in this calculation involves the quantity of reserves (interpretation, standard bank economics).

**Two common errors the evidence rules out** (interpretation, required by the brief's "avoid" list):

- *"Banks are unlimited money creators."* False: each loan is constrained by the price and risk factors above, and system-wide credit has fallen sharply when they bind (US commercial bank loans and leases fell about 10.6% from USD 7,020bn on 15 October 2008 to USD 6,276bn on 24 March 2010 [V, FRED TOTLL weekly], just before accounting rules FAS 166/167 brought securitised loans back onto bank balance sheets [E]; Indonesia's credit growth collapsed after 1998 [E]).
- *"Money creation is wealth creation."* False: a loan creates a matching asset and liability. Wealth is created by production and, in market value terms, by revaluation. Credit can *finance* wealth creation, or bid up existing asset prices, or finance consumption, with very different consequences (R3, R4).

## G. Indonesia snapshot: who creates money in Indonesia today (for the Part XII stream)

(fact unless marked; all [V] from BI Monetary Policy Report Q3 2026, published 23 October 2026, unless noted)

- **Bank credit is the main source of new rupiah money.** Bank credit grew 13.26% year-on-year in September 2026 (investment loans +23.20%, working capital +11.89%, consumer +4.86%), while third-party funds (DPK, deposits) grew 8.22%. Credit growing faster than deposits means banks were drawing down liquid assets and relying on BI liquidity incentives; liquid assets/DPK rose from 23.04% (August) to 24.08% (September).
- **Undisbursed loans** (committed but unused credit lines) were IDR 2,562.57 trillion, 21.70% of total credit ceilings: a reservoir of potential deposit creation that depends on borrower demand, not bank reserves.
- **Banking soundness**: capital adequacy ratio 24.10% and gross NPL 2.11% (August 2026), i.e., capital is not the binding constraint for the system as a whole.
- **Central bank money**: currency in circulation (UYD) IDR 1,341 trillion, +11.77% year-on-year in Q3 2026.
- **BI's tools**: BI-Rate 5.75% (held at the 20-21 October 2026 meeting), deposit facility raised to 5.00%, lending facility 6.50%; GWM cut from 9% to 7% in two steps (1 Nov and 1 Dec 2026); KLM incentive (cumulative IDR 458.5 trillion by early October 2026) being restructured to 5% of DPK (financing channel up to 3%, money-market deepening up to 2%) from 1 December 2026; RIM 84-94%; macroprudential liquidity buffer PLM 4%; CCyB 0%; LTV up to 100% and 0% minimum down payment for vehicles extended to end-2027.
- **Fiscal-monetary interface**: 2020-2022 burden sharing (about IDR 1,105 trillion of primary-market SBN purchases) [E]; Article 36A P2SK crisis power [V via press]; 2025 interest-cost sharing and IDR 200 trillion SBN purchases through August 2025 [V].
- **Exchange-rate pressure**: rupiah IDR 17,985 per USD on 20 October 2026; reserves USD 146.3bn at end-September 2026. This external constraint is a key reason Indonesia cannot treat domestic money creation as purely a domestic-demand question (interpretation; see R-international).

Interpretation: Indonesia combines (i) a standard endogenous-money banking system, (ii) a corridor-type interest-rate framework, (iii) actively used reserve requirements and credit-steering incentives of a kind largely abandoned in the US and euro area, and (iv) a history and live practice of fiscal-monetary coordination that tests the boundary of monetary financing. It is therefore a good case for the monograph's comparative argument that "how money is created" depends on institutional design, not on a single universal model.

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## H. Disputes and uncertainties

1. **How binding are capital requirements on aggregate money creation?** Mainstream view: capital raises the cost of lending and constrains undercapitalised banks; Werner (2016) [E] argues capital-adequacy regulation is ineffective at controlling credit booms because banks can create the money that buys their own equity indirectly and because profits in booms build capital. Evidence is mixed; the answer likely differs between boom and bust phases. Unresolved.
2. **Does QE raise bank lending?** Theory: weak/indirect (McLeay et al. [V]). Evidence: some studies find bank-lending effects for banks with more QE-induced deposits (e.g., Rodnyansky and Darmouni 2017 on US MBS purchases [E]); others (Butt et al. 2012 for UK [E]) find none. Unresolved; heterogeneous by program and country.
3. **Is the floor system a subsidy to banks?** Normative and empirical dispute about IORB pass-through to depositors and the fiscal cost (D.4). The ECB's choice to pay 0% on minimum reserves shows the policy is a design choice, not a necessity.
4. **Where is the line between QE and monetary financing?** Legal tests (Art. 123 TFEU; BI law) vs economic equivalence (D.1). Indonesia's 2025 arrangement is a live test case. Unresolved and normatively loaded.
5. **Werner's single-bank experiment: decisive or redundant?** The bookkeeping is not disputed by central banks after 2014; the dispute is about what it implies for regulation.

6. **Endogenous money and inflation**: endogenous money theory explains the *mechanics* of money supply; it does not by itself say whether money growth causes inflation or vice versa. The 2020-2023 episode (M2 +40%, then -4.7% year-on-year, alongside an inflation surge and decline) is consistent with several stories (demand-pull from fiscal transfers financed via QE; supply shocks; expectations). This is handled in R3; this chapter's data should not be read as evidence that money is irrelevant to inflation.
7. **Private credit and stablecoin data quality**: FSB says private credit data are severely limited [V]; stablecoin supply statistics come from private aggregators with different coverage (USD 290-306bn range) [E]. Treat as indicative.
8. **Some figures are [E] pending direct verification**: Indonesian burden-sharing totals by year (BI annual reports should be opened directly); BI Act article on primary-market prohibition; Fed 2021 remittance figure (USD 107.4bn vs 109bn depending on definition); China RRR level; private credit size; Werner's experiment details (bank, amount, date).
9. **Measurement breaks**: US M1 redefinition in May 2020; FRED reserve and required reserve series discontinued or redefined after 2020; TOTLL excludes non-bank lenders. Long-run comparisons of multiplier ratios must handle these.
10. **Effective dates still in motion**: GENIUS Act effective date and final rules (some agencies finalised, others proposed) as of November 2026 [E]; BI GWM changes take effect 1 Nov and 1 Dec 2026 [V]; Fed reserve management purchases variable month to month [E].

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# I. Implications for other streams and hypothesis register

- **H-R2-1 (supported)**: In ample-reserve systems, reserve quantity is not a binding constraint on bank lending. Evidence: Figures R2-1, R2-2; Fed zero RR [V]; McLeay et al. [V]; Borio and Disyatat [E]. Counter-evidence to watch: EM systems with active RR (Indonesia, China).
- **H-R2-2 (supported, conditional)**: Central bank asset purchases from non-banks increase broad money one-for-one on impact; purchases from banks do not. Evidence: T-accounts; McLeay et al. Figure 3 [V]; UK evidence (Butt et al. [E]).
- **H-R2-3 (supported)**: Normal bond-financed deficits leave broad money unchanged unless banks or the central bank buy the bonds. Evidence: sectoral accounting D.1.
- **H-R2-4 (open)**: Legal prohibitions on monetary financing are effective at preventing fiscal dominance. Indonesia 2020-2025 and euro area OMT/PSPP case law suggest prohibitions shape *form* more than *substance*; requires comparative evidence (R-cases).
- **H-R2-5 (open)**: Stablecoins under GENIUS-style 1:1 rules are economically narrow banks and will not expand private credit, but could reduce bank deposit funding. Evidence will come only after 2027 implementation.
- For **R3 (inflation)**: use the D.1 decomposition to separate deposit creation by bank lending, by QE-financed deficits and by bank purchases of government debt in 2020-2022.
- For **Part VII (fairness)**: the IORB transfer, the deferred-asset dynamics and Indonesia's quasi-fiscal interest sharing are concrete distributional channels of money creation, more tractable than generic "Cantillon" claims.
- For **Part X (CBDC)**: the T-account logic of deposit-to-stablecoin conversion (E.7) applies almost unchanged to deposit-to-CBDC conversion: reserves and bank deposits fall unless the central bank recycles funding to banks.

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## Data and code produced by this stream

- `data/r2/*.csv` and `data/r2/README.md` (FRED pulls; `r2_monthly_panel.csv`; `r2_reserve_regimes.csv`; `r2_seigniorage_fiscal.csv`).
- `work/r2_charts.py` (builds `research/figs/r2_fig1..4`).

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title: “R3 - Inflation and the Distribution of Purchasing Power (BRIEF Part V)”
project: “The Architecture of Money (Project III)”
authors: “M. Bagus Prabowo and Hark (he conceived and directs; Hark’s agents research and draft)”
stream: R3
status: “Research draft, Phase 4 input. Data pulled 7 Nov 2026 (WIB).”
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# R3 - Inflation and the Distribution of Purchasing Power

Research stream R3 for *The Architecture of Money*. Scope: BRIEF Part V (the twelve drivers of inflation; A. monetary vs consumer price inflation; B. distributional effects of inflation; C. the Cantillon effect; D. asset inflation and financial inequality), with an Indonesia sub-section. Authors: M. Bagus Prabowo and Hark (co-authored; he conceived and directs, Hark's agents research and draft).

Evidence tags (FRAMEWORK.md): **[V]** = page or dataset opened and the claim is on it; **[E]** = search excerpt or established knowledge, not opened. Claim types are labelled where it matters: (fact), (empirical finding: Author year), (interpretation: school/author), (normative), (counterfactual). Every chart and derived dataset in this file was produced from raw pulls saved under `data/r3/`; the scripts are `work/r3_*.py`, so every number computed here can be re-run.

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## Key findings

1. **Money growth and inflation are tightly linked only when inflation is high.** (empirical finding: this stream, replicating McCandless and Weber 1995; Teles, Uhlig and Valle e Azevedo 2016; Borio, Hofmann and Zakrajsek 2023) Using World Bank data for 136 countries over 1995-2023, the cross-country correlation between average broad-money growth and average CPI inflation (in logs) is 0.92 with a slope of 0.84. Among the 25 countries whose average inflation was 10% or more, the correlation is 0.95 and the slope 0.92, close to one-for-one. Among the 72 countries whose average inflation was below 5%, the correlation falls to 0.56 and the slope to 0.19: a country with 5 points more money growth per year had, on average, only about 1 point more inflation. Over 1980-2023 the same split gives slopes of 0.89 above 10% and 0.15 below 5%. The quantity-theory relationship is real in the long run and at high inflation. It is a poor guide to year-to-year inflation in low-inflation, inflation-targeting economies.
2. **In the United States the link depends on the time horizon and the decade.** Over 1960-2024 the correlation between annual M2 growth and CPI inflation is 0.11 at one-year horizons, 0.53 at five-year averages and 0.71 at ten-year averages (the last has only six observations). Over 1991-2019 the correlation was negative at every horizon tested (-0.30 annual, -0.62 for five-year averages). Velocity was not stable in that period, so the relation broke down.
3. **The 2020-23 episode was a joint demand-and-supply event, not a pure monetary or a pure supply story.** US M2 rose 40.5% from February 2020 (USD 15.49 trillion) to its March 2022 peak (USD 21.77 trillion), with year-on-year growth peaking at 26.8% in February 2021. It then *fell* 4.9% to October 2023, the first sustained nominal M2 contraction since the series began in 1959. CPI inflation peaked at 9.0% (seasonally adjusted, June 2022), with the money peak leading the price peak by about 16 months. The San Francisco Fed's Shapiro decomposition attributes the June 2022 peak in headline PCE inflation (7.1%) to 3.4 points supply-driven, 2.8 points demand-driven and 1.0 point ambiguous. Compared with the 2017-19 average, the extra supply-driven inflation was about 2.6 points and the extra demand-driven inflation about 2.1 points. Bernanke and Blanchard (2023; 2024 for 11 economies) find that the surge came mainly from energy, food and sectoral supply shocks hitting strong demand, with tight labour markets becoming more important later. Our reading: supply shocks, plus fiscal and monetary stimulus that let

those shocks pass through to the price level, explain the episode jointly. Neither "money printing" nor "supply chains" alone fits the data.

4. **Monetary aggregates, consumer prices, producer prices and asset prices can diverge sharply.** From December 2008 to December 2019, US M2 rose 87%, the NASDAQ Composite rose 475%, the Case-Shiller national house price index rose 39%, average hourly earnings of production workers rose 30% and the CPI rose 22%. Most of the money created by quantitative easing (QE) sat as bank reserves and as deposits held for portfolio reasons rather than for spending, and its main effect ran through lower real interest rates and higher asset valuations. The BRIEF's warning not to confuse asset inflation with CPI inflation is well founded: the two are different phenomena with different winners and losers.
5. **Inflation redistributes mainly through nominal balance sheets and consumption baskets, and its incidence depends on the type of inflation.** (empirical finding: Doepke and Schneider 2006; Pallotti et al. 2024) Unexpected moderate inflation transfers wealth from holders of nominal assets (older, richer bondholders, and cash holders) to nominal debtors (young, middle-class mortgagors) and to the government. Inflation through the consumption basket is regressive: the BLS research index by income quintile shows prices rising 22.5% for the poorest fifth and 20.5% for the richest fifth from February 2020 to June 2024 [V]. Hyperinflation and high chronic inflation are regressive on almost every channel, because the poor hold cash and lack inflation hedges (Easterly and Fischer 2001; Erosa and Ventura 2002).
6. **The Cantillon effect is a valid mechanism with weak direct evidence and strongly conditional size.** It says that new money changes relative prices and distribution depending on where it enters the economy. It requires the economy to adjust slowly, markets to be segmented, and money to be injected unevenly. Under QE, money entered through asset markets as a swap of reserves for bonds, which is consistent with a Cantillon-type asset-price channel. In 2020-21, a large share of new money entered through fiscal transfers to households at the bottom of the distribution. The bottom-50% share of US household net worth rose from 1.7% (2019Q4) to 2.4% (2021Q4) in Federal Reserve data. The injection point can therefore run in either distributional direction. Treating the Cantillon effect as a general explanation of wealth inequality is not supported.
7. **Low interest rates raise asset valuations, but the long rise in wealth concentration began before QE and has largely structural causes.** In the Federal Reserve's Distributional Financial Accounts, the top 1% share of US net worth rose from 22.8% (1989Q3) to 27.6% (2008Q4), before QE, then to 30.4% (2019Q4) and 32.5% (2026Q2). The top 1% holds 50.9% of household corporate equity and mutual fund shares; the bottom 50% holds 0.6%. Monetary policy shapes the *cyclical* path and amplifies valuation gains through equities. Household micro-studies, however, find its effect on standard inequality measures to be small or mixed (Bunn, Pugh and Yeates 2018; Lenza and Slacalek 2018/2024), or regressive through capital income (Andersen et al. 2023). The secular forces are technology, globalisation, market power, falling real rates driven partly by the "saving glut of the rich" (Mian, Straub and Sufi), demographics, tax policy and housing-supply constraints. For housing, Rognlie (2015) shows that the post-war rise in the net capital share comes entirely from housing, which points to land scarcity and regulation as much as to monetary policy.
8. **Housing affordability is driven by rates in the short run and by supply in the long run.** For the US median new house, principal and interest at 80% loan-to-value fell to about 21% of median household income in 2012-2021 as mortgage rates fell. They jumped to 33% in 2023 when rates rose, even as the price-to-income ratio fell from 5.8 (2022) to 4.75 (2025). Low rates made houses easier to finance and pushed prices up, so their net effect on affordability largely cancelled. They raised existing owners' wealth relative to prospective buyers. This is an intergenerational transfer, clarified by Fagereng et al. (2025): higher asset prices benefit net sellers and hurt net buyers.
9. **Indonesia fits the regime pattern.** In 1998, broad money growth (62.8%) and CPI inflation (58.5%) coincided during the currency collapse. Outside 1998 the annual correlation is only 0.25, and over 2020-21 broad money grew 12-14% while inflation stayed below 2%. Indonesian inflation is dominated by food, fuel and administered prices. In October 2026 headline inflation was 3.23% year on year. Food, beverages and tobacco rose 4.82% and contributed 1.40 points; transport rose 4.87% and contributed 0.59 points [V for contributions]. Food makes up 74.7% of the national poverty line (March 2026) [E], so food-led inflation falls hardest on poor households.
10. **For the monograph's central question 3 (how much of inflation each mechanism explains):** the honest answer depends on the regime and the horizon. Over decades and across high-inflation countries, money growth (usually fiscally driven) explains most of the variation. In low-inflation regimes, short-run inflation is dominated by supply shocks, energy, demand pressure relative to capacity, and expectations,

with money growth adding forecasting information at best. Bank credit creation matters mainly through asset prices and financial cycles rather than through CPI.

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# 1. Framing: what "inflation" is, and what kind of claim each theory makes

## 1.1 Definitions used in this stream

- **Consumer price inflation** (fact, definitional): the rate of change of a price index for a fixed or chained basket of household consumption, e.g. US CPI-U (BLS, Laspeyres-type with monthly updating), US PCE price index (BEA, chained Fisher, broader scope including third-party payments), euro-area HICP, Indonesia IHK (BPS, base 2022=100, 11 expenditure groups, 38 provinces). The index chosen matters: PCE inflation usually runs about 0.3-0.5 points below CPI because of formula and weights [E].
- **Producer price inflation**: change in prices received by domestic producers (US PPI final demand, PPI all commodities). It is more volatile and closer to commodity prices.
- **Monetary growth**: change in a monetary aggregate (base money, M1, M2, M3). This is a *quantity* of liabilities of the banking system, not a price.
- **Credit expansion**: growth of loans and securities on bank balance sheets (and, in broader measures, non-bank credit). Bank lending creates deposits, as set out in R2/Part IV (McLeay, Radia and Thomas 2014), so credit growth and money growth are linked but not identical: banks also create deposits when they buy securities, and destroy them when loans are repaid or when depositors buy bank equity or bonds.
- **Asset price inflation**: rise in the prices of claims on future income (equities, bonds, housing, land). These prices are not in the CPI, apart from owners' equivalent rent, which is a rental *flow* and not a house *price*.
- **Currency depreciation**: fall in the external value of the currency. It feeds CPI through import prices with incomplete and variable pass-through.

## 1.2 The equation of exchange and why it does not settle the argument

The identity MV = PY (money times velocity equals the price level times real output) holds by definition, because V is measured as PY/M. In growth rates:

money growth + velocity growth = inflation + real output growth

The *quantity theory* turns the identity into a theory by assuming that (a) velocity is stable or predictable, (b) real output is determined by real factors in the long run, and (c) money is exogenous, so causation runs from M to P (interpretation: Friedman 1956; Friedman and Schwartz 1963). Post-Keynesian endogenous-money theory reverses (c). Money is created when banks lend to meet demand for credit, and central banks set the price of reserves rather than their quantity, so money adapts to nominal spending rather than causing it (interpretation: Moore 1988; Lavoie; Goodhart). New Keynesian models make the price level depend on the interest-rate rule and on expectations, with money left in the background (interpretation: Woodford 2003; Gali). Austrian theory accepts that money causes price changes but rejects the aggregate price level as the relevant object. It focuses on *relative* price distortions from the point and timing of injection (interpretation: Mises 1912; Hayek 1931; Sieron 2019). These are not just different emphases. They are different causal claims, and the evidence below lets us judge where each is strong.

## 1.3 Why distribution belongs in the inflation chapter

Inflation is never a uniform shrinking of all money values at once, as if every price, wage and contract were multiplied by the same factor overnight. In practice it is uneven across goods (relative prices), across contracts (nominal versus indexed), across balance sheets (net nominal creditors versus debtors) and across time (who gets the new spending power first). Even a purely monetary inflation redistributes whenever contracts are fixed in nominal terms. That is why Part V of the BRIEF joins "inflation" and "the distribution of purchasing power", and why this stream treats the two together.

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# 2. The twelve drivers

Each driver is treated in the same way: mechanism, evidence, where it is strong or weak, and how the schools differ. A summary table closes the section.

## 2.1 Money supply growth

**Mechanism.** If the public's demand for real money balances is stable, sustained growth in nominal money beyond the growth of real output must eventually show up as a higher price level (interpretation: quantity theory, monetarism). The sharpest form is "inflation is always and everywhere a monetary phenomenon" (Friedman 1963, in *Inflation: Causes and Consequences*) [E].

**Long-run cross-country evidence (computed for this stream).** We pulled World Bank WDI broad-money growth (FM.LBL.BMNY.ZG) and CPI inflation (FP.CPI.TOTL.ZG) for every country with at least 75% of annual observations, and computed compounded averages per country. Correlations and slopes are in logs (log(1+x)), so that hyperinflation does not mechanically dominate.

**Table 1. Average broad-money growth vs average CPI inflation across countries**

| Period | Sample | n | Correlation | OLS slope (inflation on money growth) | Correlation, money growth minus real GDP growth vs inflation |
|---|---|---|---|---|---|
| 1980-2023 | all | 106 | 0.96 | 0.95 | 0.98 |
| 1980-2023 | average inflation >= 10% | 30 | 0.96 | 0.89 | 0.97 |
| 1980-2023 | average inflation < 5% | 47 | 0.43 | 0.15 | 0.53 |
| 1995-2023 | all | 136 | 0.92 | 0.84 | 0.96 |
| 1995-2023 | average inflation >= 10% | 25 | 0.95 | 0.92 | 0.96 |
| 1995-2023 | average inflation < 5% | 72 | 0.56 | 0.19 | 0.58 |
| 2000-2019 | all | 146 | 0.86 | 0.63 | 0.91 |
| 2000-2019 | average inflation < 5% | 83 | 0.55 | 0.16 | 0.63 |

Source: World Bank WDI, last updated 2026-10-08, pulled 7 Nov 2026 [V]; computation `work/r3_crosscountry.py`; output `data/r3/crosscountry_money_cpi_*.csv`. Chart: `research/r3-charts/fig4_crosscountry_money_inflation_199`

Limitations: (i) splitting the sample by average inflation conditions on the outcome variable, which lowers measured correlations in the low group (range restriction). The pattern is still the one the literature reports with better designs. (ii) Euro-area members have no national broad-money growth series after 1999 and drop out. (iii) Argentina, Venezuela and Zimbabwe drop out under the coverage rule, so the very-high-inflation tail is thinner than in McCandless and Weber. (iv) National definitions of "broad money" differ.

The results replicate the established literature:

- McCandless and Weber (1995), "Some Monetary Facts", *Federal Reserve Bank of Minneapolis Quarterly Review* 19(3): over 30 years and 110 countries, money growth and inflation are "highly correlated for all three money definitions", while money growth and real output growth are generally uncorrelated (except in the OECD subsample) [V, abstract at https://www.minneapolisfed.org/research/quarterly-review/some-monetary-facts]. The often-quoted correlation coefficients of about 0.92-0.96 are from the paper's tables [E, not reopened].
- Teles, Uhlig and Valle e Azevedo (2016), "Is Quantity Theory Still Alive?", *Economic Journal* 126(591): 442-464: "for countries with low inflation, the raw relationship between average inflation and the growth

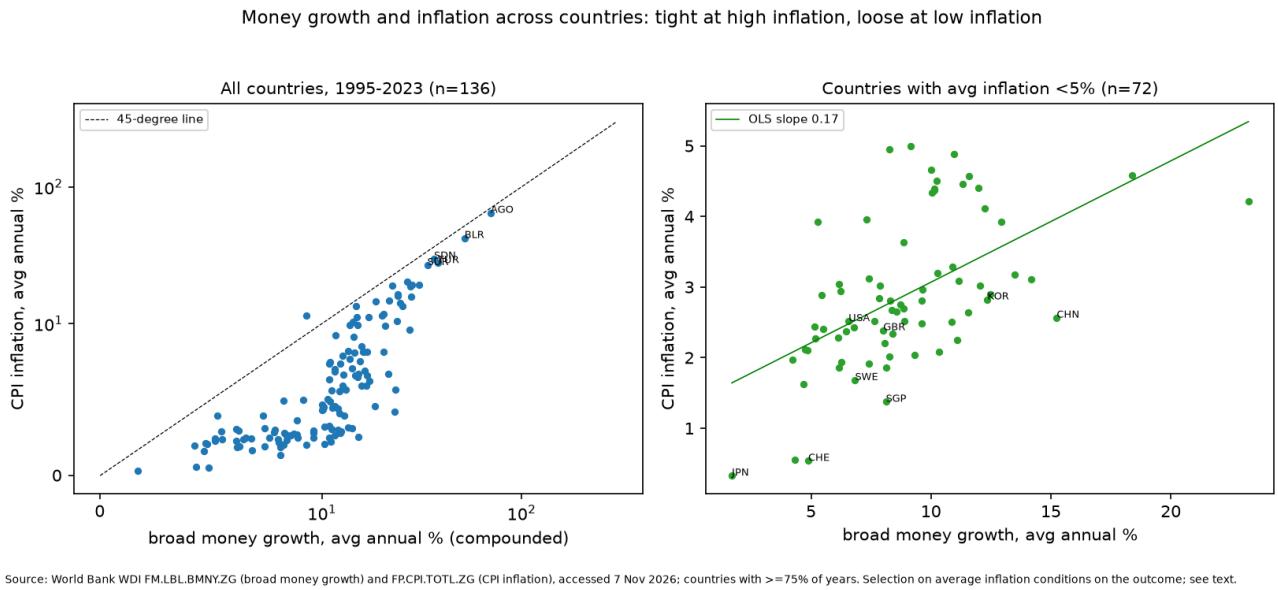



Figure 5: Fig 4

rate of money is tenuous at best". The fit "markedly improves" after correcting for output growth and the opportunity cost of money, while inflation-targeting subsamples worsen the one-for-one fit [E, abstract via search excerpt; DOI 10.1111/ecoj.12336]. Our own correction for output growth (last column of Table 1) improves the low-inflation correlation from 0.56 to 0.58 (1995-2023) and from 0.43 to 0.53 (1980-2023), which is consistent with their first correction.

- Borio, Hofmann and Zakrajsek (2023), BIS Bulletin 67: the link "is one-to-one when inflation is high and virtually non-existent when it is low", but money growth helped predict the 2021-22 surge. Their 2024 update finds the cross-country coefficient of 2020 excess money growth on 2021-22 inflation drops to 0.34 when Argentina and Turkey are excluded, though it stays statistically significant [E, BIS page and 2024 speech excerpt].

**US time-series evidence by horizon (computed).**

**Table 2. Correlation of US M2 growth and CPI inflation by horizon (annual averages, non-overlapping windows)**

| Horizon | 1960-2024 | 1960-1990 | 1991-2019 | 1991-2024 |
|---|---|---|---|---|
| 1 year | 0.11 (n=64) | 0.00 (n=30) | -0.30 (n=28) | -0.17 (n=33) |
| 2 years | 0.24 | 0.11 | -0.47 | -0.33 |
| 5 years | 0.53 (n=12) | 0.65 (n=6) | -0.62 (n=5) | -0.36 (n=6) |
| 10 years | 0.71 (n=6) | - | - | - |

Lead structure, 1960-2024: correlation of inflation with M2 growth one year earlier is 0.31, two years earlier 0.47; for 1991-2019 the corresponding values are -0.40 and -0.20. Source: FRED M2SL, CPIAUCSL [V]; `data/r3/us_m2_cpi_horizon_correlations.csv`. Limitation: windows at 5 and 10 years have very few observations, and the M2 definition changed in May 2020 (savings deposits reclassified into M1; M2 total continuous) [V, Fed H.6, see phase1/verification_log.md].

**Interpretation.** The quantity relation is a long-run and high-inflation regularity (empirical finding). In 1991-2019 the US had large, persistent swings in M2 velocity driven by financial innovation, interest-rate changes and, after 2008, reserve-heavy balance sheets. Money growth was then *negatively* associated with inflation over five-year windows. Over that period, the money stock mostly adjusted to portfolio demand rather than driving spending (interpretation: endogenous money; New Keynesian). The 2020-22 episode partly restored the link (Section 3), which is the core of the BIS argument that money's information value "may have been neglected".

**Where each school stands.** Monetarists are right about the long run and about high inflation. Post-Keynesians are right that in modern operating frameworks money is mostly endogenous in the short run. The large-sample correlations do not show which way causation runs: in high-inflation countries, money growth is usually the *means* by which fiscal deficits are financed (Section 2.9), and the deeper cause is fiscal and institutional (Sargent 1982; Cagan 1956) [E].

## 2.2 Bank credit expansion

**Mechanism.** When a commercial bank lends, it credits the borrower's deposit account, creating broad money. Repayment destroys it (McLeay, Radia and Thomas 2014) [V, BoE QB page, see phase1 log]. Credit expansion therefore adds to money growth. Whether it adds to CPI inflation depends on what the credit finances. Credit for consumption or for investment beyond capacity raises demand for goods. Credit for buying existing assets (houses, land, equities) raises asset prices and does not directly raise CPI (interpretation: Werner's "quantity theory of credit"; Bezemer; Ryan-Collins et al.) [E].

**Evidence.** US commercial bank credit rose from USD 13.98 trillion (February 2020) to USD 16.48 trillion (December 2021), +17.9%, and to USD 17.50 trillion by December 2022 (FRED TOTBKCR [V]). Much of the 2020 rise was banks absorbing deposits from fiscal transfers and buying securities, not new loans to households. The historical evidence connects credit booms more strongly with *financial crises and asset-price booms* than with CPI inflation. Schularick and Taylor (2012, *American Economic Review*) show that credit growth predicts financial crises in 14 advanced economies over 1870-2008 [E]. Jorda, Schularick and Taylor document that the rise of mortgage credit after 1950 shifted bank lending towards real estate [E]. In the 2000s US housing boom, mortgage credit surged while CPI inflation stayed near 2-3%.

**Assessment.** Credit creation is a major driver of *asset* inflation and financial fragility. It drives CPI inflation only when it finances spending on currently produced goods beyond capacity. The BRIEF's caution against "simplistic claims that all inflation is caused by bank lending" is supported. So is the opposite caution: in the 2020-21 US episode, bank lending was not the main source of the money surge. Fiscal deficits financed partly through Fed purchases were.

## 2.3 Aggregate demand

**Mechanism.** When total nominal spending (consumption, investment, government purchases, net exports) grows faster than real output can grow at given costs, firms raise prices (interpretation: Keynesian and New Keynesian; also Austrian, which emphasises demand financed by credit expansion).

**Evidence.** US real GDP rose above the Congressional Budget Office's estimate of potential in 2021, with an output gap of +2.1% in 2021Q4 (FRED GDPC1/GDPPOT [V], computed). The unemployment rate fell to 3.4% in 2023 (FRED UNRATE [V]). The Shapiro decomposition (Section 3) attributes roughly 2.1 points of the extra 2022 PCE inflation to demand. The composition of demand mattered as much as its level: spending shifted from services to goods during the pandemic, and goods supply is less elastic in the short run. Bernanke and Blanchard (2023) find that fiscal policy contributed "primarily through its effects on consumer demand for commodities and goods in limited supply rather than through the labor market" [V, Brookings summary].

## 2.4 Production capacity

**Mechanism.** Inflation results from demand *relative to* supply capacity, so falling capacity raises inflation at an unchanged level of demand. The Phillips curve may be non-linear: when the economy is near capacity, prices respond more to demand (empirical finding: Benigno and Eggertsson 2023, on labour-market tightness measured by vacancies over unemployment) [E].

**Evidence.** In 2021-22 capacity was constrained by pandemic closures, port congestion and semiconductor shortages, and by labour supply (early retirements, lower immigration). Global supply-chain pressure indices peaked in late 2021 [E]. The Shapiro method classifies a category as supply-driven when its price and quantity surprises move in opposite directions. It found the supply component peaking at 3.4 points in June 2022 (Section 3) [V, data].

## 2.5 Supply shocks

**Mechanism.** An adverse supply shock raises costs and prices while lowering output. With credible monetary policy it changes the price *level* once rather than raising inflation persistently. It turns into persistent inflation if it feeds into wages and expectations, or if policy accommodates it (interpretation: New Keynesian; the 1970s lesson).

**Evidence.** The 1973-74 and 1979-80 oil shocks, combined with accommodative policy and drifting expectations, produced double-digit inflation. The 2021-22 shocks (energy, food after the Russian invasion of Ukraine, sectoral shortages) produced a sharp but less persistent burst. Bernanke and Blanchard (2024) attribute the lower persistence across 11 economies "in part" to the credibility of inflation targets [E, NBER w32532 abstract]. **The 2026 test case:** the US-Israeli war with Iran that began at the end of February 2026 restricted shipping through the Strait of Hormuz. WTI crude averaged about USD 58 in December 2025 and USD 97 in September 2026 (FRED DCOILWTICO monthly means [V]). US gasoline prices were 34.2% higher year on year in September 2026 and energy prices 20.4% higher [E, BLS via BBC/CNBC]. Headline CPI rose to 3.7% (BLS, not seasonally adjusted) while core CPI held at about 2.8% (FRED CPILFESL seasonally adjusted, computed [V]). In the UK, the Bank of England attributed 0.7 points of the 1.1-point overshoot of its 2% target in August 2026 to direct energy effects [E, BoE Governor's letter, 17 Sep 2026]. M2 growth during this episode was a modest 4.1-5.7% year on year in January-September 2026 (FRED [V]). This is a clean example of supply-driven inflation without a preceding money surge.

## 2.6 Energy prices

Energy is the single most important supply-shock channel because it enters directly (fuel and utilities in the CPI basket) and indirectly (transport and production costs). Its weight in the basket is larger for poorer households and for households in colder or more car-dependent places, which makes energy shocks distributionally regressive (Section 5.4). In Indonesia, energy enters mostly through *administered* prices. The September 2022 fuel price increase (Pertalite from IDR 7,650 to IDR 10,000 per litre on 3 September 2022) [E, BPS Jambi article] raised administered-price inflation to 13.28% and headline CPI to 5.95% year on year in September 2022 [E, BI/TPIP analysis excerpt and BI press release]. Policy here sets the size and timing of the shock directly.

## 2.7 Wage growth

**Mechanism.** Wage increases raise unit labour costs. In services, where labour is the main input, they are a key driver of persistent inflation. Whether wages *cause* inflation or *follow* it is disputed. Bernanke and Blanchard (2023) find that in 2021-22 "most of the rise in inflation ... was driven by developments that directly raised prices rather than wages". Labour-market tightness (vacancies per unemployed worker) "began to play a more significant role" later [V, Brookings summary].

**Evidence.** US average hourly earnings of production and non-supervisory workers peaked at 7.0% growth year on year in March 2022 (FRED AHETPI [V]). In real terms (deflated by CPI) these earnings were 0.7% above their February 2020 level in June 2022 and 4.7% above it by September 2026 (computed). The early-2021 real gain (+3.1% by March 2021) mainly reflects *composition*: low-wage workers lost jobs, which raised the average. This is a standard measurement trap. A wage-price spiral of the 1970s kind did not occur: real wages fell first and recovered later.

## 2.8 Exchange rates

**Mechanism.** Depreciation raises import prices in domestic currency. Pass-through to the CPI is larger in small open economies, in economies where trade is invoiced in dollars (Gopinath's dominant-currency paradigm) [E], and where inflation expectations are poorly anchored. In emerging markets, depreciation is often the *trigger* that converts monetary or fiscal imbalances into inflation.

**Evidence.** Indonesia 1998 is the paradigm case. The rupiah collapse and the emergency liquidity support to banks (Bank Indonesia Liquidity Assistance, BLBI) [E] coincided with broad money growth of 62.8% and CPI inflation of 58.5% while real GDP fell 13.1% (WDI [V], `data/r3/indonesia_money_cpi_gdp_wdi.csv`). Part of the 1998 "money growth" was the revaluation of foreign-currency deposits included in M2 when the rupiah fell.

So the depreciation inflated *both* the measured money stock and prices, and correlation is not causation here. For the US, dollar depreciation matters less because of dollar invoicing.

## 2.9 Fiscal deficits

**Mechanism.** Deficits raise demand directly. They also become inflationary through money creation if financed by the central bank (monetary financing, seigniorage), or through the fiscal theory of the price level (FTPL) if the public does not expect future surpluses to back the debt (interpretation: Sargent and Wallace 1981, "unpleasant monetarist arithmetic"; Leeper, Sims, Cochrane) [E]. MMT holds that deficits are inflationary only when spending exceeds real resource capacity, and that taxes, not bond sales, are the main inflation-control tool (interpretation: Kelton 2020; Wray) [E].

**Evidence.** US federal deficits were USD 3.13 trillion in fiscal year 2020 and USD 2.77 trillion in fiscal year 2021, against USD 0.98 trillion in fiscal year 2019 (FRED FYFSD [V]). Over the same period the Federal Reserve's balance sheet grew from USD 4.16 trillion (February 2020) to USD 8.97 trillion (April 2022) (FRED WALCL [V]), buying a large share of the new Treasury issuance in the secondary market. This was not *direct* monetary financing: the Fed bought Treasuries in the open market, did not buy them directly from the Treasury, and later reversed part of the purchases through quantitative tightening (QT). Economically, though, it delivered the combination that the fiscal and monetarist accounts both predict will be inflationary: a large deficit, absorbed by central-bank reserves, and paid out as household deposits. Hyperinflations (Weimar, Zimbabwe, Venezuela) are, without exception in the literature this stream reviewed, fiscal-monetary events: deficits monetised when bond markets closed (Sargent 1982; Cagan 1956; Reinhart and Rogoff 2009) [E]. For Indonesia, Bank Indonesia's 2020-22 "burden sharing" purchases of government bonds in the primary market [E] coincided with low CPI inflation (1.6-4.2%). Monetary financing in a slack economy with credible institutions need not be immediately inflationary. The risk is to the *regime*, not to the next year's CPI.

## 2.10 Monetary policy

**Mechanism.** Policy rates affect inflation through credit costs, asset prices, the exchange rate, expectations and, in heterogeneous-agent models, mainly through general-equilibrium effects on labour income rather than through intertemporal substitution (empirical finding: Kaplan, Moll and Violante 2018, *American Economic Review* 108(3): 697-743) [E, AER abstract].

**Evidence.** The federal funds rate rose from 0.08% (January 2022) to 5.33% (August 2023) (FRED FEDFUNDS [V]). The 10-year TIPS real yield went from -0.91% (2021 average) to 1.68% (2023 average) [V, computed]. Inflation then fell without a recession, which the "immaculate disinflation" literature attributes to the unwinding of supply shocks combined with anchored expectations [E]. In September 2026 the Federal Reserve raised its target range to 3.75%-4.00% in response to the energy shock [E, CNBC]. Bank Indonesia held the BI-Rate at 5.75% in October 2026 and cut the rupiah reserve requirement for conventional banks from 9% to 8% (1 November 2026) and to 7% (1 December 2026) [V, BI press release 28/233/DKom; see phase1 log].

## 2.11 Inflation expectations

**Mechanism.** Firms and workers set prices and wages partly on expected inflation. Anchored expectations make shocks transitory; unanchored ones make them persistent (interpretation: Friedman 1968; Phelps; rational expectations; New Keynesian).

**Evidence.** Market-based 10-year breakeven inflation peaked at a monthly average of 2.88% in April 2022 and was 2.26% in late 2026 (FRED T10YIE [V]). Household survey expectations (University of Michigan, one year ahead) peaked at 5.4% in March 2022. They reached a higher 6.6% in May 2025, during tariff announcements, before actual inflation had moved much, and stood at 4.6% in September 2026 (FRED MICH [V]). Survey and market measures diverged, and recent research documents partisan splits in survey answers [E]. Expectations are therefore not a single observable. Anchoring of *long-run* market expectations is the best explanation for why 2021-23 did not become the 1970s.

## 2.12 Asset prices

**Mechanism.** Asset prices affect CPI inflation in three ways: through wealth effects on consumption; through housing costs (rents, and owners' equivalent rent in the US CPI, which follows market rents with a lag); and through collateral values that relax credit constraints. In the other direction, monetary conditions affect asset prices *more strongly and faster* than consumer prices. This divergence is what Section 4 documents.

**Evidence.** The Case-Shiller national index rose 44.5% from February 2020 to June 2022 and 58.0% to August 2026 (FRED CSUSHPINSA [V], computed), while CPI rose 29.7% from February 2020 to September 2026. Shelter inflation in the CPI lagged the market-rent surge by roughly a year, which kept core inflation elevated in 2023-24 [E].

## 2.13 Summary: the twelve drivers compared

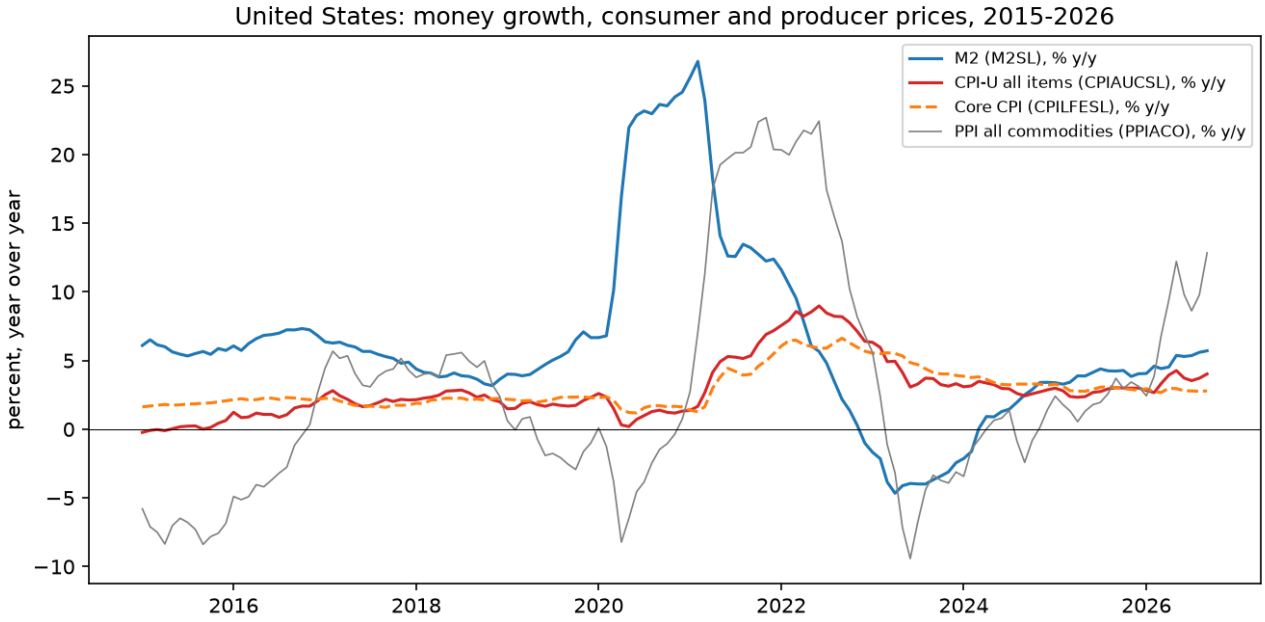
**Table 3. The twelve drivers of inflation: mechanism, strength and evidence**

| Driver | Main mechanism | Strongest when | Weakest when | Evidence quality | School most associated |
|---|---|---|---|---|---|
| Money growth | excess real balances raise spending | high inflation, long horizons, fiscal dominance | low inflation, unstable velocity, reserve-rich regimes | strong long-run cross-country; weak short-run | Monetarist, quantity theory |
| Bank credit | deposit creation financing demand or asset purchases | credit finances consumption or investment beyond capacity | credit finances existing assets | strong for asset booms and crises; weak for CPI | Post-Keynesian, credit theory, Austrian |
| Aggregate demand | spending beyond output | near capacity (non-linear) | slack | strong | Keynesian, New Keynesian |
| Production capacity | falling supply at given demand | pandemics, wars, sectoral bottlenecks | elastic supply | moderate to strong | New Keynesian, supply-side |
| Supply shocks | cost shocks to the price level | accommodated or with unanchored expectations | credible policy | strong (1970s, 2021-22, 2026) | New Keynesian |
| Energy | direct and indirect cost channel | oil-importing, energy-intensive economies | hedged or subsidised economies | strong | all |
| Wages | unit labour costs, services | tight labour markets | slack, weak bargaining | moderate; direction of causation disputed | Keynesian, post-Keynesian (conflict inflation) |
| Exchange rate | import-price pass-through | small open economies, emerging markets | dollar-invoiced economies | strong for emerging markets | open-economy models |
| Fiscal deficits | demand plus financing | monetised; fiscal dominance | credible future surpluses | strong for hyperinflations | FTPL, monetarist (Sargent-Wallace), MMT (capacity) |
| Monetary policy | rates, credit, asset prices, labour income | credible, forward-looking | zero lower bound, fiscal dominance | strong | New Keynesian |

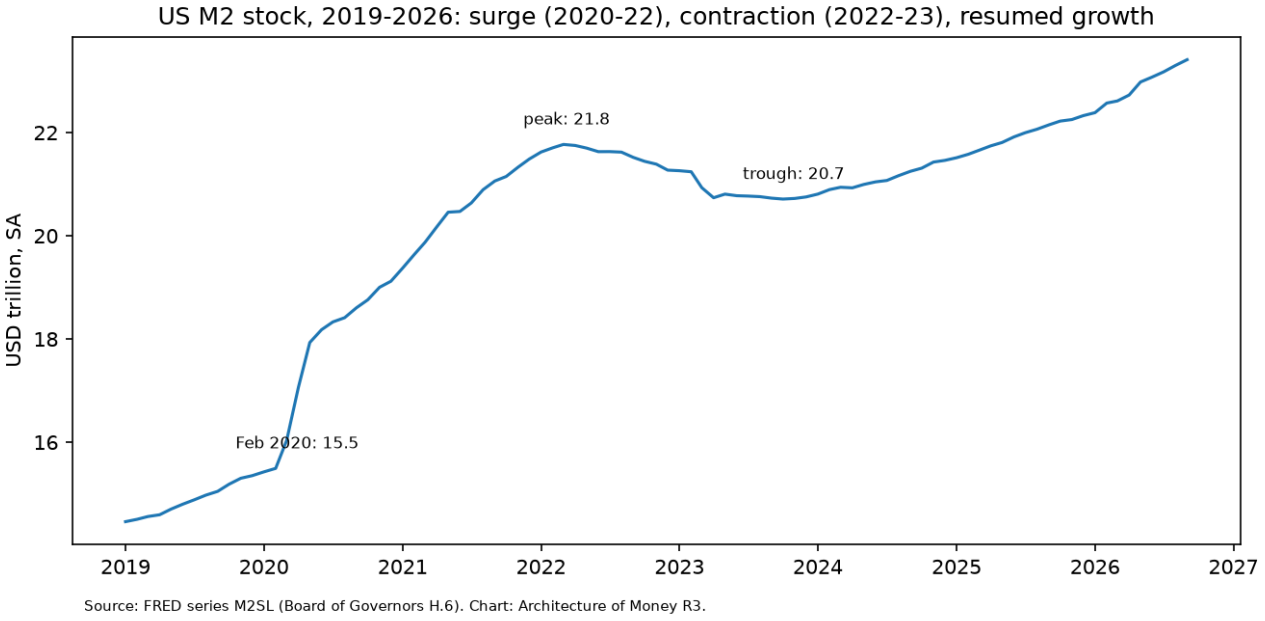
| Driver | Main mechanism | Strongest when | Weakest when | Evidence quality | School most associated |
|---|---|---|---|---|---|
| Expectations | price and wage setting | after repeated shocks | anchored regime | strong but measurement-dependent | rational expectations, New Keynesian |
| Asset prices | wealth, collateral, housing costs | housing-heavy CPI, leveraged households | low pass-through | moderate | Minsky, BIS financial-cycle view |

# 3. Case study: the 2020-2023 inflation surge (and the 2026 re-acceleration)

## 3.1 The facts in sequence (United States)




Source: FRED (Federal Reserve Board H.6; BLS). Seasonally adjusted monthly except PPI (NSA). Chart: Architecture of Money R3.




Source: FRED series M2SL (Board of Governors H.6). Chart: Architecture of Money R3.

**Table 4. US 2020-2026 key series (all FRED [V], computed in `work/r3_us.py`, `work/r3_charts1.py`)**

| Indicator | Value | Date | Note |
|---|---|---|---|
| M2 stock | USD 15.49 trillion | Feb 2020 | seasonally adjusted |
| M2 stock, peak | USD 21.77 trillion (+40.5%) | Mar 2022 | |

| Indicator | Value | Date | Note |
|---|---|---|---|
| M2 stock, trough | USD 20.71 trillion (-4.9% from peak) | Oct 2023 | first sustained nominal decline since 1959 |
| M2 stock, latest | USD 23.41 trillion | Sep 2026 | |
| M2 growth y/y, peak | 26.8% | Feb 2021 | |
| M2 growth y/y, low | -4.7% | Apr 2023 | |
| Fed total assets | USD 4.16 trillion to 8.97 trillion | Feb 2020 to Apr 2022 | WALCL; USD 6.75 trillion on 4 Nov 2026 |
| CPI-U y/y, peak | 9.0% (seasonally adjusted; BLS NSA 9.1%) | Jun 2022 | |
| Core CPI y/y, peak | 6.6% | Sep 2022 | |
| PCE price index y/y, peak | 7.2% | Jun 2022 | |
| PPI all commodities y/y, peak | 22.7% | Nov 2021 | not seasonally adjusted |
| Federal deficit | USD 3.13 trillion; 2.77 trillion | FY2020; FY2021 | FYFSD |
| Fed funds rate | 0.08% to 5.33% | Jan 2022 to Aug 2023 | |
| Cumulative change Feb 2020 to Dec 2023 | M2 +34.0%; CPI +19.1% | | |
| Cumulative change Feb 2020 to Sep 2026 | M2 +51.1%; CPI +29.7% | | |

Limitation: CPI for October 2025 was not published (federal shutdown), so the FRED seasonally adjusted series has a gap. For September 2026, FRED's seasonally adjusted year-on-year rate computes to 4.0%, while BLS's published not-seasonally-adjusted headline is 3.7% [E]. The two differ because of seasonal factors. This report quotes BLS headlines where available.

## 3.2 Where the money came from (balance-sheet view)

The 2020-21 money surge came mainly from fiscal transfers financed by bonds that the central bank largely absorbed, not from bank lending. A stylised sequence, in USD for a single USD 1,200 household payment, follows. Reserves and the Treasury General Account (TGA) are both Federal Reserve liabilities.

**T-account 1. A stimulus payment financed by a Treasury bond that the Fed buys in the secondary market**

Step 1: the Treasury sells a USD 1,200 bond to a primary dealer, whose bank account is debited.

| Fed: Assets | Fed: Liabilities |
|---|---|
| | Reserves of dealer's bank -1,200 |
| | TGA +1,200 |

Step 2: the Treasury pays the household; the household's bank is credited.

| Fed: Assets | Fed: Liabilities |
|---|---|
| | TGA -1,200 |
| | Reserves of household's bank +1,200 |

| Household's bank: Assets | Household's bank: Liabilities |
|---|---|
| Reserves +1,200 | Household deposit +1,200 |

Step 3: the Fed buys the bond from the dealer under QE.

| Fed: Assets | Fed: Liabilities |
|---|---|
| Treasury bond +1,200 | Reserves of dealer's bank +1,200 |

Net result: the public holds USD 1,200 more in deposits (M2 +1,200), banks hold USD 1,200 more in reserves, and the Fed holds the bond. The dealer's deposit falls in Step 1 and is restored in Step 3. The deposit money was created by the *fiscal* transfer. Without Step 3 the deposits would still exist, but the private sector would hold the bond instead of reserves. QE changed the *composition* of private portfolios (reserves instead of bonds) and lowered yields.

This decomposition matters for distribution (Section 6). In 2020-21 the first recipients of new deposits were households receiving transfers and firms receiving Paycheck Protection Program loans, not only financial-market sellers of bonds.

## 3.3 Published decompositions

**(a) Shapiro supply-demand decomposition (Federal Reserve Bank of San Francisco).** Each PCE category is classified month by month as demand-driven when price and quantity surprises move the same way, and as supply-driven when they move in opposite directions; the remainder is "ambiguous". We downloaded the full series (1969m1-2026m9) [V, https://www.frbsf.org/research-and-insights/data-and-indicators/supply-and-demand-driven-pce-inflation/].

**Table 5. Contributions to headline PCE inflation, percentage points (year on year)**

| Date | Supply | Demand | Ambiguous | Total |
|---|---|---|---|---|
| 2017-19 average | 0.88 | 0.62 | 0.26 | 1.76 |
| Jun 2021 | 1.58 | 1.54 | 1.24 | 4.36 |
| Jun 2022 (peak) | 3.43 | 2.76 | 0.95 | 7.14 |
| Dec 2022 | 2.69 | 2.42 | 0.48 | 5.59 |
| Dec 2023 | 1.31 | 1.03 | 0.53 | 2.87 |
| Dec 2024 | 0.80 | 1.10 | 0.74 | 2.64 |
| Dec 2025 | 1.43 | 0.92 | 0.42 | 2.77 |
| Sep 2026 | 1.28 | 1.17 | 1.20 | 3.65 |

Source: `data/r3/us_pce_supply_demand_decomposition_shapiro.csv`; chart `fig5`.

At the peak, *excess* inflation over the 2017-19 baseline was 2.55 points supply, 2.14 points demand and 0.69 points ambiguous (computed). Limitations: the method identifies shocks by sign restrictions at the category level and cannot separate the *source* of demand (fiscal, monetary, reopening). A demand surge into supply-constrained sectors can register partly as "supply".

**(b) Bernanke and Blanchard (2023, 2024).** Their US model, later applied with ten central banks to 11 economies, concludes that the surge "was due primarily to supply disruptions and sharp increases in the prices of food and energy". It finds that labour-market tightness became relatively more important as those effects faded, and that in several countries returning to target may require "a period of modestly higher unemployment" [V for 2023 summary; E for 2024 abstract]. Critique: the model treats commodity prices as exogenous, so a demand boom that raised commodity prices is booked as a supply shock. Ball, Leigh and Mishra (2022, *Brookings Papers*) give more weight to labour-market tightness and to the pass-through of headline shocks into core inflation [E].

**(c) Fiscal and demand accounts.** Di Giovanni et al. (2022-23) estimate that fiscal stimulus, *interacting with* supply constraints, explains a substantial part of the US-euro area inflation gap [E, not opened; treat the magnitude as unverified]. ECB analyses (e.g. Goncalves and Koester 2022, *Economic Bulletin*) found euro-area core inflation in 2022 more supply-driven than US core inflation [E].

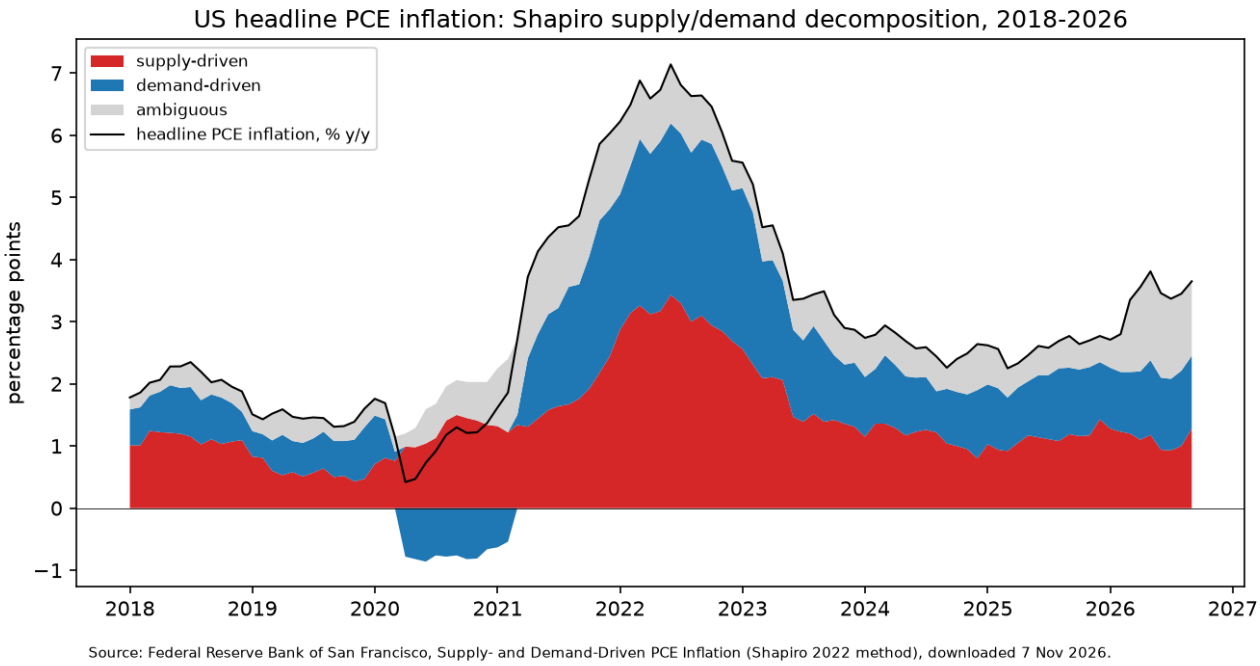



Figure 6: Fig 5

**(d) Money-based accounts.** Borio, Hofmann and Zakrajsek (2023) show that countries with larger 2020 excess money growth had higher 2021-22 inflation, and that money growth would have improved forecasts [E]. ECB Executive Board member Isabel Schnabel (Thuenen Lecture, 25 September 2023) called money growth "an underappreciated harbinger" [E]. The counter-argument: much of the excess money was the counterpart of fiscal transfers, so "money" and "fiscal" are two descriptions of the same event. The IMF (WP/23/137, June 2023) cautions that the money-inflation link is time-varying and rises with trend inflation [E].

## 3.4 What the M2 contraction tells us

M2 fell 4.9% from March 2022 to October 2023. A strict short-run monetarist reading predicts deflationary pressure, but CPI inflation stayed positive throughout (3-4% in 2023). Velocity rose as excess pandemic balances were spent down and as higher rates made holding deposits costlier. This is evidence for *velocity endogeneity*: the money stock contracted because QT and the shift of deposits into money market funds and Treasury bills drained deposits, while spending continued. Treating a rise in M2 in 2020-21 as decisive while ignoring the 2022-23 fall would be selective use of evidence. Both episodes fit a model in which money demand moves with interest rates and in which the 2020-21 money surge raised spending *because* it was delivered as income to households with high marginal propensities to consume.

## 3.5 Verdict: apportioning 2021-23 inflation

(interpretation, this stream; counterfactual ranges, not estimates)

- **Supply shocks, energy and food:** the largest single block of the 2022 peak, roughly 40-55% of the excess, by both the Shapiro and the Bernanke-Blanchard decompositions.
- **Demand (fiscal transfers, reopening, monetary accommodation):** roughly 35-50% of the excess at the peak, with a larger share of the *persistent* core component in the US than in the euro area.
- **Money growth** is not a separate block. It measures the fiscal-monetary demand impulse, and it carried genuine forecasting information in 2021.
- **Bank credit creation** was a minor direct contributor to the 2021-22 CPI surge. It was a major contributor to the house-price surge through mortgage refinancing at record-low rates.
- **Expectations** limited persistence: long-run market expectations stayed anchored.

The 2026 re-acceleration (an energy-led supply shock, with M2 growth of 4-6% and core CPI near 2.8%) is the cleanest contrasting case: inflation without a preceding money surge.

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# 4. Part A - Monetary inflation vs consumer price inflation

## 4.1 Six things that can each be called "inflation"

**Table 6. Six measures that are each called "inflation"**

| Concept | What it measures | US example | Why it can diverge from CPI |
|---|---|---|---|
| Monetary-aggregate growth | quantity of deposits and currency | M2 +87% Dec 2008 to Dec 2019 | money held as a store of value; velocity falls |
| Credit expansion | loans and securities of banks | total bank credit +17.9% Feb 2020 to Dec 2021 | credit funds asset purchases or refinancing |
| Consumer price inflation | consumption basket | CPI +22% Dec 2008 to Dec 2019 | excludes asset prices; global goods competition |
| Producer price inflation | prices received by producers | PPI all commodities +22.7% y/y Nov 2021 | commodity-heavy; margins absorb part |
| Asset price inflation | equity, bond, house prices | NASDAQ +475%; house prices +39% Dec 2008 to Dec 2019 | discount rates; risk premia; scarcity of land |
| Currency depreciation | external value | rupiah 1997-98 | capital flight; risk premia |

Sources: FRED series as listed [V]; `data/r3/us_asset_vs_consumer_index_2008_2019.csv`.

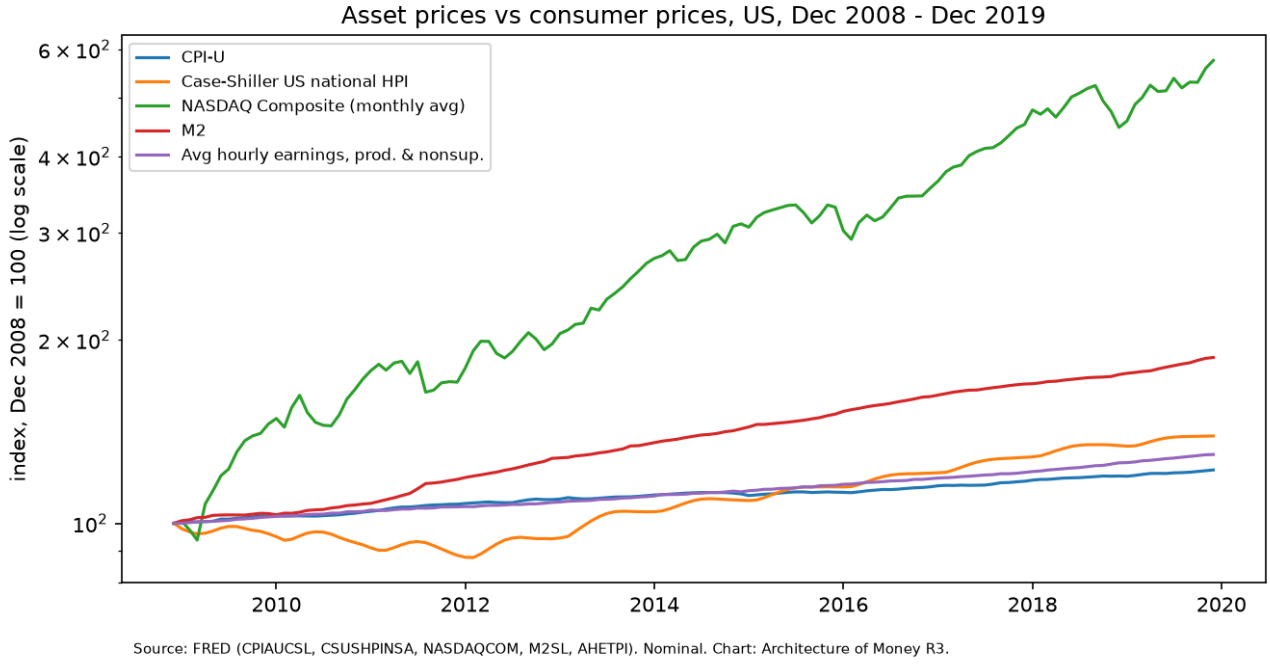



Figure 7: Fig 3

Limitations: the NASDAQ is technology-heavy and starts near a cyclical low, which overstates the general equity gain. The S&P 500 roughly quadrupled over the same period [E]. The Case-Shiller national index is a repeat-sales index. Nominal figures are used throughout.

## 4.2 Why the measures moved apart in 2009-2019

1. **Where new money sits.** QE created reserves, which only banks can hold. The deposits it created ended up with institutional investors and households who sold bonds. Those sellers rebalanced into other assets rather than into consumption (the "portfolio balance channel"; Gagnon et al. 2011 [E]).

**T-account 2. QE purchase from a pension fund (USD 100)**

| Fed: Assets | Fed: Liabilities |
|---|---|
| Treasury bonds +100 | Bank reserves +100 |

| Commercial bank: Assets | Commercial bank: Liabilities |
|---|---|
| Reserves +100 | Pension fund deposit +100 |

| Pension fund: Assets | Pension fund: Liabilities |
|---|---|
| Treasury bonds -100; Deposit +100 | (unchanged) |

The pension fund's net worth is unchanged at the moment of the swap. The price effect comes afterwards: the fund wants duration and yield, so it buys corporate bonds or equities from other investors. That raises their prices and lowers their yields, while the deposits keep circulating among asset holders. Nothing in this chain requires CPI to rise unless asset-price gains spill into consumption (wealth effects) or investment.

2. **Velocity collapsed.** With interest rates near zero, holding money cost almost nothing, and broad money demand rose. M2 velocity fell to historic lows by 2020 [E, FRED M2V not pulled].
3. **Global goods competition and technology** held down tradable-goods prices. Product innovation lowered effective prices, especially for higher-income baskets (Jaravel 2019, *Quarterly Journal of Economics*) [E].
4. **Discount-rate arithmetic.** For an asset paying a cash flow growing at g and discounted at r, price = cash flow / (r - g). A fall in r from 5% to 3%, with g at 2%, *triples* the price (1/0.03 versus 1/0.01). Long-duration assets (growth equities, land) are most sensitive. This is why falling real rates (the 10-year TIPS yield was -0.48% in 2012 and -0.91% in 2021 [V, computed]) produced large asset gains without any matching rise in consumer prices.
5. **Asset prices are not in the CPI basket.** House prices enter the US CPI only through rental equivalence, which follows rents, not prices.

## 4.3 How the schools read the divergence

- *Austrian* (interpretation: Mises, Hayek, Rothbard; Sieron 2019): asset inflation **is** inflation in the relevant sense. Credit expansion distorts relative prices, and the CPI understates the monetary distortion. Strength: it predicted that asset booms would accompany "price stability". Weakness: no reliable way to say *ex ante* which valuations are distortions rather than responses to lower real rates or higher expected growth.
- *BIS financial-cycle view* (interpretation: Borio; Borio and Lowe 2002): credit and property-price cycles are longer than business cycles and build vulnerabilities while CPI stays low. Policy should "lean against the wind" [E].
- *Mainstream New Keynesian* (interpretation: Bernanke and Gertler 1999): central banks should respond to asset prices only insofar as they affect the inflation and output forecast [E].
- *Post-Keynesian/Minskyan* (interpretation: Minsky 1986): stability breeds instability. Asset inflation financed by credit drives fragility and reflects financial structure more than "too much money".

Assessment: the divergence is a measured fact. Whether it is "hidden inflation" (an Austrian *normative* framing) or a rational repricing of long-duration assets to lower real rates (mainstream) cannot be settled by price data alone. It depends on why real rates fell, and Section 7 shows that much of that fall had non-monetary causes.

# 5. Part B - The distributional effects of inflation

Inflation redistributes through five channels, and its incidence depends on which channels dominate in a given episode:

1. **The inflation tax on money holdings** (currency and non-interest-bearing deposits).
2. **The nominal-position channel**: unexpected inflation lowers the real value of fixed nominal claims, which helps debtors and hurts creditors.
3. **The consumption-basket channel**: different households buy different baskets, so they face different inflation rates.
4. **The income-indexation channel**: wages, pensions and benefits adjust to prices with different lags.
5. **The tax-system channel**: unindexed tax brackets and taxation of nominal capital gains raise effective tax rates ("bracket creep").

## 5.1 Inflation as an implicit tax

(fact, accounting) A household holding currency C loses C x pi/(1+pi) in purchasing power per year at inflation rate pi. Example: IDR 1,000,000 held in cash for one year at 10% inflation buys what about IDR 909,000 bought a year earlier, a real loss of about IDR 91,000 (9.1%). The issuer gains seigniorage: it can issue the currency (or, in modern systems, earn interest on the assets that back it) at no interest cost. Who bears this tax depends on cash intensity. Poorer, less banked and informal-sector households hold a larger share of their liquid wealth in currency. That is why the inflation tax is regressive at high inflation (empirical finding: Erosa and Ventura 2002, *Journal of Monetary Economics* 49(4): 761-795, "inflation ... is effectively a regressive consumption tax" [E, abstract via search]). At 2% inflation with interest-bearing deposits widely available, the tax is small. In a hyperinflation it can destroy the entire cash wealth of the poor within months.

## 5.2 Nominal positions: who are the debtors and creditors?

(empirical finding: Doepke and Schneider 2006, *Journal of Political Economy* 114(6): 1069-1097) For the US, "the main losers from inflation are rich, old households, the major bondholders in the economy. The main winners are young, middle-class households with fixed-rate mortgage debt. Besides transferring resources from the old to the young, inflation is a boon for the government and a tax on foreigners" [V, JPE abstract].

**Numerical illustration (stylised).** Before an unexpected 10% rise in the price level:

| Household | Assets | Liabilities | Net worth |
|---|---|---|---|
| Young mortgagor | House 300 (real asset) | Fixed-rate mortgage 240 | 60 |
| Retired saver | Government bonds 240 | 0 | 240 |

After the shock, the house is worth 330 in nominal terms, and the mortgage and the bonds are still 240 nominal. In prices of the original period, the mortgage is now worth 218.2. The young household's real net worth rises from 60 to 81.8 (+36%). The retiree's real net worth falls from 240 to 218.2 (-9.1%). The transfer is 21.8 real units, which is exactly the real loss on the bonds. No output was created or destroyed; wealth moved. With adjustable-rate debt, or if the shock is anticipated and priced into interest rates, the transfer shrinks.

**Table 7. Who gains and who loses, by inflation regime** (interpretation, this stream; synthesising Doepke and Schneider 2006, Easterly and Fischer 2001, Erosa and Ventura 2002, Pallotti et al. 2024, Cagan 1956)

| Group | Moderate *unexpected* inflation (e.g. 2021-23) | Moderate *anticipated* inflation (stable 3-5%) | High/hyperinflation |
|---|---|---|---|
| Fixed-income households (nominal pensions, annuities) | lose (until indexation catches up) | small loss if not indexed | heavy loss |
| Cash holders | lose (inflation tax) | small, predictable loss | heavy loss; currency substitution |

| Group | Moderate *unexpected* inflation (e.g. 2021-23) | Moderate *anticipated* inflation (stable 3-5%) | High/hyperinflation |
|---|---|---|---|
| Debtors (fixed-rate, long maturity) | gain | neutral (priced into rates) | gain on existing debt; new credit disappears |
| Creditors (bondholders, depositors at below-inflation rates) | lose | neutral if rates adjust | heavy loss |
| Asset owners (equity, real estate) | mixed: real assets hedge partly; equities often fall in real terms during surprise inflation | broadly hedged | real assets the main hedge; financial assets lose |
| Wage earners | lose if wages lag prices; gain if labour market is tight | neutral to small loss | heavy loss; wage indexation lags |
| Pensioners (indexed, e.g. US Social Security COLA) | short lag loss, then protected | protected | protection breaks down |
| Governments | gain (real debt erosion, bracket creep, seigniorage) | modest gain | short-run gain, then fiscal collapse (Olivera-Tanzi effect: tax receipts lose value during collection lags) |
| Financial institutions | mixed: banks gain on deposit margins as rates rise; lose on fixed-rate assets (SVB 2023) | neutral | intermediation collapses |
| Foreign holders of domestic-currency debt | lose | priced in | heavy loss |

**Evidence for the 2021-23 euro-area episode.** Pallotti, Paz-Pardo, Slacalek, Tristani and Violante (2024, *Journal of Monetary Economics* 148) measure first-order welfare effects across households. Average welfare costs, as a share of three years' income, were "around 3% in France and Spain, 7% in Germany, and 9% in Italy". Retirees tended to lose while "roughly half of the 25-44 year-old gained". Nominal net positions were "the key driver of heterogeneity across households", and "the counterpart of this household-sector loss is a significant gain for the government" [E, journal highlights and SSRN/NBER abstract excerpts]. The cumulative euro-area price increase over 2021-23 was about 18% [E, conference slides].

## 5.3 Inflation inequality through consumption baskets

(empirical finding: Jaravel 2021, *Annual Review of Economics* 13: 599-629) "Inflation rates decline with household income in the United States". Measuring this accurately requires granular data because of aggregation bias, and the direction of product innovation, which favours higher-income baskets, is a significant driver [E, abstract].

**Table 8. US cumulative inflation by equivalised income quintile (BLS research index R-CPI-I)**

| Quintile | Feb 2020 to Jun 2024 |
|---|---|
| Lowest 20% | 22.5% |
| Second | 22.3% |
| Middle | 21.9% |
| Fourth | 21.5% |
| Highest 20% | 20.5% |

Source: Federal Reserve Bank of Minneapolis, "Lower income, higher inflation? New data bring answers at last" (7 Oct 2024), citing BLS R-CPI-I [V]. The same source reports that since 2005 prices rose 64% for the lowest-income

households, 57% for the highest and 58% on average [V]. Klick and Stockburger (BLS *Monthly Labor Review*, 2024) find the lowest quintile faced inflation about 0.28 points per year higher on average [E]. The *source* of the gap changed during the episode. Higher-income households faced higher inflation in parts of 2021 (vehicles, travel). From mid-2022 the gap reversed as food and especially shelter, which take a larger share of low-income budgets, drove inflation [V/E, Minneapolis Fed; BLS Spotlight 2022].

Caveats: (i) these indices use expenditure shares from the Consumer Expenditure Survey, which under-represents the very top. (ii) A BLS working paper using national-accounts weights finds smaller and more variable gaps [E, BLS EC250040]. (iii) The CBO found that by 2023 the share of income needed to buy a 2019 basket had *fallen* for every quintile, because nominal incomes grew faster than prices [E, CBO May 2024]. Inflation inequality is real but modest in the US. It is larger where food and energy shares differ more across households, as in emerging markets.

### 5.4 Energy and food shocks are regressive by construction

Poorer households spend more of their budgets on food and energy, so a supply shock concentrated in those goods (2022 in Europe; 2026 energy) raises their personal inflation rate more. In Indonesia the food share of the national poverty line was 74.70% in March 2026. Rice alone was 20.52% of the urban poverty line and filter kretek cigarettes 10.83% [E, BPS Profil Kemiskinan Maret 2026, via reporting]. A food-led inflation of 4.82% (October 2026) is therefore a larger real-income shock for households near the poverty line than for the median household. Fiscal offsets matter: Pallotti et al. find that "unconventional fiscal policies helped shield households" in Europe [E]. Indonesia paired the 2022 fuel price increase with expanded social assistance (bantuan sosial) for lower-income groups [E, BI monetary policy review, Sep 2022].

### 5.5 Perceptions and politics

(empirical finding: Easterly and Fischer 2001, *Journal of Money, Credit and Banking* 33(2): 160-178) Using polling data for 31,869 households in 38 countries, "the poor are more likely than the rich to mention inflation as a top national concern". Improvements in the poor's income share, poverty decline and real minimum wages are negatively correlated with inflation [E, abstract]. This finding links Part V to Parts VI and VIII: perceived unfairness of inflation is concentrated where its burden is heaviest, which helps explain the political weight of price stability and the legitimacy central banks gain from delivering it.

### 5.6 Balance of the evidence on Part B

- Moderate *surprise* inflation redistributes mainly by **age** (old creditors to young debtors) and to **governments**. Its net effect by *income* is ambiguous, because poorer households hold few nominal assets but also have few nominal debts.
- Through the consumption basket, inflation is mildly **regressive** in advanced economies and more so in emerging markets with food-heavy budgets.
- High and chronic inflation is **regressive on nearly every channel**.
- Calling inflation a "tax on the poor" (normative framing) is accurate for high inflation and for food and energy shocks, but overstated for moderate demand-driven inflation with tight labour markets. In 2021-23 the lowest-wage US workers saw the fastest nominal wage growth [E, Autor, Dube and McGrew 2023].

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## 6. Part C - The Cantillon effect

### 6.1 Cantillon's argument

Richard Cantillon's *Essai sur la nature du commerce en general* (written c. 1730, published 1755) contains the first systematic account of how an increase in money raises prices *unevenly* (fact, bibliographic [E]). In Part II (chapters 6-8) he traces what happens when money increases through a domestic mine, a trade surplus, or subsidies and loans. Mine owners and their suppliers spend first, raising the prices of what they buy. Those who receive money late, especially people on fixed incomes, face higher prices before their incomes rise. The outcome depends on *who* spends the new money on *what*, so a rise in the money stock does not raise all prices in proportion

[E, standard reading; text not reopened]. Hume's "Of Money" (1752) gave a similar but more optimistic account of a transitional stimulus [E].

Modern statement (Sieron 2019, *Money, Inflation and Business Cycles: The Cantillon Effect and the Economy*, Routledge, 172 pp.): the Cantillon effect is "a distribution effect resulting from uneven changes in money supply", with consequences for relative prices and the structure of production [E, Routledge description and Sciendo article quoting Sieron 2019: 15].

## 6.2 Modern interpretations

| Tradition | Reading of the Cantillon effect |
|---|---|
| Austrian (Mises 1912; Hayek 1931; Rothbard; Sieron 2019) | Money is never neutral, even in the long run, because injection is always uneven. Credit expansion through banks lowers interest rates, favours long-horizon and asset-market activities, and distorts the structure of production (Austrian business cycle theory). Sieron classifies channels (stock market, Treasury bond market, bank loans, government spending) and argues that outcomes differ by channel [E]. |
| Mainstream "first-round effects" | Uneven injection matters for short-run distribution but washes out as prices adjust; long-run neutrality holds approximately (interpretation: Bernanke 2015, "monetary policy is 'neutral' or nearly so in the longer term" [V, Brookings blog excerpt]). |
| Segmented-markets monetary theory | In models where only some agents trade in asset markets when money is injected (Alvarez, Atkeson and Kehoe 2002; Williamson 2008) [E], injections redistribute towards participating agents and generate real effects. This formalises a Cantillon channel in modern terms. |
| Portfolio-balance and asset-price redistribution | QE raises asset prices; holders gain wealth on paper, but welfare gains accrue to *net sellers* of assets (Fagereng et al. 2025) [E]. |
| Public-choice political economy | Who receives new money first is a political choice, and interest groups seek early access ("The redistributive politics of monetary policy", *Public Choice*, via PMC; case studies of 2008, COVID and the euro) [E]. |
| Post-Keynesian | Money enters through credit to borrowers; distribution follows *credit allocation* (who can borrow), more than the money stock itself. |

A debate within the Austrian literature is relevant. Sieron (2020, *Quarterly Journal of Austrian Economics*) rebuts critics who called the Cantillon effect "a fundamental evil unique to fiat central bank systems". He agrees that "every monetary system includes Cantillon effects", including gold mining, and argues that the effects differ by channel, for example market versus non-market injection [E, QJAE note]. On the Austrians' own terms, then, Cantillon effects are not a special feature of fiat money.

## 6.3 Conditions under which the effect is large

The effect requires all of the following, in increasing order of importance:

1. **Non-neutral injection**: new money reaches some agents before others.
2. **Slow price adjustment**: prices and wages adjust with lags, so early spenders buy at old prices.
3. **Market segmentation**: recipients spend on a particular class of goods or assets.
4. **Size relative to the market**: the injection is large relative to the markets it enters.

5. **Persistence**: the relative-price change is not reversed later.

## 6.4 Empirical evidence

Direct tests that trace "first receivers" of money are scarce, for three reasons. Money is fungible; the counterfactual distribution is unobservable; and injection effects overlap with the macroeconomic effects of the policy that caused them. The relevant evidence is therefore indirect:

- **QE and asset holders.** In the UK, Bunn, Pugh and Yeates (2018, BoE Staff Working Paper 720) find the overall effect of 2008-14 monetary easing on relative inequality measures "small". In cash terms, though, gains were far larger for wealthier households, and households near retirement gained most [E, abstract and Bank Underground summary]. The Resolution Foundation (2019) reports that, on that framework, a large share of wealth gains went to the top decile, reflecting the existing skew of wealth [E]. *This is the Cantillon asset channel at work: whoever holds the assets whose prices rise gains in cash terms.*
- **Euro area.** Lenza and Slacalek (ECB WP 2190, 2018; *Journal of Applied Econometrics* 39(5), 2024) find that QE "compresses the income distribution since many households with lower incomes become employed", with "negligible effects on wealth inequality". House-price gains, which are widely shared among owners, offset equity gains [E, abstract]. The ECB's research bulletin is titled "Quantitative easing did not increase inequality in the euro area" [E].
- **Denmark.** Andersen, Johannesen, Jorgensen and Peydro (2023, *Journal of Finance* 78(5): 2945-2989) use the entire Danish population over 1987-2014, with the currency peg providing exogenous variation in monetary policy. They find that "gains from softer monetary policy in terms of income, wealth, and consumption are monotonically increasing in ex ante income", driven by leverage and risky assets, and that "softer monetary policy increases income inequality" [E, abstract].
- **United States.** Coibion, Gorodnichenko, Kueng and Silvia (2017, *Journal of Monetary Economics* 88: 70-89) find that "contractionary monetary policy systematically increases inequality in labor earnings, total income, consumption and total expenditures" [E, abstract]. Easing therefore *reduces* earnings inequality through employment, the opposite of a pure Cantillon reading.
- **2020-21: transfers before assets.** The pandemic injection reached households first, as stimulus payments, enhanced unemployment insurance and child tax credits. In the Fed's DFA, the bottom-50% share of net worth rose from 1.7% (2019Q4) to 2.4% (2021Q4), while the top-1% share rose only from 30.4% to 30.9% [V, computed from DFA]. Pandemic QE, which entered through asset markets, ran in parallel. The two injection points worked in opposite distributional directions. Their net effect favoured the bottom half's *share* (from a very low base), and the top 1% received the largest *dollar* gains through equities.

## 6.5 Assessment

**Table 9. Cantillon effect: claims and evidence**

| Claim | Status |
|---|---|
| Where new money enters affects relative prices and distribution in the short run | supported in principle and in segmented-markets models; plausible for QE (asset channel) and for fiscal transfers (consumption channel) |
| The effect systematically favours financial institutions | partly: dealers and asset holders benefit from QE in cash terms. Banks' net interest margins were *compressed* by low rates, and bank equity underperformed after 2008 [E]. "Financial institutions" are not a single winner. |
| The effect systematically favours governments | yes for seigniorage and real debt erosion (Section 5); this is a fiscal channel more than a Cantillon sequence |
| The effect favours borrowers | depends on who can borrow; post-Keynesian credit-allocation argument; supported by Andersen et al. (leverage channel) |
| The effect explains the long-run rise in wealth inequality | **not supported** as a general explanation. Top wealth shares rose 1989-2008 before QE (Section 7), and across countries with very different monetary regimes. |

| Claim | Status |
|---|---|
| The effect is unique to fiat or central-bank money | **no**: Cantillon's own case was a gold and silver mine. Sieron (2020) agrees. |

**Conclusion (interpretation, this stream).** The Cantillon effect is a sound *mechanism* and a useful lens on *how* monetary operations are designed: injection through asset purchases, through bank credit, or through transfers. The design choice has distributional consequences that the institutions making it should weigh openly (normative). The effect's size is an empirical question, and evidence from countries with different methods ranges from small (UK, euro area) to clearly regressive in income (Denmark). It should not be presented as *the* cause of wealth inequality.

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# 7. Part D - Asset inflation and financial inequality

## 7.1 Who owns what: why asset inflation is unequal by construction

**Table 10. US household balance sheets by net-worth group (Fed Distributional Financial Accounts)**

| Group | Share of net worth 1989Q3 | 2008Q4 | 2019Q4 | 2021Q4 | 2026Q2 | Share of all corporate equity and mutual fund shares, 2026Q2 | Share of all real estate, 2026Q2 |
|---|---|---|---|---|---|---|---|
| Top 1% | 22.8% | 27.6% | 30.4% | 30.9% | 32.5% | 50.9% | 13.4% |
| Next 9% | 38.0% | 38.8% | 39.5% | 37.2% | 36.4% | 37.2% | 30.3% |
| Next 40% | 35.7% | 32.6% | 28.4% | 29.6% | 28.8% | 11.4% | 46.6% |
| Bottom 50% | 3.5% | 1.0% | 1.7% | 2.4% | 2.3% | 0.6% | 9.7% |

**Portfolio composition, 2026Q2 (share of each group's own assets)**

| Group | Real estate | Equities and mutual funds | Pension entitlements | Liabilities / assets |
|---|---|---|---|---|
| Top 1% | 10.8% | 53.5% | 5.3% | 1.9% |
| Next 9% | 21.0% | 33.4% | 20.5% | 5.9% |
| Next 40% | 37.3% | 11.8% | 26.4% | 14.3% |
| Bottom 50% | 46.9% | 3.6% | 12.7% | 58.4% |

Source: Federal Reserve Board, DFA (zip downloaded 7 Nov 2026) [V]; `data/r3/us_dfa_*.csv`; chart `fig6`. Limitations: the DFA distributes Financial Accounts aggregates using the triennial Survey of Consumer Finances, so quarterly movements partly reflect aggregate asset prices applied to fixed portfolio shares. Groups are ranked by net worth, not income.

These tables explain most of the "asset inflation increases inequality" argument. When equity valuations rise, the top 1% gains most, because over half its assets are equities. When house prices rise, the leveraged middle (next 40%) and the bottom half gain proportionally, because housing dominates their assets and debt amplifies the effect on net worth. This is the "race between the stock and the housing market" (empirical finding: Kuhn, Schularick and Steins 2020, *Journal of Political Economy* 128(9)): "housing booms lead to substantial wealth gains for leveraged middle-class households and tend to decrease wealth inequality", while "stock market booms primarily boost the wealth of households at the top" [E, abstract].

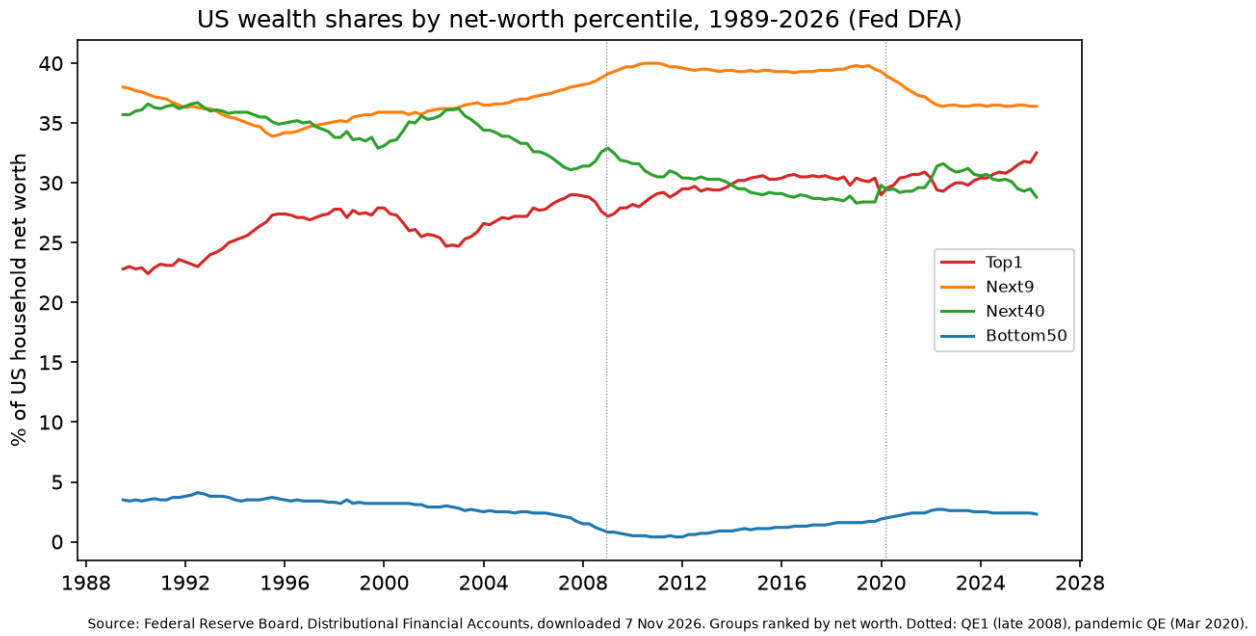



Figure 8: Fig 6

## 7.2 Low rates and valuations

Falling real interest rates raise the present value of long-duration assets (Section 4.2, point 4). US 10-year real yields fell from about 1.7-1.8% in 2008-09 to below zero in 2012 and 2020-21, then rose to about 2.2% in 2026 [V, computed]. Monetary policy influences real rates in the short and medium run. The long-run *trend* decline in neutral real rates (r*) since the 1980s is mostly attributed to non-monetary forces: demographics and longer life expectancy, slower productivity growth, a global saving glut, demand for safe assets, and rising inequality itself (empirical finding: Holston, Laubach and Williams; Rachel and Smith; Mian, Straub and Sufi) [E].

The "saving glut of the rich" (Mian, Straub and Sufi; NBER WP 26941, 2020, revised July 2025) finds that "since the 1980s, saving by the top 1% has surged, creating a saving glut of the rich comparable in scale to the global saving glut". Instead of financing investment, it has been associated with dissaving by the non-rich and by the government: rich households accumulated financial assets that are claims on household and government debt [E, NBER and authors' abstracts]. This reverses the usual causal story. Instead of low rates (monetary policy) causing inequality, rising inequality depresses rates, and low rates then raise asset values that the rich own. Monetary policy *transmits* the forces that set r* and does not create them. The case for this view: r* fell before QE and fell globally under very different central banks. The case against: central banks decide how far to push policy rates *relative* to r*, and the timing and scale of QE added to term-premium compression.

## 7.3 Housing affordability

**Table 11. US housing affordability (annual averages; computed)**

| Year | Median new-house sale price (USD) | Median household income (USD) | Price/income | 30-year mortgage rate | P&I on 80% LTV, % of median income |
|---|---|---|---|---|---|
| 1984 | 79,950 | 22,420 | 3.57 | 13.9% | 40.2% |
| 2000 | 167,550 | 41,990 | 3.99 | 8.1% | 28.3% |
| 2006 | 243,750 | 48,200 | 5.06 | 6.4% | 30.4% |
| 2012 | 244,400 | 51,020 | 4.79 | 3.7% | 21.1% |
| 2019 | 320,250 | 68,700 | 4.66 | 3.9% | 21.2% |
| 2021 | 383,000 | 70,780 | 5.41 | 3.0% | 21.8% |

| Year | Median new-house sale price (USD) | Median household income (USD) | Price/income | 30-year mortgage rate | P&I on 80% LTV, % of median income |
|---|---|---|---|---|---|
| 2022 | 432,950 | 74,580 | 5.81 | 5.3% | 31.1% |
| 2023 | 426,525 | 80,610 | 5.29 | 6.8% | 33.1% |
| 2025 | 415,400 | 87,460 | 4.75 | 6.6% | 29.1% |

Sources: FRED MSPUS (new houses sold; Census/HUD), MEHOINUSA646N (Census CPS ASEC, nominal), MORTGAGE30US (Freddie Mac) [V]; `data/r3/us_housing_affordability.csv`; chart `fig7`. Limitations: MSPUS covers *new* houses only, which skews toward new suburban stock. P&I excludes taxes, insurance and down-payment saving. Household income is not buyer income.

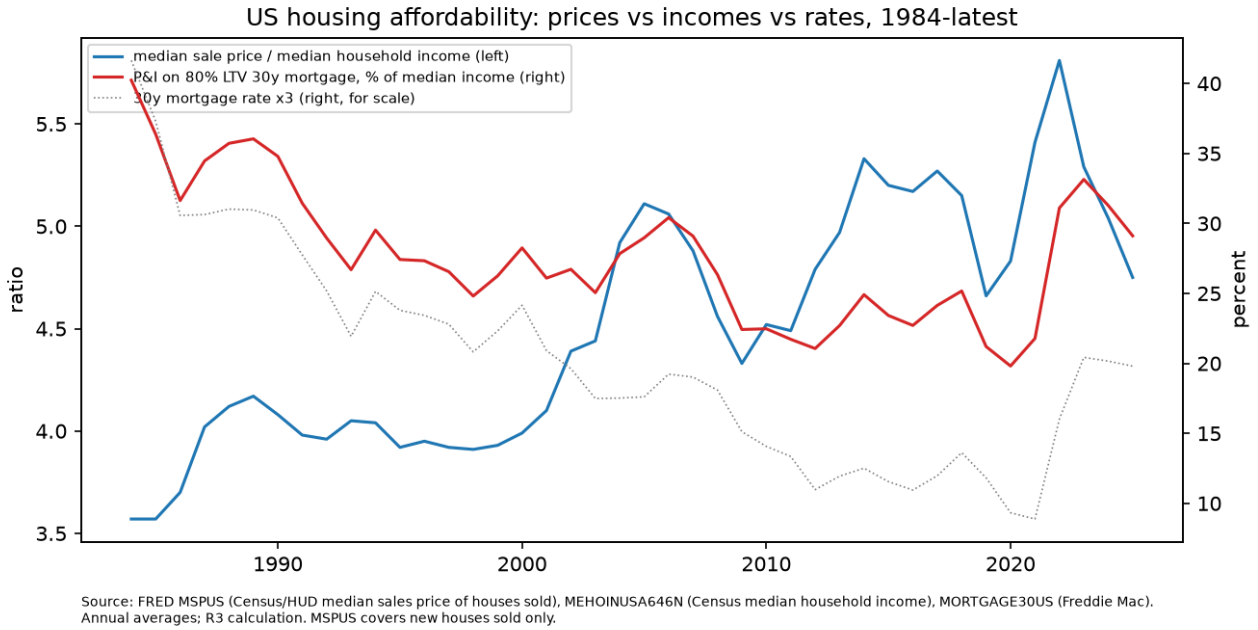



Figure 9: Fig 7

Reading: low rates in 2012-21 kept the *monthly payment* burden near 21% of income even as price-to-income rose. When rates normalised in 2022-23, the payment burden jumped to 31-33% while prices barely fell. The rise in rates hit buyers more than the earlier fall in rates had helped them, because prices had capitalised the low rates. The *down-payment* barrier (a multiple of income) rose throughout. Owners who bought or refinanced before 2022 locked in low fixed rates, a lasting transfer from future buyers to existing owners that is visible in the age profile of wealth gains.

**Monetary vs structural causes of high house prices.**

- Monetary (interpretation; empirical support in Jorda, Schularick and Taylor 2015 [E]): lower rates raise house prices, with larger effects where supply is inelastic. Mortgage credit expansion (bank money creation) finances the bidding.
- Structural (interpretation; empirical support in Glaeser and Gyourko on supply restrictions [E], and in Rognlie 2015): land-use regulation, zoning and infrastructure constraints make supply inelastic in high-productivity cities. Rognlie (2015, *Brookings Papers on Economic Activity*) shows that, once depreciation is netted out, the post-war rise in the net capital share in large developed economies "turns out to come entirely from the housing sector" [E, BPEA abstract]. That points to *scarcity* (land, permits) rather than accumulation, as the source of rising capital income.
- Implication: if housing supply were elastic, low rates would mainly raise the quantity of housing, not its

price. Monetary easing turns into house-price inflation *because of* structural supply constraints, so the two causes interact rather than compete.

## 7.4 Equity valuations, bonds and the young

- **Equities.** Lower discount rates and higher corporate profit shares both raised valuations. Greenwald, Lettau and Ludvigson attribute much of the post-1989 rise in US equity wealth to a reallocation of rewards from labour to shareholders, with lower interest rates a secondary factor [E, "How the Wealth Was Won"; magnitudes not verified here]. This structural cause, partly market power, is not monetary.
- **Bonds.** Falling yields from 1981 to 2020 produced a four-decade bond bull market. Holders of long-duration bonds (pension funds, insurers, older households) gained. The 2022 reversal produced the largest bond losses in decades and contributed to the 2023 US regional bank failures (SVB) [E].
- **Intergenerational welfare.** Fagereng, Gomez, Gouin-Bonenfant, Holm, Moll and Natvik (2025, *Journal of Political Economy* 133(11): 3494-3549) show that the welfare gain from higher asset valuations depends on "the present value of an individual's net asset sales rather than asset holdings: higher asset valuations benefit prospective sellers and harm prospective buyers". Using Norwegian transaction data for 1994-2019, they find large redistribution "from the young to the old and from the poor to the wealthy". The average individual money-metric welfare gain is about USD 10,000, ranging from -USD 185,000 at the 1st percentile to +USD 273,000 at the 99th (2011 US dollars) [E, journal abstract and DOI excerpt]. This is the most rigorous modern formalisation of the intuition behind the Cantillon and asset-inflation critiques. It shows that a rising *paper* wealth gap overstates or understates welfare inequality depending on who must buy assets in the future.

## 7.5 Structural vs monetary causes of inequality

**Table 12. Drivers of rising wealth and income concentration: monetary vs structural**

| Driver | Type | Evidence and direction | Relative weight (interpretation) |
|---|---|---|---|
| Skill-biased and automation technology; superstar firms | structural | wage dispersion, falling labour share in some sectors [E] | large |
| Globalisation, offshoring | structural | manufacturing wage effects ("China shock", Autor, Dorn and Hanson) [E] | moderate |
| Market power, rising markups | structural | De Loecker, Eeckhout and Unger [E] | moderate, contested |
| Tax policy (top rates, capital gains, estate) | policy (non-monetary) | Piketty, Saez and Zucman [E] | large in the US |
| Decline of unions and labour institutions | structural and policy | [E] | moderate |
| Housing supply constraints | structural and policy | Rognlie 2015; Glaeser and Gyourko [E] | large for housing wealth and rents |
| Demographics, saving glut of the rich, falling r* | structural | Mian, Straub and Sufi [E] | large, through asset valuations |
| Monetary policy (rates, QE) | monetary | small effect on inequality ratios (UK, euro area); regressive in income in Denmark; progressive through employment (US, Coibion et al.) | small to moderate, cyclical; larger in *dollar* gains to asset owners |
| Inflation surprises | monetary and fiscal | old to young, private to government (Doepke and Schneider) | episodic |

Piketty's r > g framework (2014, *Capital in the Twenty-First Century*) predicts rising concentration when the return on capital exceeds growth (interpretation [E]). The Rognlie critique narrows the empirical base of the rising capital share to housing. The saving-glut evidence suggests concentration lowers r rather than raising it. Both critiques leave intact the *measured* rise in wealth concentration (Table 10). They dispute its mechanism and its future path.

**Bottom line (interpretation, this stream).** Monetary policy is neither innocent nor the main culprit. Its distributional footprint is largest in *dollar* terms through asset prices (favouring the top 1% through equities and owners over renters through housing) and in *employment* terms through labour markets (favouring the bottom). The rise in concentration over 1989-2008, *before* QE, and its occurrence across many monetary regimes point to structural forces as the primary drivers. A fair normative criticism of post-2008 central banking is narrower and stronger. Its tools work *through* asset prices, it knew this, and it relied on fiscal and housing policy, which largely failed to act, to offset the distributional side-effects (normative; cf. Bernanke 2015, who argues distribution is the job of fiscal policy [E]).

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# 8. Indonesia: inflation by expenditure group and the money-price link

## 8.1 The money-price link

**Table 13. Indonesia: broad money growth, CPI inflation, real GDP growth (WDI [V])**

| Year | Broad money growth | CPI inflation | Real GDP growth |
|---|---|---|---|
| 1997 | 25.3% | 6.2% | 4.7% |
| 1998 | 62.8% | 58.5% | -13.1% |
| 1999 | 12.2% | 20.5% | 0.8% |
| 2005 | 16.3% | 10.5% | 5.7% |
| 2008 | 14.9% | 10.2% | 6.0% |
| 2015 | 9.0% | 6.4% | 4.9% |
| 2020 | 12.5% | 1.9% | -2.1% |
| 2021 | 14.0% | 1.6% | 3.7% |
| 2022 | 8.4% | 4.2% | 5.3% |
| 2023 | 3.5% | 3.7% | 5.0% |
| 2024 | 4.8% | 2.2% | 5.0% |

Annual correlation of broad money growth and CPI inflation: 0.64 over 1981-2024; 0.25 excluding 1998; 0.21 over 2000-2024 (computed). Average 1995-2023: money growth 14.5% per year, CPI 8.1%, real GDP 4.3% (from the cross-country file). Source: `data/r3/indonesia_money_cpi_gdp_wdi.csv`; chart `fig8`. Limitations: WDI broad money follows the IMF definition (money plus quasi-money) and differs from BI's published M2 (see phase1 log). In 1998, depreciation revalued foreign-currency deposits included in M2.

Reading: Indonesia shows the same regime dependence as the global sample. Money and prices moved together during the 1997-98 crisis, when the exchange rate collapsed and Bank Indonesia extended emergency liquidity. In the inflation-targeting era (Bank Indonesia formally adopted inflation targeting in 2005 [E]), money growth well above inflation has been the norm, as financial deepening raised money demand. 2020-21, when BI bought government bonds under burden sharing, saw 12-14% money growth and below-2% inflation.

## 8.2 Inflation by expenditure group, October 2026

**Table 14. Indonesia CPI (2022=100) by expenditure group, October 2026, year on year**

| Group | Inflation y/y | Contribution to headline (pp) |
|---|---|---|
| Transport | 4.87% | 0.59 |
| Food, beverages and tobacco | 4.82% | 1.40 |

| Group | Inflation y/y | Contribution to headline (pp) |
|---|---|---|
| Personal care and other services (incl. gold jewellery) | 4.52% | 0.32 |
| Food and beverage serving services/restaurants | 2.83% | n/a |
| Household equipment and routine maintenance | 2.39% | n/a |
| Health | 2.31% | n/a |
| Information, communication and financial services | 1.91% | n/a |
| Recreation, sport and culture | 1.76% | n/a |
| Education | 1.35% | n/a |
| Clothing and footwear | 1.26% | n/a |
| Housing, water, electricity and household fuel | 1.06% | n/a |
| **Headline** | **3.23%** (CPI 109.04 to 112.56) | 3.23 |

By component: core contributed 1.64 points, volatile food 0.96 and administered prices 0.63. Core was driven by gold jewellery, cooking oil, rice-with-side-dish meals, mobile phones, laptops and lubricants. Provincial year-on-year inflation ranged from 2.52% (DKI Jakarta) to 8.07% (North Sulawesi). Source: BPS press release 43/02500/HM.240/2026, 2 Nov 2026 [V, https://www.bps.go.id/id/news/2026/11/03/943/...] for the headline, contributions and component data; group rates other than food are [E], from NU Online Jatim and ANTARA reporting of the same BPS release. BI's inflation target is 2.5% +/- 1% [V, BI press release, see phase1 log]. The 2026 energy shock reaches Indonesia mainly through non-subsidised fuels (gasoline, lubricants), airfares and imported inputs, while subsidised fuel prices are administered.

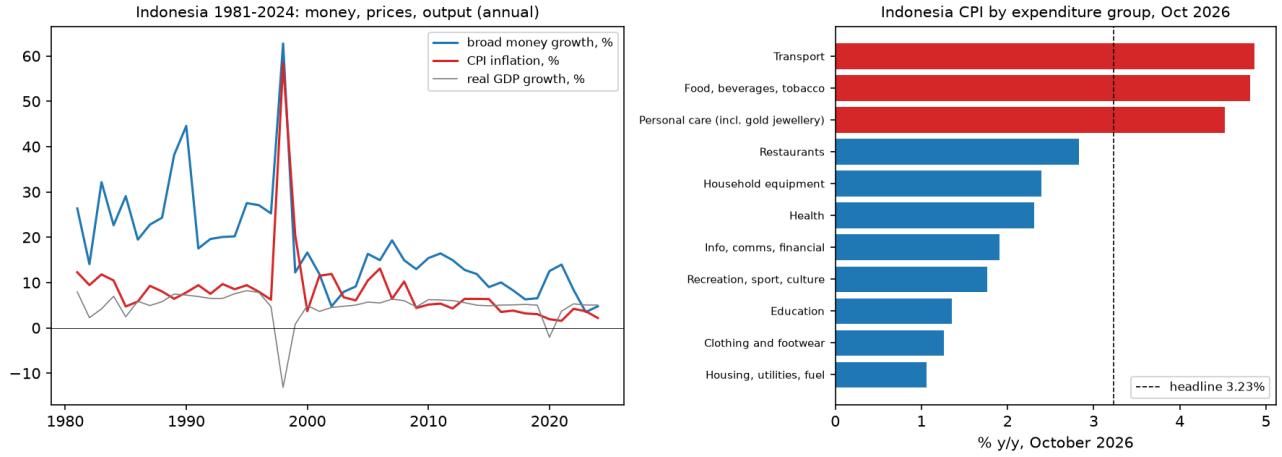
**Distributional reading for Indonesia.**

- *Food-led inflation is regressive.* Food is 74.70% of the national poverty line [E, BPS], so the poor's effective inflation rate in October 2026 was closer to the 4.8% food rate than to the 3.2% headline (inference, not a BPS measurement; BPS publishes no CPI by income group).
- *Administered prices put distribution on the policy agenda.* The 2022 fuel price increase pushed transport inflation to about 16% year on year (16.09% per BPS, as reported by Tirto [E]). Headline inflation reached 5.95%, the highest since 2015 [E], and was paired with targeted social assistance.
- *Gold jewellery in core.* Gold's large contribution to core inflation in 2026 reflects asset-price dynamics entering a consumer index, an unusual case where asset inflation is *measured* as consumer inflation. Households who own gold as a store of value gain.
- *Data gap.* Indonesia has no official inflation index by income or expenditure decile. A household-group CPI built from Susenas expenditure shares and BPS commodity prices would be a valuable contribution for the Indonesia chapter (Part XII) and is recommended as a Phase 12 task.

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# 9. Implications for the monograph's central questions

- **BRIEF Part I, Q2 (who benefits from the creation and allocation of money?).** Depends on the injection channel. QE benefits asset holders in cash terms. Fiscal transfers benefit recipients. Bank credit benefits creditworthy borrowers and, through asset prices, existing owners. Unexpected inflation benefits governments and nominal debtors. Evidence on *net* inequality effects is mixed and country-specific (Section 6.4).
- **Part I, Q3 (why does money keep purchasing power?).** Credible institutions (inflation targets, anchored expectations, fiscal backing) are why money growth well above inflation coexisted with low inflation for three decades in low-inflation regimes. Where fiscal and institutional credibility fails, the quantity relation returns at full strength (Table 1, high-inflation group). This links to R5/Part VI on trust.




Sources: left, World Bank WDI FM.LBL.BMNY.ZG, FP.CPI.TOTL.ZG, NY.GDP.MKTP.KD.ZG (accessed 7 Nov 2026); right, BPS release 2 Nov 2026 as reported (group rates partly secondary-sourced; see CSV).

Figure 10: Fig 8

- **Part XIV, Q3 (attribution of inflation).** See Key finding 10 and Section 3.5. Attribution depends on regime and horizon. A single percentage split would be false precision.
- **Part XIV, Q4 (does banking architecture favour some participants?).** Credit allocation favours collateral owners, and asset-price transmission favours existing owners over future buyers (Fagereng et al.). This is structural to how monetary policy transmits, not a hidden conspiracy, and it can be offset by fiscal, housing and tax policy (normative).
- **Part VII, D (is inflation disproportionately harmful to people with limited assets?).** Yes for high inflation and for food and energy shocks. Mildly so for moderate inflation through baskets. Ambiguous for moderate surprise inflation, which hits old creditors hardest.

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# 10. Disputes and uncertainties

1. **Causation in the money-inflation correlation.** High-inflation correlations do not show whether money causes inflation or accommodates fiscal and exchange-rate shocks. Our cross-country exercise cannot separate the two.
2. **Sample-split bias.** Splitting countries by average inflation conditions on the outcome (Table 1). The qualitative pattern agrees with studies using better designs, but our specific coefficients should not be cited as structural estimates.
3. **Decomposition dependence.** Supply versus demand shares in 2021-22 depend on the method: sign restrictions (Shapiro); a structural wage-price model with commodity prices treated as exogenous (Bernanke-Blanchard); or Phillips-curve specifications (Ball, Leigh and Mishra). Our ranges in Section 3.5 are judgements spanning these, not estimates.
4. **US CPI data gap and seasonal adjustment.** October 2025 CPI is missing. FRED's seasonally adjusted September 2026 year-on-year rate (4.0%) differs from the BLS headline (3.7%, not seasonally adjusted).
5. **Inflation inequality measurement.** Results by income group depend on expenditure weights (CE survey versus national accounts) and on product granularity (Jaravel). Gaps may be understated with coarse categories or overstated with survey weights.
6. **Cantillon effect.** No clean causal test exists. The evidence is indirect, and studies disagree (small effects in the UK and euro area; regressive in Denmark; progressive through employment in the US).
7. **DFA limitations.** Quarterly wealth shares are interpolations around triennial SCF surveys and price-driven. The 2020-21 rise in the bottom-50% share partly reflects house-price gains and deposit accumulation from transfers, some of which was later spent.
8. **Indonesia group data.** Most group-level rates for October 2026 are secondary-sourced [E] from reporting of the BPS release. The headline, the food group and contributions are [V]. The BPS table should be

re-verified before publication.

9. **Unverified items kept out of conclusions:** the magnitudes in di Giovanni et al.; the Greenwald-Lettau-Ludvigson decomposition; exact McCandless-Weber coefficients; the Indonesian transport rate for September 2022 (16.09%, secondary source).
10. **2026 context.** Facts about the Iran war, Fed hikes and energy prices come from press reports [E] and are recent. They should be re-checked against BLS, Federal Reserve and EIA primary releases at manuscript stage.

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## 11. Competing theories of inflation: strongest case, empirical record, main weakness

(Phase 4 asks to "evaluate competing inflation theories". Each school is stated at its strongest, then tested against the evidence assembled above.)

**Monetarism / quantity theory.** *Strongest case:* across countries and decades, and in every hyperinflation, sustained money growth beyond output growth goes together with inflation nearly one-for-one (Table 1, high-inflation rows; McCandless and Weber 1995). The 2020-21 money surge preceded the 2021-22 inflation, and money would have improved forecasts (BIS 2023). *Record:* excellent at long horizons and high inflation. Poor for 1991-2019 in the US (Table 2, negative correlations), when velocity drifted. *Main weakness:* the velocity assumption fails in low-inflation regimes, and in practice money is mostly endogenous to credit demand and fiscal operations, so the theory describes a correlation more reliably than a lever.

**Keynesian / New Keynesian.** *Strongest case:* inflation responds to demand pressure relative to capacity and to expectations. Anchored expectations explain why the 2021-22 shocks did not become a 1970s spiral, and why disinflation in 2023 was nearly costless (Section 2.10-2.11). *Record:* good on the role of expectations and credibility. Before 2021, Phillips-curve models underpredicted the surge because they treated the curve as flat and money as irrelevant. *Main weakness:* inflation dynamics near capacity are non-linear, and the framework neglects money and credit aggregates as signals.

**Post-Keynesian (endogenous money; conflict inflation).** *Strongest case:* banks create money when they lend, central banks set rates, and money follows nominal spending. Inflation reflects distributional conflict over income shares (profits versus wages) and cost shocks. Markups rose in some sectors in 2021-22, and the "profit-led" or "sellers' inflation" debate (Weber and Wasner 2023 [E]) puts distribution at the centre. *Record:* good on money endogeneity (Section 3.4: velocity and money adjusted to spending in 2022-23). The profit-share evidence is contested and sector-specific. *Main weakness:* less clear on what anchors the price level in the long run, and inclined to understate how fiscal-monetary demand impulses enable cost shocks to pass through.

**Modern Monetary Theory.** *Strongest case:* a currency-issuing government faces no financing constraint in its own currency, only a real-resource constraint. Inflation signals that spending exceeds capacity, and taxes and targeted policy can manage it. The 2021 US experience, where large deficits met constrained supply, fits the "resource constraint" framing. *Record:* the 2021-22 episode showed that capacity constraints bind faster and less predictably than claimed. Fiscal tightening was not used as the anti-inflation tool; central bank rate hikes did the work. *Main weakness:* it relies on political willingness to raise taxes or cut spending in time, which has little historical support.

**Fiscal theory of the price level.** *Strongest case:* the price level adjusts so that the real value of government debt equals the present value of expected surpluses. The 2020-21 deficits, financed without expected future surpluses, could explain the burst, and the fall in inflation as deficits narrowed (Cochrane) [E]. *Record:* compelling for hyperinflations and fiscal dominance (Sargent 1982). Hard to test in advanced economies because expected surpluses are unobservable. *Main weakness:* identification.

**Austrian.** *Strongest case:* credit expansion lowers interest rates below their natural level and causes uneven price changes. Asset prices and the structure of production respond first (Cantillon), so "price stability" in the CPI can coexist with large monetary distortions, as in 2009-2019 (Fig 3). *Record:* correctly emphasised asset and credit booms that CPI-focused frameworks underweighted before 2008. *Main weakness:* no operational measure of the "natural rate" or of distortion *ex ante*. Empirical evidence for a long-run Cantillon effect on inequality

is weak (Section 6). Its proposed remedies (commodity money, free banking) would move rather than remove Cantillon effects (Sieron 2020 concedes gold mining produces them).

**Supply-side / cost-push accounts.** *Strongest case:* most inflation bursts of the last 50 years began with energy and food shocks (1973, 1979, 2008, 2022, 2026). *Record:* strong for timing and composition. *Main weakness:* without accommodation or unanchored expectations, cost shocks change the price level once rather than producing persistent inflation. So supply accounts need a monetary or expectations ingredient to explain persistence.

**Synthesis (interpretation, this stream).** The schools are not interchangeable (BRIEF Part II.D), but they mostly answer *different questions*. The quantity theory explains *long-run and high* inflation. New Keynesian theory explains *how central banks keep low inflation low* and why shocks fade. Supply accounts explain the *timing* of bursts. Fiscal theory explains *regime breakdowns*. Post-Keynesian theory explains *how money is actually created and why its quantity is a poor lever* in normal times. Austrian theory explains *why CPI stability can hide asset-market and allocation effects*. A defensible monograph position uses each where its evidence is strongest and rejects each where its evidence fails.

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# 12. Sources

Tags: [V] opened in this stream (page, abstract page or dataset); [E] search excerpt or established knowledge.

## Data


- [V] FRED, Federal Reserve Bank of St. Louis, series listed in `data/r3/README.md`, via https://fred.stlouisfed.org/graph/fredg (pulled 7 Nov 2026).
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## Literature


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## 13. Files produced by this stream

Report: `/workspace/projects/b8753593-2946-449d-bfbf-95733676e121/money/research/R3-inflation-distribution.`

Charts (each checked visually):

- `research/r3-charts/fig1_us_m2_cpi_ppi_2015_2026.png`
- `research/r3-charts/fig2_us_m2_level_2019_2026.png`
- `research/r3-charts/fig3_us_asset_vs_consumer_2008_2019.png`
- `research/r3-charts/fig4_crosscountry_money_inflation_1995_2023.png`
- `research/r3-charts/fig5_us_pce_supply_demand_shapiro.png`
- `research/r3-charts/fig6_us_wealth_shares_dfa.png`
- `research/r3-charts/fig7_us_housing_affordability.png`
- `research/r3-charts/fig8_indonesia_money_cpi_and_groups.png`

Datasets (`data/r3/`, documented in `data/r3/README.md`): crosscountry_money_cpi_{1980_2023,1995_2023,2000_2019}.csv; us_m2_cpi_horizon_correlations.csv; us_asset_vs_consumer_index_2008_2019.csv; us_pce_supply_demand_decomposition_ us_dfa_wealth_shares_selected.csv; us_dfa_networth_shares_timeseries.csv; us_dfa_portfolio_composition_2019Q4.csv and _2026Q2.csv; us_housing_affordability.csv; indonesia_money_cpi_gdp_wdi.csv; indonesia_cpi_by_group_oct2026.csv; indonesia_poverty_line_composition_mar2026.csv; plus raw FRED, WDI, FRBSF and DFA pulls.

Scripts: `work/r3_crosscountry.py`, `work/r3_us.py`, `work/r3_charts1.py`, `work/r3_charts2.py`, `work/r3_shapiro.py`, `work/r3_dfa.py`, `work/r3_housing.py`, `work/r3_indo.py`.

# R4 - Trust, Philosophy and Fairness of Money (BRIEF Parts VI, VII, VIII)

Project III, *The Architecture of Money.* Authors: M. Bagus Prabowo and Hark (he conceived and directs the project; Hark's research agents researched and drafted this stream). Research stream R4. Research cut-off: November 2026. Compiled 7 November 2026 (Asia/Jakarta).

Evidence tags follow FRAMEWORK.md: **[V]** = page or dataset opened and the claim is on it; **[E]** = search excerpt or established scholarly knowledge, not opened in this session. Claim types are labelled where it matters: (fact), (empirical finding: Author year), (interpretation: school/author), (normative), (counterfactual). Raw data pulled for this stream sit in `data/r4/` (see `data/r4/README_r4.md`).

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## Key findings

1. **Trust in money is not one thing.** Survey and behavioural evidence separate at least four objects of trust: the *unit* (the currency as a measure and store of value), the *issuer* (central bank and state), the *intermediary* (the bank that holds one's deposit) and the *system* (payment rails). They move separately. In June 2026 only 28% of US adults expressed "a great deal" or "quite a lot" of confidence in banks (Gallup) [V], yet US dollar deposits and payments showed no sign of mass exit. In the euro area in late 2023, 43% "tended to trust" the ECB and 42% did not (Eurobarometer, via ECB) [V], while support for the euro itself is far higher (reported at 83% in the euro area in Eurobarometer spring 2025, [E]). Low trust in institutions coexists with high *use* of their money. This gap is the core puzzle of Parts VI and VIII.

2. **The best theoretical account of fiat acceptance is layered, not single-cause.** Search-theoretic models (Kiyotaki and Wright 1989, 1993) show that intrinsically worthless tokens can be valued in equilibrium purely through self-fulfilling expectations, but always alongside a non-monetary equilibrium in which nobody accepts them. Something must select the monetary equilibrium. The candidates are convention and history (Menger; Lewis), state demand through taxes and legal tender (Knapp; chartalism), institutional credibility (central bank reaction functions; fiscal backing) and the sheer network value of a unit already in use. These are complements. The evidence of currency collapses shows that each one can fail on its own, and that failure of credibility is the usual trigger.

3. **Behavioural economics explains *stickiness*, not *value*.** Money illusion (Shafir, Diamond and Tversky 1997), loss aversion, status quo bias, anchoring and present bias explain why households keep using a currency they distrust, why they misperceive inflation, and why exit, when it comes, is abrupt rather than gradual. Household inflation expectations are high, dispersed and only loosely tied to realised inflation. The University of Michigan median one-year expectation was 5.3% in June 2022 when CPI inflation was 9.0%, but 6.6% in May 2025 when CPI inflation was 2.4% [V, FRED MICH and CPIAUCSL]. Expectations respond to salient prices, to news and to partisanship as much as to measured inflation.

4. **People dislike inflation mainly because they believe wages lag prices.** In Stantcheva's 2023-24 US surveys, 80% believed prices systematically rise faster than wages, and people attributed their own raises to merit rather than to inflation [V]. Shiller (1997) found the same. Inflation is felt as a loss of standing and a

fairness violation, not as a change of yardstick. This matters for legitimacy: an inflation episode is a test of the monetary institution's *fairness*, not only of its competence.

5. **Distrust usually produces partial and staged exit, not abandonment.** Exit runs in order: first the store-of-value function moves (to foreign currency, real assets or gold), then pricing of big-ticket items (housing, cars) moves to a foreign unit of account, and the means of payment for daily transactions and taxes moves last, if at all. Dollarization shows hysteresis: once adopted, it persists long after stabilisation (Reinhart, Rogoff and Savastano 2003, [E]). Hirschman's exit/voice/loyalty framework fits well. Exit from a currency is costly and needs coordination with counterparties. Voice (elections, protests, referenda) is the cheaper first response. Loyalty is reinforced by legal, fiscal and infrastructural lock-in.

6. **Bank runs are the sharpest form of monetary distrust, and they are about the intermediary, not the unit.** Silicon Valley Bank lost over $40 billion of deposits on 9 March 2023 and expected over $100 billion more the next day, roughly 85% of its deposit base (Federal Reserve, Barr review, April 2023) [V]. Depositors fled *into* other dollar deposits and Treasury money market funds, not out of the dollar. The Diamond-Dybvig (1983) model predicts exactly this: a run is a coordination failure that deposit insurance and a lender of last resort can remove, at the price of moral hazard.

7. **On "who creates money", the accurate statement is shared and hierarchical.** In the US in September 2026, currency was 10.2% of M2 ($2.40 trillion of $23.41 trillion) [V, FRED]. The rest is mainly commercial bank deposits plus retail money market fund shares. Most money that households hold is private bank liabilities. These sit on a public base (reserves, currency) and depend on public backstops (deposit insurance, lender of last resort, central bank settlement). Calling this "private money creation" or "public money creation" alone misdescribes it. It is a public-private partnership in which the state licenses, backstops and constrains, and banks originate.

8. **Household debt has fallen relative to GDP in the US, UK and euro area since 2008, but has risen sharply in China, Korea, Canada and Switzerland.** BIS data on household credit as a share of GDP: US 98.4% (2007Q4) to 67.2% (2026Q1); UK 94.3% to 72.6%; euro area 58.6% to 50.2%; China 18.6% to 57.5%; Korea 66.5% to 85.1%; Switzerland 101.9% to 127.0%; Indonesia 11.6% to 15.2% [V, BIS WS_TC]. "Rising household debt everywhere" is false as a global generalisation. Distributional concentration of debt (who holds it, at what rates) matters more than aggregates.

9. **The ten thinkers disagree along four fault lines, not forty.** These are: (i) is money a *thing* (commodity, Aristotle's convention aside) or a *social relation/claim* (Simmel, Marx in part, Graeber, Minsky, Keynes)? (ii) Is credit *productive innovation* (Schumpeter, Smith on banking) or a *source of fragility and domination* (Minsky, Marx, Graeber)? (iii) Should money be *depoliticised by rule or competition* (Hayek) or *governed democratically/discretionarily* (Keynes, Polanyi, Minsky)? (iv) Is the obligation in debt *moral and sacred* or *political and renegotiable* (Graeber versus the creditor ethic Smith partly shares)? The structured matrix in Section 7.3 sets them out.

10. **The fairness debate has a strong empirical core and an irreducible normative core.** Empirically: (a) Proposals for full-reserve or sovereign money were rejected decisively when put to a vote. The Swiss Vollgeld initiative of 10 June 2018 got 24.28% Yes, won 0 of 23 cantons, on a 34.55% turnout [V, Swissvotes]. (b) Bank deposit franchises earn real rents, but these are mostly a spread on *market power over deposit rates* (Drechsler, Savov and Schnabl 2017), not "interest on money created from nothing". (c) Unanticipated inflation redistributes from nominal creditors and fixed-income holders to nominal debtors, including governments; the poor rank inflation as a top concern (Easterly and Fischer 2001). Normatively, whether the current architecture is "fair" depends on the standard used (desert, Rawlsian difference principle, non-domination, efficiency). The answer differs across these standards.

11. **Conceptual model (Part VIII).** Continued use of a currency results from four separable modes: **voluntary acceptance** (V), **practical necessity** (N), **institutional dependence** (D) and **legal obligation** (L). The ten factors in the BRIEF load onto these modes and reinforce each other. Legitimacy (V) and lock-in (N+D+L) are distinct dimensions. A 2x2 typology results: *endorsed* systems (high V, high lock-in), *captive* systems (low V, high lock-in), *competitive* systems (high V, low lock-in) and *collapsing* systems (low V, low lock-in). Continued participation alone therefore does not prove legitimacy. It proves only that exit is costlier than staying. Section 8 sets out ten testable implications to tell the modes apart.

12. **Bottom line for the monograph's question 5 ("Why do populations continue using a system they perceive as unfair?").** Because a currency is a coordination good with strong network effects, legal and fiscal anchors, and infrastructural lock-in, its use is compatible with low trust in the institutions that run it, *as long as* the unit's purchasing power stays within a tolerance band. Inside that band, dissatisfaction goes to voice. Outside it, typically above about 20-40% annual inflation sustained for years or after a deposit freeze, staged exit starts. Under hyperinflation or confiscation, exit becomes general. The tolerance band is itself shaped by memory, literacy and the availability of a credible alternative unit. The thresholds are an interpretation drawn from case evidence, not an established constant (see Disputes).

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## Conventions and scope

- This stream covers BRIEF Part VI (psychology and trust), Part VII (sociology and philosophy of fairness) and Part VIII (why societies maintain systems they criticise). Inflation mechanics and distribution (Part V) are covered in stream R3; international order (Part IX) in R5; CBDC (Part X) elsewhere. Cross-references are given where R4 depends on them.
- The stream follows the BRIEF's neutrality rule. It presents each school at its strongest. It avoids conspiracy framing about central banks. It does not treat fiat money as worthless or gold as perfect. It does not treat banks as unlimited money creators.
- Survey numbers are reported with their question wording, sample and year where available, because trust measures are very sensitive to wording ("confidence in banks" versus "trust financial services to do what is right" versus "tend to trust the ECB").

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# PART VI - THE PSYCHOLOGY OF MONEY AND MONETARY TRUST

## 6.1 The central question and why it is harder than it looks

The BRIEF asks: *why does a society collectively trust a monetary system whose value depends heavily on institutions, expectations, and the continued acceptance of other participants?*

Three clarifications sharpen the question.

**(a) "Trust" may be the wrong word for most users most of the time.** Most people do not deliberate about whether to accept their national currency. They accept it because their wage arrives in it, their rent is due in it, their taxes are assessed in it and every shop prices in it. Sociologists distinguish *trust* (an active expectation about another party's future conduct, under vulnerability) from *confidence* (a taken-for-granted expectation that a system will work, without any consideration of alternatives) (interpretation: Luhmann 1979/1988 [E]). Monetary acceptance in normal times is mostly confidence in Luhmann's sense. It turns into trust, which is conscious and conditional, only when something goes wrong. This explains why trust surveys can show low numbers while behaviour shows unbroken use: the survey invites reflection that daily life does not.

**(b) There are several objects of trust.** Table 6.1 separates them.

**Table 6.1 - Objects of monetary trust**

| Object | What the user relies on | Typical failure | Typical behavioural signature of distrust |
|---|---|---|---|
| The unit (currency) | Future purchasing power; others will accept it | High or hyper-inflation; redenomination | Currency substitution; flight to real assets; shortening of nominal contracts |
| The issuer (central bank, state) | Competent, non-opportunistic policy; fiscal solvency | Fiscal dominance; monetisation; politicisation | Rising inflation expectations; capital flight; dollarization of savings |
| The intermediary (bank) | Convertibility of deposits into central bank money at par, on demand | Insolvency; illiquidity; deposit freeze | Bank runs; cash hoarding; shift to large or state banks or money market funds |
| The system (payments infrastructure) | Payments settle; finality; privacy; availability | Outages; sanctions; fraud; exclusion | Cash retention; use of alternative rails |

(Interpretation: this stream's synthesis; the four-way split echoes the ECB's own finding that trust in the ECB co-moves with trust in EU institutions generally [V, ECB Economic Bulletin 3/2024].)

**(c) Trust in money is partly trust in *other people's* trust.** A banknote is valuable to me because I expect *you* to accept it, and you accept it because you expect a third party to. This higher-order expectation is what

search theory formalises and what Searle's social ontology describes (6.2).

## 6.2 Collective trust: convention, search, status functions and credibility

### 6.2.1 Convention and emergence (Menger, Lewis)

Carl Menger (1892, "On the Origin of Money", *Economic Journal* 2(6): 239-255) argued that money emerges without design: traders learn that some goods are more "saleable" (marketable) than others and accept them as intermediate goods. Imitation then spreads the practice until one or a few goods become general media of exchange (interpretation: Menger/Austrian school) [E]. David Lewis (1969, *Convention*) gave the general structure of such a solution: a regularity in behaviour that persists because everyone prefers to conform given that others conform, and where an alternative regularity would also have served [E].

Strongest point: convention explains why the *specific* money in use is somewhat arbitrary and very persistent. Weakness: the Mengerian story is weak as history. The anthropological and Mesopotamian record suggests that units of account were often set by temples and palaces rather than emerging from barter (Graeber 2011; cross-reference Part III stream). A convention account is compatible with state origin. It needs only that, once set, the unit persists by mutual expectation.

### 6.2.2 Search-theoretic models (Kiyotaki and Wright)

Kiyotaki and Wright (1989, "On Money as a Medium of Exchange", *Journal of Political Economy* 97(4): 927-954) model agents who produce goods they do not consume and must trade in decentralised, random bilateral meetings. Commodity monies emerge endogenously from storage costs and beliefs. In Kiyotaki and Wright (1993, "A Search-Theoretic Approach to Monetary Economics", *American Economic Review* 83(1): 63-77), intrinsically worthless fiat tokens circulate in equilibrium. Three results matter for this monograph (empirical status: theoretical results) [E]:

1. **Multiplicity.** There is always a non-monetary equilibrium (nobody accepts fiat because nobody expects others to) alongside one or more monetary equilibria. Fiat value is *self-fulfilling but not self-guaranteeing.*
2. **Value from frictions.** Money is "essential" (it improves welfare) only when there are frictions such as lack of double coincidence of wants, limited commitment and lack of record-keeping (Kocherlakota 1998, "Money is Memory", *JET* 81(2) [E]). In this view money is a substitute for a perfect public ledger of who owes what to whom.
3. **Equilibrium selection is outside the model.** The models say *that* fiat can be valued, not *which* equilibrium society lands on. Taxes, legal tender, history and credibility do the selecting.

Samuelson's (1958) overlapping-generations model makes a parallel point: fiat money can be valued as a store of value across generations as a social contract, but the equilibrium is fragile if the last generation is expected to be stuck with worthless paper (interpretation: Samuelson 1958 *JPE* 66(6)) [E]. Infinite horizons and continuity of institutions are what remove the backward-unravelling problem.

### 6.2.3 Network effects

A currency is a network good. Its value to each user rises with the number of other users who accept it, quote prices in it and hold contracts in it. Three implications follow (interpretation: network economics applied to money, e.g., Dowd and Greenaway 1993, *Economic Journal* 103 [E]):

- **Lock-in and inertia.** A rival currency must be better by more than the switching cost multiplied by the coordination problem. This explains the persistence of weak but incumbent currencies, and the persistence of the dollar's international role (cross-reference R5).
- **Tipping.** Network goods show thresholds. Once a critical mass of counterparties switches, the remaining holders switch quickly. This fits the observed abruptness of hyperinflationary currency abandonment (6.6).
- **Asymmetric hysteresis.** After a society has coordinated on a foreign currency for pricing and saving, returning is also costly. Dollarization persists after stabilisation.

### 6.2.4 Searle: money as a status function

John Searle (1995, *The Construction of Social Reality*; 2010, *Making the Social World*) gives the most precise philosophical account of how money's value can be "real" without being physical [E]. Institutional facts have the form **"X counts as Y in context C"**. Paper of a certain design (X) counts as US currency (Y) within the US legal and economic order (C). Y is a *status function*: the object can perform a function (discharging debts, paying taxes) not because of its physical properties but because it is collectively recognised as having that status. Status functions carry *deontic powers*: rights, duties, permissions and obligations. A creditor *must* accept legal tender in settlement of a debt; the bearer *may* pay taxes with it.

Three implications are important:

1. **Collective recognition, not unanimous belief.** Status functions require only that enough people recognise or accept them in practice. Recognition can be grudging, habitual or even coerced. This matches the trust/use gap in Finding 1.
2. **Electronic money is a "freestanding Y term".** Searle himself noted that much modern money has no physical X at all; it is entries in accounts that count as money (Searle 2010) [E]. The status function rests entirely on institutional records, which is why the integrity of ledgers (banks, central banks, payment systems) is the physical substrate of modern monetary trust.
3. **Status functions can be withdrawn.** Demonetisation (India's November 2016 withdrawal of legal tender status from 500 and 1,000 rupee notes, [E]) shows that the state can remove a status function by declaration. Hyperinflation shows that the public can drain it by refusing recognition in practice. Both are the same structure working in opposite directions.

Critics argue that Searle under-theorises power and conflict: whose collective intentionality counts? (Interpretation: critical social ontology, e.g., Ingham 2004 *The Nature of Money*, who stresses money as a social relation of credit and debt backed by sovereign authority [E].) For this monograph, Searle provides the form and Ingham and the chartalists provide the content: *which* institutions do the recognising and enforcing.

### 6.2.5 State anchors: taxes and legal tender

The chartalist argument (Knapp 1905; Wray; Kelton 2020) is that the state creates demand for its currency by accepting it, and only it, in payment of taxes. This anchors the monetary equilibrium, because every taxpayer needs some of the currency each period whatever their beliefs about others (interpretation: chartalism/MMT) [E]. Legal tender laws add a weaker anchor: they define what a creditor must accept to discharge a debt, but they rarely force anyone to *trade* in the currency.

Strongest evidence for the tax anchor: colonial "hut taxes" payable only in the colonial currency created demand for it among populations previously outside monetised exchange (historical interpretation, e.g., Forstater on colonial Africa [E]). Strongest evidence against it as *sufficient*: hyperinflations happen in states that still levy taxes in the domestic currency (Weimar Germany 1923; Zimbabwe 2008). The tax anchor sets a floor on demand. It does not prevent collapse if the state issues far more money than taxes withdraw. When inflation erodes the real value of tax receipts between assessment and payment (the Olivera-Tanzi effect, [E]), the anchor weakens precisely when it is needed.

### 6.2.6 Institutional credibility

Modern monetary theory in the broad (New Keynesian) sense locates fiat value in the expected *policy rule*: a central bank that predictably raises real interest rates when inflation rises (the Taylor principle) anchors the price level (Woodford 2003) [E]. The time-inconsistency literature (Kydland and Prescott 1977; Barro and Gordon 1983) explains why discretionary policy-makers are tempted to inflate and why delegation to an independent, conservative central banker (Rogoff 1985) can raise credibility [E]. The fiscal theory of the price level (Leeper 1991; Sims 1994; Cochrane 2023) adds that the price level must also be consistent with the present value of expected primary surpluses. Under this view, trust in money is ultimately trust in fiscal backing (interpretation: FTPL) [E].

These are competing accounts of what credibility *is*. They agree that credibility is about expected future institutional behaviour, not about current backing assets.

### 6.2.7 Repeated interaction and confidence in future purchasing power

Repeated interaction turns one-shot trust problems into reputational equilibria. A central bank that has kept inflation near target for decades has an asset (reputation) it is reluctant to spend. The public, through repeated experience, forms adaptive and experience-based expectations (Malmendier and Nagel 2016, "Learning from Inflation Experiences", *QJE* 131(1): 53-87, finding that individuals weight inflation experienced during their own lifetimes heavily) [E]. Trust is therefore *path-dependent and generational*: cohorts who lived through high inflation remain more sensitive to it.

**Table 6.2 - Mechanisms of collective acceptance compared**

| Mechanism | What anchors fiat value | Key authors | Failure mode | Empirical support | Main limitation |
|---|---|---|---|---|---|
| Convention / emergence | Mutual expectation, history | Menger 1892; Lewis 1969 | Coordination on another convention | Persistence of incumbents; commodity monies in POW camps (Radford 1945, [E]) | Weak as history of origins; silent on selection |
| Search / OLG theory | Self-fulfilling acceptance under frictions | Kiyotaki-Wright 1989/93; Samuelson 1958; Kocherlakota 1998 | Non-monetary equilibrium; unravelling | Laboratory experiments find fiat can emerge and collapse (e.g., Duffy and Ochs 1999, [E]) | Equilibrium selection exogenous |
| Network effects | Value rising with users | Dowd-Greenaway 1993 | Tipping to rival | Dollarization hysteresis | Explains persistence better than origin |
| Status function | Collective recognition plus deontic powers | Searle 1995/2010 | Withdrawal of recognition | Demonetisation episodes | Thin on power and conflict |
| Chartalist / tax | Tax liabilities payable only in currency | Knapp 1905; Wray; Kelton | Over-issuance relative to taxes; Olivera-Tanzi | Colonial tax monetisation | Not sufficient against hyperinflation |
| Credibility / policy rule | Expected reaction function | Kydland-Prescott; Barro-Gordon; Woodford | Fiscal dominance; time inconsistency | Inflation targeting era anchoring | Credibility hard to measure ex ante |
| Fiscal backing (FTPL) | Present value of surpluses | Leeper; Sims; Cochrane | Unbacked deficits | Some high-inflation episodes | Contested identification |
| Repeated experience | Learning from personal history | Malmendier-Nagel 2016 | Cohort trauma | Survey evidence on cohort effects | Explains expectations, not value |

**Synthesis (interpretation).** Fiat acceptance is overdetermined in stable times and underdetermined in crises. In stable times, convention, network effects, tax demand, legal tender and credibility all point the same way, so it is impossible to say which "really" holds the currency up. In crises they come apart. The order in which they fail (credibility first, then the store-of-value convention, then unit of account, and the tax anchor last) is the most informative evidence about their relative strength. Section 6.6 documents it.

## 6.3 Behavioural economics of money

### 6.3.1 Loss aversion

Prospect theory (Kahneman and Tversky 1979, *Econometrica* 47(2): 263-291) holds that people evaluate outcomes as gains and losses relative to a reference point, and that losses loom larger than gains. Tversky and Kahneman (1992, *Journal of Risk and Uncertainty* 5) estimated a loss-aversion coefficient of about 2.25 [E]. Later meta-analyses find substantial heterogeneity, with typical estimates between about 1.5 and 2.5 and some studies finding much weaker effects (e.g., Brown et al. 2024 meta-analysis in *Journal of Economic Literature*, [E]).

Monetary implications:

- **Nominal loss aversion.** People resist *nominal* wage cuts much more than equivalent real cuts through inflation. This is evidence for downward nominal wage rigidity (Akerlof, Dickens and Perry 1996, [E]). One argument for a positive inflation target ("greasing the wheels") rests on it.
- **Reluctance to realise nominal losses** on housing and assets (Genesove and Mayer 2001, *QJE*, on Boston condominiums, [E]). This reinforces the nominal frame.
- **Exit asymmetry.** Switching currencies or banks crystallises potential losses (fees, exchange rate spreads, the risk of being wrong). Loss aversion therefore biases toward staying, until the loss from staying becomes vivid. This is one micro-foundation for "dissatisfaction without abandonment".

### 6.3.2 Status quo bias and default effects

Samuelson and Zeckhauser (1988, "Status Quo Bias in Decision Making", *Journal of Risk and Uncertainty* 1: 7-59) showed that people disproportionately stick with the current option even when it is not chosen for reasons of merit [E]. Default effects in pensions (Madrian and Shea 2001, *QJE*) show how powerful inertia is in financial behaviour [E]. Monetary implications: households keep accounts at the same bank for years, often accepting deposit rates well below market (the "deposit franchise", Section 7.4.2), and keep savings in the national currency despite expected depreciation. Status quo bias is a key reason that *institutional dependence* (mode D in Section 8) can persist without voluntary endorsement.

### 6.3.3 Money illusion (Shafir, Diamond and Tversky 1997)

Shafir, Diamond and Tversky (1997, "Money Illusion", *Quarterly Journal of Economics* 112(2): 341-374) showed in survey experiments that people evaluate transactions partly in nominal terms even when they understand real terms [E]. In a classic vignette, two workers, Ann and Barbara, receive the same real raise: Ann gets a 2% nominal raise with 0% inflation, and Barbara a 5% nominal raise with 4% inflation. Most respondents judged Ann better off in economic terms, but many judged Barbara *happier* and *more likely to stay* in her job. The authors argued that people represent economic magnitudes in both nominal and real frames, and that the nominal frame is more salient because it is simpler.

Corroborating evidence: Fehr and Tyran (2001, *AER* 91(5)) found in experiments that small amounts of individual money illusion cause large aggregate nominal inertia because people expect others to suffer from it [E]. Brunnermeier and Julliard (2008, *QJE*) argued that house prices are mispriced when buyers compare nominal mortgage rates to rents without adjusting for inflation [E].

Why it matters for trust: money illusion is *pro-stability* in low inflation. It makes the unit of account feel like a fixed measuring rod, which is a precondition for the "confidence" mode of acceptance. In high inflation it breaks down, and people switch to a real frame (indexation, foreign currency pricing). The breakdown of money illusion is a marker of the transition from confidence to conscious distrust.

### 6.3.4 Anchoring

Tversky and Kahneman (1974, "Judgment under Uncertainty: Heuristics and Biases", *Science* 185) showed that numerical estimates are pulled toward arbitrary starting values [E]. In monetary life the relevant anchors are:

- **A central bank target** (2%) can serve as an anchor for expectations. Evidence suggests this works far better for professional forecasters and firms in some countries than for households, most of whom do not

know the target (Coibion, Gorodnichenko and Kumar 2018, "How Do Firms Form Their Expectations?", *AER* 108(9), on New Zealand firms' inattention [E]).

- **Remembered prices**, for example the price of petrol or eggs, anchor perceived inflation (D'Acunto, Malmendier, Ospina and Weber 2021, "Exposure to Grocery Prices and Inflation Expectations", *JPE* 129(5): frequently purchased goods dominate household inflation perceptions [E]).
- **Round-number and historical anchors** in wage bargaining.

"Anchored inflation expectations" in central-bank language is a metaphor drawn from this psychology. It means long-run expectations insensitive to short-run news. NY Fed survey data show that in September 2026 median three-year expectations rose to 3.3% and five-year expectations held at 3.0%, while one-year expectations rose to 3.9% [E, NY Fed SCE release 7 Oct 2026, corroborated by several reports]. That pattern is what "imperfect anchoring" looks like: the long end stays above a 2% target but moves less than the short end.

### 6.3.5 Present bias

Laibson (1997, "Golden Eggs and Hyperbolic Discounting", *QJE* 112(2): 443-478) and O'Donoghue and Rabin (1999, *AER*) model quasi-hyperbolic discounting, in which people overweight the present relative to all future periods [E]. Present bias explains the simultaneous holding of high-interest credit card debt and illiquid savings, and demand for commitment devices. For monetary trust it does two things. It makes credit-based purchasing power *attractive* even when people disapprove of debt in the abstract, and it makes long-run monetary erosion (2-4% a year) feel less urgent than short-run price spikes, which helps explain why moderate steady inflation is tolerated while sudden price-level jumps provoke anger.

### 6.3.6 Risk perception and experience

Risk perception is shaped by salience and availability more than by statistics (Tversky and Kahneman 1973 on availability [E]). Malmendier and Nagel (2011, "Depression Babies", *QJE* 126(1): 373-416) found that individuals who experienced low stock returns early in life take less financial risk decades later [E]. In 2016 they found the same for inflation expectations. Ehrmann and Tzamourani (2012, "Memories of High Inflation", *European Journal of Political Economy* 28(2)) found that countries with a history of high inflation show stronger public preferences for price stability [E]. The German case, with the 1923 hyperinflation in collective memory, is the standard illustration (interpretation; the strength of the "1923 trauma" as a causal factor is debated by historians).

### 6.3.7 Trust in financial institutions as a behavioural variable

Guiso, Sapienza and Zingales (2004, "The Role of Social Capital in Financial Development", *AER* 94(3): 526-556) found that in Italian regions with higher social capital (generalised trust), households use more cheques, hold less cash, invest more in stocks and have more access to institutional credit [E]. Their 2008 paper ("Trusting the Stock Market", *Journal of Finance* 63(6)) found that lack of trust explains a large part of non-participation in equity markets [E]. Trust is thus an *input* to financial deepening. Low-trust societies remain more cash- and real-asset-based. Gallup World Poll data add a time series: across 25 countries most affected by the 2008 crisis, the median share expressing confidence in "financial institutions and banks" fell from 57% before the crisis to 40% in 2009 and a low of 37% in 2012, then recovered to 63% in 2025 [V, Gallup, 23 Feb 2026]. Gallup notes that the recovery tracks households' comfort with their income closely, so trust in banks may reflect general economic optimism more than institutional reform [V].

### 6.3.8 Household inflation expectations: what the surveys show

Three major survey families measure household expectations:

- **University of Michigan Surveys of Consumers** (since 1946 in some form; median expected price change over the next 12 months, FRED series MICH).
- **New York Fed Survey of Consumer Expectations (SCE)** (since 2013; rotating panel of about 1,300 household heads; probabilistic and point forecasts).
- **ECB Consumer Expectations Survey (CES)** (since 2020; about 19,000 respondents in 11 euro area countries by 2022, per ECB [V]).

Weber, D'Acunto, Gorodnichenko and Coibion (2022, "The Subjective Inflation Expectations of Households and Firms: Measurement, Determinants, and Implications", *Journal of Economic Perspectives* 36(3): 157-184) summarise the stylised facts [E]: household expectations are (i) on average higher than realised inflation and professional forecasts, (ii) very dispersed across individuals, (iii) systematically different by gender, income and education (women, lower-income and less-educated respondents report higher expectations), (iv) driven by salient prices such as groceries and petrol, (v) weakly responsive to central bank communication in normal times, and (vi) predictive of individual spending and saving choices in some studies.

**Table 6.3 - US household expectations versus realised CPI inflation (selected months)**

| Month | Michigan median 1-yr expected inflation (%) | Realised CPI-U inflation, 12-month (%) | Gap (pp) |
|---|---|---|---|
| May 1980 | 8.6 | n/a in this table | - |
| Jul 2008 | 5.1 | n/a in this table | - |
| Dec 2019 | 2.3 | n/a in this table | - |
| Jun 2022 | 5.3 | 9.0 | -3.7 |
| Dec 2023 | 3.1 | 3.3 | -0.2 |
| Dec 2024 | 2.8 | 2.9 | -0.1 |
| May 2025 | 6.6 | 2.4 | +4.2 |
| Jun 2026 | 4.6 | 3.5 | +1.1 |
| Sep 2026 | 4.6 | 4.0 | +0.6 |

Sources: FRED MICH (University of Michigan) and CPIAUCSL (BLS), pulled 7 Nov 2026 [V]; raw files `data/r4/MICH.csv`, `data/r4/CPIAUCSL.csv`. Unit: percent. Definitions: MICH is the median expected price change over the next 12 months. CPI-U is all items, seasonally adjusted, 12-month change computed from the index. Limitation: the comparison is contemporaneous (expectation versus *current* realised inflation), not expectation versus inflation *12 months later*. Michigan methodology changed (to web-based collection) in 2024, which may affect comparability [E]. Cleveland Fed model-based one-year expectations for October 2026 were 2.98% [V, FRED EXPINF1YR]. That is a model output, not a survey, and is far below the household survey median.

Interpretation. The May 2025 reading (6.6% expected against 2.4% realised) stands out. It coincided with tariff announcements and with very large partisan gaps in reported expectations [E; partisan breakdowns are published by Michigan but were not retrieved here]. Household expectations thus act partly as a *referendum on the economic and political situation* rather than a forecast. That is evidence for the legitimacy channel. Perceptions of monetary conditions are bundled with trust in government.

### 6.3.9 Behavioural responses to inflation

Stantcheva (2024, "Why Do We Dislike Inflation?", *Brookings Papers on Economic Activity*, Spring: 1-46) surveyed representative US samples in December 2023-January 2024 (one survey with 1,500 closed-ended responses, one open-ended) [V, Brookings summary]. Findings:

- 80% of respondents believed prices systematically rise faster than wages [V].
- People attributed their own raises to job performance or career progress, not to inflation [V].
- Lower-income respondents (under $40,000) more often reported cutting quantity and quality and delaying purchases than higher-income respondents (over $125,000) [V].
- About two-thirds believed the wages of higher-income people rise faster than theirs during inflation, a "clear sense of inequity" [V].
- Inflation ranked as the most important economic and social issue across political leanings. Blame split along partisan lines [V].

This replicates Shiller (1997, "Why Do People Dislike Inflation?", in Romer and Romer eds., *Reducing Inflation*, NBER) [E]. Economists tend to model inflation costs as menu costs, shoe-leather costs and relative-price distortions. The public experiences inflation as a *real income loss and a fairness violation*. Inflation is a legitimacy problem for monetary institutions even when real wages recover on average. Brookings' summary notes that in 2023 CPI inflation was 3.4% and average hourly wages rose 4.1% [V], yet perceptions remained negative.

Behavioural responses to *high* inflation (above about 20-30%) are qualitatively different. They include shortening of contracts, indexation, a switch to foreign-currency pricing, holding foreign cash, and buying durables as stores of value (Heymann and Leijonhufvud 1995, *High Inflation*, Oxford [E]).

## 6.4 Social psychology of money

### 6.4.1 Social conformity and herding

Asch's conformity experiments (1951-56) showed that individuals often defer to a group's judgement even against clear evidence [E]. In finance, rational herding and information cascades (Banerjee 1992, *QJE*; Bikhchandani, Hirshleifer and Welch 1992, *JPE*) show that people rationally follow predecessors' actions, so that private information can be suppressed and behaviour can shift abruptly on small news [E]. Two monetary implications follow:

1. **Stability through conformity.** In normal times, everyone using the currency is itself evidence that it is safe to use. Conformity *stabilises* acceptance.
2. **Cascades in runs.** In stress, visible withdrawals (queues outside Northern Rock in September 2007; social-media-coordinated withdrawals at SVB in March 2023 [V, Fed review: "fueled by social media"]) can trigger cascades. SVB's concentrated network of venture capital firms and technology companies "withdrew uninsured deposits in a coordinated manner at an unprecedented rate" [V]. Digital communication has shortened the time-scale of monetary herding from days to hours.

### 6.4.2 Institutional dependence

People depend on monetary institutions in ways they do not choose. Wages, pensions, mortgages, taxes and benefits are denominated in the national currency and paid through banks. Being "unbanked" carries real costs. The World Bank's Global Findex 2021 found that 76% of adults worldwide had an account (71% in developing economies) [E; the Findex 2025 update was not retrieved in this session]. Dependence is a *structural* fact, not an attitude. Section 8 treats it as a distinct mode of acceptance.

### 6.4.3 Collective beliefs and narratives

Shiller (2017 AEA presidential address; 2019, *Narrative Economics*) argues that economic fluctuations are partly driven by contagious stories [E]. Monetary narratives ("the Fed is printing money", "banks create money out of thin air", "gold never fails", "Bitcoin is digital gold") spread like epidemics and shape behaviour irrespective of accuracy. Some of these narratives are partly true. For example, banks do create deposits when they lend (McLeay, Radia and Thomas 2014). But they are often attached to false inferences, such as that banks face no constraints, or that money creation equals wealth creation. A monograph that wants to inform public debate must address the *narratives* as well as the mechanics.

### 6.4.4 Cultural acceptance of financial arrangements

Cultures differ on the moral status of interest, debt and saving:

- **Interest.** Aristotle condemned usury as unnatural (Section 7.3). Medieval Christian doctrine prohibited it, then gradually relaxed. Islamic jurisprudence prohibits *riba*, and Islamic finance uses profit-and-loss sharing, leasing and mark-up structures that are economically similar in some respects (interpretation: comparative economic history) [E].
- **Debt and shame.** In German, *Schuld* means both debt and guilt, a linguistic point often cited (e.g., by Graeber 2011 and in German policy debates) to explain creditor-oriented attitudes [E].
- **Saving norms.** High household saving rates in East Asia are partly explained by cultural, demographic and institutional factors (precautionary saving, weak social insurance) (interpretation; contested).

Cultural attitudes affect *which* monetary arrangements are accepted as legitimate, for example tolerance of inflation, of bank bail-outs and of household borrowing.

### 6.4.5 Financial literacy

Lusardi and Mitchell (2014, "The Economic Importance of Financial Literacy", *Journal of Economic Literature* 52(1): 5-44) report that only about one-third of people worldwide correctly answer three basic questions on compound interest, inflation and risk diversification [E]. Low literacy implies (i) greater susceptibility to money illusion, (ii) greater reliance on heuristics and trust in intermediaries, and (iii) greater vulnerability to predatory credit. The ECB found that both extremes of trust in the ECB (zero and complete) are associated with *lower* financial knowledge [V, ECB EB 3/2024, Box 1]. Trust that is not grounded in understanding is brittle in both directions.

### 6.4.6 Intergenerational monetary habits

Monetary habits pass across generations through family practice (cash use, savings in gold, distrust of banks after a crisis) and through collective memory (Malmendier and Nagel 2011, 2016; Ehrmann and Tzamourani 2012) [E]. For example, households that experienced a bank deposit freeze (Argentina's *corralito* of December 2001 [E]; Cyprus 2013; Lebanon from 2019) show lasting preferences for holding cash or foreign currency outside the banking system (interpretation from case studies; systematic cross-country micro evidence is thinner).

## 6.5 Monetary legitimacy

### 6.5.1 Distinguishing trust in money from trust in institutions

The BRIEF requires distinguishing trust in money from trust in banks, governments and financial institutions. The data available support four propositions:

1. **Institutional trust is low in many advanced economies.** In the US in June 2026, confidence ("a great deal" or "quite a lot") in banks was 28%, in Congress 9%, in big business 17% and in the presidency 27%; small business (67%) and the military (61%) were highest [V, Gallup topline June 1-15, 2026; n of about 1,001; margin of error +/-4 pp]. US confidence in banks was 53% in May 2004, 41% in June 2007, 22% in June 2009 and 21% in June 2012 [V, same trend table]. Gallup's long series starts at 60% in 1979 [E, secondary compilation].
2. **Financial-sector trust differs by measure and region.** The 2026 Edelman Trust Barometer (online, 28 countries, 33,938 respondents, fieldwork 23 Oct-18 Nov 2025) put trust in the financial services sector "to do what is right" at 63% globally. It was 73% in developing countries and 53% in developed ones. Banks scored 65%, and crypto 41%. High-income respondents (top quartile) scored 68% against 55% for the bottom quartile [V]. Edelman's online, "do what is right" framing produces systematically higher numbers than Gallup's "confidence" framing, so the two must not be compared directly.
3. **Trust in the central bank is correlated with trust in institutions generally.** ECB analysis of its Consumer Expectations Survey found correlations above 72% between trust levels in the ECB, the European Parliament, the European Commission, the UN and national central banks. A first principal component explained 85% of the variation, approximating "trust in institutions in general" [V, ECB EB 3/2024]. Trust in the ECB is not purely a judgement on monetary policy performance.
4. **Use of the currency is far more stable than trust in any institution.** Despite the levels above, there is no evidence of significant currency substitution away from the dollar or euro by domestic residents (fact; absence of evidence noted in central bank data). Low institutional trust is therefore compatible with stable currency use, *within a tolerance band of inflation.*

**Table 6.4 - Trust survey indicators (with years)**

| Survey | Population | Measure | Year | Value | Tag |
|---|---|---|---|---|---|
| Gallup Confidence in Institutions | US adults | Banks: great deal / quite a lot | 2026 | 28% | [V] |
| same | same | same | 2025 | 30% | [V] |
| same | same | same | 2012 | 21% | [V] |

| Survey | Population | Measure | Year | Value | Tag |
|---|---|---|---|---|---|
| same | same | same | 2007 | 41% | [V] |
| same | same | same | 2004 | 53% | [V] |
| Gallup World Poll | 25 countries hit hardest in 2008 (median) | Confidence in financial institutions and banks | 2025 / 2012 / pre-crisis | 63% / 37% / 57% | [V] |
| Edelman Trust Barometer | 28-country online average | Trust financial services sector | 2026 | 63% | [V] |
| same | same | Trust banks subsector | 2026 | 65% | [V] |
| same | same | Trust crypto subsector | 2026 | 41% | [V] |
| Standard Eurobarometer (via ECB) | Euro area | Tend to trust ECB / not / don't know | Autumn 2023 | 43% / 42% / 15% | [V] |
| ECB CES | 11 euro area countries | Share with zero trust in ECB (0-10 scale) | 2020 / Dec 2023 | about 9% / 12% | [V] |

Dataset: `data/r4/trust_surveys.csv`. Limitations: question wording differs fundamentally across surveys. Gallup is US telephone. Edelman uses online panels, which under-represent offline populations, especially in developing countries. Eurobarometer is binary and face-to-face, with methodology disruptions during COVID-19 [V, ECB footnote 7]. Later Eurobarometer ECB-trust waves (2024-2026) were not retrieved in this session.

### 6.5.2 Sources of monetary legitimacy

Political science distinguishes *input* legitimacy (authorisation through democratic process), *output* legitimacy (performance: delivering price stability and financial stability) and *throughput* legitimacy (fair, transparent, accountable procedures) (Scharpf 1999, *Governing in Europe*; Schmidt 2013, *Political Studies* 61(1)) [E]. Applied to money:

| Institution / practice | Input legitimacy | Output legitimacy | Throughput legitimacy | Main legitimacy risk |
|---|---|---|---|---|
| Central bank authority | Statutory delegation by legislature; appointed leadership | Low and stable inflation; crisis management | Transparency, published minutes, testimony | Mission creep; distributional side effects of QE; inflation overshoot (2021-23) |
| Government-issued currency | Constitution or legislation; legal tender law | Stable purchasing power | Rule of law in monetary matters | Fiscal dominance; confiscation |
| Bank deposits | Bank charter and licensing; deposit insurance law | Safety, convenience, interest | Consumer protection; supervision | Runs; bail-outs perceived as unfair |
| Credit-based purchasing power | Contract law; regulation of lending | Access to housing, education, consumption smoothing | Disclosure; anti-predatory rules | Over-indebtedness; unequal access |
| Interest-bearing debt | Contract law; usury laws in some places | Allocation of capital over time | Bankruptcy law; fair collection | Moral objections to interest; debt traps |

| Institution / practice | Input legitimacy | Output legitimacy | Throughput legitimacy | Main legitimacy risk |
|---|---|---|---|---|
| Electronic payments | Payment system law; central bank oversight | Speed, low cost, reach | Privacy, data protection, dispute resolution | Exclusion; surveillance; outages; concentration |

### 6.5.3 Why people accept specific arrangements

- **Central bank authority.** It is accepted mainly through output legitimacy. Long periods of low inflation build tacit consent. The 2021-23 inflation surge was an output failure, and trust in the ECB dipped and recovered without collapse [V, ECB EB 3/2024]. Tucker (2018) argues that output legitimacy is insufficient for an unelected power and that delegation must meet explicit principles (7.4.3).
- **Government currency.** It is accepted through legal tender, taxes and network effects. Endorsement is rarely tested except at referendum moments: the euro adoption votes (Denmark rejected the euro in 2000; Sweden in 2003 [E]) and the Swiss Vollgeld vote of 2018 [V].
- **Bank deposits.** They are accepted because deposit insurance makes them, *for insured amounts*, close substitutes for central bank money. The SVB run involved overwhelmingly *uninsured* deposits [V, Fed review]. That confirms that, for insured depositors, trust in the bank is largely replaced by trust in the guarantor.
- **Credit-based purchasing power and interest-bearing debt.** These are accepted because they deliver immediate utility (present bias) and because modern life (housing, education) is organised around them. Disapproval of "debt culture" in the abstract coexists with personal borrowing. This is a classic attitude-behaviour gap.
- **Electronic payments.** These are accepted for convenience and network reach, with privacy concerns largely traded off (the "privacy paradox", e.g., Athey, Catalini and Tucker 2017 NBER WP 23488 [E]).

## 6.6 Monetary distrust

### 6.6.1 Bank runs: Diamond-Dybvig and its descendants

Diamond and Dybvig (1983, "Bank Runs, Deposit Insurance, and Liquidity", *JPE* 91(3): 401-419) model a bank that transforms illiquid long-term assets into liquid demand deposits, providing insurance against private liquidity shocks [E]. The model has two equilibria. In the good one, only depositors with genuine liquidity needs withdraw. In the run equilibrium, everyone withdraws because they expect others to, forcing costly liquidation and making the fear self-fulfilling. Deposit insurance (or suspension of convertibility, or a lender of last resort) removes the run equilibrium by making it irrational to run.

**Numerical illustration (counterfactual, stylised).**

*Bank balance sheet at t=0:*

| Assets | | Liabilities | |
|---|---|---|---|
| Long-term loans (worth 120 at t=2, only 70 if liquidated at t=1) | 90 | Demand deposits | 100 |
| Reserves (central bank money) | 10 | | |
| **Total** | **100** | **Total** | **100** |

- If 10% of depositors withdraw at t=1 for genuine needs, reserves (10) cover them. The remaining 90 depositors receive a share of 120 at t=2, about 1.33 per unit. Everyone is better off than holding cash.
- If all depositors expect a run, withdrawals of 100 at t=1 face reserves of 10 plus liquidation value of 70, so 80 is available. The bank fails; late depositors get nothing. *Running is individually rational given that others run.*

- With credible deposit insurance of 100, no depositor has a reason to run, so the run never happens and the insurance is never called. This is the logic behind the US authorities' March 2023 decision to protect all depositors of SVB and Signature Bank under the systemic risk exception [E; cross-reference R2/R3].

Later work distinguishes **fundamental runs** (driven by bad news about solvency; Gorton 1988; Allen and Gale 1998) from **panic runs** [E]. Global-games models (Goldstein and Pauzner 2005, *Journal of Finance*) remove the multiplicity: the probability of a run depends on fundamentals and the precision of private signals [E]. SVB was a fundamental-plus-panic run. It had large unrealised losses on long-duration securities after rate rises; its $1.8 billion realised loss announcement on 8 March 2023 triggered the run; and a concentrated, networked depositor base accelerated it [V, Fed review].

**SVB facts (verified).** SVB Financial Group had about $212 billion in total assets when it failed. On 9 March 2023, SVB lost over $40 billion in deposits; management expected to lose over $100 billion more on 10 March; this represented roughly 85% of the deposit base. By comparison, Wachovia saw about $10 billion of outflows over 8 days in 2008 and Washington Mutual $19 billion over 16 days [V, Federal Reserve, Review of the Federal Reserve's Supervision and Regulation of Silicon Valley Bank, April 2023, Executive Summary]. The California regulator closed SVB on 10 March 2023 [V].

What SVB tells us about trust: (i) the run was a loss of trust in *one intermediary*, not in the dollar. Funds moved to large banks and money market funds (interpretation; flows documented in Fed H.8 and ICI data, not retrieved here). (ii) It was the speed, not the logic, that was new. (iii) It revealed that the deposit franchise, the value of sticky low-cost deposits, can evaporate when deposits are uninsured and depositors are networked and attentive (Section 7.4.2).

Gallup reported in May 2023 that about half of Americans were worried about the safety of their money in banks [E, Gallup headline seen on the 2026 article page], yet aggregate deposits in the banking system did not collapse. The gap between worry and action is again visible.

### 6.6.2 Currency crises

Currency crisis models come in generations:

- **First generation** (Krugman 1979, *JMCB*): inconsistent fiscal deficits monetised under a fixed peg deplete reserves until a speculative attack, which happens at a predictable moment [E].
- **Second generation** (Obstfeld 1986, 1994): multiple equilibria. The government weighs the cost of defending the peg, and self-fulfilling attacks are possible [E].
- **Third generation / twin crises** (Kaminsky and Reinhart 1999, *AER* 89(3)): banking and currency crises reinforce each other through balance sheet mismatches, especially foreign-currency debt [E].

The 1997-98 Asian crisis, including Indonesia, is the canonical twin crisis. The rupiah fell from roughly 2,400 per US dollar in mid-1997 to above 15,000 in mid-1998 [E; exact daily lows vary by source; verify against Bank Indonesia SEKI in the Indonesia stream]. Bank runs, bank closures and a collapse of public trust followed, contributing to the end of the Suharto government in May 1998 (fact [E]; cross-reference Part XII stream).

### 6.6.3 Hyperinflation

Cagan (1956, "The Monetary Dynamics of Hyperinflation", in Friedman ed., *Studies in the Quantity Theory of Money*) defined hyperinflation as inflation exceeding 50% per month [E]. Hanke and Krus (2013, "World Hyperinflations", in Parker and Whaples eds., *Routledge Handbook of Major Events in Economic History*) list 56 episodes meeting this definition. The table has since been extended by Hanke to include Venezuela (2016-) [E]. Key features:

- **Fiscal origins.** Nearly every hyperinflation involves large deficits financed by central bank money creation, usually after war, revolution, regime collapse or loss of tax capacity (Sargent 1982, "The Ends of Four Big Inflations" [E]). Supply collapse and the Olivera-Tanzi erosion of tax revenue amplify the dynamics.
- **Abrupt stabilisation.** Sargent argued that the four interwar hyperinflations (Austria, Hungary, Poland, Germany) ended abruptly when credible *fiscal and institutional regime changes* were made, not through gradual monetary tightening [E]. Trust can be restored quickly when the institutional cause is credibly fixed.

- **Currency abandonment.** Zimbabwe abandoned its own currency for a multicurrency system (mainly the US dollar and the South African rand) in 2009 [E]. Later attempts to reintroduce a domestic currency (bond notes, RTGS dollar, ZiG in 2024) met persistent distrust (fact [E]; details not verified in this session).

### 6.6.4 Loss of purchasing power, scandals and sovereign default

- **Moderate inflation** (5-20%) erodes trust in institutions, as seen in 2021-23, without causing currency abandonment in large economies.
- **Financial scandals** (e.g., Libor manipulation 2012; the FTX collapse 2022 [E]) erode trust in *intermediaries and markets*, and sometimes push users *toward* regulated banks rather than away from them.
- **Sovereign defaults** in domestic or foreign currency (Reinhart and Rogoff 2009, *This Time Is Different*, document hundreds of external defaults and many domestic ones since 1800 [E]) reduce trust in the issuer. When the default is on domestic-currency debt or is accompanied by deposit freezes, they reduce trust in the money too.

### 6.6.5 Currency substitution and dollarization

**Official (full) dollarization** means adopting a foreign currency as legal tender. Examples include Panama (US dollar since 1904), Ecuador (2000, after a banking and currency crisis) and El Salvador (2001) [E]. **Unofficial (partial) dollarization** means residents holding deposits, loans or cash in a foreign currency while the domestic currency remains legal tender. Levy Yeyati (2006, "Financial Dollarization: Evaluating the Consequences", *Economic Policy* 21(45)) shows that financial dollarization is associated with higher banking-crisis risk and output volatility [E].

Key empirical regularities (interpretation from the literature, [E]):

1. **Ratchet effect.** Dollarization rises in crises and falls only slowly after stabilisation (Reinhart, Rogoff and Savastano 2003, "Addicted to Dollars", NBER WP 10015).
2. **Function order.** Store of value dollarizes first (deposits, savings), unit of account for large items (real estate, cars) next, and means of payment for small daily purchases last (Calvo and Vegh 1992, IMF working paper [number unverified], distinguishing currency substitution, in the means-of-payment sense, from asset substitution).
3. **Legal and fiscal anchors slow full exit.** Taxes, public wages and utility bills denominated in local currency keep demand for it alive even in heavily dollarized economies.

A cross-country panel of deposit dollarization was not assembled in this session. The IMF's financial dollarization database (Levy Yeyati; IMF staff updates) is the recommended source for the case-study stream (gap noted).

### 6.6.6 Alternative monetary arrangements

Alternatives that have grown in periods of distrust include gold and precious metals, cryptocurrencies, stablecoins (which import trust in the dollar rather than replacing it), local and community currencies, barter networks, and mobile money run by telecoms. In the 2026 Edelman survey, crypto had the lowest trust of the financial sub-sectors (41%) [V]. Of these alternatives, stablecoins have grown fastest. They are mostly *dollar claims*, so they extend dollarization rather than offering an exit from state money (interpretation; cross-reference R5 and CBDC stream).

## 6.7 Why dissatisfaction rarely becomes abandonment

### 6.7.1 Hirschman: exit, voice and loyalty

Hirschman (1970, *Exit, Voice, and Loyalty: Responses to Decline in Firms, Organizations, and States*, Harvard University Press) argued that members of a declining organisation can *exit* (leave), use *voice* (complain, protest, vote) or remain *loyal*. Loyalty raises the cost of exit and so channels dissatisfaction into voice [E]. Applied to money:

| Response | Monetary form | Cost | When used |
|---|---|---|---|
| Exit (individual) | Holding foreign currency, gold, crypto; moving deposits abroad | Moderate to high: transaction costs, legal limits, exchange risk, loss of convenience | When expected loss from staying is large and visible |
| Exit (collective) | Dollarization; regional currency; parallel currencies | Very high: coordination; legal change | After a regime has failed |
| Voice | Voting out governments over inflation; protests; referendums; litigation; public pressure on central banks | Low to moderate | First response in democracies |
| Loyalty | Continued use; trust in national institutions; patriotism about the currency | Low | When attachment and alternatives are weak |

Three features make *exit* from a currency especially costly:

1. **Exit requires counterparties.** I cannot use dollars to buy groceries if the grocer prices and takes only local currency. Exit is a *coordination* problem, not an individual choice (network effects).
2. **Some obligations cannot be exited.** Taxes, many wages and many legal obligations are fixed in the national currency (legal obligation).
3. **Exit is partial and functional.** One can exit the store-of-value function (buy dollars or gold) while remaining in the means-of-payment function. Most dissatisfaction is therefore expressed through *partial functional exit* plus *voice*, not abandonment.

### 6.7.2 Gresham versus Thiers

Gresham's law ("bad money drives out good") holds when an exchange rate between two monies is fixed by law. People spend the overvalued (bad) money and hoard the good. Thiers' law ("good money drives out bad") holds when rates float freely and acceptance is voluntary: in hyperinflation, sellers refuse the bad money (Bernholz 2003, *Monetary Regimes and Inflation* [E]). Which one prevails depends on the strength of the *legal obligation* mode. Strong legal tender enforcement and price controls favour Gresham dynamics and slow exit. Weak enforcement favours Thiers dynamics and fast exit.

### 6.7.3 A tolerance band, not a threshold

Pulling these elements together (interpretation): dissatisfaction translates into abandonment only when the expected cost of staying exceeds the cost of exit plus the coordination premium. Inflation in low single digits produces voice. Sustained high double-digit inflation produces store-of-value exit and indexation. Hyperinflation or confiscation produces general exit. Where the band's edges lie depends on (i) the availability of a credible alternative unit (usually the US dollar), (ii) legal enforcement, (iii) collective memory, and (iv) institutional trust. This becomes the formal model in Part VIII.

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# PART VII - THE SOCIOLOGY AND PHILOSOPHY OF MONETARY FAIRNESS

## 7.1 Who has the power to create money?

### 7.1.1 Allocation of monetary power

**Table 7.1 - Who creates what money, under what constraint**

| Actor | What it creates or influences | Mechanism | Main constraints | Accountability |
|---|---|---|---|---|
| Central bank | Base money: currency and reserves | Asset purchases, lending to banks, FX operations | Mandate (price stability), legal limits on monetary financing (e.g., Art. 123 TFEU), balance sheet risk, credibility | Legislature; transparency; courts in some systems |
| Commercial banks | Broad money: deposits | Lending and asset purchases credit deposits (McLeay, Radia and Thomas 2014) | Profitability; creditworthy demand; capital and liquidity rules; interest rate set by central bank; competition; deposit outflows on settlement | Supervisors; shareholders; market discipline |
| Government (Treasury) | Indirectly: deposits created when it spends, net of taxes; demand for base money through taxes | Fiscal deficits financed by bond sales; spending credits private deposits | Debt markets; legal debt limits; prohibition of direct monetary financing in most systems | Elections; parliament |
| Financial markets / non-bank financial institutions | Money-like liabilities: MMF shares, repo, stablecoins | Maturity and liquidity transformation outside banks | Run risk; regulation of MMFs and stablecoins; access to central bank facilities | Securities and payments regulators |
| International institutions | Reserve assets: SDRs (IMF); swap lines (central banks) | Allocations; swap agreements | Member consent; governance quotas | Member states |

Sources: McLeay, Radia and Thomas 2014 (BoE QB Q1) [E]; detailed mechanics in the Part IV stream.

**Composition evidence (fact).** In the United States in September 2026, currency (FRED CURRSL) was $2,397.2 billion and M2 was $23,413.9 billion, so currency was **10.2% of M2** [V, FRED, pulled 7 Nov 2026; `data/r4/CURRSL.csv`, `M2SL.csv`]. The remaining roughly 90% is deposits at depository institutions plus retail money market fund shares. Limitations: a large but uncertain share of US currency is held abroad, so domestic currency use is overstated; M2 includes retail MMF shares, which are non-bank liabilities. Reserve balances held by banks at the Federal Reserve were $2.93 trillion on 4 November 2026 [V, FRED WRESBAL]. These are central bank money that circulates only among banks and the government. For the UK, McLeay, Radia and Thomas (2014) reported that notes and coin were about 3% of broad money and bank deposits about 97% at that time [E].

**Interpretation.** The quantitative dominance of bank deposits supports the claim that *most money used by the public is private bank money.* Three qualifications are necessary to avoid the opposite error:

1. Banks create deposits only when someone borrows (or when they buy assets). Creditworthy demand, risk and profitability limit lending.
2. Deposits are valuable only because they are convertible at par into central bank money and settle through central bank money. Bank money is *derivative* of public money.
3. The state has chosen this arrangement and could change it (Chicago Plan, sovereign money). That it has not is a political choice, discussed in 7.4.1.

So the allocation is better described as **delegated, licensed, backstopped private issuance of a public unit** (interpretation: this stream; close to the "money as a public-private hybrid" view of Hockett and Omarova 2017, *Cornell Law Review* 102, "The Finance Franchise" [E]).

### 7.1.2 The "finance franchise" view

Hockett and Omarova (2017) argue that the state licenses private banks to issue the sovereign's full faith and credit, in the form of deposits backed by public guarantees, and that this is a *franchise*: a public resource leased to private actors (interpretation: legal-institutionalist) [E]. If so, franchisees owe public duties. These might include serving the public in credit allocation, accepting limits on extraction, and supporting inclusion. Critics answer that banks bear credit risk, hold capital and are already heavily regulated, and that the "franchise" rhetoric understates the private risk-bearing and screening function. The question is normative (7.4).

## 7.2 Debt and society

### 7.2.1 Historical development (brief)

Debt predates coinage. Mesopotamian temple and palace accounting recorded obligations in silver and barley units, and periodic royal debt cancellations ("clean slates", e.g., Hammurabi's edicts) restored debtors' freedom (Hudson 2018, *. . . and forgive them their debts* [E]; Graeber 2011 [E]). Usury prohibitions, debtors' prisons, bankruptcy law and, later, consumer credit regulation mark a long history in which societies have repeatedly renegotiated the balance between creditor rights and debtor protection (interpretation; cross-reference Part III stream).

### 7.2.2 Household indebtedness - data

**Table 7.2 - Household credit, % of GDP (BIS)**

| Economy | 2007Q4 | 2019Q4 | 2026Q1 (latest) | Peak since 2007 (quarter) |
|---|---|---|---|---|
| United States | 98.4 | 74.7 | 67.2 | 98.4 (2007Q4) |
| United Kingdom | 94.3 | 83.9 | 72.6 | 97.6 (2010Q1) |
| Euro area | 58.6 | 56.6 | 50.2 | 63.4 (2009Q4) |
| Germany | 60.4 | 52.1 | 47.9 | 62.8 (2007Q1) |
| France | 47.1 | 62.3 | 59.1 | 68.2 (2020Q4) |
| Japan | 59.5 | 60.8 | 60.8 | 65.9 (2021Q1) |
| China | 18.6 | 54.4 | 57.5 | 61.0 (2024Q1) |

| Economy | 2007Q4 | 2019Q4 | 2026Q1 (latest) | Peak since 2007 (quarter) |
|---|---|---|---|---|
| Korea | 66.5 | 89.6 | 85.1 | 99.1 (2021Q3) |
| Canada | 80.8 | 104.5 | 100.0 | 113.2 (2020Q4) |
| Australia | 109.0 | 118.5 | 114.1 | 123.9 (2016Q3) |
| Switzerland | 101.9 | 124.4 | 127.0 | 133.2 (2021Q1) |
| Indonesia | 11.6 | 17.0 | 15.2 | 17.8 (2020Q4) |

Source: BIS Total Credit statistics (WS_TC), credit to households and NPISHs from all sectors, market value, % of GDP, adjusted for breaks; API pull 7 Nov 2026 [V]; `data/r4/bis_hh_credit.csv` and `bis_hh_credit_summary.csv`. Limitations: GDP denominators move with nominal growth (inflation in 2021-23 mechanically reduced ratios); gross debt ignores household assets; national definitions of NPISH differ; Indonesia's low ratio partly reflects a large informal credit sector not captured.

**US composition (Q2 2026).** Total household debt $18.771 trillion, of which mortgages $13.117 trillion, HELOC $0.459 trillion, auto $1.713 trillion, credit cards $1.263 trillion, student loans $1.651 trillion and other $0.568 trillion. 4.7% of outstanding debt was in some stage of delinquency. The flow into serious delinquency for student loans was 7.83% (annualised), distorted by re-reporting of defaulted loans [V, NY Fed Quarterly Report on Household Debt and Credit, press release 11 Aug 2026]. Limitation: the data are credit bureau records (Equifax CCP), which exclude some loans. Q3 2026 data were scheduled for release on 10 November 2026, after this compilation [E].

### 7.2.3 Mortgages, student debt, sovereign debt, corporate leverage

- **Mortgages.** Mortgages are the largest household liability in most advanced economies and the main channel through which bank money creation becomes housing wealth (Jorda, Schularick and Taylor 2016, "The Great Mortgaging", *Economic Policy* 31(85): mortgage lending rose from about one-third of bank lending in 1900 to about two-thirds by the 2000s across 17 advanced economies [E]). Distributionally, mortgages let middle-class households lever into a rising asset. Renters are excluded from this and bear rising housing costs (cross-reference R3 on asset inflation).
- **Student debt.** US student debt of $1.651 trillion (Q2 2026) [V] is mostly federally owned. The debate splits. On one side, it is a human-capital investment with positive average returns. On the other, outcomes are highly dispersed, completion matters, and debt burdens delay household formation and fall disproportionately on some groups (interpretation; mixed empirical literature, e.g., Looney and Yannelis 2015 BPEA on the rise of borrowers at for-profit and non-selective institutions [E]).
- **Sovereign debt.** US total public debt was 121.2% of GDP in 2026Q2 (FRED GFDEGDQ188S, federal debt including intragovernmental holdings) [V]. Debt held by the public is lower; that series was not pulled here. Sovereign debt is the safe asset on which private finance builds (collateral, reserves, MMF holdings). It also creates distributional questions: interest payments flow to bondholders, who are disproportionately wealthier households and foreign official holders (interpretation).
- **Corporate leverage.** It funds investment but concentrates bankruptcy risk. It also connects to private credit growth (cross-reference Part IV stream).

### 7.2.4 Debt, mobility, opportunity and political relationships

**Arguments that debt enables mobility (pro):**

- Credit relaxes liquidity constraints, enabling education, housing and business formation for people without inherited wealth (fact in theory; empirical support from microfinance mixed, see Banerjee, Karlan and Zinman 2015 *AEJ: Applied* 7(1), finding modest average effects across six randomised evaluations [E]).
- Mortgage access built middle-class wealth in the 20th century (interpretation, with racial exclusion as a major caveat: redlining in the US).

**Arguments that debt entrenches inequality (con):**

- Mian, Straub and Sufi (2021, "Indebted Demand", *QJE* 136(4)) argue that rising savings at the top are lent to the bottom 90% and to governments, which depresses demand and interest rates over time, a "debt trap" in the aggregate [E].
- Mian and Sufi (2014, *House of Debt*) show that the 2007-09 US recession was deeper in counties with higher household leverage, because losses on housing fell on leveraged, lower-wealth households first [E].
- Debt imposes discipline on labour: indebted workers have less bargaining power and are less likely to strike or quit (interpretation, a political economy argument often attributed to Lazzarato 2012, *The Making of the Indebted Man* [E]; empirical support is limited).
- Streeck (2014, *Buying Time*) argues that democratic capitalism moved from inflation (1970s) to public debt (1980s-90s) to private debt (2000s) to central bank support as successive ways of postponing distributional conflict [E] (interpretation; contested by mainstream political economy).

**Political relationships.** Debt creates creditor-debtor constituencies. Creditor-heavy societies tend to favour low inflation, and debtor-heavy societies tolerate or demand more (interpretation: e.g., Posen 1995 on financial sector opposition to inflation as the basis of central bank independence [E]). Sovereign debt ties governments to bond markets, so the "bond vigilante" is a political actor (e.g., the UK gilt crisis of September-October 2022 [E]).

## 7.3 Philosophical foundations: a structured comparison of disagreements

The BRIEF asks for a comparison of *disagreements*, not summaries. Each thinker gets a one-line position, followed by a matrix along five dimensions, then the four main fault lines.

**One-line positions (as interpreted here; [E] for all classic texts):**

- **Aristotle** (*Politics* I.8-10; *Nicomachean Ethics* V.5): money (*nomisma*) exists by convention (*nomos*) to make unlike goods commensurable for just exchange. Acquisition of money for its own sake (*chrematistike*) is unnatural, and usury, "money begetting money", is the most unnatural form.
- **Adam Smith** (*Wealth of Nations* 1776, I.iv; II.ii): money arises from the division of labour to overcome the inconvenience of barter. Banking that substitutes paper for gold is a "waggon-way through the air" that frees capital for productive use, but needs prudence (real bills) and regulation (he supported limits on small notes and usury ceilings).
- **Karl Marx** (*Capital* I, 1867, ch. 1-4; III ch. 25-36): money is the general equivalent that expresses socially necessary labour time and conceals social relations (commodity fetishism). The circuit M-C-M' turns money into capital. Credit accelerates accumulation and crises, and finance capital ("fictitious capital") appropriates surplus value.
- **Georg Simmel** (*The Philosophy of Money*, 1900): money is the purest form of social relation and of trust; it is a "claim upon society". It enlarges individual freedom and impersonality while making life calculative.
- **John Maynard Keynes** (*Treatise on Money* 1930; *General Theory* 1936): money is defined by the state's choice of the money of account (chartalist influence). Money is not neutral. Liquidity preference and uncertainty make money a store of value that can stall demand. The state should manage money and credit to achieve full employment; the "euthanasia of the rentier" is desirable as capital becomes abundant.
- **Friedrich Hayek** (*Prices and Production* 1931; *Denationalisation of Money* 1976): money distorts relative prices when created by credit expansion (Austrian business cycle). The state monopoly on money is the root of inflation, and competing private currencies would discipline issuers.
- **Karl Polanyi** (*The Great Transformation*, 1944): money, like labour and land, is a "fictitious commodity". Treating it as a market good (the gold standard) produced social dislocation, which in turn provoked protective countermovements. Money must be socially embedded and politically governed.
- **David Graeber** (*Debt: The First 5,000 Years*, 2011): the barter myth is false. Credit and debt preceded coinage, and money arose with states, war and markets. The moral language of debt naturalises relations of power. Debt cancellations recur historically and are legitimate.
- **Joseph Schumpeter** (*Theory of Economic Development* 1911/1934; *History of Economic Analysis* 1954): bank credit created *ex nihilo* is the lever by which entrepreneurs command resources for innovation. The banker is the "ephor" of the exchange economy. Credit creation is central to development, and crises are part of creative destruction.
- **Hyman Minsky** (*Stabilizing an Unstable Economy*, 1986): "everyone can create money; the problem is to

get it accepted". Stability is destabilising: finance moves from hedge to speculative to Ponzi structures. A Big Government and a Big Bank (lender of last resort) are needed to contain instability, but they validate risk-taking.

**Table 7.3 - Disagreement matrix**

| Thinker | Nature of money | Credit and interest | Power | Social obligation | Role of the state in money |
|---|---|---|---|---|---|
| Aristotle | Convention (nomos) and measure for just exchange | Usury unnatural; lending at interest condemned | Wealth-getting without limit corrupts the household and polis | Exchange serves the household and community; reciprocity and justice | Money's value set by convention and law; the polis can change it |
| Smith | Instrument of circulation arising from division of labour; metallic standard as anchor | Credit and banking productive if prudent (real bills); usury ceiling supported | Warns against merchant combinations; markets disperse power | Sympathy and justice; commercial society civilises | Limited but real: regulate banks and notes, enforce contracts |
| Marx | General equivalent of value (labour); fetish masking class relations | Interest is a share of surplus value; credit fuels accumulation and crises | Money capital is power over labour; finance capital dominates | Under capitalism obligations are reduced to cash nexus | The state serves capital; money reform alone cannot end exploitation |
| Simmel | Pure social relation; claim on society; symbol of trust | Credit extends trust in time; neutral-to-positive | Money depersonalises domination but also enables freedom | Weakens personal bonds, widens impersonal interdependence | Money's value rests on the social group and its central authority |
| Keynes | State money of account; money not neutral; liquidity premium | Interest is the reward for parting with liquidity, not for saving; rentier to be euthanised | Rentier power over investment; uncertainty | Full employment as a social obligation of policy | Active management of money and credit; international clearing union |
| Hayek | Market institution; state monopoly distorts | Credit expansion beyond saving causes malinvestment | State monetary power is the main danger | Spontaneous order; no designed distributive obligation | Remove the monopoly: competing private currencies or strict rule |
| Polanyi | Fictitious commodity; token of purchasing power governed by society | Commodity-money discipline produces social damage | Haute finance and the gold standard as power | Society protects itself; embeddedness | Money must be politically governed and socially embedded |
| Graeber | Credit and debt relation; virtual credit money older than coins | Interest-bearing debt historically linked to slavery and war; cancellation recurrent | Debt morality legitimises power; states and markets co-created | Obligations are political, not sacred; "communism of everyday life" | States create money (taxes, armies); clean slates legitimate |

| Thinker | Nature of money | Credit and interest | Power | Social obligation | Role of the state in money |
|---|---|---|---|---|---|
| Schumpeter | Claims; credit as means of payment created by banks | Bank credit ex nihilo is the engine of innovation; interest is a toll on development profits | Bankers as gatekeepers ("ephors") of development | Creative destruction; obligations secondary | Central bank as background; skeptical of over-management |
| Minsky | Hierarchy of IOUs; "anyone can create money" | Credit endogenous and pro-cyclical; leads to fragility | Finance shapes the economy; validation by the state | Policy must secure employment (employer of last resort) | Big Government and lender of last resort are necessary but create moral hazard |

**The four fault lines (interpretation):**

1. **Thing versus relation.** Smith (partly) and the commodity tradition treat money as an object with a natural anchor (metal). Aristotle treats it as convention. Simmel, Graeber, Minsky, Keynes (and Marx at the level of social relations) treat it as a social relation of credit, trust or power. Hayek is distinctive: money is a market institution whose value comes from issuer discipline under competition, a relation governed by markets rather than states. *Consequence:* commodity-anchored thinkers locate monetary trust in backing assets; relational thinkers locate it in institutions and power.

2. **Credit as engine versus credit as fragility or domination.** Schumpeter and Smith see bank credit as productive (finance for innovation and circulation). Minsky sees it as productive but inherently destabilising. Marx sees it as accelerating accumulation and crisis. Graeber sees interest-bearing debt as an instrument of domination with a violent history. Hayek sees credit expansion beyond voluntary saving as distorting. *Consequence:* these positions imply opposite policies, from liberalising credit (Schumpeter-inspired), through containing it (Minsky: regulation and LOLR) and anchoring it to saving (Hayek), to cancelling it (Graeber's jubilee).

3. **Rule or competition versus democratic or discretionary governance.** Hayek wants money removed from politics through competition or rules. Keynes, Polanyi and Minsky want democratic or technocratic management for full employment and social protection. Smith sits between (prudential rules, real bills). Marx sees governance as subordinate to class power. *Consequence:* this fault line maps directly onto modern debates about central bank independence (7.4.3) and CBDC design.

4. **Moral versus political obligation in debt.** Aristotle's critique of usury and Graeber's critique of debt morality both deny that interest-bearing debt is simply just. Smith and, implicitly, the modern contract tradition treat repayment as a moral and legal duty underpinning trust. Keynes sits pragmatically: obligations should yield to full employment ("euthanasia of the rentier"). *Consequence:* whether debt relief is a moral hazard or a restoration of justice depends on this axis.

**Where they converge (interpretation).** All ten accept that money depends on something beyond the physical token: law, convention, trust, the state, or social relations. None of them, including Hayek, holds that money is "natural" in the sense of being independent of institutions. All of them see the *distribution of the power to create and allocate credit* as consequential. They differ on whether that power should be checked by markets, rules or democratic politics.

**Pairwise sharpest disagreements:**

- *Hayek vs Keynes:* rule-bound or competitive money versus managed money for employment.
- *Schumpeter vs Graeber:* credit as creative engine versus credit as domination.
- *Smith vs Polanyi:* markets as civilising versus market money as socially destructive.
- *Simmel vs Marx:* money as expanding freedom through impersonality versus money as masking exploitation.

- *Minsky vs Hayek:* both see credit-driven instability. Minsky wants a stronger state backstop, Hayek wants less state money.

## 7.4 The fairness problem

Each sub-question in the BRIEF is argued pro and con below, followed by an assessment that separates empirical findings from normative judgement.

### 7.4.1 Is money creation a public function or a private economic privilege?

**Position A - Money creation should be public (sovereign money / full reserve).**

- *Chicago Plan (1933)*: a memorandum by Henry Simons and other University of Chicago economists proposed 100% reserve requirements on demand deposits, separating money (fully backed) from lending (funded by equity or time deposits) [E]. Irving Fisher's *100% Money* (1935) argued that this would end bank runs and give the state control of the money stock [E]. Friedman (1960, *A Program for Monetary Stability*) endorsed a version [E].
- *Benes and Kumhof (2012, "The Chicago Plan Revisited", IMF WP/12/202)* simulated the Plan in a DSGE model calibrated to the US and found large output gains, reduced debt and no inflation [E]. *Caveat:* this is a model simulation with specific assumptions, and it is not an IMF policy position.
- *Positive Money* (UK, founded 2010) and the *Vollgeld* movement propose that only the central bank create money, and that new money be spent into circulation by government or distributed to citizens, with banks as intermediaries of existing money [E].
- *Iceland*: a report commissioned by the Prime Minister (Sigurjonsson 2015, *Monetary Reform: A Better Monetary System for Iceland*) recommended considering sovereign money after the 2008 banking collapse [E]. It was not implemented.

**Position B - Bank money creation is a legitimate, efficient private function under public rules.**

- Deposit creation through lending combines payment services with credit screening and liquidity provision (Diamond and Dybvig 1983; Kashyap, Rajan and Stein 2002 *JF* on synergies between deposits and loan commitments [E]).
- Full-reserve systems would push maturity transformation into *shadow banks*, recreating run risk outside regulation (Pennacchi 2012; Goodhart 1998 on endogenous money [E]). The 2008 crisis was largely a shadow-banking run, which shows that narrow banking does not eliminate runs (interpretation).
- Credit could become scarcer or more expensive, especially for small firms reliant on bank lending (counterfactual; magnitude disputed).
- Political risk: giving the state direct control of the money stock and its first spending raises fiscal-dominance concerns (interpretation: mainstream and Austrian critics agree on this point, for different reasons).

**The test case - Swiss Vollgeld initiative, 10 June 2018 (verified).** The federal popular initiative "For crisis-safe money: money creation by the National Bank only! (Vollgeld initiative)" was rejected with **24.28% Yes** and **0 of 23 cantonal votes**, on a **turnout of 34.55%** [V, Swissvotes database, https://swissvotes.ch/vote/618.00]. Federal Statistical Office figures round these to 24.3% Yes and 34.6% turnout [E, BFS search excerpt]. Parliament had opposed it (National Council 9 Yes, 169 No; Council of States 0 Yes, 42 No), as had the Federal Council. All major parties except the Greens (who gave no recommendation) recommended No [V, Swissvotes]. The highest cantonal Yes share was Geneva, about 40.3% [E].

Interpretation of the vote: it is the only national test of sovereign money by direct democracy. Voters rejected it decisively. This weighs against a claim that publics, once informed, want sovereign money. It does not settle the normative question. Low turnout, an expert consensus against, and status quo bias in referendums all shaped the result (Swiss initiatives generally fail more often than they pass).

**Assessment.** *Empirical:* full-reserve proposals would reduce runs on the banks covered but face migration of maturity transformation. Their macro effects are model-dependent. *Normative:* if money is seen as a public good (the unit and its safe form), there is a case that seigniorage-like rents should accrue to the public. If credit allocation is seen as a private screening service, private banks should be allowed to fund it with deposits under public backstops priced fairly. A middle path, already partly in place, is to tax or price the public backstop

(risk-based deposit insurance premiums; capital surcharges) and to offer a public safe money option (cash; possibly retail CBDC) alongside bank money.

### 7.4.2 Should private banks benefit from creating deposits through lending?

**The "seigniorage" claim and its correction.** A frequent claim is that banks earn "interest on money they created from nothing". The accounting is subtler. When a bank lends 100 and credits a deposit of 100, it earns the loan rate on 100. It must also (i) pay interest on the deposit, (ii) settle outflows in central bank reserves when the borrower spends, which requires attracting deposits or borrowing reserves at market rates, (iii) hold capital against credit losses, and (iv) bear operating costs. The bank's rent is the *spread*, not the full interest (cross-reference Part IV stream for T-accounts).

**The deposit franchise.** Drechsler, Savov and Schnabl (2017, "The Deposits Channel of Monetary Policy", *QJE* 132(4): 1819-1876) show that banks have market power over deposit rates. When the policy rate rises, deposit rates rise much less, especially in concentrated markets, so the deposit spread widens and deposits flow out [E]. Their 2021 paper ("Banking on Deposits: Maturity Transformation without Interest Rate Risk", *Journal of Finance* 76(3)) argues that this low "deposit beta" hedges banks' long-duration assets [E]. SVB showed the limit: when deposits are uninsured and networked, the franchise is not sticky [V, Fed review].

**Stylised example (counterfactual, illustrative).** Suppose the policy rate is 5.0%, the average deposit rate 1.5%, and operating costs of running the deposit base (branches, IT, compliance) 2.0% of deposits. The bank's net deposit franchise rent is about 1.5% of deposits a year (5.0 - 1.5 - 2.0). On deposits of $1,000 billion that is $15 billion a year. At a 0.25% policy rate and 0% deposit rate, the gross spread is 0.25%, below operating costs, so the deposit franchise *loses money* and banks earn mainly on lending and fees. *Lesson:* bank rents from deposits are highly state-dependent and come from depositor inertia and market power, which behavioural economics explains (6.3.2), not from "creation from nothing" as such.

**Pro (banks should benefit):** they provide payment services, bear credit risk, hold capital, screen borrowers and transform maturity. Rents fund these services, and competition limits them where markets are contestable.

**Con (they benefit unfairly):**

- *Implicit subsidies.* Too-big-to-fail expectations lower large banks' funding costs. Haldane (2010, "The $100 billion question", Bank of England speech) estimated very large implicit subsidies to the largest banks during the crisis [E]. Post-crisis reforms (resolution regimes, TLAC) aimed to reduce them. Estimates vary widely.
- *Exploitation of inertia.* Deposit rents fall disproportionately on less attentive, often older or less financially literate depositors (interpretation consistent with DSS and behavioural evidence).
- *Asymmetric outcomes.* Bail-outs socialise losses while profits stay private (normative critique), as seen in 2008 and partially in 2023 (uninsured depositors protected through the systemic risk exception, with costs recovered through a special assessment on banks [E]).

**Assessment.** *Empirical:* deposit franchise rents are real, cyclical and linked to market power. Implicit subsidies existed and have been reduced but not eliminated. *Normative:* the fairness case turns on whether the backstop is priced correctly and whether competition is effective. Remedies on the table include pricing deposit insurance, competition policy, transparency about deposit rates, and public options for safe money.

### 7.4.3 What is the legitimate role of central bank independence?

**Pro (the case for independence):**

- *Time inconsistency.* Delegating to an independent, conservative central banker lowers inflation bias (Kydland and Prescott 1977; Barro and Gordon 1983; Rogoff 1985) [E].
- *Empirical association.* Alesina and Summers (1993, *JMCB* 25(2)) found legal central bank independence negatively correlated with inflation in advanced economies, with no output cost [E]. Later work questions robustness: the correlation weakens for developing countries and legal versus actual independence differ (Cukierman, Webb and Neyapti 1992, *World Bank Economic Review*, using governor turnover [E]). Garriga (2016) provides an updated index [E].

- *Tucker's principles.* Paul Tucker (2018, *Unelected Power: The Quest for Legitimacy in Central Banking and the Regulatory State*, Princeton University Press) argues that delegation to independent agencies is legitimate only when (i) the goal is clear and monitorable, (ii) the public has a stable preference that elected politicians cannot credibly commit to, (iii) the delegated body does not make big distributional choices that belong to elected representatives, and (iv) there are clear operating principles, transparency and accountability to the legislature (the "Principles for Delegation") [E].

**Con (the critique):**

- *Distributional choices.* Post-2008, central banks made large distributional choices: QE raised asset prices, emergency facilities chose which markets to support, and collateral frameworks favoured some issuers. By Tucker's own criteria these stretch the delegation (Dietsch, Claveau and Fontan 2018, *Do Central Banks Serve the People?* [E]).
- *Whose preferences?* Posen (1995) argued that independence and low inflation reflect the political power of the financial sector rather than a technocratic solution [E].
- *Democratic deficit.* Goodhart and Lastra (2018, "Populism and Central Bank Independence", *Open Economies Review* 29) argue that populist backlash is a predictable reaction to visible unelected power [E].
- *Coordination.* At the zero lower bound, separating monetary from fiscal policy may be suboptimal (MMT; some New Keynesians on fiscal-monetary coordination) (interpretation).

**Assessment.** *Empirical:* independence correlates with lower inflation in advanced economies, but causality and robustness are debated. Trust in central banks correlates with general institutional trust (6.5.1). *Normative:* independence is most defensible for narrowly defined price stability with transparent accountability. It is least defensible for open-ended credit allocation or climate and industrial policy. That is why mandate creep is the main legitimacy risk. Political pressure on central banks in several countries in 2024-26 makes this a live issue (fact pattern noted; specific episodes are covered in other streams and not verified here).

### 7.4.4 Is inflation disproportionately harmful to people with limited assets?

**Yes (pro):**

- Easterly and Fischer (2001, "Inflation and the Poor", *JMCB* 33(2): 160-178) used a survey of 31,869 respondents in 38 countries. Poorer respondents were more likely to name inflation as a top national concern, and high inflation was associated with falling shares of the poor in national income [E].
- The poor hold a larger share of their wealth in cash and deposits with zero or low rates (Erosa and Ventura 2002, *JME* 49(4): inflation as a regressive consumption tax [E]).
- Jaravel (2021, "Inflation Inequality", *Annual Review of Economics* 13) documents that lower-income households often experienced higher inflation in their consumption baskets in the US over long periods, though the pattern varies by episode [E].
- Stantcheva (2024): lower-income households report more costly adjustments [V].

**Not necessarily (con):**

- Doepke and Schneider (2006, "Inflation and the Redistribution of Nominal Wealth", *JPE* 114(6)) find that unanticipated inflation redistributes from *rich, old* households (bond holders) to *young, middle-class* households with fixed-rate mortgages [E]. The biggest losers are nominal creditors, not the poorest.
- In 2021-23 US wages at the bottom rose faster than at the top. Autor, Dube and McGrew (2023, NBER WP 31010, "The Unexpected Compression") report a compression that undid a substantial part of the rise in the 90/10 wage gap since 1980 [E]. Cross-reference R3.
- Indexed transfers (social security COLA in the US) protect many low-income retirees.

**Assessment.** *Empirical:* the answer depends on the inflation regime and institutions. High and volatile inflation is regressive (cash dependence, weak indexation, limited hedging). Moderate unanticipated inflation is mixed. It erodes nominal debts (helping indebted households) while hurting holders of cash and fixed incomes. In advanced economies with indexation and tight labour markets, the poorest may not be the biggest losers. *Perceived* harm is concentrated among lower-income groups, which matters for legitimacy irrespective of measured incidence.

### 7.4.5 Do existing monetary arrangements reinforce wealth concentration?

**Yes (pro):**

- Asset price channel: low rates and QE raise equity, bond and house prices, which are held disproportionately by the wealthy. Wealth inequality measured in market values rises (cross-reference R3 on the Cantillon effect).
- Access to credit is wealth-dependent: the wealthy borrow cheaply against assets, the poor borrow expensively (credit cards, payday).
- The deposit franchise extracts more from inattentive small savers (7.4.2).

**No / mixed (con):**

- Monetary easing raises employment, which disproportionately benefits lower-income households (Lenza and Slacalek 2018, ECB WP 2190, later published: euro area QE *reduced* income inequality through employment, with small effects on wealth inequality [E]; Coibion, Gorodnichenko, Kueng and Silvia 2017, *JME* 88: contractionary monetary shocks *raise* inequality in the US [E]).
- Structural drivers of wealth concentration (technology, globalisation, tax policy, housing supply constraints, inheritance) dominate monetary effects in most decompositions (interpretation; cross-reference R3).

**Assessment.** The honest answer is that monetary arrangements *can* reinforce wealth concentration through asset prices and unequal credit access. Their effect on *income* inequality is often equalising in the short run, and they are second-order relative to structural causes. The BRIEF's caution against presenting Cantillon effects as a universal explanation is warranted.

### 7.4.6 Can monetary stability coexist with unequal access to credit?

Yes, empirically: many economies with low inflation have deeply unequal credit access. Stiglitz and Weiss (1981, "Credit Rationing in Markets with Imperfect Information", *AER* 71(3)) show that credit rationing can be an equilibrium outcome, because lenders do not raise rates to clear the market for fear of adverse selection [E]. Collateral requirements systematically favour asset holders. Policy responses include credit guarantee schemes, community reinvestment rules (US CRA 1977), development banks, financial inclusion mandates (cross-reference Part XII on Indonesia's KUR programme, not verified here) and credit information systems that let thin-file borrowers build histories.

*Normative question:* is unequal access a problem of *monetary* fairness or of *credit market* fairness? Most theories treat it as the latter. Republican (non-domination) and capability-based theories treat access to the means of economic participation as part of basic fairness (Pettit 1997; Sen 1999, [E]).

### 7.4.7 What constitutes a fair monetary system?

There is no single answer. This stream proposes that a "fair" system can be evaluated against criteria derived from competing normative frameworks:

| Framework | Fairness criterion for money | Implications |
|---|---|---|
| Libertarian / Austrian (Hayek; Nozick) | No coercive redistribution through money; sound money; freedom of currency choice | Rules or competition; no inflation tax; legal tender minimal |
| Utilitarian / welfarist | Maximise aggregate welfare: stable prices, full employment, efficient credit | Flexible inflation targeting; backstops priced for efficiency |
| Rawlsian (difference principle) | Arrangements justifiable to the least advantaged | Protect cash and small savers; inclusive access; avoid regressive inflation |
| Republican / non-domination (Pettit) | No arbitrary power over people's livelihoods; contestability | Accountability of central banks and banks; due process in account freezes; public safe-money option |

| Framework | Fairness criterion for money | Implications |
|---|---|---|
| Desert / contribution | Rewards follow productive contribution | Limit rents from market power and implicit subsidies |
| Sufficientarian / capability (Sen) | Everyone has access to basic payment and savings tools | Basic accounts; cash access; low-cost payments |

A **minimal overlapping consensus** (interpretation) that most frameworks could accept:

1. Predictable purchasing power within a publicly announced band.
2. Universal access to a safe, low-cost means of payment and store of small savings.
3. Public backstops priced, not given away.
4. Transparency and accountability of those with power to create and allocate money.
5. Due process and privacy protections in the monetary system (relevant to CBDC).
6. No hidden fiscal transfers through monetary operations without democratic authorisation.

Disagreement remains over (i) the acceptable level and trend of inflation, (ii) whether private banks should create the bulk of money, (iii) how far central banks should pursue distributional and other goals, and (iv) whether and when debt should be forgiven.

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# PART VIII - WHY SOCIETY MAINTAINS A SYSTEM IT SOMETIMES CRITICISES

## 8.1 The puzzle restated

The BRIEF lists seven behaviours that coexist: people depend on money, distrust banks, criticise inflation, disapprove of financial inequality, use credit extensively, participate in financial markets, and continue to accept the national currency. Survey data confirm the coexistence. In the US, confidence in banks is 28% [V], 80% believe prices outrun wages [V], and household debt is $18.8 trillion [V]. There is no flight from the dollar.

The question is whether continued participation shows *genuine legitimacy*, *practical necessity*, or both. The answer requires a model that separates the reasons for participation. Behaviour alone cannot tell them apart.

## 8.2 Four modes of acceptance

| Mode | Definition | Psychological state | Observable signature | Typical evidence |
|---|---|---|---|---|
| **V - Voluntary acceptance** | Use chosen because the user endorses the currency or institution and would choose it among available alternatives at low switching cost | Endorsement; trust; identification | Use persists when alternatives are cheap and legal; positive survey support; votes for the regime | Euro support in Eurobarometer; referendum outcomes; holding domestic currency when foreign currency deposits are freely available |
| **N - Practical necessity** | Use required because counterparties, prices and contracts are in the currency (network effects, lack of alternatives) | Resignation; pragmatism | Use persists despite stated dissatisfaction; exit only by those with access to alternatives | Dollar use despite low bank confidence; partial dollarization only among wealthier households |

| Mode | Definition | Psychological state | Observable signature | Typical evidence |
|---|---|---|---|---|
| **D - Institutional dependence** | Use structurally embedded in wage, pension, mortgage, tax-collection and payment infrastructures; leaving would mean leaving those institutions | Habit; inertia; status quo | Use correlates with integration in formal institutions; exit requires exit from employment or housing systems | Payroll and benefits paid only by bank transfer; mortgage in national currency |
| **L - Legal obligation** | Use compelled by law: legal tender, taxes payable in the currency, restrictions on foreign currency, capital controls | Compliance; possibly resentment | Use persists where enforcement is strong; parallel markets appear where it is weak | Taxes in national currency; foreign currency deposit bans; capital controls (e.g., Argentina "cepo", Lebanon informal controls [E]) |

**Analytical distinction.** V is a judgement of *legitimacy*. N, D and L are *constraints*. A system can score high on constraints and low on legitimacy (captive) or the reverse (endorsed but contestable). Continued use is the *sum* of these forces. Inferring legitimacy from use is a fallacy of composition (normative-methodological claim).

## 8.3 The ten interacting factors

Each factor from the BRIEF is mapped to the modes it mainly drives, with the main interactions.

**Table 8.1 - Factor-to-mode loading (strong = ++, moderate = +, weak or none = 0)**

| # | Factor | V | N | D | L | Main interaction |
|---|---|---|---|---|---|---|
| 1 | Network effects | + | ++ | + | 0 | Amplifies every other factor; generates tipping |
| 2 | Switching costs | 0 | ++ | ++ | + | Raised by 3, 7 and by behavioural biases (status quo, loss aversion) |
| 3 | Institutional dependence | 0 | + | ++ | + | Created by 6 and 7; reinforced by 5 |
| 4 | Legal obligations (legal tender, contract law, FX restrictions) | 0 | + | + | ++ | Strength depends on enforcement capacity and 9 |

| # | Factor | V | N | D | L | Main interaction |
|---|---|---|---|---|---|---|
| 5 | Tax systems | 0 | + | + | ++ | Creates baseline demand; weakened by Olivera-Tanzi under high inflation |
| 6 | Wage and employment structures | 0 | ++ | ++ | + | Wages in national currency bind workers; indexation weakens the bind |
| 7 | Financial infrastructure (payment rails, accounts) | + | ++ | ++ | + | Convenience raises V slightly; mostly lock-in |
| 8 | Collective expectations | ++ | ++ | 0 | 0 | Self-fulfilling; can flip (multiple equilibria) |
| 9 | Political legitimacy | ++ | 0 | 0 | + | Raises V; also makes L enforceable without resentment |
| 10 | Lack of practical alternatives | 0 | ++ | + | 0 | Falls when a credible foreign currency or digital alternative becomes accessible |

**Key interactions (interpretation):**

- **Network effects x collective expectations (1 x 8)** produce multiple equilibria and tipping. This is the core of fragility.
- **Tax x legal x wage structures (5 x 4 x 6)** form the *state anchor*. They keep baseline demand positive even when V falls, which explains why exit is partial.
- **Infrastructure x dependence (7 x 3)** create *path dependence*. Every additional service routed through bank accounts raises switching costs and lowers the salience of alternatives.
- **Political legitimacy x legal obligation (9 x 4)**: when legitimacy is high, legal tender is rarely felt as coercion. When legitimacy collapses, the same law is experienced as confiscation, and evasion (parallel markets) rises.
- **Lack of alternatives (10)** is the *gate*. When a credible alternative (the US dollar in Latin America; a regional currency) is cheaply available, constraints N, D and L erode quickly under stress. Digital technology (dollar stablecoins, foreign-currency fintech accounts) may be lowering this gate globally (hypothesis; cross-reference R5).

## 8.4 A simple formal model

Let household $i$ choose to keep using domestic currency $d$ or switch (partially, for a function $f$) to alternative $a$. Define:

U_i(d) = N(s_d) + L_i + D_i - pi_e,i * w_i,f - r_i,d

U_i(a) = N(s_a) - pi_a * w_i,f - r_i,a - c_i

where:

- N(s) is network value, increasing in the share $s$ of counterparties using that unit for function $f$;
- L_i is the value of legal and tax compliance (taxes payable only in $d$; penalties for using $a$);
- D_i is institutional dependence (wage, pension and mortgage denominations; bank account integration);
- pi_e,i is the household's *expected* loss of purchasing power on $d$, which is shaped by trust and behavioural biases (6.3);
- w_i,f is the household's exposure (wealth or flows) in function $f$;
- r is perceived risk (confiscation, bank failure, volatility);
- c_i is the individual switching cost (fees, access, knowledge, loss-aversion premium).

Household $i$ switches function $f$ when U_i(a) > U_i(d). Voluntary acceptance V enters through pi_e and r: higher legitimacy lowers expected inflation and perceived confiscation risk (the ECB found that no-trust respondents expect higher inflation [V]). Because N depends on the share $s$, which depends on others' switching, the aggregate is a **threshold (Granovetter 1978) coordination game** with possible tipping [E for Granovetter, "Threshold Models of Collective Behavior", *AJS* 83(6)].

**Implications of the model:**

1. *Function ordering.* For the store of value, w is large and N(s) matters little (savings need no counterparty until spent), so this function switches first. For the means of payment, N(s) dominates, so it switches last. This matches the stylised facts (6.6.5).
2. *Heterogeneity.* Wealthier and more literate households have lower c_i and higher w_i, so they exit first. Partial dollarization is regressive in access: the poor are left holding the depreciating currency. This is an inequality *produced by* the coexistence of a weak currency and accessible alternatives (interpretation consistent with Latin American evidence [E]).
3. *Hysteresis.* After tipping, N(s_a) is high, so returning requires a large fall in pi_a - pi_e or a large legal push (de-dollarization policies with mixed success).
4. *Legitimacy versus lock-in.* Observed use of $d$ is consistent with high V and low constraints, or with low V and high constraints. Only variation in alternatives, costs or law can separate them (8.6).

## 8.5 A typology: legitimacy versus lock-in

| | **High lock-in (N + D + L strong)** | **Low lock-in (alternatives cheap and legal)** |
|---|---|---|
| **High legitimacy (V high)** | **Endorsed system.** Stable; criticism through voice; reforms incremental. Examples (interpretation): Switzerland; euro area core; US in normal times. | **Competitive system.** Use reflects genuine choice; issuer disciplined by possible exit. Examples: small open economies with free FX deposits and stable money, e.g., Singapore (interpretation); historical free-banking episodes. |

| | **High lock-in (N + D + L strong)** | **Low lock-in (alternatives cheap and legal)** |
|---|---|---|
| **Low legitimacy (V low)** | **Captive system.** Continued use despite distrust; dissatisfaction builds; vulnerable to sudden tipping if lock-in weakens. Examples: high-inflation economies with capital controls (Argentina in several periods; Lebanon 2019-; Venezuela before de facto dollarization) [E]. | **Collapsing / exiting system.** Rapid currency substitution; parallel markets; possible official dollarization. Examples: Zimbabwe 2008-09; Ecuador 1999-2000 [E]. |

**Movement between cells** is the analytically interesting part:

- *Endorsed to captive*: a legitimacy shock (inflation surge, bail-out scandal, confiscation) without a fall in lock-in. Dissatisfaction rises, use stays. This describes parts of the advanced-economy experience after 2008 and 2021-23, in milder form.
- *Captive to collapsing*: lock-in fails (alternatives become accessible, enforcement weakens, wages begin to be paid in foreign currency). Tipping follows.
- *Collapsing to endorsed*: requires a credible regime change (Sargent 1982), often with a new unit (Germany's Rentenmark 1923, [E]), and hysteresis means V recovers slowly.

## 8.6 Distinguishing legitimacy from necessity: testable implications

Each implication states an observable prediction that would differ between "use reflects legitimacy" and "use reflects necessity or lock-in". Data sources are proposed. These are hypotheses for the monograph's empirical chapters, not established results.

| # | Hypothesis | Prediction if legitimacy (V) drives use | Prediction if lock-in (N/D/L) drives use | Test and data |
|---|---|---|---|---|
| H1 | Response to lifting FX restrictions | Little change in domestic currency holdings | Sharp rise in foreign currency deposits | Event studies of capital account or deposit liberalisations; IMF financial dollarization data |
| H2 | Function ordering | No systematic ordering | Store-of-value exit precedes unit-of-account exit, which precedes means-of-payment exit | Price-quotation surveys; deposit and loan currency composition in high-inflation episodes |
| H3 | Heterogeneity by access | Exit rates similar across income groups | Exit concentrated among wealthy, literate, urban households | Household surveys (e.g., Findex; national household finance surveys) |
| H4 | Trust and expectations | Trust shifts inflation expectations and holdings | Trust shifts expectations but not holdings (until a threshold) | ECB CES panel linking trust, expectations and portfolio choices; NY Fed SCE |

| # | Hypothesis | Prediction if legitimacy (V) drives use | Prediction if lock-in (N/D/L) drives use | Test and data |
|---|---|---|---|---|
| H5 | Survey-behaviour gap | Small gap between stated distrust and behaviour | Large gap: low trust, unchanged use | Gallup / Edelman versus deposit and payment data; the US 2026 combination (28% bank confidence, stable deposits) fits lock-in |
| H6 | Tax-denomination change | Little effect on currency demand | Allowing taxes in foreign currency (or indexation) accelerates exit | Natural experiments: tax indexation episodes; dollarized tax regimes |
| H7 | Wage indexation | Indexation irrelevant to holdings | Indexation or dollar wages reduce dependence and speed exit | Labour contract data in high-inflation economies |
| H8 | Availability of digital alternatives | Stablecoin and fintech FX uptake low where trust is high regardless of cost | Uptake high where inflation is high and access is easy, regardless of stated trust | Chainalysis-type adoption indices (with care); app download data; central bank surveys |
| H9 | Referendum and election behaviour | Voters endorse existing monetary regime when offered an alternative | Voters reject alternatives mainly through status quo bias and uncertainty | Swiss Vollgeld 2018 (24.28% Yes [V]); euro referendums; post-vote surveys (VOTO) |
| H10 | Recovery after crisis | Trust and use recover together | Use recovers before trust; or trust recovers but dollarization persists (hysteresis) | Gallup World Poll trust series (63% in 2025 [V]) versus deposit dollarization series |

**Falsification logic.** If lifting restrictions, offering cheap alternatives and indexing wages all leave holdings unchanged, legitimacy is the best explanation. If any of these triggers sizeable exit, lock-in is doing part of the work. Most real systems will show *both*. The model is useful because it predicts the *shares* and *conditions* rather than a binary answer.

**Preliminary reading of the evidence (interpretation, with confidence levels):**

- In large advanced economies (US, euro area), use is supported by moderate V plus strong N, D and L. Evidence: low institutional trust (Gallup 28%; ECB 43/42) [V] alongside high support for the euro as currency (83% in the euro area reported for spring 2025 [E]). This suggests that trust in the *unit* can be high while trust in the *institutions* is low. The unit enjoys legitimacy partly by *not* being identified with the institutions that run it. *Moderate confidence.*
- In high-inflation economies with accessible dollars, lock-in is the main support, and V is low. Exit is staged and regressive. *Moderate-to-high confidence* (strong case literature, limited cross-country micro data).
- In collapse cases, legal obligation fails first in practice (enforcement collapses), then the tax anchor weakens, then network effects tip. *Moderate confidence.*

## 8.7 Answer to the BRIEF's question: legitimacy, necessity, or both?

**Both, in proportions that vary by country and time, with legitimacy concentrated on the unit and necessity concentrated on the institutions** (interpretation).

1. People accept the *unit* (dollar, euro, rupiah) with a fair degree of genuine endorsement when it has been stable. That endorsement is part identity, part experience, part network value.
2. People accept *banks and credit* mostly through practical necessity and institutional dependence. Their trust is conditional and low, and their continued use reflects lock-in and deposit insurance (trust transferred to the state guarantor).
3. People accept *central bank authority* largely through output legitimacy (performance). Their trust is correlated with trust in institutions in general, and is vulnerable to inflation overshoots and visible distributional choices.
4. *Legal obligation* (taxes, legal tender) works in the background. It is rarely felt in stable systems and becomes decisive, as either anchor or source of resentment, in crises.

This explains the coexistence of criticism and participation: criticism is aimed at institutions (banks, central banks, governments, inequality), while participation is anchored in the unit and in structural lock-in. Dissatisfaction is channelled into *voice* (elections, policy debate, referendums) rather than *exit*, unless the unit itself fails.

**A normative caution.** The fact that people keep using the system is *not* evidence that it is fair. Equally, criticism is not evidence that it is illegitimate in the sense of lacking justifiable authority. Legitimacy must be assessed against the criteria in 7.4.7, using evidence on outcomes, procedures and accountability, not inferred from continued use.

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# Disputes and uncertainties

1. **Measurement of trust.** Trust surveys differ in wording, mode and sample. Gallup ("confidence"), Edelman ("trust to do what is right", online) and Eurobarometer ("tend to trust", binary) cannot be compared in levels. This stream reports each within its own series. Trust in banks, central banks and governments is highly correlated (ECB: above 72% correlations [V]), which makes it hard to isolate "monetary" trust.
2. **Euro support figure.** The 83% euro area support for the euro (spring 2025) comes from a search excerpt of a Eurobarometer dataset description [E] and should be verified against the Standard Eurobarometer 103 report before publication. Eurobarometer ECB-trust readings after autumn 2023 were not retrieved.
3. **Swiss result precision.** Swissvotes gives 24.28% Yes and 34.55% turnout [V]. BFS excerpts give 24.3% and 34.6% (or 34.5% in provisional maps) [E]. Use the Swissvotes or Federal Chancellery definitive figures; the differences are rounding and provisional-versus-final.
4. **Household inflation expectations.** Michigan methodology changed in 2024 (move to web-based interviews), which may have raised measured expectations and partisan dispersion [E]. The May 2025 spike (6.6%) should not be read as a pure inflation forecast. NY Fed SCE September 2026 figures (3.9% / 3.3% / 3.0%) come from search excerpts corroborated by several reports [E]; the release page itself was not opened.
5. **Thresholds of exit.** The "tolerance band" (roughly 20-40% annual inflation sustained for years before store-of-value exit becomes widespread) is an interpretive synthesis from case literature, not an estimated parameter. Turkey and Argentina show substantial dollarization of savings at lower inflation rates when trust is low, so the band depends on trust and access.
6. **Dollarization data.** A cross-country deposit dollarization panel was not assembled. Claims about the ratchet effect rest on literature [E].
7. **Hyperinflation count.** The Hanke-Krus figure of 56 episodes (2013) and later additions are [E]. Definitions (Cagan 50% a month) matter, and data for some episodes are reconstructed.
8. **Indonesia 1997-98 exchange rates.** Figures are [E]. Verify against Bank Indonesia SEKI in the Part XII stream.
9. **Deposit franchise example.** The 5.0 / 1.5 / 2.0 percentages are illustrative, not estimates for any bank. DSS (2017, 2021) provide empirical deposit betas and costs. Their exact operating cost estimates were not verified here.
10. **Philosophers.** The positions summarised are mainstream readings. Each is contested within its own

scholarship. Examples: whether Aristotle's *nomisma* passage implies a chartalist theory; whether Smith supported usury ceilings for the reasons given; whether Marx's theory of money requires a commodity base; and Graeber's historical claims, which some economic historians criticise for selective use of evidence.

11. **Causality in central bank independence.** The Alesina-Summers correlation is not robust in all samples, and legal independence is not actual independence. The legitimacy literature (Tucker) is normative.
12. **Inflation incidence.** Distributional incidence depends on the episode and on indexation. 2021-23 US evidence (wage compression) differs from developing-country high-inflation evidence. Cross-check with R3.
13. **Model status.** The V/N/D/L model and the 2x2 typology are this stream's analytical constructions. They organise evidence and generate hypotheses. They have not been estimated.
14. **Discrepancy noted.** The BRIEF mentions Weber et al. on inflation expectations. Weber, D'Acunto, Gorodnichenko and Coibion (2022, *JEP*) is cited from established knowledge [E]; its exact page range should be verified.

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# R5. The International Monetary Order: Hierarchy, Plumbing, Power, and Its Sustainability

Research stream R5 for *The Architecture of Money* (BRIEF Part IX; feeds Parts III.5, VII.A, XIII and XIV). Authors: M. Bagus Prabowo and Hark (he conceived and directs the project; Hark's research agents gathered sources and drafted this stream). Research cut-off: 7 November 2026. Evidence tags follow FRAMEWORK.md: [V] = page or dataset opened and the claim is on it; [E] = search excerpt, secondary report, or established knowledge not re-opened for this stream. Claim types are labelled where it matters: (fact), (empirical finding), (interpretation), (normative), (counterfactual).

Data files written by this stream (all under `money/data/`):

- `r5_international_order_indicators.csv` - every headline share and stock used below, with source URL, tag and limitation.
- `r5_bis_gli_usd_credit_history.csv` - BIS global liquidity indicator, US dollar credit to non-banks outside the United States and in emerging market economies, quarterly 2000 to Q2 2026.
- `r5_foreign_share_treasuries.csv` - foreign holdings of US Treasury securities divided by marketable Treasuries outstanding, year-ends 2010-2025 and August 2026.
- `r5/` - raw pulls (FRED, Treasury Fiscal Data, TIC Table 5, BIS SDMX).

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## Key findings

1. **The dollar is still the system's centre, but its dominance is uneven across functions (fact).** In mid-2026 the US dollar was 56.7% of allocated official foreign exchange reserves (IMF COFER, Q2 2026) [V], on one side of 89.2% of foreign exchange trades (BIS Triennial Survey, April 2025) [V], about 46% of cross-border bank claims and of outstanding international debt securities (BIS, Q2 2026, own calculation from the BIS data API) [V], 49.6% of SWIFT payment value including intra-euro traffic and 58.8% excluding it (September 2026) [V], and the denomination of USD 15.1 trillion of credit to non-bank borrowers outside the United States (BIS global liquidity indicators, Q2 2026) [V]. The United States produces roughly a quarter of world output at market exchange rates [E]. Dominance is therefore strongest where network effects are strongest (foreign exchange trading, funding, invoicing) and weakest where portfolio choice is most discretionary (official reserves).
2. **The reserve share is eroding slowly, and the erosion is concentrated (empirical finding).** The dollar's reserve share fell from a peak of about 72% (2001) to about 64% (2015) and to about 57% (2025) [V, Fed and New York Fed]. New York Fed research (Goldberg, Hannaoui and Parthasarathy, September 2026) [V] finds the post-2019 decline is driven almost entirely by a handful of large holders (China, Russia and a few others), not a broad cross-country flight. In Q2 2026 the euro's share rose to 20.6%, its highest since 2014 [V/E], mostly through active purchases; the renminbi's share was 2.1% [V].
3. **The plumbing matters more than the headline shares (interpretation).** The dollar's position

rests on deep Treasury and repo markets (USD 32.15 trillion of marketable Treasuries at 31 October 2026) [V], on US-centred clearing and settlement, and on a crisis backstop that only the Federal Reserve can provide: swap lines (peak usage about USD 585 billion in 2008-09 and about USD 450 billion in 2020) [V] and the FIMA repo facility (standing since July 2021, capped at USD 60 billion per counterparty [E]). In July-August 2026 Japan announced it would fund yen-supporting intervention through FIMA repo rather than by selling Treasuries [V]. The safety net is now openly part of geopolitics, which is both a source of strength and a new risk.

4. **Global liquidity is a dollar phenomenon with growing euro participation (fact).** Dollar credit to non-banks outside the United States rose from USD 2.5 trillion (2000) to USD 15.1 trillion (Q2 2026); USD 4.6 trillion of this is owed by borrowers in emerging market economies [V]. Euro-denominated foreign-currency credit has been growing faster than dollar credit since 2024 (EUR 5.39 trillion, +13.8% year on year) [V]. Emerging-market dollar debt transmits US monetary conditions abroad (the "global financial cycle"), and dollar appreciation tightens financial conditions for borrowers with currency mismatches.

5. **The "new Triffin dilemma" is the right frame, with an important correction (interpretation).** The old Triffin problem (1960) was about gold convertibility. The new one (Farhi, Gourinchas and Rey 2011) is about the fiscal capacity of a single sovereign to supply safe assets to a growing world. Foreign holders own about 29% of marketable Treasuries (August 2026), down from about 49% in 2014 [V, computed], so the US now relies more on domestic absorption. The US net international investment position was minus USD 22.4 trillion in Q2 2026, roughly 69% of GDP [V, FRED]. The binding constraint is less the arithmetic of foreign demand than the credibility of US fiscal and monetary institutions.

6. **Dominance is self-reinforcing through invoicing (empirical finding).** The dominant currency paradigm (Gopinath and co-authors) shows that dollar invoicing makes the dollar exchange rate, not bilateral rates, the main driver of trade prices and volumes for many countries. This makes the dollar's role persistent even when the US share of world trade falls, and it raises the cost to any country of exiting dollar use unilaterally.

7. **Weaponization is real, but it has not produced mass exit (fact and finding).** About EUR 210 billion of Bank of Russia assets remain immobilised in the EU, most at Euroclear; in December 2025 the EU made the immobilisation indefinite but did not adopt the proposed "reparations loan", financing Ukraine instead through a EUR 90 billion jointly borrowed loan [E, multiple sources]. Central-bank gold buying (1,082 t in 2022, about 1,050 t in 2023, 1,092 t in 2024, 863 t in 2025, 675 t in the first three quarters of 2026) [E/V, World Gold Council] is the clearest behavioural response; reallocation from dollars to other fiat currencies is small. Sanctions mainly raise demand for "sanction-proof" assets (gold, domestic infrastructure) among a minority of reserve holders.

8. **Alternatives are infrastructural, not currency replacements (fact).** BRICS leaders at New Delhi (12-13 September 2026) again only "acknowledged" work on payment interoperability and local-currency settlement; India stated there is "no proposal" for a BRICS currency [E]. The BIS left Project mBridge on 31 October 2024 (verified on the BIS project page: "handing the project over to the partners") [V]; Saudi Arabia disclosed in September 2026 that it had left after its proof of concept in May 2025 [E]. China's CIPS processed RMB 180 trillion in 2025, essentially flat on 2024 [E]. The renminbi remains capped by capital controls and limited legal predictability for foreign investors.

9. **2025-2026 was a stress test, not a regime change (interpretation).** After the 2 April 2025 tariff announcement, the dollar fell while long Treasury yields rose, an unusual "sell America" pattern [E: ECB, TBAC]. The Fed's broad dollar index fell about 7.4% over 2025 and has partly recovered in 2026 (121.7 at 30 October 2026, versus 129.3 at end-2024) [V]. The US-Israel war on Iran from 28 February 2026 and the disruption of the Strait of Hormuz produced only a modest dollar rise and losses on Treasuries, rather than a classic flight to US safety [E]. Kevin Warsh became Fed chair on 22 May 2026 amid open debate about Fed independence [E, Federal Reserve press release read via excerpt]. None of these events reduced dollar shares in transactions; they did revive diversification at the margin (euro, gold) and questions about the reliability of the safety net.

10. **Most likely trajectory: a slowly thinning dollar hegemony inside a more fragmented, more gold-heavy, more euro-inclusive system (counterfactual, analytical not predictive).** A sudden replacement is implausible on current evidence because no candidate offers the combination of depth,

convertibility, legal predictability and crisis backstop. The genuine risks to the dollar order are endogenous: fiscal dominance, politicisation of the Fed and of the safety net, and erosion of the rule-of-law premium. The genuine alternatives are a multipolar reserve system (euro, gold, smaller currencies), regional payment bridges, and private digital dollars that could entrench rather than replace dollar use.

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# 1. Scope, method and evidence conventions

## 1.1 What this stream covers

BRIEF Part IX asks for "the global hierarchy of money": the dollar, reserve currencies, the euro, the renminbi, settlement systems, foreign exchange markets, Treasury markets, capital flows, dollar debt, swap arrangements, global liquidity, the Triffin dilemma, why currencies become dominant, and whether the order is sustainable. The commissioning note for this stream adds: dominant currency pricing, the "new Triffin" argument, Eichengreen-Mehl-Chitu on multipolar history, sanctions, BRICS payment initiatives, mBridge, central-bank gold buying, and developments in 2025-2026.

## 1.2 Sources and their limits

The quantitative backbone comes from official statistical sources opened or queried for this stream: IMF COFER data briefs; BIS Triennial Survey 2025; BIS locational banking statistics, debt securities statistics and global liquidity indicators (queried through the BIS SDMX API on 7 November 2026); US Treasury International Capital (TIC) Table 5; Treasury Monthly Statement of the Public Debt (Fiscal Data API); FRED series for the broad dollar index and the US international investment position; World Gold Council Gold Demand Trends; Federal Reserve and New York Fed research notes; the BIS mBridge project page; the Japanese Ministry of Finance intervention statement. The SWIFT Global Currency Tracker (October 2026 edition) was also opened. CIPS totals, BRICS outcomes, Euroclear figures, and war-related market moves are tagged [E] because they were read from search excerpts or secondary reports rather than the primary document.

Every share statistic in this field has a definitional trap. Five recur:

- **Foreign exchange turnover shares sum to 200%**, because each trade involves two currencies.
- **SWIFT shares** cover messages on SWIFT only (customer and institutional payments), include intra-euro-area traffic unless stated otherwise, and exclude payments settled inside CIPS or inside a single bank's books. They measure message value, not economic importance.
- **COFER shares** are shares of *allocated* foreign-exchange reserves; they exclude gold and SDRs, and shares move mechanically with exchange rates and bond prices.
- **BIS "international debt securities"** are securities issued outside the issuer's local market; euro-area issuers selling euro bonds in London count, which inflates the euro's share relative to a "foreign currency debt" concept.
- **BIS cross-border bank claims** include intra-euro-area claims, which also inflate the euro.

For that reason this stream reports more than one measure for each function and states which one is used.

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# 2. A conceptual map: what "international money" means

## 2.1 Functions, by private and official users

The standard taxonomy (Cohen 1971; Krugman 1984; Kenen 1983) [E] crosses the three functions of money with two classes of user:

| Function | Private use | Official use |
|---|---|---|
| Medium of exchange | Vehicle currency in FX trading; settlement of trade and financial payments | Intervention currency |
| Unit of account | Invoicing of trade; denomination of debt; pricing of commodities | Anchor or peg for exchange-rate regimes |
| Store of value | Currency of deposits, bonds, cash held abroad | Reserve currency |

This grid matters for the monograph's broader argument (Parts II and VI) because the same three functions that ground domestic money reappear internationally, but without a sovereign to enforce legal tender or collect taxes. International money is therefore a pure case of acceptance through network effects, institutional trust and the availability of a backstop, which makes it a natural test of the credit, state and conventionalist theories of money discussed in R1.

## 2.2 The hierarchy of money, extended across borders

Perry Mehrling's "hierarchy of money" [E] ranks claims by their position in the settlement chain: gold or its modern equivalent at the top, then central-bank reserves, then bank deposits, then securities. Internationally the hierarchy has an extra layer. A Brazilian bank's dollar deposit at a London bank ultimately settles in reserves at the Federal Reserve held by a US correspondent. Offshore dollars ("eurodollars") are private promises to pay US-based dollars; their quality depends on access, directly or through a chain of correspondents, to Fed money. The Fed's swap lines and FIMA repo facility are the instruments through which the top of the US hierarchy is extended to selected foreign central banks in emergencies (Section 7).

A simple illustration of how offshore dollar money is created (T-accounts; numbers illustrative, in USD million):

*Step 1. A London bank lends USD 100 to an Indonesian exporter, crediting a dollar deposit (credit creation outside the US, R2 mechanism).*

| London bank: Assets | | London bank: Liabilities | |
|---|---|---|---|
| Loan to Indonesian firm | +100 | Dollar deposit of Indonesian firm | +100 |

*Step 2. The firm pays a US supplier. The London bank must deliver Fed money through its US correspondent.*

| London bank: Assets | | London bank: Liabilities | |
|---|---|---|---|
| Loan to Indonesian firm | 100 | Due to US correspondent (or reduction of its balance there) | 100 |

| US correspondent: Assets | | US correspondent: Liabilities | |
|---|---|---|---|
| Reserves at Fed | -100 | Balance owed to London bank | -100 |

The London bank can create a dollar *deposit* without Fed permission, but it cannot create the *settlement asset*; it must hold, borrow or swap for dollars at the top of the hierarchy. When dollar funding markets seize (2008, March 2020), that dependence becomes a run on offshore dollar intermediaries. This is the international analogue of the domestic point made in R2: banks create money, but they are limited by their need for settlement balances, capital and funding.

# 3. The US dollar: a dashboard of its international role

## 3.1 Headline indicators

| Function / market | Measure | Dollar | Euro | Renminbi | Period | Source | Tag |
|---|---|---|---|---|---|---|---|
| Official store of value | Share of allocated FX reserves | 56.70% | 20.60% | 2.11% | 2026 Q2 | IMF COFER data brief, 30 Sep 2026 | [V] |
| Vehicle currency | Share of FX turnover (of 200%) | 89.2% | 28.9% | 8.5% | April 2025 | BIS Triennial Survey 2025 | [V] |
| Payments | SWIFT share of value, incl. intra-euro | 49.63% | 21.69% | 3.33% (5th) | Sept 2026 | SWIFT Global Currency Tracker (Oct 2026 edition) | [V] |
| Payments | SWIFT share, excl. intra-euro | 58.83% | 13.58% | 2.43% (6th) | Sept 2026 | same | [V] |
| Trade finance | SWIFT trade-finance share (MT400/MT700) | 82.05% | 4.48% | 8.26% (2nd) | Sept 2026 | SWIFT Global Currency Tracker (Oct 2026 edition) | [V] |
| Banking | Cross-border bank claims, all instruments | 46.3% | 31.7% | n.a. | 2026 Q2 | BIS LBS via SDMX API (own calc.) | [V] |
| Banking | International and foreign-currency banking claims, loans and deposits, ex intra-euro | about 55% | about 20% | small | 2024 | Fed FEDS Note 2025 | [V] |
| Bonds | Outstanding international debt securities | 46.4% | 39.4% | 1.0% | 2026 Q2 | BIS debt securities via SDMX (own calc.) | [V] |
| Bonds | Foreign-currency debt issuance share | about 60% | about 26% | small | 2010-2024 | Fed FEDS Note 2025 | [V] |

| Function / market | Measure | Dollar | Euro | Renminbi | Period | Source | Tag |
|---|---|---|---|---|---|---|---|
| Global credit | Foreign-currency credit to non-banks outside the issuing area | USD 15.13 tn | EUR 5.39 tn | n.a. | 2026 Q2 | BIS GLI via SDMX | [V] |
| Trade invoicing | Share of export invoicing, Americas / Asia-Pacific / rest of world (excl. Europe) | 96% / 74% / 79% | Europe: euro 66% | n.a. | 1999-2019 avg | Fed FEDS Note 2025, citing Boz et al. 2020 | [V] |
| Cash abroad | Dollar banknotes held abroad | over USD 1 tn, about half of the total | n.a. | n.a. | 2025 Q1 | Fed FEDS Note 2025 | [V] |
| Anchor | Countries anchored to the dollar | over half of countries | n.a. | n.a. | 2019 | Ilzetzki, Reinhart, Rogoff 2021, cited by Fed | [V] |
| Digital tokens | Share of stablecoin supply in dollars | about 99% | under 1% | n.a. | Sept-Oct 2026 | CoinDesk research; Fed note 2025 | [E]/[V] |

Limitation (one line per row is in the CSV): each indicator measures a different population; none is a complete measure of "currency power". The Federal Reserve's composite index (weights: reserves 25%, FX turnover 25%, foreign-currency debt 25%, banking claims 12.5%, banking liabilities 12.5%) puts the dollar in a "narrow range between 65 to 70 since 2010", the euro at about 24, the yen about 7, sterling about 6, and the renminbi about 3 [V].

## 3.2 Official reserves (store of value)

The IMF's Q2 2026 brief reports total foreign-exchange reserves of USD 13.22 trillion, up from USD 13.10 trillion in Q1. Dollar claims were "largely unchanged", so the fall in the dollar share from 57.18% to 56.70% reflects growth in other currencies, notably a near-USD 100 billion rise in euro claims "due mostly to active purchasing" [V]. The yen share fell to 4.96% through "active selling", higher Japanese bond yields and yen depreciation, "followed by a joint U.S.-Japan intervention in Q3" [V].

Over the long run (fact): the dollar share peaked near 72% in 2001, was about 64% in 2015 and about 57-58% in 2024-2025 [V: Fed note; New York Fed]. The IMF's 2026 Annual Report gives "about three-fifths, at 57 percent in 2025" and a euro share of 20.2% in 2025 [E].

Interpretation. Three readings compete:

- *Stealth erosion* (Arslanalp, Eichengreen and Simpson-Bell, IMF WP 2022) [E]: the dollar's loss has gone mainly to "non-traditional" currencies (Australian and Canadian dollars, Korean won, Nordic currencies) rather than to the euro or renminbi, made possible by electronic trading and automated market making.
- *Concentration* (Goldberg, Hannaoui and Parthasarathy 2026) [V]: "roughly equal numbers of countries increased and decreased their dollar holdings"; for 62 countries with complete data the "preferences channel"

was slightly positive (+0.3 percentage point) over 2019-2023, and the aggregate decline is "almost entirely attributable" to China, Russia, Mexico and Morocco under assumptions consistent with COFER. Their conclusion: "there is little evidence of a widespread official diversification away from dollars."
- *Geopolitical diversification* (2022 onward): sanctions on Russia led a subset of holders to shift from all Western currencies toward gold (Section 12).

These readings are compatible. Most reserve holders still hold dollars for intervention and liquidity; a few large ones are reallocating for strategic reasons; and the tail of the portfolio is diversifying into smaller currencies.

## 3.3 Foreign exchange markets (vehicle currency)

The BIS 2025 Triennial Survey (fact) [V]: turnover averaged USD 9.6 trillion per day in April 2025, up 28% from 2022; FX swaps remained the most traded instrument but fell to 42% of turnover from 51%; the dollar was on one side of 89.2% of trades, up from 88.4%; the euro fell to 28.9% from 30.6%; sterling fell to 10.2%; the renminbi rose to 8.5% and the Swiss franc to 6.4%. Banxico's tabulation shows the dollar share was 87.6% in 2016, 88.3% in 2019 and 88.4% in 2022 [E]. Sales desks in the UK, US, Singapore and Hong Kong handled 75% of trading [V].

Limitation. The BIS itself notes that the April 2025 survey "was conducted amidst elevated FX volatility and a surge in trading activity that followed trade policy announcements", and that dollar depreciation "appears to have led many institutional investors and asset managers with dollar asset exposures to limit further FX losses ... by selling dollars forward" [V]. Part of the rise in the dollar's turnover share therefore reflects hedging *against* the dollar, a reminder that turnover measures use, not confidence. The renminbi's rise to fifth place is genuine but partly reflects onshore and Hong Kong hedging related to China's own trade.

Interpretation. The vehicle-currency role is the most network-driven function: if most pairs trade most cheaply through the dollar, a trader converting Indonesian rupiah into Brazilian real will usually go rupiah-dollar-real. This is why dollar dominance in FX has *increased* while the reserve share fell. A shift would require a liquid cross market that is cheaper than two dollar legs, which no currency other than the euro (in Europe) provides at scale.

## 3.4 Payments

SWIFT renamed its RMB Tracker the "Global Currency Tracker" in February 2026 [V]. SWIFT's own disclaimer states that the tracker "does not represent complete market or industry statistics" [V]. September 2026 data put the dollar at 49.63% of global payment value, the euro at 21.69%, sterling at 6.89%, the yen at 3.40% and the renminbi at 3.33% (fifth) [V]. Excluding intra-euro-area payments, the dollar's share was 58.83%, the euro's 13.58% and the renminbi's 2.43% (sixth, behind the Canadian dollar) [V, SWIFT Global Currency Tracker, October 2026 edition]. In trade finance (letters of credit and collections, MT 400/700), the dollar held 82.05%, the renminbi 8.26% and the euro 4.48% [V]. Monthly shares are volatile: the dollar was 52.09% in August 2026 [E].

PBOC Governor Pan Gongsheng argued on 19 October 2026 that if payments through CIPS and banks' internal systems are included, the renminbi's share of international payments rises to "6 to 7 percent" and it ranks third [E, speech reproduced on the BIS central bankers' speeches site]. This is a broader population than SWIFT's and the methodology has not been published, so it should be treated as an official claim rather than a comparable statistic.

## 3.5 International banking and debt

From the BIS locational banking statistics (Q2 2026, all reporting banks, all counterparties, cross-border positions, all instruments) [V, own calculation]: total cross-border claims USD 49.47 trillion; dollar USD 22.92 trillion (46.3%); euro USD 15.70 trillion (31.7%); yen USD 2.29 trillion; sterling USD 2.15 trillion; Swiss franc USD 0.60 trillion. The BIS commentary for Q2 2026 reports cross-border bank *credit* (loans plus debt securities) of USD 40.7 trillion, growing 11% year on year, with euro credit accounting for USD 424 billion and dollar credit for USD 383 billion of the USD 877 billion quarterly increase [V]. That is a rare quarter in which the euro contributed more than the dollar to the expansion.

From the BIS debt securities statistics (Q2 2026, international market, all issuers) [V, own calculation]: outstanding international debt securities USD 35.10 trillion; dollar USD 16.27 trillion (46.4%); euro USD 13.82 trillion (39.4%); sterling 7.5%; renminbi 1.0%; yen 0.9%. As noted, the euro figure is inflated by euro-area issuers using international

markets in their own currency; the Fed's "foreign currency debt" measure, which excludes home-currency issuance, puts the dollar near 60% and the euro near 26% [V].

## 3.6 The Treasury market: the reserve asset itself

Size (fact) [V, Treasury Fiscal Data, Monthly Statement of the Public Debt]: marketable Treasury securities outstanding were USD 32.15 trillion on 31 October 2026 (bills 7.38, notes 16.34, bonds 5.56, TIPS 2.16, floating-rate notes 0.71). Total gross federal debt including intragovernmental holdings was USD 39.46 trillion in Q2 2026 (FRED GFDEBTN) [V].

Foreign holdings (fact) [V, TIC Table 5]: USD 9.332 trillion in August 2026, of which foreign official holdings were USD 3.756 trillion. The largest holders were Japan (USD 1,068 billion), the United Kingdom (USD 1,010 billion; above USD 1 trillion for the first time), mainland China (USD 591 billion), Belgium (USD 491 billion), Canada (USD 465 billion), the Cayman Islands (USD 463 billion) and Luxembourg (USD 448 billion). Total foreign holdings peaked at USD 9.487 trillion in February 2026.

Foreign share (computed) [V]: foreign holdings divided by marketable debt fell from about 50% (2010-2014) to 43% (2016), 40% (2018), 34% (2020), 30% (2022-2025) and 29.3% (August 2026). See `r5_foreign_share_treasuries.csv`.

| Year-end | Foreign holdings (USD tn) | Marketable Treasuries (USD tn) | Foreign share |
|---|---|---|---|
| 2010 | 4.47 | 8.86 | 50.4% |
| 2014 | 6.16 | 12.52 | 49.2% |
| 2018 | 6.27 | 15.62 | 40.1% |
| 2020 | 7.07 | 20.98 | 33.7% |
| 2022 | 7.19 | 23.94 | 30.1% |
| 2025 | 9.27 | 30.27 | 30.6% |
| Aug 2026 | 9.33 | 31.83 | 29.3% |

Limitations. TIC attributes holdings by custodian location, so Belgium (Euroclear), Luxembourg and the Cayman Islands stand in for owners elsewhere, including possibly Chinese and hedge-fund holdings; TIC totals include small non-marketable holdings; the ratio mixes a market-value-based TIC estimate with face-value debt.

Interpretation. Foreigners still buy Treasuries in absolute terms, but their share has fallen because debt issuance has outrun foreign demand, the Fed absorbed debt during quantitative easing, and official holders (especially China) have reduced holdings. The marginal buyer is increasingly domestic and price-sensitive (hedge funds through basis trades, money market funds, and, from 2025, stablecoin issuers). This is directly relevant to the "new Triffin" debate (Section 9).

Market functioning matters as much as size. The Treasury Borrowing Advisory Committee told the Treasury in April 2025 that "the highly unusual simultaneous decline in valuations for US equities, Treasuries, and the US dollar drove some investors to question the 'safe haven' " status of US assets [E]. The ECB's November 2025 Financial Stability Review found the ten-year Treasury yield "surged by almost 50 basis points between 4 and 11 April, the third largest weekly increase since 1986" [E]. The Treasury market's liquidity during stress depends on dealer balance sheets, which are constrained by leverage rules, and ultimately on the Fed as buyer of last resort (March 2020).

## 3.7 The external balance sheet of the United States

FRED (BEA data) [V]: US external assets were USD 46.97 trillion and liabilities USD 69.39 trillion in Q2 2026, a net international investment position of minus USD 22.42 trillion, about 69% of annualised GDP (USD 32.56 trillion). The position was minus USD 1.3 trillion in Q4 2007 and minus USD 7.0 trillion in Q4 2014. Much of the deterioration since 2014 reflects valuation (US equities outperforming foreign equities, raising the value of foreign-owned US stocks) as well as current-account deficits.

Interpretation. Gourinchas and Rey (2007) [E] characterised the US as a "world venture capitalist": it issues safe, low-yielding liabilities (Treasuries, deposits) and holds risky, high-yielding assets (FDI, equities), earning an "exorbitant privilege" return differential in normal times and paying an "exorbitant duty" (wealth transfers to foreigners through valuation losses) in global crises. Whether the privilege persists when liabilities are increasingly equity rather than debt, and when a large share of the negative position reflects US equity valuations, is an open empirical question.

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# 4. The euro: the incomplete challenger

## 4.1 Position

The euro is the clear second international currency by every measure: about 20% of reserves (20.60% in Q2 2026) [V], 28.9% of FX turnover [V], about 22% of SWIFT payments (13.6% outside the euro area) [V], about 26% of foreign-currency debt issuance [V], and EUR 5.39 trillion of credit to non-banks outside the euro area (Q2 2026, +13.8% year on year) [V]. The ECB's June 2026 report on the international role of the euro found that its composite index "increased by 0.2 percentage points at constant exchange rates and 0.9 percentage points at current exchange rates in 2025", that the euro's share across indicators was "around 20%", and that since 2014 it has risen by about 1.5 percentage points at constant exchange rates [E, ECB press release and report excerpts]. The ECB also warned that "there is no room for complacency, as fragmentation is growing in the international monetary system" [E].

The Federal Reserve's 2025 note observed that EU jointly-backed debt had reached about USD 700 billion by May 2025, "small compared to the over $28 trillion of U.S. Treasuries outstanding" [V]. In December 2025 the EU added a EUR 90 billion joint borrowing programme for Ukraine (Section 12), which marginally enlarges the stock of supranational euro safe assets [E].

## 4.2 Why the euro has not displaced the dollar

- **No single safe asset of scale (fact).** The euro area issues national government bonds with different credit risk; the closest to a common safe asset is EU (NextGenerationEU) and ESM/EFSF debt, still well under USD 1 trillion. A reserve manager wanting USD 50 billion of AAA paper can buy it in minutes in Treasuries; the euro equivalent is dispersed across German Bunds (limited supply), French, Dutch and supranational bonds.
- **Fiscal and political incompleteness (interpretation).** The Fed's note states that "political uncertainty arising from the diverse set of countries who have adopted the euro will likely continue to be a force holding back more widespread adoption" [V]. The 2010-2012 sovereign crisis showed that euro-area government debt can carry default and redenomination risk.
- **Capital markets union unfinished (fact).** Securities settlement and insolvency regimes remain partly national.
- **Strengths (fact and interpretation).** The euro area is open to capital, has strong property rights and independent courts, and the ECB is credible. From 2025 its relative attractiveness rose because of US policy uncertainty. Euro credit growth to emerging markets and the active purchases in COFER Q2 2026 show that it is the main beneficiary of any diversification among Western-aligned holders.
- **Constraint from geopolitics (interpretation).** The euro is issued by close US allies who participated in sanctions on Russia; to a reserve holder fearing Western sanctions, the euro is not an escape. Its rise is therefore a diversification *within* the Western currency bloc.

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# 5. The renminbi: internationalisation under capital controls

## 5.1 Indicators

- Reserves: 2.11% of allocated reserves (Q2 2026) [V].

- FX turnover: 8.5% of 200% (April 2025), fifth most traded [V]. A BIS Quarterly Review article (December 2025) reports renminbi turnover at 8.8% of global turnover on a broader measure and describes it as propelling growth in emerging-market currency trading [E].
- Payments: 3.33% of SWIFT value (September 2026), 2.43% outside the euro area [V]; PBOC estimate of 6-7% including CIPS and in-house settlement [E].
- Trade finance: second to the dollar at 8.26% versus 82.05% (SWIFT, September 2026) [V].
- International bonds: 1.0% of outstanding international debt securities (Q2 2026) [V].
- China's own cross-border receipts and payments: the PBOC reports RMB 35 trillion in H1 2025, with goods-trade settlement at 28% of China's cross-border transactions in domestic and foreign currencies; the renminbi "is the largest settlement currency for China's external receipts and payments" [E, PBOC interview October 2025 and Governor Pan October 2026]. Shanghai's financial office cites RMB 70.6 trillion of cross-border RMB receipts and payments in 2025 [E].

## 5.2 Infrastructure: CIPS

China's Cross-border Interbank Payment System (CIPS), launched in 2015, is the clearing and settlement system for cross-border renminbi payments. CIPS's own historical table [E, cips.com.cn PDF]: 2015 RMB 0.48 trillion; 2019 RMB 33.9 trillion; 2022 RMB 96.7 trillion; 2023 RMB 123.1 trillion; 2024 RMB 175.5 trillion; 2025 RMB 180.2 trillion (8.44 million transactions). Participants at end-2025: 193 direct and 1,573 indirect, reaching over 5,000 banking institutions in about 190 countries and regions [E]. Growth in value slowed from 42.6% (2024) to about 3% (2025) [E, computed from the CIPS table].

Limitations. CIPS values are gross settlement values, include a large share of flows tied to Chinese trade and Hong Kong offshore activity, and a substantial share of CIPS traffic is still messaged over SWIFT (one Taiwanese press estimate puts it near 80%) [E, unverified]. CIPS therefore reduces China's dependence on US *clearing* (CHIPS) but not necessarily on SWIFT *messaging*.

## 5.3 Swap lines

At end-2025 the PBOC had bilateral local-currency swap agreements with central banks or monetary authorities of 32 jurisdictions, totalling more than RMB 4.5 trillion; foreign central banks had RMB 94.2 billion outstanding under them [E, PBOC table]. The largest lines: Hong Kong RMB 800 billion; Korea and Indonesia RMB 400 billion each; ECB and UK RMB 350 billion each; Singapore RMB 300 billion; Russia RMB 150 billion; Argentina RMB 130 billion. In September 2025 the PBOC renewed lines with the ECB (RMB 350 billion / EUR 45 billion, three years), the Swiss National Bank and Hungary's central bank [E].

Interpretation. PBOC swap lines differ from Fed swap lines in purpose and use. Fed lines supply the world's funding currency in crises and were drawn in hundreds of billions. PBOC lines mainly support trade settlement and offshore renminbi liquidity, and have also been used as quasi-balance-of-payments support (Argentina, Pakistan, Sri Lanka) [E]. Their low utilisation (RMB 94 billion drawn against RMB 4.5 trillion of capacity) shows that renminbi demand is limited by the narrow range of things one can buy with renminbi outside China.

## 5.4 The binding constraint: capital controls and legal predictability

China maintains capital-account restrictions to preserve monetary autonomy with a managed exchange rate (the open-economy trilemma) and to protect the domestic banking system. Since 2015-2016, when capital outflows drained about USD 1 trillion of reserves [E], Beijing has prioritised stability over openness. A reserve currency must be usable in size and at short notice, including in a crisis; a holder must trust that it can sell assets and repatriate proceeds. The Fed's 2025 note summarises: "the renminbi is not freely exchangeable, the Chinese capital account is not open, and investor confidence in Chinese institutions is relatively low" [V].

Interpretation. China's strategy is better described as building a *parallel* system (CIPS, swap lines, offshore centres, mBridge, e-CNY, commodity contracts settled in renminbi) that makes the renminbi usable for trade with China and sanction-resistant for counterparties who need that, rather than as an attempt to replace the dollar as the world's reserve asset. That strategy is consistent with China's own preference for control over capital flows, and it is succeeding on its own terms (China's trade settlement) while failing on dollar-replacement terms (reserves, bonds).

# 6. International settlement systems: the plumbing

| System | Currency | Function | Notes |
|---|---|---|---|
| Fedwire Funds | USD | Real-time gross settlement in central-bank money | Final settlement of dollar payments |
| CHIPS | USD | Private large-value netting system | Clears most cross-border dollar payments among large banks [E] |
| TARGET (T2) | EUR | Eurosystem RTGS | Central-bank money settlement of euro payments |
| CLS | 18 currencies [E] | Payment-versus-payment settlement of FX trades | Eliminates Herstatt risk for covered trades |
| SWIFT | All | Messaging, not settlement | Belgian cooperative; subject to EU law, which is why EU sanctions could disconnect Russian banks |
| CIPS | CNY | Cross-border RMB clearing and settlement | Own messaging standard, still often uses SWIFT |
| SPFS | RUB | Russian financial messaging | Domestic alternative to SWIFT, limited foreign use [E] |
| mBridge | Multi-CBDC | Wholesale CBDC platform | BIS exited October 2024; partners continue (Section 13) |

Two analytical points follow.

First, **messaging and settlement are different chokepoints**. Disconnection from SWIFT impedes communication; denial of access to US correspondent banks impedes settlement of dollars, which is the more powerful lever because dollar payments must ultimately settle in the US. Farrell and Newman's "weaponized interdependence" (International Security, 2019) [E] identifies these network hubs as the source of the "panopticon" (information) and "chokepoint" (denial) effects that give the US and the EU coercive power.

Second, **the hidden dollar debt in FX swaps**. Because FX swaps and forwards are off-balance-sheet, non-US banks and institutional investors carry large short-dated dollar obligations that are not recorded as debt. Borio, McCauley and McGuire (BIS Quarterly Review, September 2017) [E] estimated tens of trillions of such obligations. FX swaps were 42% of daily FX turnover in April 2025 (about USD 4.0 trillion per day) [V]. This hidden leg is why dollar funding stress hits foreign banks and insurers even when their balance sheets look matched, and why Fed swap lines are needed.

# 7. The dollar safety net: Fed swap lines and FIMA repo

## 7.1 History

(Fact) [V, Fed FEDS Note 2025 unless noted]:

- **December 2007 - 2010**: temporary swap lines first with the ECB and SNB, expanded in 2008 to 14 central banks; peak usage about USD 585 billion.
- **October 2013**: lines among the Fed, ECB, Bank of Japan, Bank of England, Swiss National Bank and Bank of Canada made standing and unlimited.

- **March 2020**: daily operations on standing lines; temporary lines reopened with nine central banks (Australia, Brazil, Korea, Mexico, Singapore, Sweden, Denmark, Norway, New Zealand); usage peaked near USD 450 billion. The temporary lines expired at end-2021 [E].
- **31 March 2020**: FIMA repo facility created, letting foreign central banks with Treasury holdings at the New York Fed borrow dollars against them instead of selling; made permanent in July 2021.
- **March-April 2023**: daily swap operations during the Credit Suisse rescue.
- **2025**: "usage of U.S. dollar central bank liquidity swap lines was modest" and FIMA usage "limited" [E, New York Fed annual report on open market operations 2025]. Fed quarterly financial reports show central bank liquidity swaps of USD 180 million at 31 March 2026 and USD 250 million at 30 June 2026, and zero FIMA repo outstanding at both dates [E].
- **31 July 2026**: Japan's Ministry of Finance bought yen "in coordination with the U.S. Department of the Treasury", the first joint intervention since 2011, and stated that "Japan also plans to utilize the Federal Reserve's ... FIMA Repo Facility in the future" [V, MOF statement 3 August 2026]. Japan spent JPY 15.4 trillion between 30 July and 26 August 2026 [V, MOF], and Treasury Secretary Bessent publicly urged that the FIMA facility "be upsized" [E].

## 7.2 Balance-sheet mechanics of a swap line

Illustrative, USD billion, exchange rate 1 EUR = 1.10 USD.

*Fed lends USD 50 billion to the ECB against EUR 45.45 billion.*

| Federal Reserve: Assets | | Federal Reserve: Liabilities | |
|---|---|---|---|
| Central bank liquidity swap (EUR claim) | +50 | Reserves / ECB's dollar account at the New York Fed | +50 |

| ECB: Assets | | ECB: Liabilities | |
|---|---|---|---|
| Dollars at New York Fed | +50 | EUR deposit of the Fed at the ECB | +45.45 (= USD 50) |

*ECB lends the dollars to a euro-area bank against collateral (the ECB bears the credit risk).*

| ECB: Assets | | ECB: Liabilities | |
|---|---|---|---|
| Dollar loan to bank | +50 | EUR deposit of the Fed | 45.45 |
| Dollars at New York Fed | 0 (transferred to the bank's correspondent) | | |

On maturity the trades reverse at the original exchange rate, so the Fed bears no exchange-rate risk and no credit risk on the euro-area bank; its counterparty is the ECB. FIMA repo differs: the Fed buys Treasuries from the foreign central bank under a repurchase agreement, so the collateral is the dollar asset itself.

## 7.3 Interpretation: public good or instrument of power?

- **Stabilising function (empirical finding).** Studies of 2008 and 2020 (for example Bahaj and Reis; Goldberg, Kennedy and Miu) [E] find swap lines reduced covered-interest-parity deviations and dollar funding stress, which also protected US markets from fire sales of dollar assets.
- **Selective membership (fact and interpretation).** Only five foreign central banks have standing, unlimited access. Most emerging markets rely on their own reserves, IMF facilities, regional arrangements (Chiang Mai Initiative Multilateralisation) or FIMA repo, which presupposes they already hold Treasuries. This is an *exclusive* lender of last resort: a hierarchy within the hierarchy.

- **Politicisation risk (interpretation, 2025-2026).** In September 2025 Adam Posen of the Peterson Institute urged central banks to pool dollar reserves because Fed help "cannot be guaranteed" given political attacks on the Fed [E, Reuters]. A PIIE policy brief (June 2026) warned that "geopolitically motivated US swap lines" would harm both the dollar and Fed independence [E]. A US congressman urged the Fed in October 2026 to reconsider Hong Kong's FIMA access [E]. The 2026 yen episode shows the safety net being used to support an ally's currency, with the Treasury rather than the Fed in the lead.

The normative question for the monograph (Part VII) is whether a global public good (lender of last resort in the world's funding currency) can remain credible when access is discretionary and the issuing state is willing to link it to foreign policy. A safety net whose availability depends on politics is worth less, which raises the precautionary demand for reserves and gold, which in turn is costly for the reserve-holding countries (Section 9).

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# 8. Global liquidity, the global financial cycle and dollar-denominated emerging-market debt

## 8.1 What the BIS global liquidity indicators measure

The BIS global liquidity indicators (GLIs) "track credit to non-bank borrowers, covering both loans extended by banks and funding from global bond markets through the issuance of international debt securities", focusing on credit "denominated in the three major reserve currencies (US dollar, euro and Japanese yen) to non-residents" [V, BIS release 29 October 2026]. They are a measure of the *offshore* use of a currency as a funding currency.

## 8.2 The numbers

From the BIS SDMX API (series WS_GLI, Q.USD.3P/4T.N.A.I.B.USD) [V]:

| End-period | USD credit to non-banks outside the US (USD tn) | of which to non-banks in emerging market economies (USD tn) |
|---|---|---|
| 2000 Q4 | 2.49 | 0.80 |
| 2007 Q4 | 5.53 | 1.44 |
| 2014 Q4 | 9.38 | 2.94 |
| 2019 Q4 | 12.12 | 3.80 |
| 2021 Q4 | 13.47 | 4.25 |
| 2022 Q4 | 12.81 | 4.06 |
| 2024 Q4 | 13.13 | 4.07 |
| 2025 Q4 | 14.26 | 4.33 |
| 2026 Q2 | 15.13 | 4.60 |

Of the USD 15.13 trillion at Q2 2026, USD 8.20 trillion was international debt securities and the rest bank loans; for emerging market economies, USD 2.56 trillion was debt securities [V]. Selected country stocks of dollar credit to non-banks at Q2 2026 [V]: Mexico USD 389 billion; China USD 303 billion; Saudi Arabia USD 265 billion; Turkey USD 234 billion; Indonesia USD 218 billion; Brazil USD 200 billion; Chile USD 170 billion; India USD 115 billion; Argentina USD 102 billion.

Euro and yen credit (Q2 2026) [V]: euro credit to non-banks outside the euro area EUR 5.39 trillion (+13.8% y/y), of which EUR 0.90 trillion to emerging market economies (+14.1%); yen credit outside Japan JPY 70.0 trillion (+7.1%). The BIS notes that euro-denominated international debt securities grew 13% year on year, "almost double the growth rate of their US dollar-denominated counterparts (6.7%)", and that "the euro's share of foreign currency credit has increased since 2022" at current and constant exchange rates [V].

Limitations. GLI stocks are valued in dollars and so rise when the dollar depreciates (2025) and fall when it appreciates (2022); the BIS publishes adjusted growth rates to strip this out. Coverage of non-bank lenders (private credit, direct lending) is incomplete.

### 8.3 Interpretation: the global financial cycle

- **Rey (2013, Jackson Hole) [E]**: a global financial cycle in capital flows, asset prices and credit co-moves with the VIX and US monetary policy; for countries with open capital accounts, the "trilemma" becomes a "dilemma": floating exchange rates do not fully insulate domestic financial conditions unless capital flows are managed.
- **Bruno and Shin (2015) [E]**, the risk-taking channel: a weaker dollar strengthens the balance sheets of dollar borrowers, encourages banks to lend more in dollars, and loosens global financial conditions; a stronger dollar does the reverse. The dollar exchange rate is therefore a barometer of global risk appetite (Avdjiev, Du, Koch and Shin 2019) [E].
- **Original sin and its redux.** Eichengreen and Hausmann (1999) [E] described the inability of emerging economies to borrow abroad in their own currencies. Many have since developed local-currency bond markets, but Carstens and Shin (2019) [E] note "original sin redux": foreign investors in local-currency bonds hedge or bear currency risk, so depreciations trigger outflows and yield spikes, transferring the mismatch to the lenders' side without removing it.

This links to the monograph's discussion of who benefits from money creation (Part VII). The Fed's policy rate is set for the US economy, yet through dollar debt and the global financial cycle it is also, in effect, a policy rate for a large share of the world's borrowers who have no voice in setting it. That asymmetry is a central fairness issue of the international order (normative question, not a conclusion).

### 8.4 Indonesia as an illustration (cross-reference to R-Indonesia stream)

Indonesian non-bank borrowers owed about USD 218 billion of dollar credit at Q2 2026 [V, BIS GLI]. Indonesia also holds one of the PBOC's largest swap lines (RMB 400 billion) [E] and is an mBridge observer (BIS project page) [V]. During the 1997-98 crisis, unhedged dollar borrowing by Indonesian corporates converted a currency collapse into a solvency crisis (Part XI case). The current exposure is smaller relative to GDP and better hedged [E, not quantified here], but the mechanism is the same.

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## 9. The Triffin dilemma, old and new

### 9.1 The classic version

Robert Triffin (*Gold and the Dollar Crisis*, 1960) [E] argued that under Bretton Woods the world's demand for reserves could be met only if the United States ran balance-of-payments deficits that supplied dollars abroad; but growing dollar liabilities relative to US gold reserves would undermine confidence in the USD 35 per ounce convertibility promise. Either liquidity would be scarce (deflationary bias) or confidence would erode (run on gold). The dilemma was resolved by the end of gold convertibility in August 1971 (see R1 on Bretton Woods and the Nixon Shock).

### 9.2 The "new Triffin dilemma"

Farhi, Gourinchas and Rey (*Reforming the International Monetary System*, CEPR, 2011) [E] restated the problem for a fiat world. Demand for safe, liquid stores of value rises with world output and with emerging markets' precautionary reserve accumulation. If one sovereign supplies the bulk of the world's safe assets, its fiscal capacity, which grows with its own economy, will fall behind world demand as its share of world output shrinks. The issuer then faces a choice: limit the supply of safe debt (creating a global safe-asset shortage, low real rates and a deflationary bias) or keep issuing until the debt's safety is doubted (risking a run on the reserve asset). The analogy to gold is the limited fiscal capacity backing the debt.

Related arguments:

- **Safe-asset shortage** (Caballero, Farhi and Gourinchas 2017) [E]: excess demand for safe assets lowers the natural rate and can push economies to the zero lower bound.

- **Convenience yield** (Krishnamurthy and Vissing-Jorgensen 2012; Jiang, Krishnamurthy and Lustig 2021) [E]: foreigners pay a premium for Treasuries' safety and liquidity; a fall in the convenience yield signals erosion of privilege.
- **Obstfeld (2013) and others** [E] question whether a current-account deficit is necessary for supplying safe assets, since the US can issue safe liabilities while acquiring risky foreign assets (gross, not net, positions matter).

## 9.3 Evidence in 2025-2026

- **Fiscal trajectory (fact).** Marketable Treasuries rose from USD 16.7 trillion at end-2019 to USD 32.15 trillion in October 2026 [V], almost doubling in under seven years. Moody's removed the last AAA rating in May 2025 [E, Fed note cites the downgrade].
- **Foreign absorption (fact).** The foreign share of marketable Treasuries fell from about 49% (2014) to about 29% (2026) [V]. Foreign official holdings were USD 3.76 trillion in August 2026, below their February 2026 level of USD 4.01 trillion [V].
- **Convenience erosion (empirical finding, contested).** Several analyses report a decline in Treasury convenience measures in the 2020s and a weakening of the safe-haven correlation during stress, including in April 2025 [E, secondary summaries]. The ECB's November 2025 FSR documented the unusual co-movement of a falling dollar and rising long yields after 2 April 2025 [E].
- **Safe-haven behaviour during the 2026 Iran war (empirical observation).** Reuters reported in August 2026 that "none of the assets investors usually pick in times of trouble ... gold and the dollar have played the traditional safe-haven role consistently"; the dollar index rose only 1.4% in the first six months of the war, much of it reflecting yen weakness, while Treasuries lost 3.5% on a total-return basis as inflation rose [E]. An AEI analysis attributed the early dollar rise to both safe-haven demand and the US position as a net oil exporter [E].

Interpretation. The new Triffin logic is visible in slow motion: rising issuance, falling foreign share, a shrinking convenience premium, and less reliable safe-haven behaviour. But there is no evidence of a run: foreign holdings were near record levels in 2026, and the dollar's transactional shares rose. The tension is being absorbed by price (higher term premia) rather than by quantity (foreign selling).

## 9.4 Who pays? A simple reserve-accumulation T-account

An emerging-market central bank (EMCB) intervenes to prevent appreciation, buying USD 10 billion from domestic exporters and investing in Treasuries.

| EMCB: Assets | | EMCB: Liabilities | |
|---|---|---|---|
| US Treasuries | +10 (USD) | Bank reserves (local currency equivalent) | +10 |
| | | Sterilisation bills issued (if sterilised) | +10, reserves -10 |

The EMCB earns the Treasury yield (say 4%) and pays the domestic sterilisation rate (say 6%), a carrying cost of 2 percentage points a year, plus currency risk. Rodrik (2006) [E] estimated the social cost of excess reserves at about 1% of developing-country GDP. The reserves are self-insurance against sudden stops, precisely because the global safety net (Section 7) is incomplete. In effect, poorer countries lend to the richest sovereign at low yields to insure themselves against the instability of a system centred on that sovereign. This is the strongest distributive criticism of the dollar order (normative), and it is also why a more reliable multilateral safety net would reduce the demand for dollar reserves.

# 10. The dominant currency paradigm

## 10.1 The argument

Traditional open-economy models assumed producer-currency pricing (exports priced in the exporter's currency, Mundell-Fleming) or local-currency pricing (priced in the importer's currency). Gita Gopinath ("The International Price System", Jackson Hole 2015) and Gopinath, Boz, Casas, Diez, Gourinchas and Plagborg-Moller ("Dominant Currency Paradigm", *American Economic Review* 2020) [E] documented that most trade is invoiced in a small number of currencies, above all the dollar, regardless of whether the US is a party to the trade, and that prices are sticky in the invoicing currency.

## 10.2 Implications (empirical findings)

- **Pass-through and trade volumes depend on the dollar exchange rate.** A depreciation of the rupiah against the dollar raises rupiah import prices for goods invoiced in dollars, even from Korea; but Indonesian exports invoiced in dollars do not become cheaper abroad in the short run, so the expenditure-switching benefit of depreciation is weak.
- **A stronger dollar depresses world trade.** Boz, Gopinath and Plagborg-Moller and later BIS work (Bruno, Kim and Shin) [E] find a 1% dollar appreciation against all currencies predicts a decline in rest-of-world trade volumes, through both pricing and financial (dollar credit) channels.
- **Invoicing and financing reinforce each other.** Firms that invoice in dollars prefer dollar debt to match revenues; banks that fund in dollars prefer dollar loans. This is a complementarity that sustains equilibrium (Gopinath and Stein 2021, "Banking, Trade, and the Making of a Dominant Currency") [E].
- **Policy implication.** For dollar-invoicing economies, exchange-rate flexibility stabilises less than textbook models imply; central banks therefore intervene, hold reserves and watch the Fed. This is the micro foundation for "fear of floating" (Calvo and Reinhart 2002) [E].

Interpretation. The dominant currency paradigm explains why the dollar's role in trade is larger than the US share of trade, and why tariffs do not straightforwardly reduce dollar usage: even when the US trades less with the rest of the world, third-country trade continues to be invoiced in dollars. It also explains why a de-dollarisation campaign must change invoicing conventions, finance and hedging markets together, not one at a time.

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# 11. Why some currencies become internationally dominant

## 11.1 The determinants

The BRIEF lists seven: institutional credibility, market depth, financial infrastructure, trade networks, geopolitical power, legal predictability and convertibility. The literature adds two: network externalities (the more a currency is used, the cheaper it is to use) and the availability of a lender of last resort in that currency.

Comparative assessment (interpretation; scores are qualitative judgements by this stream, not measurements; 5 = strongest):

| Determinant | USD | EUR | CNY | JPY | GBP | Gold (non-currency reserve) |
|---|---|---|---|---|---|---|
| Institutional credibility (central bank, inflation record) | 4 (Fed record strong; independence questioned 2025-26) | 4 | 3 | 4 | 4 | n.a. (no issuer) |
| Market depth and liquidity of safe assets | 5 | 3 (fragmented) | 2 (closed) | 4 | 3 | 3 (deep spot, thin official) |

| Determinant | USD | EUR | CNY | JPY | GBP | Gold (non-currency reserve) |
|---|---|---|---|---|---|---|
| Financial infrastructure (settlement, custody, hedging) | 5 | 4 | 3 (CIPS growing) | 4 | 4 | 2 (physical, custody) |
| Trade networks / invoicing | 5 | 4 (Europe-centred) | 3 (China-centred) | 2 | 2 | 1 |
| Geopolitical power and alliances | 5 | 3 | 4 | 2 | 2 | n.a. |
| Legal predictability for foreign holders | 4 (sanctions risk for adversaries) | 4 (same) | 2 | 4 | 4 | 5 if held at home; low if custodied abroad |
| Convertibility / open capital account | 5 | 5 | 2 | 5 | 5 | 5 (but not a means of payment) |
| Lender of last resort in the currency | 5 (Fed swaps, FIMA) | 4 (ECB repo lines) | 2 | 4 | 4 | 1 |

Limitations. The scores compress contested judgements and should be read as a structured argument, not data. Gold is included because central banks increasingly treat it as the alternative to all fiat reserve currencies.

## 11.2 Theory and history

- **Network externalities and inertia (Krugman 1984; Matsuyama, Kiyotaki and Matsui 1993) [E].** If the benefits of using a currency rise with the number of users, there can be multiple equilibria and strong persistence: an incumbent can stay dominant long after its fundamentals weaken. The classic example is sterling's long dominance after the US economy overtook Britain's in the late nineteenth century.
- **Eichengreen, Mehl and Chitu, *How Global Currencies Work: Past, Present, and Future* (Princeton University Press, 2017) [E].** Using new data on reserve composition and international bond issuance in the interwar period, they challenge the "single dominant currency with long inertia" view. Their findings: in the 1920s the dollar and sterling shared reserve and debt functions; the dollar overtook sterling in reserves by the mid-1920s, sterling regained ground after the 1933 dollar devaluation, and the transition was faster and more reversible than the inertia story assumed (Chitu, Eichengreen and Mehl 2014 on bond markets; Eichengreen and Flandreau 2009 on reserves) [E]. They argue that modern technology lowers switching costs, so a multipolar system with two or three international currencies is historically normal, not unstable.
- **The "new view" applied to today (interpretation).** If Eichengreen-Mehl-Chitu are right, the dollar's dominance is less locked in than network models suggest, and a shift could occur relatively quickly once a credible alternative safe asset exists. If the "old view" is right, the dollar will remain dominant for decades even with weaker fundamentals. The evidence of 2015-2026 (slow reserve diversification to smaller currencies; rising dollar shares in FX and payments) supports a mixed reading: the store-of-value function behaves like the new view, the medium-of-exchange and funding functions behave like the old view.
- **Hegemonic stability (Kindleberger 1973) [E].** Kindleberger argued the interwar collapse reflected Britain's inability and the United States' unwillingness to act as stabiliser: maintain open markets for distress goods, provide countercyclical lending and act as lender of last resort. The dollar order after 1945 was the US accepting those roles. The 2025-2026 period, with high tariffs and a politicised safety net, is a partial withdrawal from the first and a conditionalisation of the third. This is the most serious structural worry about the dollar order (interpretation).

- **Safety and depth, not just size.** Farhi and Maggiori (2018) [E] model a hegemon issuing safe debt that is tempted to inflate it away; multipolarity can be more stable or less stable depending on whether competitors discipline each other or face coordinated runs.

### 11.3 Summary judgement

The dollar's dominance rests on the joint presence of all determinants. Each challenger lacks at least two: the euro lacks a unified safe asset and full geopolitical autonomy; the renminbi lacks convertibility and legal predictability; the yen and sterling lack scale; gold lacks yield and payment use. That combination, not any single factor, explains persistence.

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## 12. Sanctions and the "weaponization" of money

### 12.1 Instruments

Financial sanctions use the network hubs described in Section 6: blocking property under US jurisdiction (OFAC), denying correspondent accounts, secondary sanctions on third-country banks, disconnection from SWIFT (an EU-law decision because SWIFT is Belgian), and immobilisation of central-bank reserves held in custody abroad. Precedents: Iran's banks were cut from SWIFT in 2012 [E]; Afghanistan's central-bank reserves in the US were frozen in 2021 [E]; Venezuela's gold at the Bank of England became unusable by the Maduro government after 2019 litigation [E].

### 12.2 The 2022 freeze of Russian reserves

(Fact, [E] unless noted.) In late February 2022, the G7 and EU prohibited transactions with the Central Bank of Russia, immobilising roughly half of its reserves. The European Parliament's research service puts immobilised Russian sovereign assets worldwide at "nearly EUR 300 billion" by February 2024, of which about EUR 210 billion in the EU, mainly at Euroclear [E, EPRS October 2026]. Euroclear reported EUR 202 billion of sanctioned Russian assets on its balance sheet at end-June 2026 [E, Euroclear H1 2026 release excerpt] and, per secondary reports, EUR 204 billion at end-September 2026.

Use of the assets has advanced in steps:

- **Windfall profits.** Since 2024 the EU has captured the interest income on cash balances arising from immobilised assets. The EEAS reported a fifth transfer of EUR 1.4 billion on 3 August 2026, bringing the total to EUR 8 billion; 95% goes to the Ukraine Loan Cooperation Mechanism that services the G7 Extraordinary Revenue Acceleration loans [E, EEAS].
- **Indefinite immobilisation.** On 12 December 2025 EU governments agreed to keep EUR 210 billion immobilised "for as long as needed" instead of renewing every six months by unanimity, removing the risk that one member state could veto the rollover [E, Reuters; BBC; Carnegie].
- **Reparations loan not adopted.** The Commission proposed (3 December 2025) either a "reparations loan" funded by cash balances linked to the immobilised assets (up to EUR 210 billion) or joint EU borrowing [E, EPRS]. In December 2025 the European Council chose a EUR 90 billion Ukraine Support Loan financed by joint borrowing; the first tranche was released in June 2026 [E, Carnegie]. Belgium opposed the reparations loan over legal and liquidity risks concentrated at Euroclear. In August 2026 Sweden, the Netherlands, Spain and Poland asked the Commission to revisit the option [E].
- **Russian counter-measures.** The Bank of Russia sued in Russian courts; secondary reports describe a roughly EUR 200 billion judgment against Euroclear confirmed on cassation on 22 October 2026, which Euroclear says is unenforceable in the EU [E, unverified primary].

### 12.3 Effects on reserve behaviour

- **Gold (fact).** Central-bank net purchases (World Gold Council estimates, which include unreported buying and are revised): 1,082 t (2022), about 1,050 t (2023), 1,092 t (2024, revised), 863 t (2025), 675 t in Q1-Q3 2026, including 321 t in Q3 2026, the highest quarter since Q4 2024 [V for 2026; E for earlier years]. The

2010-2021 average was 473 t a year [E]. Reported buying in Q3 2026 was only 120 t; "unreported activity accounted for around 63% of the quarter's demand", a pattern evident since 2022 [V]. The largest reporters were the PBOC (63 t in Q3) and the National Bank of Poland (40 t); Russia reported a 19 t decline [V].

- **Fiat reallocation (finding).** The Fed's 2025 note found the dollar share "basically unchanged since 2022", "suggesting that U.S. sanctions on Russia ... have not led to fears of dollar 'weaponization' causing a notable reallocation of reserves out of dollars"; it adds that about three-quarters of foreign government holdings of safe US assets are held by countries with military ties to the United States (Weiss 2022) [V]. The same note reports that the share of gold in official reserves more than doubled from below 10% in 2015 to over 23%, mostly through price, with physical holdings up less than 10% [V].
- **Gold versus the euro (computation, approximate).** At the Q3 2026 average gold price (USD 4,262/oz) [V] and an assumed official gold stock of about 36,000 t [E], official gold would be worth about USD 4.9 trillion, more than the USD 2.72 trillion of euro reserves [E, MNI from COFER] and about 27% of the sum of FX reserves plus gold. This is a valuation statement, not evidence of a large shift in holdings.

### 12.4 Interpretation: legal predictability cuts both ways

The determinants table in Section 11 lists legal predictability as a source of dollar and euro strength. Sanctions do not contradict this for most holders: the rule of law that protects a Norwegian or Korean reserve manager is the same legal system that can lawfully freeze an aggressor's assets under sanctions statutes. But for countries that anticipate future conflict with the West, the 2022 precedent converts Western legal systems from a guarantee into a risk. Gold held at home, renminbi assets, and payment systems outside Western jurisdiction become insurance. The price of that insurance is high (no yield on gold, illiquidity and capital controls in renminbi), which is why the response has been partial.

A second-order effect deserves attention: every use of reserves as leverage strengthens the incentive of non-aligned countries to build alternative plumbing, even if they never use it. Alternatives that are idle in normal times can matter in a crisis; their option value is the point.

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## 13. Structural alternatives in development

### 13.1 BRICS payment initiatives

(Fact, [E].) Since the 2023 Johannesburg summit, BRICS declarations have emphasised local-currency settlement, correspondent-banking links and payment interoperability. The 2024 Kazan declaration launched the BRICS Cross-Border Payments Initiative (BCBPI) and discussed a BRICS Clear depository and a reinsurance company; the 2025 Rio declaration (6 July 2025) "welcome[d] the 'Technical Report: BRICS Cross-border Payments System' " and tasked finance ministers to continue discussions [E, BRICS Brazil PDF excerpt]. The New Delhi declaration (18th summit, 12-13 September 2026) "acknowledge[d] the work done to study the cross-border interoperability of payment and messaging channels, and the discussions on promoting trade settlements and investments using BRICS local currencies, while respecting national priorities and acknowledging that there is no one-size-fits-all approach" [E, TASS quoting the declaration]. India's Secretary (Economic Relations) Sudhakar Dalela said "There is no proposal in the BRICS for BRICS currency as of now" [E, multiple Indian outlets]. A commentary in BusinessLine summarised: "There is no BRICS payment system, no common settlement currency and no timeline" [E].

Interpretation. BRICS members have divergent interests: India does not want a China-centred system; Brazil, India, South Africa and the Gulf members want to keep access to Western finance; Russia and Iran want sanction-proof channels. The common denominator is interoperability of national systems (UPI, Pix, CIPS, SPFS) and bilateral local-currency trade, which reduce dollar intermediation at the margin without creating a rival reserve asset. Local-currency settlement also runs into the problem that bilateral trade surpluses accumulate in currencies the surplus country cannot easily use (for example Russia's rupee balances from oil sales to India, 2023 [E]).

## 13.2 Project mBridge

(Fact.) mBridge began in 2021 as a BIS Innovation Hub project with the Hong Kong Monetary Authority, the Bank of Thailand, the PBOC's Digital Currency Institute and the Central Bank of the UAE; a real-value pilot ran in 2022; the Saudi Central Bank joined in 2024; the platform reached minimum viable product in mid-2024. The BIS project page states: "The BIS announced in October 2024 that it was handing the project over to the partners" and lists its status as "Concluded"; observers at that time included Bank Indonesia, the ECB, the New York Fed's Innovation Centre, the IMF and the World Bank [V]. On 31 October 2024 BIS General Manager Agustin Carstens said the BIS was leaving "not because it was a failure and not because of political considerations", and that "mBridge is not the 'BRICS bridge' " [E, BIS speech page excerpt and press].

Developments since (all [E]): SAMA told the Financial Times (reported 20 September 2026) that it ceased participating after completing its proof of concept on 13 May 2025, as planned; the Monetary Authority of Macao joined and went live in 2026; the remaining partners are reported to be moving towards commercial rollout under a Hong Kong-based entity. Reported cumulative volumes vary: about USD 55.5 billion across more than 4,000 transactions with about 95% settled in digital yuan (Forbes, May 2026), and about USD 69 billion as of June 2026 (Atlantic Council, cited by AGBI). These figures are unverified against primary sources and small relative to global payments (CIPS alone settled about USD 25 trillion in 2025).

Interpretation. The BIS's exit "verified" in the commissioning brief is confirmed. mBridge shows that multi-CBDC wholesale settlement is technically feasible; its governance has shifted from a neutral international organisation to a China-centred consortium, which limits its appeal to countries wary of Chinese leverage just as SWIFT and CHIPS raise concern about Western leverage. The BIS's own alternative, Project Agora with G7-aligned central banks and private banks, is exploring tokenised correspondent banking [E].

## 13.3 Gold

Gold's advantages as a reserve asset are no counterparty risk (if held domestically), no issuer who can sanction it, and a long record as a crisis hedge. Its disadvantages: no yield, storage and transport costs, price volatility (it more than tripled from 2015 to 2026), and no use as a means of payment in modern settlement. Gold is a store of value that complements the dollar system rather than replacing its transactional core (interpretation). R1's analysis of the gold standard applies: a return to gold as a monetary anchor would reintroduce deflationary bias and the instability documented in the interwar period (Eichengreen, *Golden Fetters*, 1992) [E].

## 13.4 The SDR and a supranational reserve asset

Keynes's 1944 bancor plan and Zhou Xiaochuan's 2009 essay [E] proposed a supranational reserve currency. The IMF's Special Drawing Right exists (general allocation of SDR 456 billion in August 2021 [E]), but it is a claim on IMF members' currencies, not a currency used by private actors; there are no deep private SDR markets, no SDR-denominated invoicing and no SDR lender of last resort. Expanding its role would require a political transfer of monetary authority that no major issuer currently supports (interpretation).

## 13.5 Stablecoins and the digital dollar

(Fact.) The GENIUS Act (P.L. 119-27), signed on 18 July 2025, requires US payment-stablecoin issuers to hold at least one dollar of permitted reserves (coins and currency, insured deposits, short-dated Treasury bills, Treasury repo and similar) per dollar of tokens; the Bank Policy Institute notes it takes effect on 18 January 2027 [E, CRS; BPI]. Stablecoin supply was about USD 305 billion in October 2026, roughly 99.5% dollar-denominated, with Tether (USD 183 billion) and USDC (USD 72 billion) holding about 85% [E, CoinDesk]. Supply peaked in May 2026 and has since been flat [E]. The Treasury Secretary described stablecoins as a "revolution" that would "buttress the dollar's status as the global reserve currency" [E, Treasury statement]. One estimate puts issuers' holdings of bills, bill repo and fund shares at about USD 185 billion, 2.6% of bills outstanding [E, unverified].

Interpretation. Dollar stablecoins extend dollar use to households and firms abroad through private tokens, potentially accelerating "digital dollarisation" in countries with weak currencies (Cowen 2025, cited by the Fed) [V]. This could entrench dollar dominance and add a new source of demand for Treasury bills, while weakening monetary sovereignty and bank deposit bases in emerging markets. It also creates a run-prone private money layer whose safety rests on reserve quality and redemption rules (see R2 on money-like liabilities and R-CBDC on

digital money). The dollar's main digital competitor is therefore not a BRICS coin but the absence of a regulated digital euro or renminbi at comparable scale outside China.

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## 14. Developments in 2025-2026: a chronology with evidence

| Date | Event | Monetary relevance | Source / tag |
|---|---|---|---|
| 13 Jan 2025 | Fed broad dollar index peaks at 130.04 | Start of a large dollar decline | FRED DTWEXBGS [V] |
| 2 Apr 2025 | "Liberation Day" tariffs announced; 9 Apr partial 90-day pause | Dollar fell while ten-year yields rose almost 50 bp in a week (third-largest weekly rise since 1986); equities, Treasuries and dollar fell together | ECB FSR Nov 2025 [E]; TBAC letter [E] |
| May 2025 | Moody's downgrades the US sovereign | Fiscal credibility | Fed FEDS Note 2025 [V] |
| 30 Jun 2025 | Dollar index (DXY) down 10.8% in H1, worst first half since 1973 | Valuation boost to non-dollar reserve shares | Press reports [E]; broad index 119.41, -7.6% vs end-2024 [V] |
| 6 Jul 2025 | BRICS Rio declaration | Technical report on cross-border payments; no currency | BRICS PDF [E] |
| 18 Jul 2025 | GENIUS Act signed | Regulated dollar stablecoins | CRS [E] |
| Sept 2025 | US-Japan finance ministers' joint statement on FX | Basis for 2026 joint intervention | MOF statement [V] |
| Sept 2025 | PBOC renews swap lines with ECB, SNB, MNB | Renminbi safety net | PBOC/embassy [E] |
| 30 Sept 2025 | BIS Triennial: USD 89.2% of FX trades | Vehicle role intact | BIS [V] |
| 12 Dec 2025 | EU immobilises EUR 210 billion of Russian assets indefinitely | Sanctions durability | Reuters [E] |
| Dec 2025 | EU agrees EUR 90 billion joint-borrowing loan for Ukraine; reparations loan set aside | Legal limits of weaponisation | Carnegie; EPRS [E] |
| End-2025 | Dollar index down over 9% in 2025 (worst since 2017); broad index 119.75 (-7.4%) | | NPR [E]; FRED [V] |
| 29 Jan 2026 | Broad dollar index low of 117.44 | | FRED [V] |
| Feb 2026 | SWIFT renames RMB Tracker "Global Currency Tracker" | | SWIFT [V] |
| 28 Feb 2026 | US and Israel launch strikes on Iran; Iran disrupts Strait of Hormuz | Oil shock; inflation; test of safe-haven behaviour | Reuters; BBC [E] |

| Date | Event | Monetary relevance | Source / tag |
|---|---|---|---|
| 8 Apr 2026 | Temporary US-Iran ceasefire; later breakdown and US blockade of Iranian ports from 13 April | Brent briefly above USD 120 in April; average about USD 90 in 2026 to late August | Wikipedia summary; Reuters [E] |
| May 2026 | US CPI rising at fastest pace in three years (April data) | Rate-hike expectations | Reuters [E] |
| 22 May 2026 | Kevin Warsh sworn in as Fed chair (nominated 4 Mar; confirmed 13 May) | Fed independence debate; swap-line governance | Federal Reserve press release [E] |
| 2 Jun 2026 | ECB: euro's international role rose moderately in 2025 | | ECB [E] |
| 31 Jul 2026 | US-Japan joint yen-buying intervention; Japan to use FIMA repo | Safety net used for an ally; Treasury-led | Japan MOF [V] |
| Aug 2026 | Reuters: dollar +1.4% since the war began; Treasuries -3.5% total return | Weak safe-haven performance | Reuters [E] |
| 2 Sept 2026 | NY Fed: reserve diversification concentrated in a few holders | | Liberty Street [V] |
| 12-13 Sept 2026 | BRICS New Delhi summit: no common currency | | Indian and Russian press [E] |
| 30 Sept 2026 | COFER Q2 2026: USD 56.70%, EUR 20.60% | Euro gains through active buying | IMF [V] |
| 29 Oct 2026 | BIS: euro share of foreign-currency credit rising | | BIS [V] |
| 30 Oct 2026 | Broad dollar index 121.67 | Partial recovery in 2026 | FRED [V] |
| 3 Nov 2026 | WGC: central banks bought 321 t of gold in Q3 2026 | | WGC [V] |
| Early Nov 2026 | Renewed US strikes on Iranian targets near Hormuz; Brent near USD 98-104 | Unresolved energy shock | CNBC; WSJ [E] |

Interpretation of the period:

1. **The 2025 tariff shock revealed a contingent safe-haven status.** Corsetti, Lloyd and Ostry (VoxEU, September 2025) [E] argue that dollar depreciation after tariffs met with retaliation is consistent with past evidence; the spike in long yields was the unprecedented part. The episode suggests investors began to price policy risk specific to the United States.
2. **The 2026 war showed the dollar's energy-hedge character more than its safe-haven character.** As a net oil exporter, the US gains in relative terms from oil shocks; the dollar's modest rise is consistent with terms-of-trade effects as much as flight to safety (AEI, June 2026) [E]. Treasuries suffered from the inflation shock.
3. **Transactional dominance was untouched.** FX turnover, payments, dollar credit and Treasury foreign holdings all reached or stayed near record levels.

4. **Diversification gained at the margin.** Euro reserves through active purchases (Q2 2026), euro credit growth, gold purchases rising again in Q2-Q3 2026.
5. **The safety net became more political.** The yen intervention combined Treasury FX operations, FIMA repo and an explicit alliance rationale. This may reassure allies while signalling to others that access is conditional.

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# 15. Is the existing order sustainable? Evaluation and structural alternatives

## 15.1 Hypothesis register (adversarial tests)

| # | Hypothesis | Evidence for | Evidence against | Status at Nov 2026 |
|---|---|---|---|---|
| H1 | De-dollarisation is under way across all functions | Reserve share down from 72% (2001) to 56.7%; euro and gold gains; CIPS growth | FX share up to 89.2%; SWIFT payments share near 50%; dollar credit at record USD 15.1 tn; reserve decline concentrated in few holders | Rejected as stated; partial erosion only in reserves |
| H2 | Sanctions on Russia caused a broad flight from dollar reserves | Gold buying doubled after 2022; China reduced Treasuries | Dollar share roughly flat 2022-2024 (Fed); diversification concentrated (NY Fed) | Weak support; effect operates via gold and a few holders |
| H3 | The renminbi will become a major reserve currency within a decade | Rising FX and trade-finance shares; swap network; CIPS | 2.1% of reserves; 1% of international bonds; capital controls; CIPS growth stalled in 2025 | Not supported on current policies |
| H4 | The euro is the main beneficiary of any diversification | COFER Q2 2026 active euro buying; euro credit growth; ECB index up | Fragmented safe assets; euro also a sanctions currency | Supported, but gains are incremental |
| H5 | The new Triffin dilemma is binding | Debt doubled since 2019; foreign share 29%; convenience erosion; weak safe-haven behaviour | Foreign holdings near record; no run; demand from domestic and stablecoin buyers | Partly supported: tension expressed in term premia, not quantities |
| H6 | Dollar dominance mainly reflects US geopolitical power | Alliance holders own about three-quarters of official safe US assets (Weiss) | Private network effects (FX, invoicing, funding) operate independent of alliances | Both; geopolitics more important for official, networks for private use |
| H7 | Stablecoins will entrench the dollar | 99% of supply in dollars; GENIUS Act; Treasury endorsement | Supply stalled near USD 300 bn; regulatory and run risks; foreign restrictions | Plausible, unproven |

| # | Hypothesis | Evidence for | Evidence against | Status at Nov 2026 |
|---|---|---|---|---|
| H8 | BRICS will create an alternative currency | Repeated declarations on payments | Explicit Indian denial; no institution, timeline or common asset | Rejected for the foreseeable horizon |
| H9 | Politicisation of the Fed and safety net will reduce dollar use | Posen, PIIE warnings; Warsh transition; swap-line debate | No measurable shift in usage yet; allies (Japan) leaning in | Open; key risk to monitor |

## 15.2 Sources of strength (why the order persists)

- **Network complementarities** among invoicing, funding, FX vehicle use and reserves (Sections 3, 10).
- **The largest safe-asset market** in the world, with deep repo and derivatives markets and a central bank willing to backstop it (March 2020).
- **The safety net** that only the Fed can provide in dollars.
- **Lack of a complete alternative** (Section 11).
- **Alliance structure**: most large official holders are US security partners (Weiss 2022, cited by the Fed) [V].
- **Private choice**: dollar dominance is reproduced daily by millions of firms, banks and investors choosing the cheapest, deepest instrument, not by decree. This is the international analogue of the domestic argument in Part VIII that continued participation reflects both genuine utility and lack of practical alternatives.

## 15.3 Sources of fragility (what could undermine it)

- **Fiscal dominance**: if debt dynamics force the Fed to accommodate deficits, inflation would erode the dollar's store-of-value function. US inflation rose again in 2026 under the oil shock, which tested this.
- **Institutional erosion**: perceived loss of Fed independence, unpredictable trade and sanctions policy, or weakening of legal protections for foreign investors would reduce the rule-of-law premium, which is the hardest asset to rebuild.
- **Market-structure fragility**: Treasury liquidity in stress depends on dealer capacity and leveraged basis trades; a disorderly episode would damage the safe-asset role.
- **Geopolitical fragmentation**: more sanctions and counter-sanctions push a minority of countries into parallel systems, reducing network benefits for all.
- **Distributive legitimacy**: the costs borne by emerging economies (spillovers, reserve carrying costs, exclusion from the safety net) feed political demand for alternatives even when economic alternatives are inferior.

## 15.4 Structural alternatives, compared

| Alternative | Feasibility (2026-2036) | Institutional requirements | Stability properties | Who gains / loses | Assessment |
|---|---|---|---|---|---|
| A. Dollar hegemony with slow erosion (status quo) | High | Continued US fiscal and institutional credibility | Efficient in normal times; US-centred spillovers; exclusive safety net | US privilege; network benefits for all; EM bear spillovers | Most likely baseline |
| B. Multipolar reserve system (dollar, euro, renminbi, gold, smaller currencies) | Medium | Euro safe asset; Chinese capital-account opening; broader swap networks | Diversified safe assets; risk of runs between currencies (Farhi-Maggiori) | Reduces US privilege; spreads seigniorage | Plausible over a long horizon, led by the euro and gold |

| Alternative | Feasibility (2026-2036) | Institutional requirements | Stability properties | Who gains / loses | Assessment |
|---|---|---|---|---|---|
| C. Fragmented blocs (Western dollar-euro system vs China-centred system) | Medium | Parallel plumbing (CIPS, mBridge), bloc safety nets | Lower efficiency; sanction resilience within blocs; higher transaction costs | Non-aligned countries face higher costs and choices | Growing at the margin; full bifurcation unlikely given China's own dollar exposure |
| D. Supranational reserve (expanded SDR, bancor) | Low | Political delegation of monetary authority; private SDR markets | Theoretically symmetric; removes Triffin problem | Spreads privilege; constrains issuers | Normatively attractive to some; politically remote |
| E. Return to a gold anchor | Very low | Convertibility commitments; fiscal discipline | Deflationary bias; interwar experience | Gold holders; creditor countries | Rejected on historical evidence (R1) |
| F. Digital-dollar entrenchment (stablecoins, tokenised Treasuries) | Medium | Regulation (GENIUS), interoperability, foreign tolerance | New run risks; capital-flow volatility; weaker EM monetary sovereignty | US Treasury financing; tech issuers; EM savers seeking safety; EM banks and central banks lose | Real possibility; could deepen dollar dominance |
| G. Interoperable multi-CBDC and fast-payment bridges (mBridge, Agora, Nexus, BRICS interlinking) | Medium for payments; low for reserves | Common standards, legal frameworks, liquidity in FX pairs | Faster, cheaper payments; settlement still needs liquid FX and safe assets | Remittance senders; trading firms; reduces correspondent-bank rents | Changes plumbing, not the store-of-value hierarchy |

## 15.5 Overall evaluation

(Interpretation, with the reasoning stated so that it can be challenged.)

The international monetary order is **sustainable in its transactional core and less sustainable in its institutional underpinnings**. The network economics of invoicing, funding and FX trading make the dollar very hard to displace, and no alternative currently offers the full package. The weaknesses lie in the conditions that made the package credible: fiscal capacity, central-bank independence, predictable openness, and a lender of last resort perceived as neutral. These are political variables, which is why the 2025-2026 period matters more than its modest effect on statistics suggests.

The most plausible structural change is **a gradual move towards a system with one dominant and several secondary international monies**, in which the euro gains as the Western-aligned alternative, gold gains as the non-aligned and sanction-proof reserve, the renminbi gains as a regional and trade-settlement currency under capital controls, and dollar stablecoins extend the dollar's reach among private users. Eichengreen-Mehl-Chitu's historical evidence that multipolar arrangements have existed before (the 1920s) supports the feasibility of such a system; Kindleberger's warning that the interwar multipolar system collapsed without a stabiliser cautions that multipolarity is not automatically stable.

What would change this judgement: (i) a disorderly Treasury market event or an inflationary episode that the Fed is seen to tolerate for fiscal reasons; (ii) issuance of a large, unified euro safe asset and an ECB swap network

open to more countries; (iii) Chinese capital-account liberalisation with credible legal protections; (iv) a sanctions shock affecting a large non-adversary holder.

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## 16. Disputes and uncertainties

1. **Measurement disputes.** Which indicator defines "dominance"? SWIFT payments, COFER reserves, FX turnover and debt shares give answers ranging from about 47% to 89% for the dollar. PBOC estimates of renminbi payment shares (6-7%) are not comparable with SWIFT data and lack published methodology.
2. **Valuation effects.** COFER and GLI shares move with exchange rates; the 2025 dollar depreciation mechanically raised non-dollar shares. The IMF's Q2 2026 note says valuation effects were limited that quarter, but Q4 2025 and Q1 2026 shares were heavily influenced by them.
3. **Unreported gold buying.** About two-thirds of estimated central-bank gold demand since 2022 is unreported; buyer identities and motives are inferred.
4. **Custodial attribution in TIC.** Belgian, Luxembourg and Cayman holdings mask true owners, including possibly Chinese official and hedge-fund positions; country-level conclusions about "China selling Treasuries" are uncertain.
5. **Convenience-yield erosion.** Estimates depend on the benchmark (swap spreads, AAA corporates, foreign government bonds), all affected by balance-sheet regulation; the evidence is suggestive, not settled.
6. **Old view versus new view of currency competition.** Network-inertia models and the Eichengreen-Mehl-Chitu multipolar view make different predictions about the speed of transition; the 2015-2026 data are consistent with both depending on the function examined.
7. **Status of mBridge.** Reported volumes (USD 55.5 billion to USD 69 billion) and the governance of the Hong Kong-based successor were not verified against primary documents.
8. **Russian assets.** Euroclear's end-September 2026 figure and the Russian court judgment come from secondary reports.
9. **War-related market moves.** Oil-price peaks, dollar and Treasury performance during the 2026 Iran war are from news reports; the conflict was ongoing at the cut-off, so effects are provisional.
10. **Normative disagreement.** Whether the dollar order is a global public good or a source of unfair privilege depends on weighting efficiency gains (network benefits, deep markets) against distributive costs (spillovers, reserve costs, exclusion, coercive use). This stream does not settle that question; Part VII of the monograph should present both cases.

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## 17. Connections to other parts of the monograph

- **Part II (theories of money):** international money is accepted without legal tender or tax obligation, supporting conventionalist and network accounts while showing that state power (alliances, sanctions, the Fed's backstop) still matters, partly vindicating chartalist emphasis on authority.
- **Part III.4-5 (Bretton Woods and after):** the classic Triffin dilemma, the eurodollar market and the post-1971 dollar standard are the historical background for Sections 2, 9 and 11.
- **Part IV (money creation):** offshore banks create dollar deposits but depend on settlement in Fed money (Section 2.2); swap lines are central-bank money creation across borders (Section 7.2).
- **Part V and VII (distribution and fairness):** global financial cycle spillovers and reserve carrying costs are distributive effects of the international hierarchy (Sections 8-9).
- **Part VIII (why systems persist):** the dollar is the clearest case of acceptance driven by network effects and lack of alternatives rather than by formal legitimacy.
- **Part X (CBDC):** mBridge, Agora and stablecoins (Section 13) show digital infrastructure reshaping payments faster than reserve hierarchies.
- **Part XII (Indonesia):** Indonesian dollar debt (USD 218 billion), the PBOC-BI swap line and Bank Indonesia's mBridge observer status (Section 8.4).
- **Part XIII (scenarios):** Section 15.4 maps directly onto Scenarios A (continuity), C (tokenised infrastructure) and D (fragmented order).

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# R6 - Central Bank Digital Currencies and Digital Money

Research stream R6 for the monograph *The Architecture of Money* by M. Bagus Prabowo and Hark. Covers BRIEF Part X (A-F) and the digital-payments, QRIS and Project Garuda parts of Part XII (Indonesia).

Research cut-off: 7 November 2026. Compiled by Hark's research agent under FRAMEWORK.md rules.

Companion dataset: `data/cbdc_status.csv` (status classification and adoption figures, one row per metric, with source URL, evidence tag and limitation).

Evidence tags: [V] = the page was fetched and the claim appears on it. [E] = from a search excerpt or established knowledge; not opened. Claim types are marked where they matter: (fact), (empirical finding: author year), (interpretation: school/author), (normative), (counterfactual).

Status vocabulary used strictly throughout (BRIEF Part X.B): **research** (papers, consultations, design work, no live system); **experimentation** (proofs of concept or sandbox tests, usually simulated or tiny real value); **pilot** (real money, real users, restricted scope or geography, no general launch decision); **legislative preparation** (a bill or legal mandate is being written or negotiated); **issued** (formally launched as general-use central bank money). One extra category is needed in 2026: **prohibited by statute**.

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## Key findings

1. **Only three of the ten jurisdictions in the brief have actually issued a retail CBDC: the Bahamas (Sand Dollar, October 2020), Nigeria (eNaira, October 2021) and Jamaica (JAM-DEX, 2022).** All three remain tiny relative to cash. The Sand Dollar stood at BSD 2,820,335 on 31 July 2026, 0.39% of notes and coins in circulation [V]. JAM-DEX issue was JMD 294.7 million at end-July 2026 [V]. Nigeria's central bank itself now concedes that "adoption has been slow" and that the eNaira has processed only about NGN 22 billion (about USD 16 million) in transactions since launch [V, quoted via TechCabal]. (fact)

2. **China and India run the two largest pilots, and neither is a formal general launch.** China's e-CNY reached about 4.3 billion cumulative transactions worth about CNY 31 trillion by end-September 2026 [V]. India's e-rupee in circulation fell 24% in a year, from INR 1,016.5 crore (31 March 2025) to INR 771.7 crore (31 March 2026), about 0.02% of banknote value [V]. (fact)

3. **China has quietly changed what the e-CNY is.** From 1 January 2026, e-CNY balances held in commercial bank wallets are treated as bank deposit liabilities, pay interest under deposit-rate rules, are covered by deposit insurance and count in the reserve-requirement base [V]. The People's Bank of China governor described this as a move "from digital cash to digital deposit money" [V]. (fact) In balance-sheet terms the main retail e-CNY is now closer to a tokenised bank deposit backed by required reserves than to a direct central-bank liability held by the public. (interpretation: this report)

4. **The euro area is in legislative preparation, not issuance.** The Council adopted its position on 17 December 2025; the European Parliament voted 416-169-22 on 9 July 2026 to open trilogues; negotiations had met at least five times by late October 2026 [V]. The ECB says it would be ready for a potential

first issuance in 2029 if the regulation is adopted in 2026, with a 12-month pilot with 36 payment service providers from the second half of 2027 [V]. Nothing is issued and no holding limit has been set. (fact)

5. **The EUR 3,000 figure is an illustrative upper bound of an ECB technical exercise, not a decided holding limit.** At Parliament's request the ECB tested hypothetical limits of EUR 500 to EUR 3,000 across 2,025 banks. At EUR 3,000, ignoring deposit inflows from the decline of cash, the aggregate liquidity coverage ratio would fall from 166% to 163%; in an extreme flight-to-safety scenario 13 banks (0.3% of banking assets) would reach 100% LCR and 9 (0.1%) would risk falling below it. The ECB states "the figures are illustrative and do not represent a decision" [V]. (empirical finding: ECB 2025/2026 model estimate)

6. **The United States has moved from "no CBDC" by executive order to "no Fed CBDC" by statute, but only until 31 December 2030.** The 21st Century ROAD to Housing Act (P.L. 119-101) became law without the President's signature on 11 July 2026; it bars the Federal Reserve Board and Reserve Banks from issuing or creating a CBDC directly or through intermediaries, and sunsets on 31 December 2030 [V, via Cooley legal summary]. The broader Anti-CBDC Surveillance State Act (H.R. 1919) passed the House 219-210 in July 2025 but has not been enacted [V]. At the same time the US has a statutory framework for private dollar stablecoins (GENIUS Act, P.L. 119-27, 18 July 2025) [V], and the New York Fed participates in wholesale tokenisation work (Project Agora) [V]. (fact)

7. **The United Kingdom has not decided.** The Bank of England and HM Treasury say a decision and design blueprint will come by end-2026; primary legislation would be required before any launch; the Bank's provisional view is that holding limits of GBP 10,000-20,000 would be appropriate at launch; the Bank confirmed "programmable money is not in consideration" [V]. (fact)

8. **Sweden, the earliest advanced-economy mover, is back at the research stage.** The e-krona technical pilot ended in 2023; a 2023 state inquiry found insufficient social need; in 2026 the Riksbank asked the Riksdag to prepare legislation in case a digital euro is issued and estimated implementation would take at least five years [V/E]. (fact)

9. **Indonesia's Digital Rupiah (Project Garuda) is at the experimentation stage, wholesale first.** Milestones: white paper (30 Nov 2022), consultative paper on a wholesale cash ledger (31 Jan 2023), proof of concept (Jul 2023-Aug 2024), PoC report (13 Dec 2024) [V]. On 28 September 2026 the Governor told DPR Commission XI that development had entered an experimental stage for cross-border transactions and that BI had obtained a DSN-MUI fatwa on the permissibility of the central bank's digital currency and stablecoin [V, ANTARA]. No issuance date has been published. (fact)

10. **Wholesale projects are moving faster than retail ones, and they are where the geopolitics now sits.** Project Agora (BIS/IIF, seven central banks including the New York Fed and Bank of England, 40+ firms) published a prototype report on 27 May 2026 showing atomic cross-border settlement with tokenised reserves and tokenised deposits, and is moving to real-value tests [V]. mBridge is now run by its member central banks (China, Hong Kong, Thailand, UAE, Saudi Arabia) after the BIS handed it over in October 2024 [V]. Switzerland's Helvetia pilot has provided live wholesale CBDC on the SIX Digital Asset Platform since December 2023 and will run until at least June 2028 [V]. (fact)

11. **The disintermediation risk is real in principle but small under the designs actually proposed.** Every advanced-economy design on the table (euro area, UK) is unremunerated, capped and distributed by banks; the principal stress channel is a run in a crisis, which caps and waterfall links are designed to blunt. (interpretation: consensus of BIS/ECB/BoE work, contested by some academics who argue caps protect bank profitability rather than stability; see Disputes.)

12. **Privacy commitments are design choices, not technical inevitabilities.** The ECB, the European Parliament and the Bank of England have each committed to data minimisation and to the central bank not seeing personal identities; the Bank of England rules out "programmable money" while allowing user-chosen "programmable payments" [V]. China's design is account-based with "controllable anonymity" and real-name tiers [E]. Whether a CBDC is more or less private than bank money depends on law and architecture, not on the label. (interpretation)

13. **On monetary power, CBDCs as designed in 2026 are mainly an evolution of payment infrastructure, not a structural revolution** - with two exceptions that would be structural: a remunerated,

uncapped retail CBDC (not proposed anywhere among the big economies), and multi-CBDC cross-border platforms that bypass correspondent banking and dollar clearing (mBridge, CBETS). (interpretation: this report)

14. **Indonesia's retail payments are already highly digital without a CBDC.** QRIS reached 70.6 million users and 49.4 million merchants in Q3 2026, with transaction volume up 75.8% year on year; BI-FAST processed 1,631 million transactions worth Rp 4,049 trillion in Q3 2026 [V]. Meanwhile currency in circulation still grew 11.77% year on year to Rp 1,341 trillion [V]. The strongest case for a Digital Rupiah is therefore wholesale (tokenised securities settlement, monetary operations, cross-border), not retail inclusion. (interpretation)

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# 0. Method, scope and how to read this file

## 0.1 Sources and verification

Every CBDC status in this report was checked against an official or primary page fetched on 6-7 November 2026 where possible: ECB speeches and press releases, the European Parliament Legislative Train, the Chinese government portal (gov.cn), the Bank of England, the Reserve Bank of India Annual Report 2025-26, the Central Bank of The Bahamas monthly statement of assets and liabilities, the Riksbank, the Swiss National Bank, the BIS Innovation Hub, Bank Indonesia press releases, and the US Congress (CRS) and congressional offices. Where only a newspaper or law-firm summary was available (Jamaica, Nigeria's PSV 2028, the US housing act), the claim is tagged [V] with the intermediary named, because the page was opened and the claim is on it, but the limitation is stated.

## 0.2 What counts as a "CBDC"

A CBDC is a digital form of money, denominated in the national unit of account, that is a **direct liability of the central bank** (fact; standard BIS/CPMI definition, CPMI-MC 2018 [E]). This excludes: commercial bank deposits (liability of a bank), e-money and wallet balances (liability of a licensed issuer), stablecoins (liability of a private issuer, usually backed by assets), and cryptoassets such as bitcoin (no issuer liability at all).

Two 2026 developments blur this definition and must be flagged:

- China's January 2026 reclassification makes wallet-held e-CNY a **bank** deposit liability [V]. A strict reader would say the retail e-CNY is now a hybrid: central-bank money at the wholesale layer (operators hold reserves) and bank money at the retail layer.
- Switzerland's wholesale CBDC is, in the SNB's words, "in both economic and legal terms ... a representation of sight deposits held at the SNB in a technically different form" [V]. Wholesale CBDC is therefore not new money in substance; it is reserves on a new ledger.

## 0.3 What the evidence can and cannot say

Adoption data for CBDCs are thin, inconsistent and often reported as cumulative flows (transactions since launch) rather than stocks (balances outstanding) or active users. Cumulative transaction values are not comparable with M1 or with currency in circulation. Where possible this report uses **stocks reported on a central-bank balance sheet** (Bahamas, India, Jamaica) because they are the cleanest measure of how much CBDC people actually hold.

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# Part A - CBDC fundamentals

## A.1 Definition and the liability test

(fact) Money in a modern two-tier system comes in three layers:

| Layer | Issuer | Holder | Example |
|---|---|---|---|
| Central bank money, physical | Central bank | Public | Banknotes and coins |
| Central bank money, digital, wholesale | Central bank | Banks and some FMIs | Reserves (settlement balances) |
| Commercial bank money | Commercial bank | Public and firms | Current and savings deposits |
| (Private non-bank money) | E-money institution, stablecoin issuer | Public | GoPay/OVO balances, USDC |

A **retail CBDC** adds a fourth cell: central bank money, digital, held by the public. A **wholesale CBDC** is a new technical form of the second row (reserves), usually on a distributed ledger, often accessible to a wider set of financial institutions than classic RTGS.

The liability test matters because it determines credit risk and convertibility. A banknote and a retail CBDC carry no credit risk in domestic currency (the central bank cannot default on its own liabilities in its own currency in the operational sense, though it can lose value through inflation). A bank deposit carries the bank's credit risk, mitigated by deposit insurance (in Indonesia up to Rp 2 billion per depositor per bank under LPS [E]). A stablecoin carries the issuer's credit and liquidity risk, mitigated by reserve rules (in the US, 1:1 reserves under the GENIUS Act [V]).

## A.2 Balance-sheet mechanics: what happens when someone converts a deposit into CBDC

The cleanest way to see what a CBDC does is a T-account. Take a household that converts 100 (any currency unit) of bank deposits into retail CBDC.

**Before conversion**

| Household - Assets | Household - Liabilities |
|---|---|
| Bank deposit 100 | |

| Commercial bank - Assets | Commercial bank - Liabilities |
|---|---|
| Reserves at central bank 150 | Deposits (household) 100 |
| Loans 850 | Other deposits 800 |
| | Equity 100 |

| Central bank - Assets | Central bank - Liabilities |
|---|---|
| Government bonds 1,000 | Banknotes 600 |
| | Reserves (banks) 400 |

**After conversion of 100 into CBDC**

| Household - Assets | Household - Liabilities |
|---|---|
| CBDC 100 | |

| Commercial bank - Assets | Commercial bank - Liabilities |
|---|---|
| Reserves at central bank 50 | Other deposits 800 |
| Loans 850 | Equity 100 |

| Central bank - Assets | Central bank - Liabilities |
|---|---|
| Government bonds 1,000 | Banknotes 600 |
| | Reserves (banks) 300 |
| | CBDC 100 |

(fact; accounting identity) The central bank's balance sheet does not grow; its liabilities change composition from reserves to CBDC. The commercial bank loses 100 of deposits and 100 of reserves. Its liquidity ratio falls (reserves/deposits from $150/900 = 16.7\%$ to $50/800 = 6.25\%$). This is exactly the same mechanism as a cash withdrawal - which is why central banks describe CBDC as "digital cash" and why the disintermediation debate is really a debate about **how much** and **how fast**, not about a new mechanism.

If the bank's reserves were insufficient, it would replace them by borrowing from the central bank against collateral:

| Commercial bank - Assets | Commercial bank - Liabilities |
|---|---|
| Reserves 150 (restored) | Other deposits 800 |
| Loans 850 | Central bank borrowing 100 |
| | Equity 100 |

| Central bank - Assets | Central bank - Liabilities |
|---|---|
| Government bonds 1,000 | Banknotes 600 |
| Loans to banks (collateralised) 100 | Reserves 400 |
| | CBDC 100 |

Now the central bank's balance sheet has grown by 100 and it has a new credit exposure to banks. This is the channel that critics (e.g., Bindseil 2020 [E]) and the ECB's own model analysis [V] discuss: a large shift into CBDC would turn the central bank into a large lender to banks, raising questions about collateral, pricing and who decides credit allocation (see Part C).

## A.3 Retail versus wholesale

| Feature | Retail CBDC | Wholesale CBDC |
|---|---|---|
| Holders | Households, firms | Banks, FMIs, sometimes non-bank PSPs |
| Closest existing analogue | Banknotes | Central bank reserves |
| Main policy purpose | Public access to central bank money as cash declines; payment competition; inclusion; sovereignty | Settlement of tokenised assets (DvP), cross-border PvP, 24/7 settlement, programmability |
| Main risk debated | Disintermediation, privacy, cost, low adoption | Fragmentation, governance of shared ledgers, legal finality |
| Examples in this report | Sand Dollar, eNaira, JAM-DEX, e-CNY, e-rupee, digital euro (proposed), digital pound (proposed) | Helvetia, Agora (tokenised reserves), mBridge, Project Garuda w-Digital Rupiah, ECB Pontes |

## A.4 Account-based versus token-based

(fact; Auer and Boehme 2020, BIS Quarterly Review [E]) The distinction is about how ownership is verified:

- **Account-based**: the system checks the *identity* of the payer against an account register. Privacy depends on who sees the register. Most real designs, including the e-CNY ("account-based hybrid", in the words of PBOC Deputy Governor Lu Lei [V]) and the digital pound's centralised settlement ledger [V], are account-based at the core.
- **Token-based**: the system checks the *validity of the object* (a digitally signed token), like checking a banknote for counterfeits. Offline payments are typically token-like, with value stored on a secure device.

In practice, "account versus token" is less important than (a) who runs the ledger, (b) what data the ledger holds, and (c) whether intermediaries can see identities. The Bank of England's design keeps personal identifiers out of the core ledger [V].

## A.5 Direct, intermediated and hybrid architectures

| Architecture | Who holds the claim | Who runs customer relations | Examples |
|---|---|---|---|
| Direct (one-tier) | Central bank | Central bank | None at scale |
| Indirect / synthetic | Intermediary (backed by central bank reserves) | Intermediary | Arguably China after 1 Jan 2026 for bank-wallet e-CNY; Hong Kong's e-HKD early concepts [E] |
| Intermediated / two-tier ("platform model") | Central bank | Banks and PSPs (PIPs/ESIPs in the UK; PSPs in the euro area) | Digital euro (proposed), digital pound (proposed), Sand Dollar, JAM-DEX |
| Hybrid | Central bank holds the claim, intermediaries run services, central bank keeps a backup copy of holdings | Intermediaries | e-CNY original design [E] |

The ECB states plainly that "the ECB would not provide customer-facing services. Banks and other regulated payment service providers would distribute the digital euro" [V]. The Bank of England calls its approach a "platform model" with the digital pound "acting as shared infrastructure rather than a single product" [V].

## A.6 Two-tier monetary systems and the "singleness of money"

(interpretation: ECB, BoE, BIS) The policy rationale offered in Europe and the UK is not that bank money is failing but that, as cash use declines, the public could lose its only direct claim on the central bank, and with it the everyday anchor for the par convertibility of private money. Cipollone: "trust in commercial bank money rests on its convertibility at par into central bank money. This convertibility ensures the singleness of money" [V].

A counter-view appears even inside the Bank of England's own academic advisory group: "retail CBDC is not strictly required to preserve uniformity", because what really anchors par value is wholesale convertibility via reserves; retail CBDC "may play a valuable supporting role" [V]. (interpretation: Milne, Skeie, Niepelt subgroup) This is a genuine and unresolved disagreement (see Disputes).

## A.7 Offline payments

Offline capability is a design goal for the digital euro (Parliament's position includes both online and offline digital euro [V]), for the digital pound (BoE offline payments design note, 2025 [V, referenced]), and for the e-CNY ("on/off-chain, online/offline" [V]). India's RBI is testing offline e-rupee [E]. Offline matters for three reasons: resilience when networks or power fail, inclusion in poor connectivity areas, and privacy (offline transactions can be designed so that only payer and payee see them, as with cash). The ECB has said offline digital euro would offer cash-like privacy, with data known only to payer and payee [E, ECB privacy page; not re-fetched in this run].

### A.8 Interoperability

Interoperability means a CBDC can be paid into and out of existing accounts and schemes. The digital euro design includes "waterfall" (payments beyond the balance drawn from a linked bank account) and "reverse waterfall" (inflows above the limit swept to the bank account) [V]. The ECB has signed agreements to reuse European card and API standards (European Card Payment Cooperation, nexo standards, Berlin Group) [V]. The Bank of England has published an interoperability design note (2025) [V, referenced in AAG minutes]. In cross-border settings, interoperability is the whole point of Agora and mBridge.

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## Part B - International developments (strict status classification)

### B.0 Summary table

| Jurisdiction | Instrument | Strict status (Nov 2026) | Key date | Best adoption measure | Tag |
|---|---|---|---|---|---|
| Bahamas | Sand Dollar | Issued | Launched 20 Oct 2020 | BSD 2.82m in circulation, 31 Jul 2026 (0.39% of notes+coins) | V |
| Nigeria | eNaira | Issued; being repositioned | Launched 25 Oct 2021; PSV 2028 June 2026 | about NGN 22bn cumulative transactions (CBN) | V (via TechCabal) |
| Jamaica | JAM-DEX | Issued | 2022 | JMD 294.7m CBDC issue, end-Jul 2026 | V (via Jamaica Observer) |
| China | e-CNY | Pilot (very large) | Pilots since 2019-20; deposit reclassification 1 Jan 2026 | 4.3bn cumulative transactions, about CNY 31tn, end-Sep 2026 | V |
| India | e-rupee | Pilot | Wholesale 1 Nov 2022, retail 1 Dec 2022 | INR 771.7 crore outstanding, 31 Mar 2026 (-24% y/y) | V |
| Euro area | Digital euro | Legislative preparation (+ technical preparation; pilot planned H2 2027) | Council 17 Dec 2025; EP 9 Jul 2026 | Not issued | V |
| United Kingdom | Digital pound | Research / design (decision due end-2026) | Consultation 2023 | Not issued | V |
| Sweden | e-krona | Research (pilot ended 2023) | Riksbank letter 7 Oct 2026 | Not issued | V |
| Indonesia | Digital Rupiah | Experimentation (stage 3, cross-border) | PoC report 13 Dec 2024; DPR 28 Sep 2026 | Not issued | V |

| Jurisdiction | Instrument | Strict status (Nov 2026) | Key date | Best adoption measure | Tag |
|---|---|---|---|---|---|
| United States | Fed CBDC | Prohibited by statute to 31 Dec 2030; prohibited by EO | P.L. 119-101, 11 Jul 2026 | Not issued | V (via Cooley) |
| Switzerland (wholesale) | Helvetia wCBDC | Pilot (live wCBDC) | Since Dec 2023, to at least Jun 2028 | 10 eligible institutions | V |
| BIS/IIF (wholesale) | Agora | Experimentation, moving to real-value tests | Report 27 May 2026 | about CHF 800k in 30 test transactions (Jul 2026) | V/E |
| PBOC, HKMA, BOT, CBUAE, SAMA | mBridge | Pilot / MVP run by members | BIS exit Oct 2024 | 4,047 transactions, CNY 387.2bn (to Nov 2025) | V/E |

## B.1 China - e-CNY

**Status: pilot (large-scale).** (fact) The PBOC has never announced a nationwide "launch" with an end to pilot status; official communication continues to describe pilot regions and the "pilot" programme, while volumes are now large.

**Adoption data.**

- Cumulative to end-September 2026: about 4.3 billion transactions worth about CNY 31 trillion [V, China Daily, 28 Oct 2026]. Limitation: cumulative gross flows since the pilot began; not a stock.
- September 2026 alone: 99.67 million transactions, CNY 1.87 trillion, up 12% (volume) and 123% (value) year on year [V]. Average transaction value in September 2026 was therefore about CNY 18,760, far above typical retail purchases (calculation from the two figures). (interpretation: this suggests heavy corporate, government or interbank-adjacent use, not mainly consumer spending; unverified which segment drives value.)
- Cumulative to end-November 2025: 3.48 billion transactions, CNY 16.7 trillion [V, gov.cn]. Value therefore almost doubled in ten months.
- Wallets: about 230 million personal wallets and 18.84 million institutional wallets opened, end-November 2025 [E, Tsinghua PBCSF via 21jingji]. Limitation: wallets opened, not active users.
- Outstanding e-CNY balance: **unverified** in this run. PBOC has in the past included e-CNY in M0; a current stock figure was not found on an opened official page.

**The 2026 redesign.** The gov.cn notice of 29 December 2025 [V] states that from 1 January 2026:

- e-CNY held in commercial bank wallets counts as a deposit liability of the bank;
- it earns interest under deposit-interest regulations;
- it is covered by deposit insurance;
- it enters the reserve-requirement base;
- non-bank payment institutions offering e-CNY must hold 100% reserves.

Governor Pan Gongsheng described the shift as from "digital cash" to "digital deposit money"; Deputy Governor Lu Lei described the system as an "account-based hybrid" design [V, China Daily]. New infrastructure named: the Digital Currency Electronic Payment System (DCEPS), an e-CNY SMART Contract framework, and CBETS for cross-border settlement, which signed its first 26 direct participants in June 2026 [V]. ICBC reported handling over CNY 11 trillion in e-CNY over six years [V].

T-account of the post-2026 retail structure (illustrative, CNY 100 held in a bank wallet):

| Household |
|---|
| Assets: e-CNY wallet balance at Bank A 100 |

| Bank A - Assets | Bank A - Liabilities |
|---|---|
| Required reserves at PBOC (share r of 100)<br>Other assets (100 - r) | e-CNY deposit liability 100 |

| Non-bank PSP - Assets | Non-bank PSP - Liabilities |
|---|---|
| Reserves at PBOC 100 (100% requirement) | e-CNY wallet balances 100 |

(interpretation: this report) The bank-wallet e-CNY now looks like a tokenised, interest-bearing, insured bank deposit on a central-bank-run rail; the non-bank wallet looks like a narrow-bank or fully reserved stablecoin. Both preserve two-tier banking and reduce the disintermediation risk that an unremunerated direct central-bank liability would pose. They also remove the most cash-like feature of the original design.

**Privacy model.** The PBOC's stated principle since its 2021 white paper is "controllable anonymity", with tiered wallets: small-value wallets openable with a phone number, higher tiers requiring full real-name verification [E, PBOC 2021 white paper]. Limitation: the 2021 white paper was not re-fetched; the 2026 design notice does not discuss privacy tiers.

**Cross-border.** mBridge processed 4,047 transactions worth CNY 387.2 billion to end-November 2025, about 95.3% of it in e-CNY [E]. CBETS is a separate PBOC cross-border system [V].

## B.2 Euro area - digital euro

**Status: legislative preparation, with ECB technical preparation running in parallel.** (fact)

Timeline [V unless marked]:

- ECB investigation phase 2021-2023; preparation phase from 1 November 2023 (ECB) to 30 October 2025.
- 28 June 2023: Commission proposal for a regulation (procedure 2023/0212(COD)) [E date; procedure number V].
- 30 October 2025: ECB Governing Council decided to move to the next phase; if legislation is adopted in 2026, pilot from 2027 and readiness for potential first issuance in 2029; development cost about EUR 1.3 billion to first issuance, operating cost about EUR 320 million per year from 2029 [V per notes from ECB press release].
- 17 December 2025: Council negotiating position: ECB sets holding limits within a ceiling defined by the Council; free basic services; compensation regime with capped fees during a transition, then cost-based [V, EP Legislative Train].
- 23 June 2026: ECON committee report adopted 43-14-1 (rapporteur Fernando Navarrete Rojas, EPP) [V].
- 9 July 2026: plenary confirmed the negotiating mandate 416-169-22 [V].
- 14 July 2026: ECB selected 36 PSPs for the pilot from more than 50 applicants [V, ECB speech footnote].
- 21 September 2026: Eurosystem launched Pontes (wholesale settlement of tokenised assets in central bank money) [V].
- Late October 2026: trilogue teams had met five times [V]; a sixth trilogue was scheduled for 5 November 2026 [E].

Parliament's position [V]: online and offline digital euro; distribution by all PSPs; free basic services; capped individual holdings; cash-access obligations; data minimisation, privacy by design and privacy-enhancing technologies; roll-out over at least 24 months.

Main open issues [V]: compensation model for PSPs (Council's national exemptions versus Parliament's "no worse off" principle), the funding of distribution, and holding limits, which "had not yet been discussed at political level".

Design commitments repeated by the ECB in October 2026 [V]: no remuneration; individual holding limits; waterfall and reverse-waterfall links to bank accounts; ECB provides no customer-facing services; Eurosystem charges no scheme or processing fees; legal tender status; pan-European acceptance. Cash remains legal tender.

Stated motivation (interpretation: ECB): "Two-thirds of euro area card payments rely on international schemes. Of the 21 euro area countries, 13 have no domestic card scheme" [V]. The digital euro is framed as sovereignty and competition infrastructure, as much as a cash substitute.

## B.3 United States

**Status: prohibited by statute (through 31 December 2030) and by executive order.** (fact)

- 23 January 2025: Executive Order "Strengthening American Leadership in Digital Financial Technology" prohibits agencies "from undertaking any action to establish, issue, or promote CBDCs" and terminates any agency plans [V via TFCNN quotation of Section 5; the White House page itself was fetched earlier in this stream but the CBDC text was truncated - re-check recommended].
- 17 July 2025: House passed H.R. 1919, Anti-CBDC Surveillance State Act, 219-210 [V, Emmer press release].
- 18 July 2025: GENIUS Act signed, P.L. 119-27, creating a federal regime for payment stablecoins: 1:1 permitted reserves (cash, insured deposits, short T-bills, T-bill repos, government MMFs, central bank reserves), no interest paid by issuers to holders, BSA/AML coverage, holder priority in bankruptcy, not securities or commodities, not federally insured; banks permitted to issue tokenised deposits [V, CRS IN12553, updated 20 Aug 2026].
- December 2025: the final FY2026 NDAA dropped the anti-CBDC provision that the House version had carried [E, The Hill].
- 29 April 2026: House attached H.R. 1919 to a FISA-related bill (S. 1318) and sent it to the Senate [V]; the Senate later rejected a motion to proceed 47-52 [E].
- 1 June 2026: the Senate Banking Committee's substitute for the CLARITY market-structure bill reportedly struck the anti-CBDC title [E].
- 11 July 2026: the 21st Century ROAD to Housing Act (P.L. 119-101) became law without the President's signature. It bars the Federal Reserve Board and any Federal Reserve bank from issuing or creating a CBDC or "any digital asset substantially similar", directly or through intermediaries; defines CBDC as a dollar digital asset that is US currency, a direct Fed liability and widely available to the public; sunsets 31 December 2030; does not foreclose future congressional authorisation [V, Cooley; TFCNN gives section reference Title XI Sec. 1101 and an exemption for "open, permissionless, and private" dollar currency - V via TFCNN, statute PDF not opened].

(interpretation: Cooley) "The question of a US CBDC is deferred rather than resolved." The US has therefore chosen a clear path: no retail CBDC, regulated private stablecoins and tokenised bank deposits for retail digital dollars, and participation in wholesale tokenisation via the New York Fed in Agora [V].

## B.4 United Kingdom - digital pound

**Status: research / design phase; no decision; primary legislation would be needed.** (fact)

- 2023: BoE-HMT consultation; four commitments in response, including that Parliament would decide on any launch through primary legislation, that the Bank and Government would not access personal data via the core infrastructure, and that the state would not program the digital pound [V partially; fourth commitment text truncated on fetch].
- 2025: design notes on product strategy, interoperability, alias service and offline payments [V, AAG minutes].
- January 2026 (published 2 April 2026): CBDC Academic Advisory Group minutes record the Bank's "provisional view ... that holding limits for the digital pound in the GBP 10-20k range are appropriate for launch, with scope for recalibration"; "remuneration is not part of the proposed digital pound design"; "programmable money is not in consideration"; "a centralised settlement ledger remains the current proposal" [V].
- 4 March 2026 progress update (page updated 21 September 2026): no decision has been made; design phase ends in 2026; blueprint, joint assessment and decision "later this year"; Digital Pound Lab Phase 2 ran to July 2026 [V].
- 21 October 2026: Treasury answer to a parliamentary question: decision by end-2026 [E].

- 12 October 2026: a BoE deputy governor reportedly said there is no urgent need for a retail digital pound, pointing to tokenised deposits and sterling stablecoins [E, The Paypers via TFCNN].

(interpretation) The UK is drifting from "likely to be needed" (2023 language) towards "keep the option open", with growing emphasis on tokenised deposits and regulated sterling stablecoins.

## B.5 India - e-rupee (e-INR)

**Status: pilot (retail and wholesale).** (fact)

- Wholesale pilot from 1 November 2022, retail pilot from 1 December 2022 [E, RBI press releases].
- e-rupee in circulation: INR 771.7 crore on 31 March 2026 versus INR 1,016.5 crore on 31 March 2025 [V, RBI Annual Report 2025-26, ch. VIII, para VIII.12]. Change: -24.1%. As a share of banknotes in circulation (INR 41,23,995 crore, 31 March 2026 [V, Table VIII.1]) the e-rupee is about 0.019%. Limitation: the RBI paragraph gives a single figure "e-rupee in circulation"; split between retail and wholesale not stated on the fetched page (wholesale balances are reportedly near nil due to auto-redemption [E]).
- Earlier: INR 16.39 crore (FY23) and INR 234.12 crore (FY24) [E].
- April 2026: Deputy Governor T. Rabi Sankar cited about 1 crore (10 million) users and about 15 crore (150 million) transactions worth about INR 34,000 crore [E].
- Features under test: programmability for targeted transfers (e.g., DBT), offline payments, interoperability with UPI QR codes, planned cross-border pilots [E].
- The same RBI chapter reports a household payments survey finding "continued strong preference for usage of cash" [V].

(interpretation) India is the clearest case of a CBDC competing against an extremely successful fast-payment system (UPI). The fall in outstanding e-rupee in FY2025-26 suggests that, absent incentives, users see little reason to hold it.

## B.6 Nigeria - eNaira

**Status: issued (launched 25 October 2021); being repositioned toward infrastructure and government payments.** (fact)

- CBN Payments System Vision 2028, unveiled 1 June 2026, states the eNaira has "millions of wallets" and has processed about NGN 22 billion (about USD 16 million) in transactions; "Adoption has been slow, barriers include limited stakeholder engagement and buy-in in design and implementation, limited adoption and integration drive, limited resources and capacities for retail CBDC implementation . . . " [V, quoted in TechCabal 9 Jun 2026; PSV 2028 PDF at cbn.gov.ng not opened].
- PSV 2028 positions the eNaira within cross-border payments, open banking and digital identity priorities rather than as a stand-alone wallet [V, TechCabal].
- IMF (2023) found about 98.5% of eNaira wallets inactive in the early period [E, IMF WP/23/104]. CBN Financial Stability Report (June 2024): 28.46 million customers onboarded but about 197,041 active wallets [E]. In Q1 2024 the eNaira equalled about 0.36% of currency in circulation [E].
- During the 2023 naira-redesign cash shortage, the eNaira did not become a meaningful substitute [V, TechCabal citing IMF WP].

(empirical finding: IMF 2023; CBN 2026) Issuing a CBDC does not create demand for it when bank apps, USSD and mobile money already serve users, and when onboarding requires identity documents the target unbanked population lacks.

## B.7 The Bahamas - Sand Dollar

**Status: issued (launched 20 October 2020)** - the world's first nationwide retail CBDC. (fact)

Balance-sheet evidence (Central Bank of The Bahamas, Statement of Assets and Liabilities, 31 July 2026) [V]:

| CBOB demand liabilities, 31 Jul 2026 | BSD |
|---|---|
| Notes in circulation | 690,648,004 |

| CBOB demand liabilities, 31 Jul 2026 | BSD |
|---|---|
| Coins in circulation | 36,786,629 |
| Sand Dollars in circulation | 2,820,335 |
| Commercial banks' deposits | 2,225,039,377 |
| Other demand liabilities | 213,536,737 |
| Total demand liabilities | 3,168,831,082 |

Sand Dollars equal 0.39% of notes and coins and 0.09% of total demand liabilities (calculation). Earlier 2026 monthly statements show BSD 2,977,658 (30 April) and BSD 2,468,049 (30 June) [E, search excerpts of the same official series]. Limitation: the stock is small and volatile; the 2024 Annual Report cites about 133,481 consumer wallets and 2,034 merchants [E].

(interpretation) After nearly six years the Sand Dollar is functioning, legally robust and almost unused relative to cash - a useful counterexample to both fears ("CBDCs will replace cash") and hopes ("CBDCs will transform inclusion").

## B.8 Jamaica - JAM-DEX

**Status: issued (2022).** (fact)

From the Bank of Jamaica as reported by the Jamaica Observer, 23 August 2026 [V]:

- CBDC issue JMD 294.7 million at end-July 2026 (JMD 290.7 million a month earlier).
- Transactions JMD 51.7 million in January-July 2026, versus JMD 19.3 million in all of 2025 and about JMD 3 million in 2024 (2026 seven-month value is 2.7 times all of 2025).
- About 320,000 subscribers.
- Only two wallet providers live; a third institution testing, wallets expected Q1 2027; BOJ expects "roughly about maybe 40 per cent of the points of sale across the island" to accept JAM-DEX by H1 2027.
- JAM-DEX was not used to distribute grants after Hurricane Melissa; BOJ is discussing disaster-relief use with Government and the World Bank.

BOJ Annual Report 2025 figures (305,026 registered users at end-2025; JMD 276.0 million minted; JMD 144.4 million held by the public) are [E]. A reported cost dispute with the Jamaica Bankers Association over integration is [E].

(interpretation) Jamaica shows that merchant acceptance and wallet-provider economics, not technology, are the binding constraint. Growth from a tiny base is real but the CBDC is around 0.1% of currency [E].

## B.9 Sweden - e-krona

**Status: research.** The technical pilot (2020-2023) is finished; no decision to issue; no legislative inquiry yet appointed. (fact)

- Project started 2017; pilot phases 1-4 (2021-2024 reports) in an "isolated test environment"; "The project ended in 2023, and the focus is now on monitoring the global development of digital currencies" [V, Riksbank e-krona page].
- March 2023: Payments Inquiry ("The State and Payments") "assessed then that there was insufficient social need for the Riksbank to issue an e-krona", while noting global changes may alter the assessment [V].
- 12 March 2026: Payments Report 2026 recommends an inquiry into the legislative amendments needed, to be set up once the digital euro regulation is finalised and the ECB decides to issue; an e-krona could, by agreement with the ECB, use the digital euro platform [E, Riksbank page excerpt].
- 7 October 2026 (published 15 October): letter to the Riksdag Finance Committee; under ch. 4 s. 9 of the Sveriges Riksbank Act (2022:1568) the Riksbank may issue digital means of payment only with Riksdag consent; "The Riksbank has not taken a position on whether a digital krona should be issued"; implementation would take at least five years [E, Riksbank PDF excerpt and statsskuld.se].
- Only 5% of Swedes used cash for their latest in-store purchase in 2025 (10% in 2023) [E, Riksbank survey via TokenPost].

## B.10 Indonesia - Digital Rupiah (Project Garuda)

**Status: experimentation (stage 3: cross-border experiments); wholesale first; not issued.** (fact)

Milestones [V unless noted]:

- 30 November 2022: White Paper "Project Garuda: Navigating the Architecture of Digital Rupiah"; two forms envisaged, w-Digital Rupiah (wholesale, for banks and FMIs) and r-Digital Rupiah (retail, distributed via wholesalers) [V, BI page].
- 31 January 2023: Consultative Paper Stage I, "Wholesale Rupiah Digital Cash Ledger" (immediate state: issuance, transfer, redemption of w-Digital Rupiah); public consultation 31 January-15 July 2023 [V].
- July 2023-August 2024: proof of concept on two permissioned DLT platforms; report published 13 December 2024 [V].
- December 2025: BI said 2026 experimentation would cover tokenised securities for monetary operations, followed by an end-to-end stage covering both wholesale and retail [E, Warta Ekonomi].
- 28 September 2026: in a working meeting with DPR Commission XI, the Governor said BI had obtained a DSN-MUI (National Sharia Council) fatwa on the permissibility of the digital currency and stablecoin being prepared by the central bank, and that Digital Rupiah development had entered an experimental stage for cross-border transactions [V, ANTARA 3 Oct 2026 op-ed by Lucky Akbar]. Search excerpts identify the Governor as Destry Damayanti and report BI weighing a CBDC versus a BI-issued stablecoin model [E].
- 27 August 2026: a deputy-governor candidate told the DPR the Digital Rupiah is "not urgent" [E, CNBC Indonesia].

Legal basis: Law No. 4 of 2023 (P2SK, financial sector development and strengthening) amended the BI Law so that the rupiah may take the form of paper, metal and digital rupiah, with BI as sole issuer [E; not opened in this run]. BSPI 2030 (Indonesia Payment System Blueprint 2030) lists five initiatives - Infrastructure, Industry, Innovation, International, and Rupiah Digital (the "4I-RD" framework) - to be implemented 2025-2030 [E/V from BI pages in notes].

ANTARA frames it as BI's position that the Digital Rupiah is a CBDC, "a direct claim on the central bank", "designed as a complement, not a substitute, for cash", distinct from crypto and stablecoins [V].

(interpretation) Indonesia's sequencing - wholesale first, tokenised securities and monetary operations next, cross-border experiments, retail last and undecided - matches the global pattern where wholesale use cases are clearer than retail ones. See Part G for the Indonesian institutional implications.

## B.11 Wholesale projects

**Project Agora (BIS + IIF).** Status: experimentation, moving to real-value testing. (fact) Participants: Bank of England, Federal Reserve Bank of New York, Banque de France (for the Eurosystem), Bank of Japan, Bank of Korea, Bank of Mexico, Swiss National Bank, and more than 40 private institutions; Bank of Canada joined in 2026 [V]. Findings of the 27 May 2026 report [V]: atomic, all-or-nothing multi-currency settlement using tokenised central bank reserves and tokenised commercial bank deposits is achievable with finality across seven jurisdictions; a layered architecture lets each central bank keep autonomy over its currency; privacy can be protected at balance and transaction levels; "tokenisation ... does not alter the legal characterisation of ... central bank reserves and commercial bank deposits"; conditional and always-on payments become possible. Real-value tests in July 2026: 28 banks and central banks, about CHF 800,000 in 30 transactions across 17 scenarios in CHF, EUR, GBP, JPY, KRW and USD, average about 80 seconds to settle [E, Redbridge; BIS project page cited by others]. China's central bank and banks are not involved [E, Reuters].

**mBridge.** Status: MVP/pilot operated by member central banks; BIS withdrew. (fact) Started 2021 (BIS Innovation Hub, Bank of Thailand, CBUAE, PBOC Digital Currency Institute, HKMA); real-value pilot 2022; Saudi Central Bank joined 2024; MVP mid-2024; "The BIS announced in October 2024 that it was handing the project over to the partners" [V]. Observers in 2024 included Bank Indonesia, the ECB, the IMF, the Reserve Bank of India and the New York Fed's innovation centre [V]. Volumes: 4,047 transactions, CNY 387.2 billion to end-November 2025, about 95% in e-CNY [E]. Macao reportedly joining in 2026 [E].

**Project Helvetia (SNB).** Status: pilot with live wholesale CBDC. (fact) Phases I-II (2020-2021, with BIS and SIX) were experiments; since end-2023 the SNB has issued wholesale CBDC on the SIX Digital Asset Platform; the project "will run until at least June 2028"; ten institutions are currently eligible (including UBS, Zuercher

Kantonalbank, Raiffeisen, Commerzbank and SIX SIS); the SNB also explores synchronised settlement via an RTGS link used in production by SeturionX; participation "does not ... constitute a commitment on the SNB's part to introduce wholesale CBDC on a permanent basis" [V].

**ECB Pontes and Appia.** Pontes, launched 21 September 2026, connects market DLT platforms with Eurosystem settlement in central bank money; full implementation with 24/7 service expected by mid-2028. Appia is a blueprint for an integrated European tokenised ecosystem due in 2028 [V].

(interpretation) Wholesale tokenisation has split into two families: **Western "unified ledger / tokenised deposits plus reserves"** (Agora, Pontes, Helvetia) that keeps commercial banks central, and **multi-CBDC central-bank-to-central-bank platforms** (mBridge, CBETS) that can bypass correspondent banks and dollar clearing. The second family is where geopolitical stakes concentrate (see Part E).

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# Part C - CBDC and banking

## C.1 CBDC and commercial bank deposits: substitutes, complements or both

(fact) A retail CBDC is a close substitute for the transactional part of bank deposits (current accounts) and for cash. It is a weak substitute for savings deposits unless it is remunerated. Every one of the three issued retail CBDCs (Bahamas, Nigeria, Jamaica) is unremunerated and capped by wallet tier; the digital euro and digital pound designs are unremunerated and capped [V]. China is the exception: since January 2026 bank-wallet e-CNY pays deposit interest, but it is a bank liability [V].

Theory gives four distinct results that the monograph should keep separate:

1. **Equivalence (interpretation/empirical model: Brunnermeier and Niepelt 2019 [E]).** If the central bank automatically lends back to banks the funding they lose when deposits move into CBDC, at the same terms, nothing real changes: the "pass-through" keeps bank lending constant. The result shows that disintermediation is a policy choice about central bank lending, not a law of nature.
2. **Market-power result (empirical model: Andolfatto 2021; Chiu, Davoodalhosseini, Jiang and Zhu 2023 [E]).** Where banks have deposit market power and pay below-competitive rates, an interest-bearing CBDC acts as an outside option that forces banks to raise deposit rates. Deposits and even lending can rise rather than fall, within a range of CBDC rates.
3. **Disintermediation result (model: Keister and Sanches 2023; Fernandez-Villaverde, Sanches, Schilling and Uhlig 2021 [E]).** If CBDC is attractive enough, it crowds out deposits; under some conditions the central bank becomes a "deposit monopolist" and has to intermediate credit itself or through banks, with efficiency and governance costs.
4. **Run result (Diamond-Dybvig logic applied to CBDC; ECB and BoE stress work [V]).** In normal times flows are small; in a crisis a risk-free, frictionless digital asset makes a run faster and larger than a run into cash.

(interpretation) The policy designs in Europe and the UK are built to sit in result 1's spirit (keep banks funded) while guarding against result 4 (caps, no interest, waterfall).

## C.2 Disintermediation in numbers: what the ECB actually estimated

The ECB's technical analysis for Parliament (October 2025, restated by Cipollone on 6 October 2026) is the most detailed public estimate by any major central bank [V]:

| Item | Value | Notes |
|---|---|---|
| Banks modelled | 2,025 | Significant and less significant institutions |
| Hypothetical holding limits tested | EUR 500 to EUR 3,000 per person | Illustrative, not a decision |
| Deposit inflow from digitalisation (cash to deposits) by 2034 | about EUR 127 billion | = 0.4% of banking assets, 1.5% of retail sight deposits |

| Item | Value | Notes |
|---|---|---|
| Aggregate effect, business-as-usual, limits up to EUR 3,000, counting the digitalisation inflow | No aggregate outflow | Inflow exceeds digital-euro outflow |
| Aggregate LCR at EUR 3,000, ignoring digitalisation inflow | 166% falls to 163% | Liquidity coverage ratio |
| Aggregate NSFR at EUR 3,000, ignoring inflow | 128% falls to 127% | Net stable funding ratio |
| Flight-to-safety scenario at EUR 3,000: banks reaching 100% LCR | 13 banks, 0.3% of banking assets | Extreme tail, no policy response assumed |
| Of which at risk of falling below 100% | 9 banks, 0.1% of assets | |
| Profitability effect (technical annex) | -9 to -18 basis points of ROE | Business-as-usual; profitability rises when digitalisation trend included [E, annex excerpt] |
| Bank investment cost | about 3.4% of significant banks' annual IT upgrade budgets over four years | [E, Cipollone Nov 2025 speech excerpt] |

Limitations: these are ECB model results under ECB assumptions about adoption and behaviour; the flight-to-safety scenario assumes no central bank response; the ECB itself says "these findings must be interpreted within the scope and assumptions of the analysis" [V].

**On the EUR 3,000 figure (verification required by the brief):** It is verified [V] that EUR 3,000 is the **upper end of a range of hypothetical limits (EUR 500-3,000) that the ECB modelled at the European Parliament's request**. It is not a proposal, a Council ceiling, or an agreed limit. The ECB had earlier (2023-2024) floated "around EUR 3,000" as an illustrative calibration [E]. The Council wants the ECB to set the limit within a Council-defined ceiling; Parliament's position on who sets the ceiling differs [V/E]; holding limits had not been discussed at political level in trilogues as of late October 2026 [V].

## C.3 Holding limits: the UK comparison

The Bank of England's provisional view is a **GBP 10,000-20,000** holding limit at launch, potentially relaxed later [V]. That is roughly four to seven times higher than the upper end of the ECB's modelled range (at an assumed GBP/EUR rate around 1.15-1.20 [E]). The BoE also published a Financial Stability Paper (2025) on holding limits for both systemic sterling stablecoins and a potential digital pound, using a "severe illustrative stress scenario" that AAG members questioned as possibly unrepresentative of real runs [V, AAG minutes].

(interpretation) The ECB-BoE gap reflects different starting points: the UK limit was originally set to allow most salaries to flow through a digital pound account; the euro-area range is designed for a "digital cash" use case with automatic sweeps. It also reflects political pressure: European banks lobbied hard for low limits [E], while one academic critic argues the low euro limit protects "bank funding costs more than it" protects stability [E, Helmich, Rosa and Roubini, Sept 2026].

## C.4 Remuneration

(fact) No major advanced-economy proposal remunerates retail CBDC. The BoE: "remuneration is not part of the proposed digital pound design" [V]. The ECB: "It would not be remunerated" [V].

The BoE academic subgroup argued that remuneration is "a key driver of adoption", that an unremunerated CBDC "may struggle to compete with potential interest-bearing stablecoins and tokenised deposits", and suggested tiered or state-contingent remuneration as "a more flexible tool than static limits" [V]. (interpretation: Levin, Romelli, Markose, Ibikunle) Tiered remuneration (zero or positive up to a threshold, penalty rate above) was first proposed by Bindseil (2020) [E] and is the main alternative to hard caps.

Note the interaction with the GENIUS Act: US payment stablecoin issuers may not pay interest, but "holders are not defined, and there is no restriction against exchanges paying interest to customers" [V, CRS]. A yield-bearing

private digital dollar can therefore emerge indirectly. If private digital money pays and public digital money does not, the public option may simply lose (interpretation).

## C.5 Run risk and financial stability

T-account: a run in a crisis, without and with a holding limit (illustrative, bank with 1,000 deposits).

**No limit: 30% of deposits flee to CBDC in days**

| Bank - Assets | Bank - Liabilities |
|---|---|
| Reserves 100 -> 0 | Deposits 1,000 -> 700 |
| Liquid securities 200 -> 0 (sold or repoed) | Central bank emergency borrowing 0 -> 0 |
| Loans 800 | Equity 100 |
| Shortfall: 300 outflow vs 300 liquid assets; next euro of outflow needs ELA | |

**Limit of 3% of deposits per head (illustrative, cap binds)**

| Bank - Assets | Bank - Liabilities |
|---|---|
| Reserves 100 -> 70 | Deposits 1,000 -> 970 |
| Liquid securities 200 | Equity 100 |
| Loans 800 | |

(interpretation) The cap converts an open-ended run channel into a bounded one. It does not eliminate runs into other safe assets (government bills, money market funds, stablecoins, deposits at "safe" banks), which already exist; Cipollone makes exactly this point: "people can already easily and instantly transfer their deposits from bank accounts to other entities, where - unlike with the digital euro - they may be remunerated, unlimited or denominated in foreign currencies" [V].

(empirical finding: 2023 US bank failures [E]) The March 2023 run on Silicon Valley Bank (about USD 42 billion requested in one day [E]) happened without any CBDC, through instant bank transfers to other banks and to Treasury MMFs. Digital run speed is already a property of the existing system.

## C.6 Central bank balance sheets

(fact; accounting) Issuing CBDC changes the composition of central bank liabilities; it expands the balance sheet only if the central bank lends to replace bank funding or if the public converts cash into CBDC and the central bank buys assets (see A.2).

(interpretation) Three balance-sheet issues follow:

1. **Seigniorage.** An unremunerated CBDC funds the central bank at zero cost, like banknotes. The ECB says digital euro costs (about EUR 1.3 billion to first issuance, about EUR 320 million per year operating) would be offset by seigniorage [V per ECB press release noted]. The BoE AAG subgroup noted that "an unremunerated currency could generate potentially significant backing asset income", which may itself influence launch decisions where central banks face losses after quantitative tightening [V].
2. **Collateral and credit allocation.** If CBDC drains bank funding, central bank lending against collateral replaces it, and the central bank's collateral framework then shapes which assets (and indirectly which borrowers) get cheap funding (interpretation: Bindseil 2020; Positive Money/new-economics advocates welcome this as a tool, while BIS and most central banks reject directing credit [E]).
3. **Exposure concentration.** In China's post-2026 model the reserve requirement on e-CNY and 100% reserves for non-banks keep the PBOC balance-sheet effect similar to reserves rather than to a large direct retail liability [V].

### C.7 Credit allocation

(interpretation) Under all designs actually proposed, commercial banks continue to originate credit; the CBDC changes funding mix at the margin. A structural change in credit allocation would require either (a) a large, remunerated, uncapped CBDC that shrinks deposits so much that the central bank must lend directly to the economy or decide which banks receive funding, or (b) a "sovereign money" or narrow-banking reform (Positive Money, Chicago Plan revisited by Benes and Kumhof 2012 [E]). Neither is on any official agenda in the jurisdictions studied.

### C.8 Monetary policy transmission

(interpretation: BIS/IMF literature [E]) Three possible effects:

- **Neutral to slightly stronger pass-through** if CBDC is unremunerated: it behaves like cash; the policy rate still works through bank funding and lending rates.
- **A new direct instrument** if CBDC is remunerated: the CBDC rate becomes a floor for deposit rates, potentially strengthening pass-through (Bordo and Levin 2017 [E]) and in theory enabling negative rates on CBDC if cash were abolished (a scenario no central bank in this study proposes; cash remains legal tender in every case).
- **Weaker control** if CBDC demand is volatile and large, complicating reserve management. The ECB explicitly calibrates limits to preserve "the smooth implementation and transmission of monetary policy" [V].

The US statute and House bill explicitly target this channel: H.R. 1919 would bar the Fed from using any CBDC to implement monetary policy [V, Emmer press release].

### C.9 The BIS and IMF view in brief

- **BIS Annual Economic Report 2021, ch. III** ("CBDCs: an opportunity for the monetary system") argued CBDCs could preserve the two-tier system and the singleness of money if designed with intermediaries [E].
- **BIS Annual Economic Report 2023, ch. III** proposed a "unified ledger" combining tokenised central bank money, tokenised deposits and tokenised assets [E]. Agora is the practical test of that idea [V].
- **BIS Annual Economic Report 2025, ch. III** argued stablecoins perform poorly on singleness, elasticity and integrity and that tokenised central bank reserves, tokenised deposits and government bonds on unified ledgers are the better path [E].
- **IMF** work (e.g., Soderberg et al. 2023, "How should central banks explore CBDC? A dynamic decision-making framework"; IMF CBDC Virtual Handbook) frames CBDC as a country-specific choice, stresses legal frameworks and cyber resilience, and documents adoption problems in issued retail CBDCs, including Nigeria [E].
- **Group of seven central banks + BIS (2020-2021) reports** set three foundational principles: "do no harm" to monetary and financial stability, coexistence with cash and other money, and promoting innovation and efficiency [E].

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## Part D - CBDC and individual freedom

The brief requires examining both legitimate public-policy benefits and potential abuses, without presuming that every CBDC is programmable or designed for transaction control. This section is organised by the brief's list.

### D.1 Financial privacy and transaction surveillance

(fact) Privacy in a CBDC is set by three choices: (1) what data the core ledger holds; (2) what intermediaries see and keep; (3) what legal limits govern access by the central bank, government and law enforcement.

Commitments on the record:

| Authority | Commitment | Tag |
|---|---|---|
| European Parliament (negotiating position) | Data minimisation, privacy by design, privacy-enhancing technologies; offline digital euro | V |
| ECB | Eurosystem would not be able to identify users from online transactions (pseudonymised data); offline payments with cash-like privacy, details known only to payer and payee | E (ECB privacy page, not re-fetched) |
| Bank of England | Bank and Government would not access personal data via the core infrastructure; "privacy from whom?" framing; core ledger holds no personal identifiers; intermediaries carry AML duties under strong regulation | V |
| Bank of England | "Programmable money is not in consideration" | V |
| US Congress (P.L. 119-101) | Ban on Fed CBDC, with an exemption for dollar currency that is "open, permissionless, and private" | V via TFCNN |
| PBOC | "Controllable anonymity"; tiered wallets; account-based hybrid | E (2021 white paper); V (account-based hybrid) |
| Bank Indonesia | Payment-system data regulation (PBI) being issued in Q4 2026 under the Personal Data Protection Act (UU PDP) as legal basis for data exchange | V (BI press release 21 Oct 2026) |

(interpretation) Compared with existing bank money, a well-designed intermediated CBDC need not reduce privacy: banks and card networks already see every card and transfer. Compared with cash, every online CBDC design reduces privacy, because some intermediary records the transaction. The honest comparison is therefore CBDC versus cards and bank transfers for online payments (similar or better, if the law limits access) and offline CBDC versus cash (close, but with device-level limits and caps).

(normative, labelled) Privacy protections written into statute are stronger than those written into central bank design notes, because design can change but statutory limits require legislative change. This is the core argument of both privacy advocates and US CBDC opponents, and it is a fair one.

## D.2 Programmable money versus programmable payments

(fact; BoE definitions) **Programmable money** means the money itself carries rules restricting how, where or when it can be spent (e.g., an expiry date, a merchant whitelist). **Programmable payments** means a user or service provider sets up conditional instructions (e.g., pay on delivery, split a bill, pay a subscription) using ordinary money.

- The BoE rules out programmable money and allows programmable payments chosen by users [V].
- The BoE academic subgroup warned that programmable money "could introduce economically distinct 'tiers' of money if not designed carefully" [V] - i.e., it would break the singleness of money.
- The ECB's innovation platform workstreams cover "conditional payments" and AI-enabled payments [V]; the ECB has said the digital euro would never be programmable money [E].
- China has released an e-CNY "SMART Contract framework" [V]; India's RBI tests programmability for targeted transfers [E]. These are the clearest cases of programmability being used for government transfers,

where money could be restricted to certain uses (as food stamps already are).

(interpretation) Restricted-purpose government transfers already exist without CBDCs (food vouchers, Indonesia's social assistance cards, US SNAP). Programmability makes them cheaper and more precise. The risk is scope creep: tools built for targeted benefits being applied to general money. The safeguard is legal: limit programmability to opt-in payments and named benefit programmes.

## D.3 Data ownership and access

(interpretation) In a two-tier model the user's transaction data are held by the intermediary, subject to data-protection law (GDPR in the EU, UK GDPR, UU PDP in Indonesia, China's Personal Information Protection Law). The central bank holds, at most, pseudonymised settlement records. The policy questions are: can law enforcement obtain data without a court order; can the government query the central ledger; is there a retention limit; can users port their data. None of the issued retail CBDCs studied here has published a comprehensive, statute-level answer to all four [unverified for Bahamas/Jamaica/Nigeria; not checked in this run].

## D.4 Cybersecurity and operational resilience

(fact) The BoE academic subgroup noted that low-cost transactions "broaden the attack surface", recommended throttling or cool-off measures, and a pathway to post-quantum cryptography [V]. Offline functionality is cited as a resilience feature [V]. A central ledger is a single point of failure in a way that distributed cash is not; conversely, a public CBDC rail can serve as a backup when private card schemes fail (ECB argument on dependence on non-European schemes [V]).

(empirical finding: Jamaica 2025-26 [V]) After Hurricane Melissa damaged banking infrastructure in western Jamaica, the BOJ relied on temporary branches and simplified due diligence in a "cash-dominant ecosystem"; JAM-DEX was not used for relief. This is a reminder that digital resilience depends on power and connectivity, and that cash remains the resilience instrument in disasters.

## D.5 Financial inclusion

(empirical finding) The inclusion promise of retail CBDCs is not supported by the evidence so far. Nigeria's identity requirements (BVN or NIN) excluded many unbanked people the eNaira aimed to reach [V, TechCabal]; India's e-rupee balances fell [V]; the Sand Dollar is tiny [V]. Inclusion gains in the countries studied came instead from fast-payment systems and QR standards (India's UPI [E], Brazil's Pix [E], Indonesia's QRIS: 70.6 million users, 49.4 million merchants, 97% of merchants MSMEs [V/E]).

## D.6 Legal protections, due process and institutional accountability

(normative, labelled; widely shared across schools) A legitimate CBDC regime should have: (1) a statute authorising issuance (the EU regulation, UK primary legislation, Sweden's Riksdag consent, Indonesia's P2SK amendment); (2) statutory privacy limits; (3) judicial authorisation for data access; (4) a ban or strict limits on programmable money; (5) cash preserved as legal tender; (6) parliamentary oversight of holding limits and fees.

How the jurisdictions score on (1) and (5), by verified sources:

| Jurisdiction | Statute authorising issuance? | Cash protected? |
|---|---|---|
| Euro area | Regulation under negotiation [V] | Parliament position includes cash-access obligations; ECB "cash remains legal tender" [V/E] |
| UK | Primary legislation required before launch [V] | Not tested in this run |
| Sweden | Riksdag consent required (Riksbank Act 4:9) [E] | Separate cash law debate (not checked) |
| Indonesia | P2SK 2023 recognises digital rupiah form [E] | Rupiah notes and coins remain legal tender (Currency Law 7/2011) [E] |

| Jurisdiction | Statute authorising issuance? | Cash protected? |
|---|---|---|
| US | Statute prohibits Fed CBDC to 2030 [V] | n/a |
| China | PBOC Law revision to cover digital RMB drafted 2020 [E]; status unverified | Cash remains legal tender [E] |

## D.7 Balanced assessment

| Legitimate benefits (evidence) | Potential abuses (mechanism) | Safeguard |
|---|---|---|
| Public digital means of payment as cash declines (Sweden 5% cash at POS [E]) | Surveillance if ledger holds identities and access is unrestricted | Statutory data minimisation; intermediary model; judicial access |
| Resilience and sovereignty against foreign card schemes (EU two-thirds card share [V]) | Programmable restrictions on spending | Ban programmable money; opt-in programmable payments only |
| Lower merchant costs (Eurosystem charges no scheme fees [V]) | Exclusion of people without ID or devices | Offline tiers; cards; cash preserved |
| Targeted transfers in disasters (Jamaica plans [V]) | Freezing of funds without due process | Same legal standards as bank accounts |
| Cross-border efficiency (Agora 80-second settlement in tests [E]) | Capital-control enforcement or sanctions evasion | Governance of multilateral platforms; FATF standards |

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## Part E - CBDC and monetary power

The brief asks whether CBDCs shift the balance of power among central banks, commercial banks, governments, fintech providers and citizens, and whether they are a structural change or an evolution of payment infrastructure.

### E.1 A framework: four margins of power

(interpretation: this report) Monetary power can be split into four margins:

1. **Issuance power** - who creates the money people use (today: mostly commercial banks via lending; see McLeay, Radia and Thomas 2014 [E]).
2. **Settlement power** - who runs the final settlement layer (central banks, via RTGS; increasingly also card schemes and big-tech wallets for retail).
3. **Infor-

###
E.2
Actor
by
actor

| Actor | Effect of CBDC as designed in 2026 (capped, unremunerated, intermediated) | Effect of a counterfactual remunerated, uncapped CBDC |
|---|---|---|
| Central banks | Gain a retail settlement layer and seigniorage on CBDC balances; little change in issuance power; more operational and reputational risk | |

(counterfactual) The structural scenario in the right-hand column is not proposed by any of the ten jurisdictions; China's interest-bearing e-CNY is a bank deposit, not a central bank account [V].

### E.3 International monetary power

(interpretation)
The
most
conse-
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tial
shift
is not
do-
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tic
but
cross-
border:

- **Multi-CBDC platforms** (mBridge, CBETS) let central banks settle trade and FX bilaterally in central bank money without correspondent banks or dollar clearing. mBridge volume to late 2025 was about CNY 387 billion, 95% e-CNY [E] - small relative to global FX turnover (BIS Triennial 2025 about USD 9.6

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### E.4 Structural change or infrastructure evolution? (interpretation: this report, weighing evidence)

- **Retail CBDC in advanced economies: evolution.** Caps, no interest, bank distribution and legal privacy limits make the digital euro and digital pound a public payment rail plus a cash-like instrument. They change payment market structure (scheme fees, sovereignty) more than monetary structure. - **Is-**

# Part F - Alternative digital monetary systems: comparison matrix

## F.1 Definitions used

- **CBDC (retail)**: digital central bank liability held by the public.
- **Wholesale settlement (reserves / wCBDC / RTGS)**: central bank liability held by banks and FMIs.
- **Existing electronic bank money**: deposits, transferred via cards, BI-FAST, UPI, SEPA, etc.
- **E-money**: prepaid balances issued by licensed non-banks (in Indonesia, server-based e-money such as GoPay, OVO, DANA, ShopeePay), usually with safeguarded float.
- **Tokenised deposits**: bank deposits recorded on a programmable ledger, still bank liabilities (Agora finding: tokenisation does not change legal character [V]).
- **Stablecoins**: private tokens claiming a fixed value, backed by assets (fiat-backed) or algorithms; in the US, "payment stablecoins" under the GENIUS Act [V].
- **Cryptocurrencies**: native tokens with no issuer liability (bitcoin, ether).

## F.2 The matrix

| Dimension | Retail CBDC | Wholesale settlement (reserves, wCBDC) | Bank money (deposits) | E-money | Tokenised deposits | Fiat-backed stablecoins | Cryptocurrencies |
|---|---|---|---|---|---|---|---|
| **Issuer / liability** | Central bank | Central bank | Commercial bank | Licensed e-money issuer | Commercial bank | Private issuer (bank subsidiary or non-bank) | None (protocol) |
| **Trust rests on** | State, central bank independence, law | Same | Bank solvency + deposit insurance + supervision + LOLR | Safeguarding of float + licensing | Same as deposits | Reserve quality, redemption rights, attestation, regulation | Protocol rules, decentralisation, market belief |
| **Governance** | Statute + central bank; legislature sets limits | Central bank; FMI rules | Banking law, supervisors | Payment regulators (BI in Indonesia) | Banking law; shared-ledger governance unresolved | GENIUS Act in US; MiCA in EU; varies | Developers, miners/validators; no legal governor |
| **Backing** | Central bank assets (but no redemption needed: it is the base money) | Same | Bank loans and securities; partial reserves | Float held in bank accounts / safe assets | Bank assets | Cash, T-bills, repos, MMFs (US 1:1 permitted reserves [V]) | None |
| **Convertibility at par into central bank money** | Is central bank money | Is central bank money | Yes, on demand, via bank (backed by insurance and LOLR) | Yes, via issuer redemption; regulated | Yes (deposit) | Contractual; can break in stress (secondary market depegs) | No par; floating price |

| Dimension | Retail CBDC | Wholesale settlement (reserves, wCBDC) | Bank money (deposits) | E-money | Tokenised deposits | Fiat-backed stablecoins | Cryptocurrencies |
|---|---|---|---|---|---|---|---|
| **Interest** | None in proposed designs; China bank-wallet e-CNY pays deposit rate (bank liability) | Reserves may be remunerated (IORB, deposit facility) | Yes on many deposits | Usually none | Possible | Issuer barred from paying in US; exchanges may pay [V] | None intrinsic; staking yields |
| **Privacy** | Set by law/design; intermediated; offline can be cash-like | Institutional, not personal | Bank sees everything; law limits sharing | Issuer sees everything | Bank + ledger participants | Public blockchains: pseudonymous but traceable; issuer can freeze | Pseudonymous, publicly traceable |
| **Financial stability profile** | Run channel in crises (mitigated by caps); no credit risk | Core of system stability | Run risk mitigated by insurance, LCR, LOLR | Small; run risk if float mismanaged | As deposits; possibly faster runs | Run and fire-sale risk on reserve assets; depeg risk; BIS: fails "singleness" test [E] | Price volatility; leverage; exchange failures (FTX 2022 [E]) |
| **Legal tender** | Yes in EU proposal [V]; yes for Sand Dollar/eNaira [E] | n/a | No (but universally accepted) | No | No | No | No (El Salvador experiment largely reversed 2025 [E]) |
| **Programmability** | Programmable payments possible; programmable money ruled out in UK [V] | Yes (atomic DvP/PvP) | Via APIs | Via app logic | Native | Native (smart contracts) | Native |

## F.3 What the matrix shows

(interpretation)

1. **Only CBDC and wholesale reserves are central bank money.** Everything else is a private claim whose par value depends on convertibility into central bank money. This is why central banks argue that tokenised finance needs a tokenised central bank settlement asset (Pontes, Agora, Helvetia) [V].
2. **Tokenised deposits are the incumbent's answer.** They keep bank credit creation and deposit insurance intact while adding programmability. Agora is designed around them [V], and the GENIUS Act explicitly permits banks to issue them [V].

3. **Stablecoins compete with deposits, e-money and CBDC simultaneously.** Their stability depends on reserve rules and redemption rights; the BoE's proposal to apply holding limits to "systemic stablecoins" as well as to a digital pound shows regulators view them as the same run channel [V].
4. **Cryptocurrencies are not money in the same functional sense** (no issuer, volatile unit, not a unit of account for prices in the jurisdictions studied). They are best treated as assets.

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# Part G - Indonesia: digital payments, QRIS, BI-FAST and the Digital Rupiah (BRIEF Part XII)

## G.1 Payment system data, Q3 2026

| Indicator | Value | Period | Definition | Source | Tag | Limitation |
|---|---|---|---|---|---|---|
| Digital payment transactions | 18.09 billion; +39.24% y/y | Q3 2026 | Mobile banking + internet banking + server-based e-money transactions | BI press release 21 Oct 2026 https://www.bi.go.id/en/publikasi/ruang-media/news-release/Pages/sp_2823326.aspx | V | Volume only; BI footnote 3 limits scope to mobile banking, internet banking and server-based e-money, so card payments are excluded |
| Mobile payments volume growth | +45.55% y/y | Q3 2026 | as above | same | V | Growth rate only |
| Internet payments volume growth | +14.26% y/y | Q3 2026 | as above | same | V | Growth rate only |
| QRIS transaction volume growth | +75.80% y/y | Q3 2026 | QR Code Indonesian Standard transactions | same | V | Level not given in release |
| QRIS users | 70.6 million | end-Q3 2026 | Registered users (consumers) | same | V | Users, not active users; may double-count across apps |
| QRIS merchants | 49.4 million | end-Q3 2026 | Registered merchants; about 97% MSMEs per Governor | same; ANTARA 26 Oct 2026 | V / E (97%) | Registered, not active |
| BI-FAST volume | 1,631 million; +31.78% y/y | Q3 2026 | Retail credit transfers via BI-FAST | same | V | Excludes intrabank transfers |

| Indicator | Value | Period | Definition | Source | Tag | Limitation |
|---|---|---|---|---|---|---|
| BI-FAST value | Rp 4,049 trillion; +33.91% y/y | Q3 2026 | as above | same | V | Average about Rp 2.48 million per transaction (calculation) |
| BI-RTGS value | Rp 53,817 trillion; -4.62% y/y; 2.79 million transactions | Q3 2026 | Large-value system | same | V | |
| Currency in circulation | Rp 1,341 trillion; +11.77% y/y | Q3 2026 | Notes and coins outside BI | same | V | Cash demand still rising |
| QRIS transactions, H1 2026 | 12.55 billion; about double H1 2025 | Jan-Jun 2026 | QRIS volume | BI news release 17 Sep 2026 (E); ANTARA op-ed 3 Oct 2026 (V) | V/E | **Value disputed**: ANTARA says Rp 1.12 quadrillion (+93.92% y/y); BI's 17 Sep 2026 release excerpt says about Rp 600.7 trillion. Unresolved; see Disputes |
| QRIS users / merchants, June 2026 | 65.77 million / 44.86 million (96.68% MSMEs) | end-Jun 2026 | as above | ANTARA 3 Oct 2026 | V | |
| Digital payments, August 2026 | 6.11 billion; +40.36% y/y; QRIS +67.22%; BI-FAST 549 million, Rp 1,346 trillion | Aug 2026 | as above | BI MPR Sept 2026 (E) and ANTARA (V) | V | |

(fact) BI's October 2026 policy package under BSPI 2030 [V]: QRIS cross-border sandboxes for Timor-Leste and Hong Kong corridors from Q4 2026; expansion of the Indonesia Credit Card (KKI); a PBI on payment-system data as legal basis for data exchange consistent with the Personal Data Protection Act; rules on cooperation between payment system providers and supporting providers.

(fact) BI plans to use QRIS transaction data as an input to MSME credit scoring [V/E, ANTARA 26 Oct 2026, Governor statement].

## G.2 What the data imply for a retail Digital Rupiah

(interpretation)

1. **Retail payment digitisation in Indonesia is being achieved through interoperability standards**

**on top of bank and e-money balances**, not through a new form of central bank money. QRIS (a QR standard) and BI-FAST (an instant-payment rail operated by BI) together deliver what retail CBDCs elsewhere promised: cheap, instant, interoperable payments with mass merchant acceptance.
2. **Cash demand is still rising** (+11.77% y/y [V]), so the "cash is disappearing" rationale used in Sweden and the euro area does not apply to Indonesia in 2026.
3. **The inclusion gap is literacy and trust, not rails.** BI cites a 2026 national survey with a financial literacy index of 69.57% against an inclusion index of 93.61% [E, BI 17 Sep 2026].
4. **A retail Digital Rupiah would compete with bank deposits, e-money and QRIS-linked wallets.** Under the experience of Nigeria and India, a retail CBDC launched into such an ecosystem without a distinct use case is likely to see low adoption (empirical finding by analogy; not a forecast).

### G.3 What a wholesale Digital Rupiah could mean for Indonesia's institutional structure

(interpretation)

- **Monetary operations and securities settlement.** BI's stated 2026 experiments with tokenised securities for monetary operations [E] point to a w-Digital Rupiah settling SRBI, SBN and repo transactions atomically (DvP) on a shared ledger. This could reduce settlement risk and collateral friction, as Helvetia aims to in Switzerland [V].
- **Cross-border.** Experiments for cross-border use [V] and mBridge observer status [V] suggest interest in faster rupiah-partner currency settlement, complementing LCT arrangements with Japan, South Korea, Malaysia, Singapore, Thailand, China and the UAE [V].
- **Bank role.** Under the w-Digital Rupiah/r-Digital Rupiah design, banks would be wholesalers distributing r-Digital Rupiah, preserving the two-tier structure [V, BI page].
- **Sharia compliance.** The DSN-MUI fatwa [V] removes one adoption barrier in a majority-Muslim economy; an unremunerated CBDC avoids interest (riba) questions by construction.
- **CBDC versus BI-issued stablecoin.** If BI chooses a "stablecoin-like" token on public or permissioned chains [E], the legal character (direct BI liability or not) and privacy model would need to be specified; this is **unverified** and should not be treated as decided.

### G.4 Who creates money in Indonesia, and would a CBDC change it?

(fact; standard) As elsewhere, most broad money (M2) in Indonesia is bank deposits created when banks lend; BI controls the price and quantity of reserves (BI-Rate 5.75%, deposit facility 5.00%, lending facility 6.50% as of 21 Oct 2026; reserve requirement being cut from 9% to 7% for conventional banks by 1 December 2026 [V]). (interpretation) A capped, unremunerated r-Digital Rupiah would not change this; a w-Digital Rupiah would change the technology of reserves, not who creates money. See R-stream on Indonesia (Part XII) for the full credit-creation analysis.

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## Disputes and uncertainties

1. **Is retail CBDC necessary to preserve the singleness of money?** ECB: yes, as cash declines [V]. BoE academic subgroup: not strictly; wholesale convertibility is the key anchor [V]. Unresolved; depends on whether public trust needs a retail-held public money.
2. **Do holding limits protect stability or bank profits?** ECB model: limits up to EUR 3,000 contain outflows [V]. Critics (Helmich 2026 [E]; some academics) argue low limits mainly protect bank funding costs and cripple usefulness, especially zero limits for businesses. The BoE's GBP 10-20k range [V] shows how wide reasonable calibrations are.
3. **What exactly is the EUR 3,000 number?** Verified as the upper bound of a modelled range [V]; widely misreported as the planned limit. No legal limit exists.
4. **Is China's e-CNY still a CBDC?** After the January 2026 reclassification, bank-wallet e-CNY is a bank deposit liability [V]. Whether to classify the retail e-CNY as CBDC, tokenised deposit or a hybrid is a

definitional dispute; this report calls it a hybrid and keeps it in the "pilot" status because the PBOC still labels it a pilot.

5. **US legal status details.** The ROAD to Housing Act CBDC section was verified through a law-firm summary and a fact-checking site, not through the govinfo PDF; the precise section number and the "open, permissionless, and private" exemption should be checked against the statute text before publication.
6. **UK direction.** Official line: decision by end-2026 [V/E]; reported deputy-governor remark of "no urgent need" [E]. The final decision may be to pause.
7. **Indonesia's choice between CBDC and a central-bank stablecoin** is reported only in search excerpts [E]; ANTARA's opinion piece describes the Digital Rupiah as a CBDC [V]. Treat as open.
8. **QRIS H1 2026 value:** Rp 1.12 quadrillion (ANTARA, V) vs about Rp 600.7 trillion (BI release excerpt, E). The difference may reflect different definitions (e.g., including or excluding certain QRIS modes such as TUNTAS/transfer, or a typo). Use volume, users and merchants, which agree, until the BI SPIP statistics table is checked.
9. **Adoption metrics are not comparable.** e-CNY reports cumulative flows; India, Bahamas and Jamaica report stocks; Nigeria reports cumulative value and "millions of wallets". Cross-country rankings of "CBDC success" built on these are unreliable.
10. **mBridge volumes** come from Chinese academic/press excerpts [E] and are not on a BIS or member-central-bank page opened in this run.
11. **Agora real-value test figures** (CHF 800,000, 30 transactions, 80 seconds) are [E]; the 27 May 2026 report findings are [V].
12. **Privacy commitments are mostly design statements, not yet law** (except where statute exists). The EU regulation's final privacy text is not agreed [V].
13. **Counterfactual adoption**: no evidence exists on how a large advanced-economy retail CBDC would be used, because none has launched. All advanced-economy impact numbers are model outputs.

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# R7 - Comparative Case Studies (BRIEF Part XI) and Indonesia as a Dedicated Case (BRIEF Part XII)

Project III, "The Architecture of Money". Authors: M. Bagus Prabowo and Hark (he conceived and directs the project; Hark's agents research and draft). Research stream R7, compiled 7 November 2026 (Asia/Jakarta). CBDC status for Indonesia (Project Garuda, digital rupiah) is covered by stream R6 and is referenced here only where the institutional analysis needs it.

Evidence tags follow FRAMEWORK.md: [V] = the page or dataset was opened and the claim is on it; [E] = search excerpt, secondary source or established knowledge not re-opened in this session. Claim types are labelled where it matters: (fact), (empirical finding), (interpretation), (normative), (counterfactual).

Datasets produced by this stream (all under `money/data/`):

- `cases.csv` - 19 cases coded on the nine BRIEF fields plus "supports", "contradicts", key numbers, sources, tags.
- `r7/wdi_case_panel.csv` - World Bank WDI panel for 12 countries, 1960-2025 (CPI inflation, real GDP growth, central government debt/GDP, private credit/GDP, broad money/GDP), raw JSON in `r7/raw/` [V].
- `indonesia_inflation_history.csv`, `indonesia_msme_credit.csv`, `indonesia_burden_sharing.csv`, `indonesia_key_indicators_2026.csv`, `indonesia_institutional_timeline.csv`.

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## Key findings

1. **No single theory survives the full case set.** Every school is confirmed somewhere and embarrassed somewhere. Monetised deficits explain Weimar, Zimbabwe, Venezuela, Argentina and Sukarno-era Indonesia well (fiscal-dominance / quantity-theory family). But Japan (1999-2021), Switzerland (2011-2021) and the post-2008 advanced economies expanded base money massively with low or negative CPI inflation, which contradicts the mechanical quantity-theory and Austrian predictions of the period and supports endogenous-money and liquidity-trap accounts. Conversely, 2021-23 showed that large fiscal transfers financed partly by central bank balance sheets *can* lift inflation when they meet supply constraints, which embarrasses the "deficits never matter while there is slack" reading of MMT. (interpretation, built on the coded cases in `cases.csv`)

2. **What separates stable from unstable systems is institutional, not metallic.** None of the six stable systems (US, Switzerland, Singapore, Japan, Germany/Bundesbank, Sweden) relies on commodity convertibility. What they share is (a) a fiscal authority that does not need the central bank to fund recurrent deficits, (b) a central bank with legal and political room to refuse, (c) deep domestic-currency debt markets, and (d) public expectations formed by a long record. Germany's own stability was built after two currency collapses (1923, 1948), so credibility is learned and can be rebuilt. (interpretation; empirical support from

WDI CPI medians: CH 1.64%, SG 1.73%, JP 1.96%, DE 2.21%, SE 2.92%, US 3.02% over 1960-2025, computed from `r7/wdi_case_panel.csv` [V])

3. **Crises are usually fiscal or banking crises that become monetary crises.** In the crisis set, inflation follows a breakdown of public finance (Weimar reparations and Ruhr occupation; Zimbabwe quasi-fiscal deficits; Venezuela oil collapse; Argentina chronic deficits; Indonesia 1960s, where budget receipts fell from about 10% to about 4% of national income [V]) or a banking/peg collapse (Lebanon; Indonesia 1997-98). Fractional-reserve lending on its own does not appear as the primary trigger in any of these cases; connected lending, currency mismatch and pegs do. (interpretation)

4. **Distributional damage is systematic.** In every high-inflation case, holders of local-currency deposits, pensions and wages lose; holders of real assets, foreign currency and debt gain; insiders with early information gain most (Lebanon's pre-freeze transfers, Indonesia's BLBI). Low-inflation regimes have subtler distributional channels (asset-price effects of QE, negative rates and housing). (interpretation, supported by case evidence)

5. **Indonesia's money is created mostly by commercial banks, but the state shapes it heavily.** Bank credit was Rp9,245.1 trillion in September 2026 (+13.26% yoy) while third-party funds grew only 8.22% [V for growth rates, BI 21 Oct 2026; level E, OJK]. Currency in circulation was Rp1,341 trillion [V]. Credit growth in 2026 is concentrated in corporate and investment lending (corporate +22.18% vs MSME +2.38% yoy, Aug 2026 [E]); the MSME share of bank credit fell from 21.02% (2021) to 16.50% (Aug 2026) [V, BI SSKI Table 20]. The state steers credit through state-owned banks (which received Rp200 trillion of government cash in September 2025 [E]), Bank Indonesia's macroprudential liquidity incentives (Rp458.5 trillion as of October 2026 [V]) and KUR subsidised lending.

6. **Indonesia's monetary history is a cycle of fiscal dominance, crisis, and institutional repair.** Sukarno-era deficit financing produced inflation of 639% in 1966 on the Djakarta CPI (12 months to June 1966: over 1,500%) [V, IMF 1970]; the 1997-98 crisis produced 77.63% (Dec-Dec, BPS) [E] and a bank-restructuring bill of roughly Rp640 trillion [E]; each crisis was followed by a new legal settlement (1968 central bank law; Law 23/1999 independence). The 2020-22 burden-sharing (BI bought about Rp1,104.85 trillion of government securities in the primary market under SKB I-III, computed from BI annual figures [E]) and the June 2026 amendment of the P2SK Law (Law 4/2026), which adds a mandate to create conditions for real-sector growth and job creation and gives the DPR a performance-evaluation role, mark a new swing toward coordination with government. Governor Perry Warjiyo resigned on 25 July 2026; Destry Damayanti became governor on 2 September 2026 [E/V]. Whether this is accountable coordination or creeping fiscal dominance is the central open question for Indonesia. (interpretation)

7. **Inflation hurts Indonesia's poor through food.** The poverty line in March 2026 was 74.70% food, with rice the largest single item [E, BPS]; in September 2026 volatile-food inflation was 5.03% yoy against core 2.84% [V]. An illustrative calculation (counterfactual weights, see Section 12.12) suggests the effective inflation rate of a food-heavy poor household can run around 1 percentage point above headline in such months. BPS does not publish an official CPI by income group; this is a data gap.

8. **Financial inclusion is wide but shallow.** SNLIK 2025 measured financial inclusion of 80.51% and literacy of 66.46% (continuity method), but capital-market inclusion of only 1.34%, and inclusion among people without completed primary schooling of 56.95% versus 99.10% for university graduates [V, OJK-BPS]. Private credit was only 36.28% of GDP in 2025 [V, WDI], low for an economy of Indonesia's income level. Inclusion in payments (QRIS: 70.6 million users, 49.4 million merchants in Q3 2026 [V]) has outrun inclusion in credit.

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# Method

**Case selection.** The BRIEF names 19 cases in three buckets. To avoid cherry-picking, each case is coded on all nine fields with the same template, and each row in `cases.csv` must list at least one theory the case *supports* and one it *contradicts*. Stable cases were deliberately checked for evidence against credibility theories (e.g., the Bundesbank routinely missed its money targets), and crisis cases for evidence against fiscal explanations (e.g., Lebanon's crisis began in the banking system under a peg, not with monetised deficits).

**Quantitative backbone.** The WDI API panel (`r7/wdi_case_panel.csv`) gives comparable annual-average CPI inflation; coverage is uneven (Venezuela stops in 2016, Argentina's WDI series starts only in 2018 because of the 2007-2015 INDEC credibility gap, Zimbabwe starts in 2010). Where WDI is missing, national statistics (INDEC, BCV, BPS) are cited with their own definitions. Annual-average and December-on-December rates differ sharply during high inflation (Indonesia 1998: 58.45% annual average vs 77.63% Dec-Dec), so every figure names its definition.

**Limitations.** Many narrative facts for older cases are tagged [E] because they rest on standard histories (Friedman and Schwartz 1963; Eichengreen 1992, 2011; Sargent 1982; Cagan 1956; Feldman 1993) rather than re-opened primary documents. They are well established but should be footnoted to the books in the final manuscript.

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# PART XI - COMPARATIVE CASE STUDIES

Each case gives the nine BRIEF fields compactly, then an "adversarial reading" stating what it supports and contradicts.

## 11.1 Relatively stable systems

### 11.1.1 United States (Federal Reserve era)

- **Monetary structure (fact).** Fiat dollar since 1971 (convertibility for foreign official holders suspended 15 August 1971) [E]. Federal Reserve created 1913; statutory "dual mandate" of maximum employment and stable prices since the 1977 amendment; explicit 2% PCE inflation target adopted January 2012 [E]. Since 2008 the Fed runs an "ample reserves" floor system, paying interest on reserves [E; see R2].
- **Banking architecture.** Several thousand banks, deposit insurance (FDIC since 1933), and unusually large capital markets and non-bank finance; mortgage credit routed largely through government-sponsored enterprises.
- **Fiscal conditions.** Persistent federal deficits; debt financed in the world's deepest bond market, with foreign official demand for Treasuries (Triffin/"exorbitant privilege", see R5).
- **Inflation history [V, WDI].** Annual CPI inflation peaked at 13.55% in 1980, fell under Volcker's tightening to low single digits, and stayed low through the Great Moderation; 8.0% in 2022, 4.12% in 2023, 2.95% in 2024.
- **Institutional credibility.** High; long-term inflation expectations stayed near target through 2022 according to market and survey measures [E]. Political pressure on Fed independence recurs.
- **Public confidence.** Use of the dollar at home and abroad is near universal, even though surveyed trust in banks and government is low [E]. This is the BRIEF's Part VIII paradox in its purest form: dependence without warm trust.
- **Distribution.** Post-2008 QE and near-zero rates raised equity and house prices, benefiting wealthier households, while also supporting employment and mortgage refinancing for middle-income households; empirical studies disagree on the net effect (Coibion et al. 2017 find contractionary shocks raise inequality; Bernanke 2022 argues the net effect of easing was not regressive) [E].
- **Political responses.** Banking Act 1933, Dodd-Frank 2010, emergency facilities 2020.
- **Long-term consequences.** The dollar's international role outlived gold by more than five decades.

**Adversarial reading.** Supports institutional-credibility and expectations-anchoring theories (the 2023-24 disinflation without deep recession is hard to square with a purely monetarist or purely Phillips-curve account). Contradicts strict monetarism in timing: broad money surged in 2020, inflation followed in 2021-22, but velocity moved so much that a stable M-to-P link would have mispredicted both the 2009-19 decade and the size of the 2021-22 burst. Contradicts fiscal-crisis pessimism (no spiral despite debt above 100% of GDP), but the case does not prove debt is costless; it shows that a reserve-currency issuer has more room than others.

### 11.1.2 Switzerland (Swiss National Bank)

- **Monetary structure.** Floating franc, except for the EUR/CHF 1.20 minimum rate from September 2011 to 15 January 2015; negative policy rate from 2015 to 2022 (-0.75%) [E]. The SNB accumulated foreign-currency assets of several hundred billion francs, i.e. it created very large amounts of base money to buy euros and dollars [E].
- **Banking.** Universal banks of global size relative to GDP; Credit Suisse was absorbed by UBS in March 2023 with SNB liquidity assistance and a federal guarantee, under emergency law [E].
- **Fiscal.** The federal "debt brake" (since 2003) keeps public debt low [E].
- **Inflation [V].** Median CPI inflation 1.64% (1960-2025); peak 9.77% in 1974; 2.84% in 2022; 0.15% in 2025.
- **Credibility and confidence.** Very high; the franc is a safe-haven currency, which is itself a problem (appreciation pressure).
- **Distribution.** Negative rates hurt savers and pension funds and favoured property owners; the strong franc hurt exporters and tourism.
- **Political responses.** The 2015 floor removal was a shock; the 2023 bank rescue raised a too-big-to-fail debate.

**Adversarial reading.** The SNB balance sheet is perhaps the cleanest refutation of the claim that central bank money creation must produce domestic CPI inflation: base money rose many-fold while inflation averaged near zero for a decade. It supports demand-for-money and safe-haven explanations (when the world wants your money, you can create a lot of it). It contradicts any claim that credibility immunises a banking system from failure: Credit Suisse collapsed under a credible central bank.

### 11.1.3 Singapore (currency board, then exchange-rate-centred policy)

- **Monetary structure.** From 1967 the Board of Commissioners of Currency, Singapore issued currency fully backed by foreign assets, a currency-board arrangement [E]. The Monetary Authority of Singapore (MAS, 1971) conducts monetary policy since 1981 by managing the Singapore-dollar nominal effective exchange rate (S$NEER) within an undisclosed policy band - "basket, band and crawl" - rather than setting a short-term interest rate; interest rates are largely imported [E]. Currency issuance was merged into MAS in 2002 [E].
- **Banking.** Strongly capitalised domestic banks, open capital account, large official reserves managed partly by GIC and Temasek.
- **Fiscal.** Structural surpluses; Singapore Government Securities are issued to develop markets and invest CPF balances, not to finance deficits [E].
- **Inflation [V].** Peak 22.37% (1974); median 1.73% (1961-2025); 6.13% (2022), 4.83% (2023), 0.9% (2025).
- **Credibility and confidence.** Very high.
- **Distribution.** Compulsory saving (CPF) and public housing are the main redistributive channels; a stronger S$NEER shifts real income from exporters to consumers of imports.
- **Political responses.** In 2022 MAS tightened repeatedly by re-centring and steepening the band.

**Adversarial reading.** Supports the view that "the right monetary instrument depends on economic structure": for a small, very open economy the exchange rate dominates the price level. It contradicts universalist prescriptions that every country should target an overnight interest rate, and it contradicts the MMT-adjacent idea that a sovereign currency issuer should treat fiscal deficits as the normal state; Singapore runs surpluses by choice and has very high monetary credibility. But it is not evidence for gold or commodity anchors either: the currency board held foreign financial assets, not metal.

### 11.1.4 Japan

- **Monetary structure.** After the asset bubble burst (1990-91), the Bank of Japan pioneered zero rates (1999), quantitative easing (2001-06), quantitative and qualitative easing (April 2013), a negative rate (-0.1%, January 2016) and yield-curve control (September 2016); it ended negative rates in March 2024 and raised rates further in 2024-25 [E].
- **Banking.** Main-bank system; non-performing loans in the 1990s were recognised slowly; consolidation into megabanks.
- **Fiscal.** Gross public debt above 200% of GDP, mostly held domestically, with the BoJ holding a very large share of government bonds [E; exact share to verify against BoJ flow of funds].

- **Inflation [V].** 23.22% in 1974; near zero or negative for most of 1995-2012; -0.23% in 2021; 2.5% (2022), 3.27% (2023), 2.74% (2024), 3.17% (2025).
- **Confidence.** High; Japan remains cash-intensive.
- **Distribution.** Two decades of near-zero deposit rates transferred income from savers (often older households) to borrowers, including the government; post-2013 equity gains accrued to asset holders.

**Adversarial reading.** The strongest single case against mechanical quantity-theory and Austrian predictions: base money multiplied without CPI inflation for two decades. It supports liquidity-trap Keynesian, balance-sheet-recession (Koo) and expectations-trap New Keynesian accounts. However, it also contradicts naive MMT optimism in one respect: when inflation arrived in 2022-25 from imported energy and a weak yen, the BoJ's huge bond holdings complicated exit and raised the fiscal cost of higher rates.

### 11.1.5 Germany: the Bundesbank model (1948-1998)

- **Monetary structure.** The currency reform of 20 June 1948 replaced the Reichsmark with the Deutsche Mark at steep conversion ratios for financial claims [E]; the Bank deutscher Laender (1948) became the Deutsche Bundesbank under the Bundesbank Act of 1957, with a statutory duty to safeguard the currency and independence from federal instructions [E]. From 1975 it announced monetary-growth targets [E].
- **Banking.** Three-pillar system of private universal banks, public savings banks (Sparkassen) and cooperative banks.
- **Fiscal.** Moderate deficits; the reunification shock of 1990 raised spending and rates.
- **Inflation [V].** Peak 7.03% (1973); 1960-2025 median 2.21%.
- **Credibility.** Among the highest of any central bank; often attributed to collective memory of 1923 and 1948 (interpretation).
- **Distribution.** The 1990 monetary union converted East German wages and small savings at 1:1 and larger savings at 2:1, a deliberate redistribution toward East Germans [E].
- **Political responses / consequences.** High German rates after 1990 transmitted through the EMS and contributed to the 1992-93 ERM crisis; the Bundesbank template shaped the ECB statute.

**Adversarial reading.** Supports credibility and central-bank-independence theories. But it contradicts monetarist mechanism claims: the Bundesbank missed its own monetary targets roughly half of the time (a commonly cited finding of von Hagen and Bernanke-Mihov; [E]), yet inflation stayed low; the targets functioned as communication and discipline rather than as a stable causal lever. It also shows stability was *built after* collapse, which contradicts fatalist claims that a currency, once destroyed, cannot recover trust.

### 11.1.6 Sweden: the Riksbank, the world's oldest surviving central bank

- **Monetary structure.** Johan Palmstruch's Stockholms Banco issued Europe's first banknotes (1661), over-issued and failed; the Estates founded the Riksens Standers Bank in 1668, ancestor of the Riksbank [E]. Modern regime: floating krona since November 1992; inflation target of 2% announced 1993 and applied from 1995 [E]; negative repo rate 2015-2019 [E]; e-krona investigated but not issued (status in R6).
- **Banking.** Concentrated large banks. The 1990-93 crisis was resolved with a blanket guarantee, nationalisation of weak banks and asset-management companies (Securum, Retriva), often cited as a model [E].
- **Fiscal.** After the 1990s crisis, a fiscal framework with a surplus target and expenditure ceilings.
- **Inflation [V].** Peak 13.7% (1980); 8.37% (2022), 8.55% (2023), 0.68% (2025).
- **Confidence.** High; one of the world's lowest cash-use societies.
- **Distribution.** Negative rates coincided with rapid house-price and household-debt growth, favouring owners over renters and younger entrants.

**Adversarial reading.** The 1661-1668 episode is an early natural experiment showing that note issue without credible backing fails quickly, which supports credit-theory and institutional accounts as much as commodity accounts. Modern Sweden supports inflation-targeting and crisis-resolution institutions. It contradicts the assumption that disappearing cash automatically requires a retail CBDC (Sweden has not issued one), and it shows that a credible target does not protect against asset-price and household-debt cycles.

## 11.2 Monetary and financial crises

### 11.2.1 Weimar Germany (1914-1924)

- **Structure.** Gold convertibility suspended at the outbreak of war in 1914; the war was financed by borrowing and Reichsbank discounting of Treasury bills rather than taxes [E]. After 1919, reparations obligations, a fragile coalition fiscal base and, in January 1923, the Franco-Belgian occupation of the Ruhr (with government-financed passive resistance) produced deficits covered by the press [E].
- **Inflation.** Cagan (1956) dates the hyperinflation August 1922-November 1923, with a peak monthly rate of roughly 29,500% in October 1923 [E].
- **Stabilisation.** The Rentenmark (November 1923; 1 Rentenmark = 1 trillion paper marks), a balanced budget, an end to discounting of government bills, and the 1924 Dawes Plan with a new, more independent Reichsbank [E]. Sargent (1982) uses this as a canonical "regime change" in which inflation stopped abruptly once the fiscal regime changed, with little output cost [E].
- **Distribution.** Holders of savings, mortgages, bonds and fixed nominal pensions were expropriated; debtors (including industrialists and the state, for its domestic debt) gained; the middle class experienced the episode as a moral as well as financial betrayal (Feldman 1993) [E].
- **Consequences.** Enduring German inflation aversion; contested contribution to political radicalisation (the deflation of 1930-32 under Bruening is at least as relevant to 1933 - interpretation).

**Adversarial reading.** Supports fiscal-theory and monetarist explanations of hyperinflation (deficits monetised) and Sargent's emphasis on regime credibility. Contradicts simple "supply-shock" explanations (the Ruhr occupation was a shock, but it was financed by printing). It cautions against a common Austrian/populist narrative that blames "central banking" in general: the Reichsbank of 1923 was the instrument of a government that could not or would not tax.

### 11.2.2 Zimbabwe (2000-2026)

- **Structure and fiscal.** After land seizures from 2000 cut agricultural output and exports, the Reserve Bank of Zimbabwe financed government and quasi-fiscal programmes (subsidised loans, exchange-rate subsidies) by money creation [E].
- **Inflation.** The official annual rate reached 231 million percent in July 2008 [E]; Hanke and Kwok (2009) estimated about 79.6 billion percent month-on-month in mid-November 2008 and 89.7 sextillion percent year-on-year [E]. The economy dollarised in 2009 (multicurrency regime). Re-introduction attempts followed: bond notes (2016), RTGS dollar (2019), Zimbabwe dollar, and Zimbabwe Gold (ZiG) in April 2024 [E]. WDI records 255% (2019), 557% (2020), 98.6% (2021), 104.7% (2022) [V].
- **Confidence and distribution.** Deposit holders in local currency lost repeatedly, including when USD balances were converted at official rates; households and firms with physical dollars or real assets were protected.
- **Consequences.** The population repeatedly chose foreign money when allowed.

**Adversarial reading.** Supports fiscal-dominance theories strongly. It is a hard case for pure chartalism: tax obligations in local currency did not prevent mass abandonment of the Zimbabwe dollar once confidence broke; the state's power to *require* its money in tax payments sustained some demand but not value. Chartalists respond that the state lost real productive capacity and fiscal control, so the case does not refute the claim that taxes create demand for money but shows that taxes alone cannot create *value* (interpretation).

### 11.2.3 Venezuela (2013-2026)

- **Structure and fiscal.** Oil-revenue collapse from 2014, price and exchange controls, and central bank financing of the state and PDVSA [E]. The bolivar was redenominated three times: 2008 (3 zeros), 2018 (5 zeros), 2021 (6 zeros) [E].
- **Inflation.** The BCV, after a publication gap, reported 274% (2016), 863% (2017), 130,060% (2018), 9,586% (2019) [E]; the opposition-controlled National Assembly estimated much higher 2018 figures [E]. After a relative lull (WDI series ends 2016), BCV reported 475% for 2025 (first release in over a year, 6 March 2026, via AFP) [E], and monthly inflation of 6.2% in October 2026, the lowest of the year [E, Bloomberg Linea, 4 November 2026]. AFP reported the ouster of Nicolas Maduro in this period; political details are outside

R7's scope and must be verified separately.
- **Confidence and distribution.** De facto dollarisation; mass emigration; wage and pension earners in bolivars devastated.

**Adversarial reading.** Supports fiscal-dominance and political-economy explanations (statistical opacity is itself part of the credibility collapse). Contradicts the belief that price controls can suppress monetary-financed inflation. It also contradicts any claim that hyperinflation ends a regime quickly: high inflation persisted for over a decade.

### 11.2.4 Argentina, including the Milei-era disinflation (2023-2026)

- **History.** The 1991 Convertibility Law fixed the peso 1:1 to the dollar with near currency-board backing; it ended in December 2001 with a deposit freeze (corralito), default and "pesification" of dollar deposits at unfavourable rates in 2002 [E]. Inflation then returned, with official statistics discredited during 2007-2015 [E].
- **Milei period [V for 2026 data].** After the December 2023 devaluation and a sharp fiscal adjustment, INDEC's December-on-December CPI fell from 211.4% (2023) to 117.8% (2024) and 31.5% (2025) [E, Bloomberg Linea table of INDEC data]. INDEC's September 2026 release shows monthly inflation of 1.9%, year-on-year 33.4%, and 23.6% cumulative in 2026, with core at 1.7% m/m and regulated prices 2.3% m/m [V, INDEC IPC September 2026]. Year-on-year inflation has thus *stalled* around 31-34% since late 2025 (31.3% in October 2025 to 33.4% in September 2026 [V, same release]). Commentators describe "stagflation" in 2026, with industry and construction contracting [E]. Urban poverty was reported at 32.3% in H1 2026 [E].
- **Distribution.** The 2024 adjustment initially pushed poverty above 50% (2024 H1, reported [E]); subsequent disinflation helped real wages. Regulated prices (utilities, transport) rose fastest, a deliberate removal of subsidies whose burden fell on households (49.1% yoy for housing and utilities in September 2026 [V]).

**Adversarial reading.** Supports the fiscal theory of the price level and regime-change accounts (the step from 200%+ to about 30% followed a fiscal anchor). Contradicts the claim that eliminating the deficit alone delivers rapid convergence to low inflation; inertia, indexation, exchange-rate policy and credibility gaps keep monthly inflation near 2%. It also contradicts simple "printing" narratives: in 2026 the drivers are regulated-price catch-up and food, not monetary financing of the Treasury. Disinflation under Milei is real but incomplete (fact, as of the September 2026 release).

### 11.2.5 Lebanon (1997-2026)

- **Structure.** The lira was pegged at 1,507.5 per US dollar from 1997 [E]. Banque du Liban sustained the peg and the state's financing through "financial engineering", attracting dollar deposits at high rates from commercial banks, which had in turn attracted them from depositors (interpretation of IMF/World Bank analyses) [E].
- **Collapse.** From October 2019 banks restricted withdrawals; the state defaulted on Eurobonds in March 2020 [E]. The IMF's 2023 Article IV recorded that GDP had fallen by about 40%, the parallel exchange rate had lost 98% of its value, and the central bank had lost two-thirds of its reserves [E]. Annual inflation was 221.3% in 2023, 45.2% in 2024 [V WDI for 2023: 221.34; 2024: 45.24] and 14.6% in 2025 [V]. A unified official rate of 89,500 per dollar was adopted in early 2024 [E]. Financial-sector losses are estimated at about USD 72 billion [E, World Bank/State Department]. Customer deposits fell from USD 172 billion (2009) to USD 88 billion (2024) [E, MEI]. A bank resolution law was enacted in July 2025 and amended in August 2026; the "financial gap" law allocating losses was still before Parliament in late 2026 [E].
- **Distribution.** Losses fell on depositors through withdrawal limits and conversions of "lollars" at below-market rates; politically connected depositors who moved money abroad early avoided losses (widely reported, details contested) [E].

**Adversarial reading.** Lebanon refutes the intuition that a bank deposit is the same as central bank money. It supports bank-run, peg-collapse and political-economy accounts, and is the clearest case in the set where *commercial bank money* failed while physical dollars kept value. It contradicts monocausal fiscal-printing narratives at the start (the peg held while the system borrowed dollars), though money printing in lira after 2019 then drove inflation.

### 11.2.6 Indonesia 1997-98 (summary; full treatment in Part XII, Section 12.6)

The crisis combined a fixed-to-floating exchange-rate collapse, unhedged dollar corporate debt, a banking system enlarged after 1988 deregulation with connected lending, and political crisis. Inflation reached 77.63% Dec-Dec in 1998 [E BPS] (58.45% annual average [V WDI]); GDP contracted 13.13% [V WDI]; Bank Indonesia extended Rp144.54 trillion of liquidity support (BLBI) to 48 banks [E]. It supports twin-crisis and "original sin" (foreign-currency debt) theories and contradicts the belief that a sound budget protects a country from a currency crisis: Indonesia had near-balanced budgets before 1997.

## 11.3 Major institutional transformations

### 11.3.1 The United States after the Great Depression

- **Structure.** Under the gold standard, the US money stock fell by roughly one-third between 1929 and 1933 (Friedman and Schwartz 1963) amid waves of bank failures [E]. Executive Order 6102 (April 1933) required private gold holdings to be delivered to the Federal Reserve; the Gold Reserve Act (January 1934) revalued gold from $20.67 to $35 per ounce, a devaluation of the dollar by about 41% in gold terms [E].
- **Banking.** The Banking Act of 1933 created federal deposit insurance and separated commercial from investment banking; the Banking Act of 1935 centralised open-market policy in the FOMC [E].
- **Inflation.** Consumer prices fell about one-quarter in 1929-33 and recovered after 1933 [E].
- **Distribution.** Deflation raised the real burden of debt; farm and household foreclosures were widespread; unemployment peaked near a quarter of the labour force [E].

**Adversarial reading.** Supports the monetarist account (Fed failure to prevent monetary contraction), Eichengreen's "golden fetters" account (gold standard transmitted deflation; countries that left gold earlier recovered earlier) and Bernanke's credit-channel account (bank failures destroyed information capital). It contradicts the Austrian "liquidationist" view that letting the system clear would have been beneficial, and it contradicts the claim that gold backing guarantees stability. Deposit insurance shows that confidence in bank money can be *engineered* by the state.

### 11.3.2 Bretton Woods (1944-1971)

- **Structure.** Agreed in July 1944 by 44 countries: fixed but adjustable par values against the dollar; the dollar convertible into gold for official holders at $35/oz; IMF and IBRD (World Bank) established; capital controls tolerated [E].
- **Banking.** Nationally regulated, low-competition banking ("financial repression"), later eroded by the Eurodollar market [E].
- **Fiscal / inflation.** War debts were reduced by growth, inflation and repressed interest rates; US inflation rose from the late 1960s.
- **Credibility.** Undermined by the Triffin dilemma: the system needed growing dollar liquidity, which required US deficits that eroded confidence in convertibility (Triffin 1960) [E]. The London Gold Pool (1961-68) failed to hold the market price.

**Adversarial reading.** Supports the open-economy trilemma (capital controls bought domestic policy autonomy) and Triffin's logic. Contradicts nostalgic claims that a gold anchor provides automatic stability; the anchor was managed politically and abandoned when it bound.

### 11.3.3 The Nixon Shock and the move to floating (1971-1973)

On 15 August 1971 the US suspended official gold convertibility, with a temporary surcharge on imports and wage-price controls; the Smithsonian Agreement (December 1971) tried to re-peg, and generalised floating followed in March 1973 [E]. US inflation then rose, peaking at 13.55% in 1980 [V], until the Volcker disinflation.

**Adversarial reading.** Supports network-effect and institutional theories of reserve-currency status: the dollar's dominance survived the end of gold, contrary to widespread predictions at the time. It also shows the costs of losing a nominal anchor without building a new one (the 1970s), which supports credibility theories and partly vindicates commodity-standard critics on *that* decade.

### 11.3.4 The creation of the euro (1992-2026)

Maastricht Treaty (1992); ECB established June 1998; the euro introduced on 1 January 1999 for 11 states, banknotes in 2002 [E]. Monetary policy is single; fiscal policy remains national under the Stability and Growth Pact. The sovereign debt crisis (2010-12) exposed the bank-sovereign "doom loop" and the absence of a lender of last resort for governments; Draghi's "whatever it takes" (July 2012) and the OMT programme ended acute redenomination risk [E]. Banking union began with the Single Supervisory Mechanism (2014); the Transmission Protection Instrument was added in 2022 [E]. Membership has continued to expand (Bulgaria's accession on 1 January 2026 is reported [E - verify in R5]).

**Adversarial reading.** Supports optimum-currency-area critiques (Mundell) and chartalist critiques (money separated from a fiscal sovereign is fragile). Contradicts confident predictions of break-up. Shows that credibility can be supplied by institutional commitment even without a single treasury, but at the price of periodic crisis politics.

### 11.3.5 China's financial system

The People's Bank of China became a dedicated central bank in 1983 and received its law in 1995 [E]. The system is bank-dominated, with large state-owned banks, regulated deposit rates (liberalised formally in 2015), capital controls, a managed renminbi (dollar peg to 2005; reform in 2005 and August 2015) [E], and since 2021 a property-developer crisis and local-government debt problems. CPI inflation peaked at 24.26% in 1994 and has hovered near zero in 2023-2025 (0.23%, 0.22%, 0.06%) [V]. e-CNY status is covered by R6.

**Adversarial reading.** Supports endogenous credit-money and state-capacity theories: China combined very rapid credit growth with low inflation and few classic bank runs because the state owns the banks and controls the capital account. Contradicts market-liberal predictions that repression must lead quickly to crisis, but also contradicts simple claims that credit growth is harmless: the property and local-government debt overhang, and deflation risk, are the deferred cost. Distributionally, suppressed deposit rates transferred income from household savers to borrowers (SOEs, local governments, developers) - a Chinese analogue of the Cantillon channel operating through regulation rather than money issue (interpretation).

### 11.3.6 Post-2008 monetary policy

Advanced-economy central banks cut rates to the zero lower bound, launched large-scale asset purchases, moved to floor systems and (in some cases) negative rates; quantitative tightening began in 2022 [E; data in R2/R4]. For a decade CPI inflation undershot targets despite a many-fold increase in reserves; then 2021-23 saw the largest inflation surge in four decades, driven by a combination of fiscal transfers, reopening demand, supply bottlenecks and energy shocks [E].

**Adversarial reading.** Supports the endogenous-money view that reserves do not mechanically become loans (McLeay, Radia and Thomas 2014) and contradicts the textbook money multiplier and the many 2009-13 forecasts of high inflation. Partially supports Cantillon-type asset-channel concerns: QE operated by raising asset prices, so first beneficiaries were asset holders, but the employment channel complicates any simple regressive verdict. 2021-23 shows that the "inflation is dead" consensus of the 2010s was also wrong.

### 11.3.7 The rise of digital payments: M-Pesa, Pix, UPI, QRIS

These are *payment rails* for existing money, not new money (fact). M-Pesa (Safaricom, Kenya, 2007) is mobile e-money backed by trust-account balances in banks [E]. India's UPI (2016, NPCI) processed 24.07 billion transactions worth Rs 29.37 lakh crore in September 2026 [E, NPCI table via search; Business Standard 1 Oct 2026]. Brazil's Pix (central-bank-owned, November 2020) handled almost 80 billion transactions worth over R$35 trillion in 2025 according to the BCB management report [E]. Indonesia's QRIS (standard launched 2019) had 70.6 million users and 49.4 million merchants with transactions growing 75.80% yoy in Q3 2026 [V, BI 21 Oct 2026].

**Adversarial reading.** Supports a "public infrastructure" theory: when a central bank owns or mandates an interoperable rail, inclusion and competition rise quickly without issuing a retail CBDC. It contradicts claims that only crypto-assets or CBDCs can deliver fast cheap payments. It also raises new concentration and surveillance questions (payment data held by platforms and the state), which link to Part X.

## 11.4 Cross-case synthesis

**Table 11.1 - What each family of theories gets right and wrong across the 19 cases (interpretation; coded in `cases.csv`).**

| Theory family | Best-supporting cases | Most-contradicting cases |
|---|---|---|
| Quantity theory / monetarism | Weimar, Zimbabwe, Venezuela, Indonesia 1960s, US 1929-33 (money contraction) | Japan 1999-2021, Switzerland 2011-21, post-2008 AEs, Bundesbank target misses |
| Fiscal theory / fiscal dominance | Weimar, Argentina 2023-25 disinflation, Indonesia 1966-69 stabilisation | US and Japan high debt without spiral (so far) |
| Endogenous money / post-Keynesian | China, post-2008, Japan | Hyperinflations, where exogenous money finance dominates |
| New Keynesian expectations | US 2023-24 disinflation, Sweden, Singapore | Argentina 2026 stall; Japan's failure to lift expectations for years |
| MMT / chartalism | Japan and US funding capacity in own currency | Zimbabwe, Venezuela, Lebanon (exit from state money), 2021-23 inflation |
| Austrian | Credit-boom-then-bust patterns (Japan 1990, China property, Indonesia 1997) | Absence of predicted inflation after QE; US 1930s liquidationist outcome |
| Commodity / gold | Long-run stability claims for 1870-1914 (outside this set) | US 1929-33, Bretton Woods collapse, Nixon shock survival of dollar |
| Credit / institutional | Bundesbank, Indonesia 1999 reform, Lebanon (deposits as claims) | Credit Suisse failure under credible SNB |

**Three regularities.** (1) Hyperinflations end through fiscal-institutional regime change plus a new or re-anchored monetary unit, often with external support (Weimar/Dawes; Indonesia 1966-69 with IGGI aid; Argentina 2025 with the IMF). (2) Banking crises are resolved through state guarantees and recapitalisation that socialise losses (US 1933, Sweden 1992, Indonesia 1998, Switzerland 2023); the distribution of those losses, not the technique, is the main political conflict. (3) Populations remain inside the national monetary system after crises unless a foreign currency is practically available and legally tolerated (Zimbabwe, Venezuela, Lebanon, Argentina saving in dollars); where it is not (Indonesia 1998, Germany 1923 for most wage earners), dissatisfaction did not produce mass abandonment, which supports the BRIEF Part VIII distinction between legitimacy and institutional dependence.

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# PART XII - INDONESIA AS A DEDICATED CASE STUDY

## 12.1 Colonial monetary arrangements (1746-1942)

- (fact [E]) The Dutch East India Company (VOC) operated a lending bank in Batavia, De Bank van Leening (1746), as part of a company-state monetary system in which coins of many origins circulated and the company acted as both trader and quasi-sovereign.
- (fact [E, BI history page excerpt]) De Javasche Bank (DJB) was founded on 24 January 1828 under a royal charter (octrooi) as a circulation bank with note-issuing privileges for Java; it also discounted bills and did ordinary banking business. BI describes it as "the first circulation bank in Asia". Its legal basis from 1922 was the Javasche Bankwet 1922, amended in 1927 and 1930 [E, Murdadi 2013].
- (interpretation) The colonial architecture separated *money issue* (a private, Dutch-owned, profit-making shareholder company with a state charter) from *the population's monetary experience* (largely cash and informal credit, pawnshops, and moneylenders). This is a historical example of the BRIEF's "private money integrated into a national system": the colony's currency was a liability of a chartered private bank whose credibility rested on the Dutch state and the guilder's metallic standard.
- During the Japanese occupation (1942-45) occupation money issued through the Nanpo Kaihatsu Ginko replaced guilder notes [E, BI history].

## 12.2 Independence, ORI and the establishment of Bank Indonesia (1945-1953)

- **Competing currencies (1945-49).** After the Proclamation, the Netherlands-Indies Civil Administration (NICA) re-established DJB to circulate NICA money, while the Republic created the Jajasan Poesat Bank Indonesia (October 1945) and then Bank Negara Indonesia (1946), originally intended to be its central bank [E, BI history].
- **ORI.** The Oeang Republik Indonesia (ORI) entered circulation on 30 October 1946 (now commemorated as Hari Oeang). The Ministry of Finance's history page states that a law of 25 October 1946 fixed 10 ORI = 5 grams of pure gold and an exchange rate of 1 ORI to 50 Japanese-occupation rupiah in Java and Madura [E - excerpt of https://www.kemenkeu.go.id/sejarah-oeang; full page returned HTTP 500 when fetched]. Regional emergency currencies (ORIDA) also circulated.
- **Interpretation.** ORI was a sovereignty instrument first and a monetary instrument second: in a war of independence, accepting the Republic's money was a political act. This supports the state theory of money (chartalism) in its *legitimacy* dimension: people accepted ORI because they accepted the Republic, not because of the nominal gold clause, which was never convertible in practice (interpretation).
- **1949-1953.** The Round Table Conference (1949) left DJB, still Dutch-owned, as circulation bank of the federal RIS. In March 1950 the "Gunting Sjafruddin" (cutting higher-value notes in half, one half becoming a forced loan to the state) reduced money in circulation [E]. Law 24/1951 (enacted 6 December 1951, promulgated 15 December 1951) nationalised DJB; the government acquired about 97% of shares by purchase [E, BI history excerpt; law text at peraturan.bpk.go.id]. Law 11/1953 (in force 1 July 1953) created Bank Indonesia as the central bank, superseding the 1922 DJB Act; its first governor was Sjafruddin Prawiranegara, the last DJB president [E, BI].

## 12.3 Sukarno: deficit financing, monetary purges and the 1965-66 collapse

- **Erosion of central bank autonomy.** Under Guided Democracy (from 1959) the budget expanded with national projects, military spending and confrontation with Malaysia; BI financed the gap. In 1965 the "Bank Tunggal" (single bank) policy merged state banks into BI, making it an instrument of government [E].
- **Purges and redenomination.** In August 1959 the Rp500 and Rp1,000 notes were devalued to one-tenth of face value and some deposits frozen [E]. Presidential Decree (Penpres) 27/1965 of 13 December 1965 introduced a new rupiah at Rp1 new = Rp1,000 old (in West Irian at par with the IB rupiah) and authorised BI to issue all denominations [E - BI Museum page excerpt].
- **Quantities [V, IMF Finance and Development, Tomasson 1970].** Between 1960 and 1966 the ratio of budget receipts to national income fell from about 10% to about 4%; in 1966 the current deficit exceeded 90% of current receipts; the money supply increased by 760% in 1966; the Djakarta price index rose by more than 1,500% in the twelve months to June 1966 and 639% in calendar 1966; real liquidity fell by about 60% in 1960-66.
- **Which inflation figure?** The figure "635%" commonly attributed to BI/BPS, the IMF's 639% (Djakarta CPI, Dec-Dec), the Ministry of Finance's 650% (Media Keuangan, 2024 [E]) and WDI's 1,136% (national CPI, annual average [V]) describe the same episode with different indices and averaging. The BRIEF's "600%+" is therefore correct as a lower bound; the defensible statement is "inflation of roughly 600-650% on December-on-December measures in 1966, and over 1,000% on an annual-average basis".
- **Not only money.** The same IMF article stresses that inflation "was due not simply to excessive monetary expansion but also to a serious breakdown of public confidence in the currency and the emergence of critical supply bottlenecks". It reports that in late 1967 and early 1968 a rice shortage coincided with speculation: rice prices rose about 100% in one month and the general price level 82% from October 1967 to February 1968; after supplies improved, a 57% increase in money supply coincided with only 11% price growth in six months [V]. This is direct evidence against a monocausal monetary account even in a monetary-financed inflation (empirical finding, IMF 1970).
- **Distribution.** Civil servants and urban wage earners on fixed nominal pay, and holders of rupiah savings, lost most; those with access to rice, foreign exchange, import licences or state allocations gained (interpretation [E]). The economic collapse coincided with the political violence of 1965-66, in which hundreds of thousands were killed; the causal relation between economic crisis and violence is beyond this stream and should be handled with care in the manuscript (fact/caution).

## 12.4 Suharto: technocratic stabilisation and the New Order financial system (1966-1997)

- **Stabilisation 1966-69 [V, IMF 1970].** Launched in October 1966 by a group of technocrats with political backing (commonly identified with Widjojo Nitisastro, Ali Wardhana, Emil Salim, Mohammad Sadli and colleagues, the "Berkeley mafia" [E]). The programme: (i) an austere current budget and early elimination of bank financing of the deficit; (ii) public utility and fuel price increases; (iii) a *selective credit* programme directing new bank credit to production bottlenecks, while total money still grew 130% (1967) and 120% (1968); (iv) foreign aid via the Inter-Governmental Group on Indonesia and debt rescheduling, with aid counterpart funds financing development spending; (v) liberalised exchange markets culminating in unification at Rp378/USD on 17 April 1970; (vi) a deposit scheme from October 1968 paying 6% *per month* on 12-month deposits, which raised time deposits 175% in January-April 1969. Inflation fell to 113% (1967), 85% (1968) and 10% (1969).
- **Lesson (interpretation).** The IMF author's conclusion that the stabilisation "contradicts the popular belief that stabilization needs to entail economic stagnation" anticipates Sargent's regime-change thesis. But the programme was also politically unique: it relied on an authoritarian government able to raise fuel prices and suppress opposition, and on large foreign aid; it is not a template that transfers easily to democracies (interpretation).
- **The "balanced budget" rule.** From 1967 the New Order formally balanced the budget by counting foreign aid and loans as revenue, thereby prohibiting domestic monetary financing while accumulating external debt [E]. This is an early fiscal rule that later carried over to the 3%-of-GDP deficit and 60%-of-GDP debt

ceilings in Law 17/2003 on State Finances [E].

- **Central bank law 1968.** Law 13/1968 made BI the central bank but subordinate to government policy through a Monetary Board chaired by the Minister of Finance; BI's mandate included supporting development [E]. During the oil boom BI extended large volumes of subsidised "liquidity credits" (KLBI) to priority sectors through state banks (directed credit) [E].
- **Devaluations.** Major devaluations in November 1978, March 1983 and September 1986 [E].
- **Deregulation.** The June 1983 package removed credit ceilings and freed most interest rates at state banks; BI introduced its certificates (SBI) for open-market operations in 1984 [E]. PAKTO (Paket Oktober, 27 October 1988) opened bank licensing and branching to new private and joint-venture banks and sharply cut the reserve requirement (commonly cited as from 15% to 2%) [E]. The number of banks rose to 238-240 by 1997 [E, IMF Country Report 04/189]. Inflation in 1989-1996 ran around 5-10% (WDI: 6.42%-9.67%) [V].
- **Interpretation.** PAKTO is a natural experiment for the BRIEF's question about bank money creation: when entry and reserve constraints were relaxed, bank credit and deposits grew rapidly (broad money/GDP from 30.07% in 1988 to 56.0% in 1997 [V, WDI]) - endogenous money in action - but the growth was concentrated in banks owned by business groups that lent to affiliates. The constraint that mattered was not reserves but supervision of risk and connected lending.

## 12.5 Rupiah history in brief

1946 ORI; 1950 note-cutting; 1959 purge; 1965 redenomination (1,000:1); 1966-70 unification at Rp378/USD; managed peg/crawl with devaluations 1978, 1983, 1986; intervention band widened and abandoned in August 1997; free float since; post-crisis peaks above Rp16,000 (1998), stress episodes in 2008, 2013, 2015, 2018, 2020; in 2026 the rupiah traded above Rp18,000 per dollar (above Rp18,000 around the June 2026 law vote, reported as a record low [E, Reuters via Katadata]; Rp17,985 on 20 October 2026 [V, BI]; JISDOR Rp18,161 on 2 November 2026 [E, BI homepage]). Redenomination proposals (removing three zeros) have been discussed repeatedly in the 2010s-2020s without enactment [E].

## 12.6 The 1997-1998 crisis: anatomy, bank restructuring and costs

**Sequence (fact, mostly [E]).**

1. Thailand floats the baht (2 July 1997); contagion to the rupiah; BI widens and then abandons its band (14 August 1997).
2. First IMF Letter of Intent, 31 October 1997. On 1 November 1997, 16 small private banks were closed (about 2.5-3% of system assets) *without full deposit protection* [E, IMF WP/01/52; IMF CR 04/189]. Depositors ran from private banks to state and foreign banks.
3. BI provided emergency liquidity (Bantuan Likuiditas Bank Indonesia, BLBI) to banks facing runs and failing clearing balances. A blanket guarantee and the Indonesian Bank Restructuring Agency (IBRA/BPPN, Keppres 27/1998) followed in early 1998 (sources date the guarantee to late January or February 1998 - a dispute to resolve with the original decree).
4. Riots and political crisis; Suharto resigned on 21 May 1998.
5. On 13 March 1999, after audits, 38 banks were closed, 7 taken over by IBRA, and 9 declared eligible for joint recapitalisation with the government [E, BI Governor Sabirin, BIS Review 34/1999]. The number of banks fell from 238 (1997) to 138 (end-2003) [E, IMF CR 04/189].

**Quantities.**

| Item | Value | Source / tag |
|---|---|---|
| BLBI outstanding Dec 1998 | Rp147.7 trillion | UMY JESP 2002 citing BI [E] |
| BLBI agreed between MoF and BI (6 Feb 1999) | Rp144.54 trillion to 48 banks | same [E] |
| BPK investigative audit (2000): BLBI potentially lost/irregular | Rp138.4 trillion (95.78%) | Bisnis 2021 citing BPK [E] |
| Government recapitalisation bonds | about Rp422.6-430 trillion | CNBC Indonesia 2026; Pasardana 2021 [E] |

| Item | Value | Source / tag |
|---|---|---|
| Total restructuring outlay | about Rp640-643 trillion | DJKN 2021; Asianbanks 2000 [E] |
| Net fiscal cost of the blanket guarantee (estimate) | about 40% of annual GDP | Fane and McLeod 2001 (ANU) [E] |
| GDP growth 1998 | -13.13% | WDI [V] |
| CPI inflation 1998 | 77.63% Dec-Dec (BPS); 58.45% annual average | BPS [E]; WDI [V] |
| Poverty 1998 | 24.2% (49.5 million) | DJKN 2024 [E] |
| Satgas BLBI recovery by H1 2024 | Rp38.2 trillion of Rp110.45 trillion claims (34.59%) | Metro TV 2024 [E] |

**Balance-sheet anatomy of BLBI (illustrative T-accounts, Rp trillion, round numbers).** The point is to show who created which money and who bore the loss.

Step 1 - a run moves deposits from Bank X to a state bank; Bank X's reserves at BI go negative in clearing; BI lends to Bank X (BLBI):

| Bank Indonesia - Assets | | Bank Indonesia - Liabilities | |
|---|---|---|---|
| Claim on Bank X (BLBI) | +100 | Reserves of State Bank S | +100 |

| Bank X - Assets | | Bank X - Liabilities | |
|---|---|---|---|
| Reserves at BI | 0 (+100 borrowed, -100 paid out) | Deposits | -100 |
| | | BLBI owed to BI | +100 |

BI *created* base money (reserves) to replace the deposits that fled; broad money is roughly unchanged (deposits moved from X to S) but base money and BI's credit exposure rose.

Step 2 - Bank X is insolvent; the government assumes the BLBI claim and issues bonds to BI (and recapitalisation bonds to surviving banks):

| Government - Assets | | Government - Liabilities | |
|---|---|---|---|
| Claim on Bank X / IBRA assets (impaired) | +100 (recovery value perhaps 30) | Bonds to BI | +100 |

Loss allocation: if recoveries are Rp30 trillion, Rp70 trillion becomes a permanent public debt financed by future taxes. The money that was created to save deposits stays in the system; the losses on the bad loans behind it are socialised. This is the precise sense in which the 1998 crisis redistributed wealth: depositors (including large ones) were protected, shareholders who had lent to their own affiliates often escaped full liability under settlement agreements, and taxpayers bore the residual (interpretation, consistent with the BPK audit and the long Satgas BLBI recovery effort). Reports in September 2026 that the government had finished paying BLBI-related debts after about 30 years [E, CNBC Indonesia 23 Sep 2026] should be verified against Ministry of Finance statements before use.

**Adversarial reading.** Supports: original-sin and twin-crisis theories; Minsky (stability bred risky lending after 1988); lender-of-last-resort moral-hazard critiques. Contradicts: the claim that IMF-style closure of weak banks restores confidence automatically (the November 1997 closures without deposit protection triggered runs - a design error widely acknowledged since, including by IMF staff); and the claim that fiscal prudence alone prevents crises.

## 12.7 Bank Indonesia's independence: Law 23/1999 and its amendments

- **Law 23/1999 (17 May 1999)** made BI a state institution "independent ... free from interference by the Government and/or other parties, except for matters expressly regulated" (BI profile, [V]), with a single objective: to achieve and maintain the stability of the rupiah's value (prices and exchange rate). It prohibited BI from lending to the government and limited government securities purchases to the secondary market [E]. Inflation targeting was formally adopted in July 2005 [E].
- **Amendments.** Law 3/2004 (Supervisory Board, emergency financing facility), Law 6/2009 (from Perppu 2/2008 during the global crisis, widening collateral for short-term liquidity), Law 21/2011 (OJK takes over bank supervision from end-2013), Law 9/2016 (Financial System Stability Committee, KSSK), Law 2/2020 (from Perppu 1/2020: temporary authority to buy government securities in the primary market during COVID-19, until 2022) [E].
- **P2SK Law 4/2023 (12 January 2023).** An omnibus law of 27 chapters and 341 articles amending 17 financial-sector laws [E, Kemenkeu Media Keuangan]. For BI it (i) broadened the objective to "achieve rupiah stability, maintain payment system stability, and help maintain financial system stability in order to support sustainable economic growth"; (ii) confirmed BI as the sole issuer of digital rupiah; (iii) gave BI standing crisis powers to buy long-term government securities in the primary market, to buy or reverse-repo securities held by the deposit insurer (LPS) for bank resolution, and to provide repo financing to corporates via banks [E, BI crisis protocol page]; and (iv) according to BI "maintained BI's independence" [E, BI LKBI 2023].
- **Law 4/2026 amending P2SK (enacted 17 June 2026; passed by the DPR on 4 June 2026).** BI's profile page now states that Law 23/1999 was "last amended by Law No. 4 of 2026" [V]. The amended Article 7(2) states that BI, in pursuing its objective, "implements policies and a policy mix that can create an economic environment conducive to real-sector growth and job creation" [E, quoted in the BPK copy of the law and IDN Times]. The law also gives the DPR authority to evaluate the performance of BI, OJK and LPS, with results to be followed up, approves BI's budget, clarifies dismissal and replacement procedures for board members, strengthens legal protection for officials acting in good faith, and allows the sovereign wealth fund Danantara to issue special bonds [E, Katadata 5 June 2026; Mondaq 2026]. Analysts warned that targeting real-sector growth risks higher inflation; Reuters framed concerns about central bank independence amid a record-low rupiah and an equity market down more than 30% year-to-date [E, Reuters as reported by Katadata]. The Jakarta Post headline read "BI loses independence, gets job creation mandate" [E]. Government and DPR maintain that independence is preserved and that the DPR cannot direct interest-rate or intervention decisions [E, Kompas 24 June 2026].
- **Leadership change.** Governor Perry Warjiyo resigned "voluntarily for personal reasons" on 25 July 2026; Senior Deputy Governor Destry Damayanti was named acting governor and, as the President's sole nominee, approved by the DPR (27 August / 1 September) and sworn in on 2 September 2026 for 2026-2031 [E, BI press release excerpt; Antara]. Reuters noted the rupiah strengthened on news of her nomination [E].

**Assessment (interpretation).** Using the four standard dimensions of central bank independence:

| Dimension | 1999 law | After P2SK 2023 | After Law 4/2026 |
|---|---|---|---|
| Goal independence | Narrow single goal | Broadened (stability triad + sustainable growth) | Further broadened toward real-sector growth and jobs |
| Instrument independence | Full | Full in normal times; crisis primary-market purchases allowed | Formally retained; DPR barred from directing decisions (as reported) |
| Personal independence | Fixed terms, limited dismissal | Similar | Dismissal mechanism revised; legal protection strengthened |
| Financial independence | Budget autonomy with oversight | Similar | DPR budget approval and performance evaluation |

The movement is toward a Fed-like multiple mandate combined with stronger legislative accountability. Whether it amounts to fiscal dominance depends on behaviour, not text: the October 2026 decision to hold the BI-Rate at

5.75% and raise the deposit facility rate to defend the rupiah [V] shows that in practice the stabilisation function still constrains growth-oriented easing. A fair conclusion is that *de jure* independence has narrowed and *de facto* independence is being tested (interpretation; dispute flagged).

## 12.8 BI-Ministry of Finance relations, burden sharing and the 2025 cash placement

**Burden sharing 2020-2022.** Under Perppu 1/2020 (Law 2/2020), three joint decrees (SKB) between the Minister of Finance and the BI Governor governed BI's primary-market purchases of government securities (SBN):

| Year | SKB I (standby buyer at market rates) | SKB II / III (private placement) | Total |
|---|---|---|---|
| 2020 | Rp75.86 tn | SKB II Rp397.56 tn (public goods; BI bore the interest) | Rp473.42 tn |
| 2021 | Rp143.32 tn | SKB III Rp215.00 tn (at 3-month reverse repo rate; BI rebated coupons on Rp58 tn) | Rp358.32 tn |
| 2022 | Rp49.11 tn | SKB III Rp224.00 tn (coupon rebate on Rp40 tn) | Rp273.11 tn |
| **2020-22** | | | **Rp1,104.85 tn (computed)** |

Sources: BI annual financial reports 2020 and 2022, BI LPI 2021, Katadata 1 Feb 2023 [E - official figures through excerpts]; computation in `indonesia_burden_sharing.csv`. BI also bought SBN in the secondary market (e.g., Rp166.20 tn in 2020 and Rp8.62 tn in 2021) for market stabilisation [E]. Governor Warjiyo cited Rp974.09 tn "to date" in November 2022 and a projected Rp1,144 tn for three years [E] - figures that differ from the sum above because of timing and perimeter.

**What burden sharing was, in balance-sheet terms (illustrative, Rp tn).** SKB II-type direct placement:

| BI - Assets | | BI - Liabilities | |
|---|---|---|---|
| Government bonds (SBN) | +100 | Government deposit account at BI | +100 |

When the government spends, its BI deposit falls and bank reserves and private deposits rise:

| BI - Assets | | BI - Liabilities | |
|---|---|---|---|
| SBN | +100 | Government account | 0 (+100, -100) |
| | | Bank reserves | +100 |

| Commercial banks - Assets | | Commercial banks - Liabilities | |
|---|---|---|---|
| Reserves at BI | +100 | Deposits of households/firms receiving spending | +100 |

This is direct monetary financing in the technical sense of the BRIEF (Part IV.D): the central bank, not the market, supplied the funding, and base money and broad money both rose. Whether it was inflationary depended on slack: CPI inflation was 1.92% in 2020 and 1.56% in 2021 [V, WDI annual average], rising to 4.21% in 2022 mainly through fuel-price adjustments and global commodity prices (interpretation). The IMF publicly urged an end to the arrangement in early 2022 [E], and it ended as scheduled in December 2022 (fact [E]). The episode supports the view that monetary financing in a deep recession with a time limit and legal sunset need not be inflationary, while showing how politically attractive the tool is - one reason markets read the 2026 amendment nervously (interpretation).

**The Rp200 trillion cash placement (September 2025).** On 12 September 2025, Minister of Finance Purbaya Yudhi Sadewa, appointed days earlier, moved Rp200 trillion of government cash balances (SAL) from BI to five state-owned banks under KMK 276/2025: BRI Rp55 tn, BNI Rp55 tn, Mandiri Rp55 tn, BTN Rp25 tn, BSI Rp10 tn; banks were told not to use the funds to buy SBN or BI's SRBI securities [E, Kemenkeu press release excerpt; CNBC Indonesia]. Reuters reported that the government's cash at BI had reached Rp430 trillion and that the minister argued BI had kept liquidity too tight [E].

Balance-sheet mechanics (Rp tn):

| BI - Liabilities | |
|---|---|
| Government account at BI | -200 |
| Reserves of state banks | +200 |

| State banks - Assets | | State banks - Liabilities | |
|---|---|---|---|
| Reserves at BI | +200 | Government deposits (third-party funds) | +200 |

The operation created no new government debt and no new BI asset; it converted government money held at the central bank (outside the banking system) into bank reserves and bank deposits. Base money held by banks and broad money (government deposits at commercial banks count as third-party funds, though some definitions of M2 exclude central government deposits) rose by up to Rp200 trillion overnight [interpretation of mechanics; definitions vary]. It does *not* force banks to lend: lending requires creditworthy borrowers. OJK reported banks had disbursed roughly 20-70% of the placement within six weeks [E], and the liquidity ratio (AL/DPK) rose from 22.53% to 24.20% [E]. The case illustrates two BRIEF themes at once: (i) that "reserves are not loanable funds" (credit growth was 7.7% in September 2025 [E] but accelerated to 13.26% by September 2026 [V], with other drivers including government programmes and BI's KLM incentives); and (ii) that the Treasury, not only the central bank, can change the quantity of base money in the banking system through its cash-management choices - a point usually missing from textbook accounts.

## 12.9 Banking and credit creation in Indonesia

**Who creates the money.** Currency in circulation was Rp1,341 trillion in Q3 2026 (+11.77% yoy) [V, BI]. Bank credit was Rp9,245.1 trillion in September 2026 [E, OJK via CNN Indonesia]. Since new loans create matching deposits (McLeay, Radia and Thomas 2014), the bulk of the broad money stock is a by-product of bank lending and of banks' purchases of government and BI securities, net of repayments (interpretation of standard mechanics; R2 covers the theory).

**Illustrative T-account of an Indonesian bank loan (Rp million).** A KUR micro-loan of Rp50 million by BRI to a warung owner:

| BRI - Assets | | BRI - Liabilities | |
|---|---|---|---|
| Loan to borrower | +50 | Borrower's Simpedes deposit | +50 |

The deposit is new money. When the borrower pays a supplier at another bank via QRIS or BI-FAST, BRI loses reserves to the other bank:

| BRI - Assets | | BRI - Liabilities | |
|---|---|---|---|
| Loan | +50 | Deposit | 0 (+50, -50) |
| Reserves at BI | -50 | | |

The limits are therefore (a) BRI's need to hold reserves (statutory reserve requirement (GWM) of 9% of third-party funds for conventional banks, being cut to 7% by 1 December 2026, with part of it remunerated [V]) and liquidity buffers (macroprudential liquidity buffer PLM 4% [V]); (b) capital (industry CAR 24.10% in August 2026 [V]), which absorbs the expected loss on the KUR loan, partly covered by a government guarantee and interest subsidy; (c) profitability given the policy rate and funding costs; (d) borrower demand and creditworthiness. Indonesia therefore illustrates both frameworks: reserve requirements still exist and are actively used as a policy instrument (unlike the US since 2020), but they are a cost and liquidity constraint rather than a mechanical multiplier, since BI supplies reserves at its policy rate and adjusts GWM to support credit (interpretation).

**How BI influences credit (channels, with 2026 examples [V unless noted]).**

1. *Price of reserves*: BI-Rate 5.75%, deposit facility 5.00%, lending facility 6.50% (21 October 2026), and the SRBI/SVBI/SUVBI pro-market operations that set money-market rates.
2. *Quantity/cost of reserves*: GWM cut from 9% to 7% (1 November and 1 December 2026) explicitly to offset slow deposit growth (8.22% yoy) relative to credit (13.26% yoy).
3. *Macroprudential liquidity incentives (KLM)*: reductions in required reserves for banks that lend to priority sectors; Rp458.5 trillion of incentives received as of early October 2026, with the financing channel capped at 3% of third-party funds from December 2026 and "further expanded support for lending to business sectors that support job creation".
4. *Macroprudential ratios*: Macroprudential Intermediation Ratio (RIM) 84-94%; Inclusive Macroprudential Financing Ratio (RPIM) for MSME/inclusive lending; LTV up to 100% for property and 0% minimum down payment for vehicles extended to end-2027.
5. *Exchange-rate management*: spot, DNDF and offshore NDF intervention and DNDF premium incentives to attract inflows (USD 9.9 billion attracted since June 2026, per BI).
6. *Payment system*: QRIS, BI-FAST and the Indonesian Credit Card (KKI) shape the cost of transacting with bank money.

**What determines the distribution of credit (evidence).**

- *By borrower size.* MSME credit share of bank credit: 19.67% (2020), 21.02% (2021), 20.80% (2022), 20.32% (2023), 19.00% (2024), 17.31% (2025), 16.50% (August 2026) [V, BI SSKI Table 20]. MSME credit grew only 2.29% yoy in August 2026 against total credit growth of about 13.6% [V/E]; corporate credit grew 22.18% [E]. MSME NPLs are 4.53% versus 2.11% gross NPL for the system [V], and OJK reports only 3.51% of MSMEs keep financial statements [E, detik 11 Aug 2026] - a structural information problem.
- *By account count versus value.* MSMEs hold 19.38% of credit accounts but 16.50% of value; micro loans are 90.63% of MSME accounts with an average balance of Rp33.36 million [V]. Large numbers of small loans coexist with concentration of value.
- *By purpose.* Investment credit +23.20%, working capital +11.89%, consumer +4.86% yoy (September 2026) [V]. BI notes that lending to corporations "unrelated to Government programs" also increased [V], which implies that government-linked programmes are a large share of the 2026 credit surge (interpretation; programme-level data not verified here).
- *By owner.* State-owned bank credit grew 16.95% yoy in July 2026, fastest of any group [E, OJK]; state banks also received the Rp200 trillion placement.
- *By sector (MSME credit, August 2026, Rp bn [V]).* Wholesale and retail 648,618 (shrinking 3.84% yoy); agriculture 291,417 (+7.06%); manufacturing 145,971; accommodation and food 77,904.
- *Depth.* Private credit/GDP 36.28% and broad money/GDP 42.54% in 2025 [V, WDI] - financial deepening has stalled near the levels of the mid-1990s, far below Malaysia, Thailand or China (comparison [E]).

**Interpretation.** Credit allocation in Indonesia is determined by (i) collateral and information (favouring corporates and salaried consumers), (ii) state ownership and programmes (state banks, KUR, government

placements), (iii) BI's macroprudential incentives, and (iv) price. The 2023-2026 decline in the MSME share happened despite RPIM and KLM, suggesting that incentive-based steering is weaker than structural factors such as the loss of COVID restructuring relief and high MSME NPLs (interpretation, contested). Cantillon-style "first receiver" advantages are visible in a specific, measurable form: large corporates and state-linked projects obtain new bank money first and on better terms, while micro firms rely on KUR, pawnshops and online lenders (fintech lending inclusion only 4.40% [V], but with high rates).

## 12.10 Government debt and monetary policy

- Central government debt was Rp10,326.93 trillion on 30 September 2026 (SBN Rp8,962.40 tn; loans Rp1,364.53 tn), 40.30% of GDP, down from 41.26% in June 2026 [E, DJPPR via excerpts and multiple press reports]. The legal ceilings are 60% of GDP for debt and 3% of GDP for the annual deficit (Law 17/2003) [E]. The 2026 budget planned a deficit of 2.68% of GDP; mid-year outlook 2.85% [E]; the 2027 draft budget 2.40% [E].
- WDI's "central government debt" series for Indonesia shows 72.49% of GDP in 1997 and 55.2% in 1998 [V] (definition: includes post-crisis recap bonds; coverage gaps), illustrating how the 1997-98 bank rescue converted private losses into public debt.
- **Interaction with monetary policy.** BI's balance sheet holds a large SBN portfolio after 2020-22 purchases; since September 2023 BI issues SRBI (rupiah securities backed by its SBN holdings) as its main pro-market absorption instrument [E]. Higher BI rates raise government funding costs (10-year SBN yield level in 2026: unverified in this stream); foreign ownership of SBN ties the rupiah to global risk appetite. The September 2025 SAL placement and the 2026 law both point toward closer fiscal-monetary coordination (interpretation).
- External debt was 30.9% of GDP in August 2026 [E, BI SULNI October 2026].

## 12.11 Financial inclusion

SNLIK 2025 (OJK-BPS; fieldwork 22 January-11 February 2025; 10,800 respondents aged 15-79 in 34 provinces) [V]:

| Indicator | Continuity method | DNKI coverage method |
|---|---|---|
| Financial literacy | 66.46% | 66.64% |
| Financial inclusion | 80.51% | 92.74% |
| Sharia literacy / inclusion | 43.42% / 13.41% | same |
| Urban / rural inclusion | 83.61% / 75.70% | 94.48% / 90.03% |
| Inclusion: no/incomplete primary vs university | 56.95% vs 99.10% | 83.64% vs 99.77% |
| Inclusion: unemployed vs employees/professionals | 64.82% vs 95.11% | 84.04% vs 98.15% |
| Sector inclusion: banking / insurance / pawnshops / capital market / fintech lending | 70.65% / 28.50% / 8.23% / 1.34% / 4.40% | - |

The 2024 survey gave 65.43% literacy and 75.02% inclusion [V]. Measurement caveat: the DNKI method counts BPJS social insurance membership, which raises inclusion mechanically; "inclusion" means *usage* of any product, not access to credit on fair terms.

**Interpretation.** Indonesia's inclusion is payments-led and bank-account-led; credit and wealth inclusion lag. The gap by education (literacy 43.20% for those without completed primary schooling versus 90.63% for graduates [V]) suggests that inclusion policy based on product availability will keep reproducing inequality unless coupled with education and simpler, safer products.

## 12.12 Inflation by income group

**Data status.** BPS publishes a single national CPI (weights from the 2022 Cost of Living Survey, SBH [E]) and does not publish official CPIs by income decile; this is a gap for the BRIEF's question.

**Evidence.**

- The March 2026 poverty line was Rp669,235 per capita per month, of which food was 74.70% (urban 73.91%, rural 76.06% per BPS [E]); rice contributed about 20.52% (urban) and 24.60% (rural) of the poverty line, and kretek cigarettes the second-largest share [E, BPS via Investortrust/Kompas]. The poverty line rose 9.86% between March 2025 and March 2026 [E] - far faster than headline CPI (around 2-3%), because it tracks the prices of the poor's basket.
- In September 2026 headline CPI was 3.28% yoy, core 2.84%, administered prices 3.25%, volatile food 5.03% [V].
- In 1998, food-led inflation pushed poverty to 24.2% [E].

**Illustrative calculation (counterfactual weights; not an official statistic).** Take September 2026 component inflation [V]. Suppose a poor household's basket is 60% volatile food, 25% core, 15% administered prices, and a top-decile basket is 10% volatile food, 70% core, 20% administered:

- Poor household effective inflation = 0.60x5.03 + 0.25x2.84 + 0.15x3.25 = 3.02 + 0.71 + 0.49 = **4.22%**
- Top-decile effective inflation = 0.10x5.03 + 0.70x2.84 + 0.20x3.25 = 0.50 + 1.99 + 0.65 = **3.14%**

A gap of about one percentage point in a single year, compounding when food shocks recur (and reversing in years when food prices fall). The volatile-food share of the official CPI is much smaller than 60%, so the figures are illustrative only.

**Other channels.** Poor households hold more cash and fewer inflation-hedged assets; they rarely borrow at fixed nominal rates (so gain little from debtor redistribution); their incomes adjust with lags (informal wages; provincial minimum wages set annually). Government responses - rice stabilisation (SPHP), fuel subsidies, cash transfers, the regional inflation-control teams (TPIP/TPID) - matter as much as monetary policy for this group (interpretation).

## 12.13 Digital payments and QRIS

QRIS (Quick Response Code Indonesian Standard) was launched on 17 August 2019 and became mandatory for QR payments from 1 January 2020 [E]. In Q3 2026 QRIS had 70.6 million users and 49.4 million merchants, transactions grew 75.80% yoy, total digital payments reached 18.09 billion transactions (+39.24%), and BI-FAST processed 1,631 million transactions worth Rp4,049 trillion [V]. BI is extending cross-border QRIS links (sandboxes with Timor-Leste and Hong Kong from Q4 2026) and local-currency settlement with seven partners [V]. QRIS and the National Payment Gateway were raised as trade barriers by US trade authorities in 2025 [E], showing that payment rails are now geopolitical infrastructure.

**Interpretation.** QRIS is a standard and switching layer over existing bank and e-money balances; it does not create new money, but it makes bank money and e-money more usable for micro-merchants, which supports inclusion (Section 12.11) and supplies data that can inform credit scoring. CBDC questions (Project Garuda) are treated in R6.

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## 12.14 Answers to the six Indonesia questions

### Q1. Who creates money in Indonesia?

(fact/interpretation) Three creators, in order of magnitude:

1. **Commercial banks** create most broad money when they lend or buy securities, crediting deposits (stock of bank credit Rp9,245 trillion, September 2026). This creation is limited by capital, liquidity rules (GWM, PLM), BI's interest rates, profitability and borrower demand - not by a fixed multiple of reserves.

2. **Bank Indonesia** creates base money (currency Rp1,341 trillion; bank reserves) through monetary operations, FX purchases, lending facilities, and - in 2020-22 - direct purchases of government bonds (about Rp1.1 quadrillion). BI also destroys base money when it sells FX to defend the rupiah or issues SRBI.
3. **The government** affects money creation indirectly: by spending its deposits at BI (which adds reserves), by moving its cash into commercial banks (the Rp200 trillion placement), by guaranteeing loans (KUR) and by owning the largest banks. Non-bank lenders (finance companies, fintech lenders, pawnshops) extend credit but mostly re-lend bank money; e-money issuers hold customer funds backed by bank placements.

## Q2. How does Bank Indonesia influence bank credit?

Through the price of reserves (BI-Rate and corridor), the quantity and cost of required reserves (GWM changes, remuneration), targeted macroprudential incentives (KLM, RPIM, RIM), macroprudential ratios (LTV, down payments), exchange-rate stabilisation (which shapes risk premia and bank funding), and payment-system design. In 2026 BI is simultaneously *tight* on price (holding 5.75% and raising the deposit facility to defend the rupiah) and *loose* on quantity and direction (cutting GWM, expanding KLM toward job-creating sectors). This mix attempts to separate exchange-rate defence from domestic credit support - a stance that works only while capital flows and inflation cooperate (interpretation).

## Q3. What determines the distribution of credit?

Collateral, information and scale (favouring corporates and salaried borrowers), state ownership and programmes (state banks, KUR, government placements and programme lending), BI incentives, and relative returns (banks can earn 6-7% on SBN with zero capital charge). The evidence: MSME share down from 21.02% (2021) to 16.50% (Aug 2026); corporate credit +22% vs MSME +2% in 2026; MSME NPL 4.5%. Regulation shapes the margin but has not reversed the structural tilt (interpretation).

## Q4. How does inflation affect different income groups?

Disproportionately through food. Poor households' baskets are dominated by food (the poverty line is about 75% food, with rice about a fifth to a quarter), so volatile-food inflation - 5.03% in September 2026 against 2.84% core - raises their effective inflation above headline (illustratively by about 1 point). In high-inflation episodes (1966, 1998) losses were catastrophic for wage earners and cash holders, while those with real assets, foreign currency or privileged credit gained. Indonesia lacks an official CPI by income group; constructing one from Susenas and SBH microdata would be a valuable contribution for the manuscript (recommendation).

## Q5. How does Indonesia balance monetary stability and economic development?

Historically by swinging between two poles: development-first regimes that subordinated the central bank (1959-66; partly 1968-1983 directed credit) and ended in inflation or banking crisis, and stability-first regimes after crises (1966-69 stabilisation; 1999 independence). Since 2005 inflation targeting delivered single-digit inflation (2.18% in 2024, 1.91% in 2025 annual average [V]) with growth around 5%. The 2023 and 2026 laws, burden sharing, the Rp200 trillion placement and growth-oriented macroprudential policy show the pendulum moving back toward development and coordination. The test in late 2026 is that the rupiah is weak and equities fell sharply in H1 2026; BI has so far prioritised the exchange rate on its main policy rate. A balanced judgement: coordination is defensible in a democracy that wants growth and jobs, but the historical record says the safeguards that matter are legal limits on monetary financing, transparent sunset clauses, and a central bank willing to say no (normative, flagged as such).

## Q6. What would a retail or wholesale CBDC mean for Indonesia's institutional structure?

(Status of Project Garuda and the digital rupiah: see R6.) Institutionally, P2SK already makes BI the sole issuer of digital rupiah, so a CBDC would extend BI's direct liability from cash and bank reserves to digital form. A *wholesale* CBDC would mainly modernise interbank and securities settlement, reinforcing the two-tier structure. A *retail* CBDC would interact with three Indonesian specifics: (i) state banks already dominate deposits and receive government funds, so deposit substitution toward BI could weaken private banks more than state banks;

(ii) QRIS and BI-FAST already deliver fast cheap payments, weakening the payments rationale; (iii) after Law 4/2026, a retail CBDC would sit in a central bank with a broader growth mandate and stronger parliamentary oversight, raising both programmability and privacy questions (Part X.D). A CBDC would not change who creates most money unless holding limits were absent and deposits migrated at scale (interpretation).

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## Register of key numbers (value, unit, year, definition, source, limitation)

| # | Value | Unit | Period | Definition | Source | Tag | One-line limitation |
|---|---|---|---|---|---|---|---|
| 1 | 639 | % | 1966 | Djakarta CPI, Dec-Dec | IMF F&D 1970, Tomasson, https://www.elibrary.imf.org/view/journals/022/0007/0A008-en.xml | V | Capital-city index, not nationwide |
| 2 | >1,500 | % | 12 months to Jun 1966 | Djakarta price index | same | V | Peak-window measure, not calendar year |
| 3 | 760 | % | 1966 | Money supply growth | same | V | Money definition of the era, data quality weak |
| 4 | 1,136.25 | % | 1966 | National CPI, annual average | WDI FP.CPI.TOTL.ZG via api.worldbank.org | V | Annual average inflates relative to Dec-Dec in accelerating inflation |
| 5 | 378 | Rp/USD | 17 Apr 1970 | Unified exchange rate | IMF F&D 1970 | V | Official rate |
| 6 | 30.07 -> 56.0 | % of GDP | 1988 -> 1997 | Broad money/GDP | WDI FM.LBL.BMNY.GD.ZS | V | Includes revaluation of FX deposits |
| 7 | 77.63 | % | 1998 | CPI Dec-Dec, 44 cities, 1996=100 | BPS via secondary tables | E | Method changed Apr 1998 |
| 8 | 58.45 | % | 1998 | CPI annual average | WDI | V | Different averaging from #7 |

| # | Value | Unit | Period | Definition | Source | Tag | One-line limitation |
|---|---|---|---|---|---|---|---|
| 9 | -13.13 | % | 1998 | Real GDP growth | WDI NY.GDP.MKTP.KD.ZG | V | Later rebasings may revise |
| 10 | 144.54 | Rp tn | 6 Feb 1999 | BLBI agreed MoF-BI | UMY JESP 2002 citing BI | E | Other sources give 147.7 (Dec 1998) |
| 11 | 138.4 (95.78%) | Rp tn | 2000 | BLBI flagged by BPK audit | Bisnis 2021 citing BPK | E | "Potential loss/irregularity", not final loss |
| 12 | ~640 | Rp tn | 1998-2004 | Total bank restructuring outlay | DJKN 2021; Asianbanks 2000 | E | Mixes nominal values across years |
| 13 | ~40 | % of GDP | 1997-2001 | Net fiscal cost of guarantee (estimate) | Fane and McLeod 2001 | E | Model-based estimate |
| 14 | 238 -> 138 | banks | 1997 -> 2003 | Number of commercial banks | IMF CR 04/189 | E | Counts mergers and closures together |
| 15 | 1,104.85 | Rp tn | 2020-2022 | BI primary SBN purchases SKB I-III | computed from BI reports (`indonesia_burden_sharing.csv`) | E | Perimeter differs from BI's own 971.99/1,144 figures |
| 16 | 200 | Rp tn | 12 Sep 2025 | Gov cash moved BI -> 5 state banks | Kemenkeu KMK 276/2025 press release | E | Excerpt; later top-ups not tracked |
| 17 | 5.75 / 5.00 / 6.50 | % | 21 Oct 2026 | BI-Rate / DF / LF | BI press release 28/233/DKom | V | Point in time |
| 18 | 9 -> 7 | % of TPF | Nov-Dec 2026 | Conventional bank GWM | same | V | Average-based requirement |
| 19 | 13.26 / 8.22 | % yoy | Sep 2026 | Bank credit / deposit growth | same | V | Nominal; includes FX loans at rupiah value |
| 20 | 9,245.1 | Rp tn | Sep 2026 | Bank credit outstanding | OJK via CNN Indonesia | E | Press report of OJK figure |
| 21 | 1,341 | Rp tn | Q3 2026 | Currency in circulation | BI press release | V | Quarter-end |

| # | Value | Unit | Period | Definition | Source | Tag | One-line limitation |
|---|---|---|---|---|---|---|---|
| 22 | 16.50 | % | Aug 2026 | MSME share of commercial-bank credit | BI SSKI Table 20 | V | Reclassification breaks 2021, 2023 |
| 23 | 4.53 | % | Aug 2026 | MSME NPL ratio | same | V | Excludes restructured performing loans |
| 24 | 22.18 vs 2.38 | % yoy | Aug 2026 | Corporate vs MSME credit growth | OJK via CNBC Indonesia | E | Press report |
| 25 | 458.5 | Rp tn | early Oct 2026 | KLM incentives received by banks | BI press release | V | Incentive, not new lending |
| 26 | 10,326.93 | Rp tn | 30 Sep 2026 | Central government debt | DJPPR via excerpts | E | Central government only |
| 27 | 40.30 | % of GDP | 30 Sep 2026 | Gov debt/GDP | same | E | GDP denominator annualised |
| 28 | 3.28 / 2.84 / 5.03 | % yoy | Sep 2026 | CPI / core / volatile food | BI press release | V | National CPI weights (SBH 2022) |
| 29 | 74.70 | % | Mar 2026 | Food share of poverty line | BPS via press | E | Poverty-line basket, not CPI basket |
| 30 | 80.51 / 66.46 | % | SNLIK 2025 | Inclusion / literacy (continuity) | OJK-BPS release 2 May 2025 | V | Usage-based; sample 10,800 |
| 31 | 70.6 m / 49.4 m | users / merchants | Q3 2026 | QRIS | BI press release | V | Registered, not necessarily active |
| 32 | 36.28 | % of GDP | 2025 | Domestic credit to private sector | WDI FS.AST.PRVT.GD.ZS | V | Excludes non-bank and offshore credit |
| 33 | 33.4 / 1.9 | % y/y / % m/m | Sep 2026 | Argentina CPI | INDEC IPC Sep 2026 | V | Urban national index |
| 34 | 211.4 / 117.8 / 31.5 | % Dec-Dec | 2023/24/25 | Argentina CPI | INDEC via Bloomberg Linea | E | Secondary reproduction |

| # | Value | Unit | Period | Definition | Source | Tag | One-line limitation |
|---|---|---|---|---|---|---|---|
| 35 | 130,060 | % | 2018 | Venezuela CPI | BCV (published 2019) | E | Official figure contested by NA estimates |
| 36 | 475 | % | 2025 | Venezuela CPI | BCV via AFP/Le Monde 7 Mar 2026 | E | First release in over a year |
| 37 | 221.3 / 45.2 / 14.6 | % | 2023/24/25 | Lebanon CPI annual avg | WDI (221.34, 45.24, 14.6) | V | Dollarised economy distorts index |
| 38 | ~98 | % | 2019-2023 | Lebanon lira depreciation | IMF Art IV 2023 | E | Parallel market rate |
| 39 | 557.2 | % | 2020 | Zimbabwe CPI | WDI | V | Series only 2010-2022 |
| 40 | 13.55 | % | 1980 | US CPI | WDI | V | CPI-U |
| 41 | 24.07 | bn tx | Sep 2026 | UPI volume | NPCI statistics page via search; Business Standard | E | Excerpt of official table |
| 42 | ~79.8 | bn tx | 2025 | Pix transactions | BCB Relatorio de Gestao do Pix 2026 (excerpt) | E | Excerpt |

## Disputes and uncertainties

1. **1966 inflation figure.** 635% (commonly cited BI/BPS), 639% (IMF Djakarta CPI), 650% (Kemenkeu Media Keuangan) and 1,136% (WDI annual average) coexist. All are defensible for their definitions; the manuscript should state the index and averaging every time.
2. **1998 inflation.** 77.63% Dec-Dec (BPS, after an April 1998 method change) versus 58.45% annual average (WDI). Some popular sources quote "82%" - unverified here.
3. **DJB nationalisation share.** BI's history says the government bought about 97% of DJB shares; another source cited 99.4% for 8.9 million guilders. Unreconciled; use "about 97% (BI)".
4. **Blanket guarantee date (1998).** Sources say late January or February 1998; check Keppres 26/1998 text.
5. **BLBI totals.** Rp147.7 tn (Dec 1998), Rp144.54 tn (Feb 1999 agreement, 48 banks; one BPK clipping says 54 banks), Rp138.4 tn flagged by BPK. Final net loss to the state is not established in this stream; Satgas BLBI recoveries are partial. A September 2026 report that BLBI-related government debt to BI is fully paid is unverified.
6. **Burden-sharing total.** Rp1,104.85 tn (computed from annual BI figures) versus Rp974.09 tn (Warjiyo, Nov 2022) and Rp1,144 tn (projection). Different perimeters (inclusion of SKB I greenshoe, timing).
7. **Interpretation of Law 4/2026.** Whether BI "lost independence" (Jakarta Post headline) or retained instrument independence with a broader mandate (government/DPR) is contested. The text of the law (BPK database) should be read in full for the final manuscript; R7 relied on excerpts and press summaries

for Article 7(2) and DPR evaluation provisions.

8. **Reasons for the 2026 leadership change.** Perry Warjiyo's resignation is officially "for personal reasons". R7 makes no inference about other causes.
9. **Composition of 2026 credit growth.** BI says credit to corporations "unrelated to Government programs" also rose, implying a large programme-linked component. Programme-level data were not obtained.
10. **Inflation by income group.** No official series; the calculation in 12.12 is illustrative with assumed weights.
11. **Older-case narrative facts** (Weimar peak monthly rate, Zimbabwe 2008 estimates, US 1929-33 money contraction, Bretton Woods details) are [E]; standard references should be footnoted.
12. **Venezuela political events in 2026** (reported ouster of Nicolas Maduro; interim leadership) are outside R7's scope and are flagged for verification by whichever stream handles current events.
13. **Bulgaria's euro accession** (1 January 2026) is reported [E]; R5 should confirm.
14. **Japan BoJ JGB share and 2025-26 policy rate path** were not verified in this stream.
15. **Theory scoring** in Table 11.1 is an interpretive summary; reasonable scholars from each school would contest some placements (e.g., MMT proponents argue Zimbabwe/Venezuela are supply-collapse cases, not counter-examples; Austrians argue QE inflation appeared in asset prices rather than CPI).

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- [E] Kementerian Keuangan, Sejarah Oeang - https://www.kemenkeu.go.id/sejarah-oeang (fetch returned HTTP 500)
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